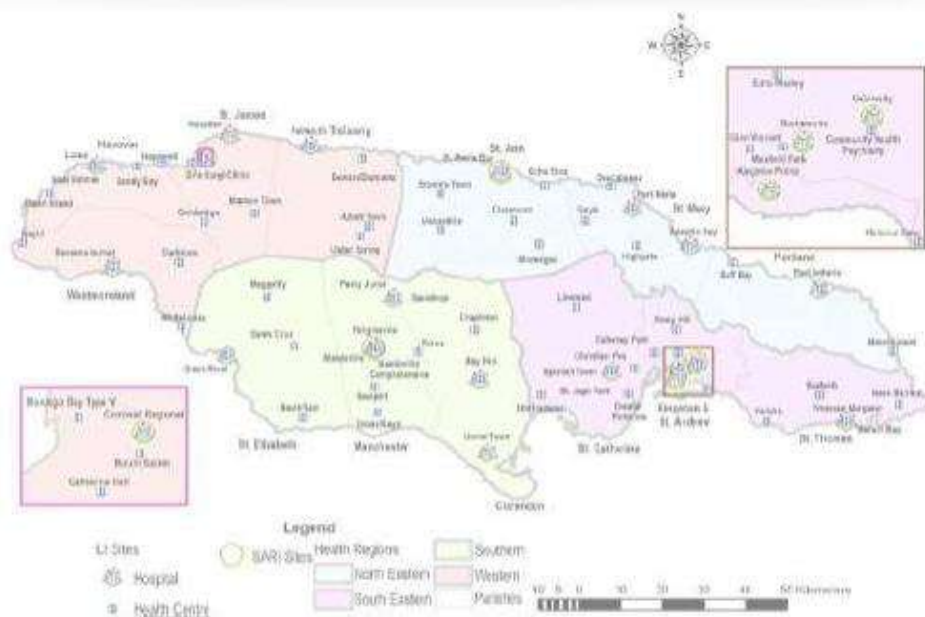




MINISTRY OF
**HEALTH &
WELLNESS**

National Surveillance Manual for JAMAICA



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National Surveillance Manual for Jamaica

Ministry of Health and Wellness

Jamaica

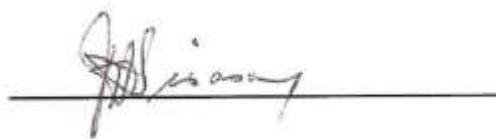
July 2026

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Ministry of Health and Wellness

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Acronyms and Abbreviations

Acronym/ Abbreviation	Meaning
AFP	Acute Flaccid Paralysis
CARPHA	Caribbean Public Health Agency
CHA	Community Health Aide
CMO	Chief Medical Officer
CRS	Congenital Rubella Syndrome
CRVS	Civil Registration & Vital Statistics
EBS	Event-Based Surveillance
EDMSSB	Emergency, Disaster Management & Special Services Branch
EOC	Emergency Operations Centre
EPI	Expanded Programme on Immunization
HD	Health Department
HEOC	Health Emergency Operations Centre
HIV/ AIDS	Human Immunodeficiency Virus / Acquired Immunodeficiency Syndrome
HMSR	Hospital Monthly Statistical Report
HPPB	Health Promotion & Protection Branch
IBS	Indicator-Based Surveillance
IHR	International Health Regulations
ILI	Influenza-Like Illness
JISS	Jamaica Injury Surveillance System
KSA	Kingston and Saint Andrew
LRTI	Lower Respiratory Tract Infection
MCSR	Monthly Clinical Summary Report
MO(H)	Medical Officer of Health
MOHW	Ministry of Health and Wellness
NCD	Non-Communicable Disease
NEOC	National Emergency Operations Centre
NIC	National Influenza Centre
NIRA	National Identification and Registration Authority
NLS	National Lab Services
NNCDSS	National Non-Communicable Disease Surveillance System
NPHL	National Public Health Laboratory
NSU	National Surveillance Unit
PAHO	Pan American Health Organization
PHEIC	Public Health Emergency of International Concern
PHN	Public Health Nurse
PMTCT	Prevention of Mother-To-Child Transmission
PS	Permanent Secretary
RD	Regional Director
RGD	Registrar General's Department

Acronym/ Abbreviation	Meaning
RHA	Regional Health Authority
RTD	Regional Technical Director
RT-PCR	Reverse Transcriptase Polymerase Chain Reaction
SARI	Severe Acute Respiratory Illness
SOP	Standard Operating Procedures
STATIN	Statistical Institute of Jamaica
TB	Tuberculosis
UHWI	University Hospital of the West Indies
WHO	World Health Organization

Preface

A well-functioning public health surveillance system is essential for identifying and responding to both ongoing and emerging health threats. Without timely and accurate surveillance, public health officials may underestimate the burden of known conditions or fail to detect new ones until they have spread widely. While notifiable diseases remain central to surveillance efforts, challenges persist. Variability in reporting practices, terminology, case investigation approaches, and data use across providers can hinder the system's effectiveness. Too often, the emphasis is placed on data collection and transfer rather than on the meaningful use of information to guide public health action at all levels.

This edition represents a major update to the 2009 National Surveillance Manual for Jamaica, reflecting Jamaica's continued commitment to strengthening its national surveillance system. It is designed to support the efficient use of public health resources by promoting standardised, practical, and responsive approaches to surveillance. The manual serves as both a foundational guide for those newly involved in surveillance and a reference for experienced practitioners seeking to stay aligned with current standards and practices.

One of the key updates in this edition is its reorganised structure, which separates the general description of the surveillance system and how it functions from specific protocols on notifiable diseases. The first eight chapters provide an overview of surveillance concepts, definitions, roles, and processes within the Jamaican surveillance system. Chapters nine to fifteen focus on specialised surveillance contexts such as disaster events, hotel and port health, and international obligations under the International Health Regulations (IHR). In addition, investigation forms, tools and other reference guidelines are included as appendices.

Several new chapters have also been introduced to reflect evolving public health priorities and ensure greater alignment with international standards and current best practices. These include guidance on mortality surveillance, event-based surveillance, and risk assessment, as well as disease-specific surveillance for new and emerging conditions. These additions acknowledge the

importance of developing a more comprehensive surveillance system that can respond to a broader range of health threats.

As a living document, this manual will continue to be refined based on user feedback, changing public health needs and knowledge. The goal is a stronger, more integrated surveillance system that not only captures data but turns it into actionable insight.

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The successful update of this manual was made possible through the dedication, expertise, and collaborative efforts of numerous individuals. The National Surveillance Unit wishes to extend its sincere gratitude to past and current staff at all levels of the surveillance system who contributed their insights and technical knowledge to ensure the accuracy and relevance of this edition. Their expertise has been invaluable in refining the content and improving its practical application. This update is thus a reflection of collective effort, and we thank everyone who played a role in its development.

Special thanks also to Ms. Zahra Miller for leading the consultancy in updating the manual, and to the Ministry of Health and Wellness staff for revising various sections of the document, ensuring it is more current and relevant.

We hope this manual serves as a valuable resource for those committed to improving surveillance and monitoring practices.



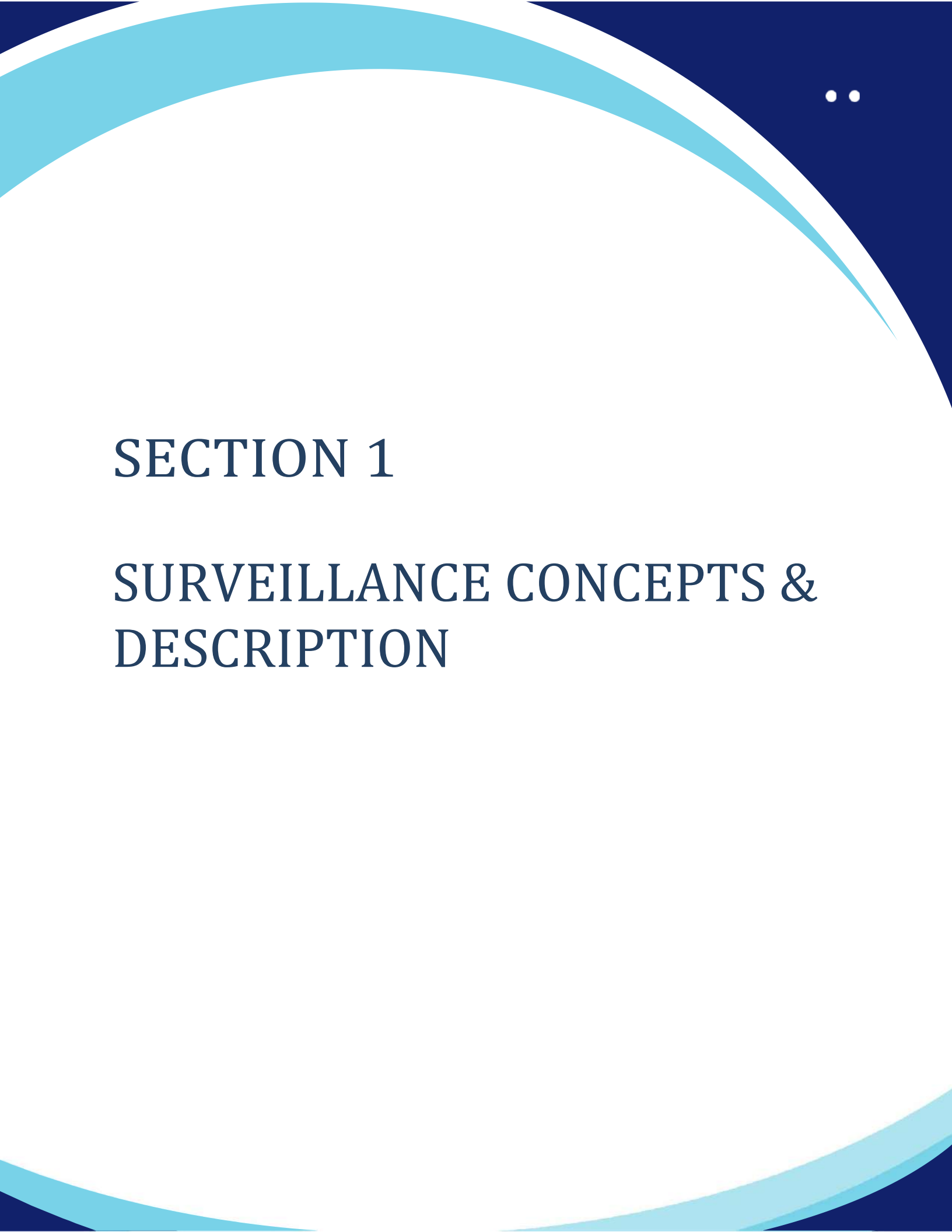
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SECTION 1

SURVEILLANCE CONCEPTS & DESCRIPTION

Chapter 1: Introduction to Surveillance

Surveillance is defined by the World Health Organization (WHO) and the US Centers for Disease Control and Prevention (CDC) as the ongoing systematic collection, analysis, and interpretation of outcome-specific data for use in planning, implementation, monitoring, and evaluation of public health practices. Surveillance systems should gather data from relevant sources, then validate and analyse this data to generate useful information for public health action (Figure 1.1).

Key message: The purpose of any surveillance system is to provide information for action. The best surveillance system in the world is worthless unless the public health professionals responsible for reviewing the data respond to what they see.

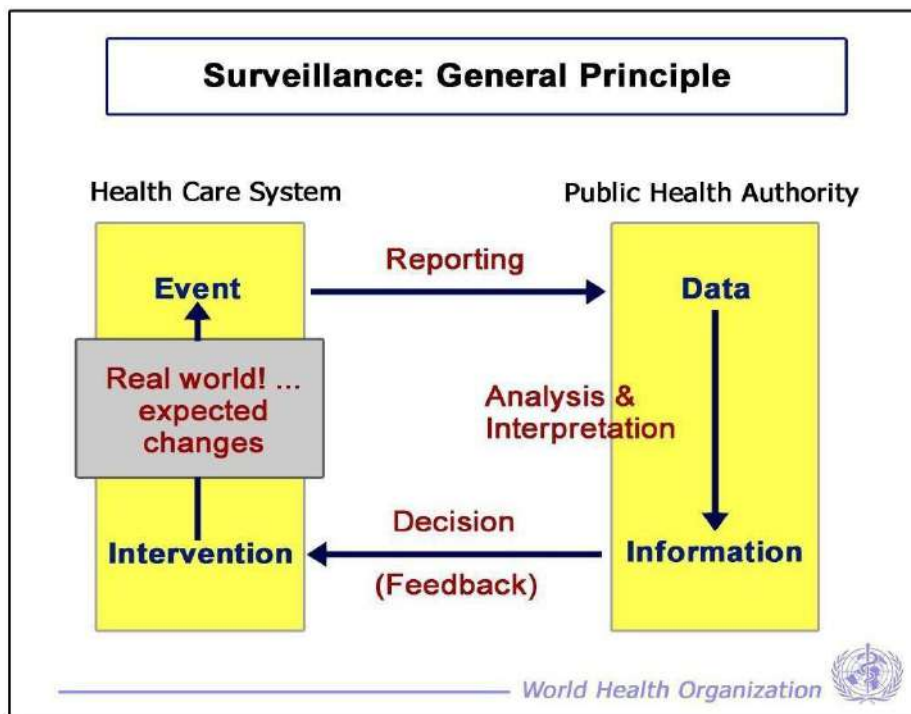


Figure 1.1. General principle of surveillance

A surveillance system aimed at timely and effective response to unusual disease situations (such as epidemics) relies on timely and accurate data reporting. If a surveillance system is to inform effective control measures, programmes must be monitored using appropriate programme indicators. Surveillance systems inform health policy by monitoring health status using

appropriate health indicators. In planning resource allocation, the administrative data in the health system must be monitored.

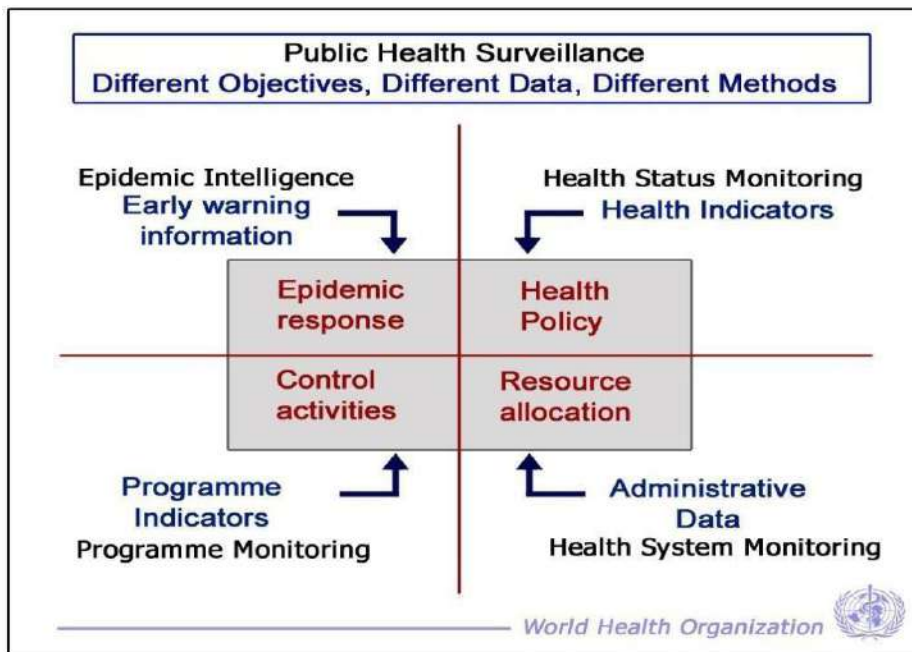


Figure 1.2. Outline of Public Health Surveillance

Objectives, Sources of Data and Methods

Surveillance facilitates the early detection of unusual events, clusters and outbreaks to initiate appropriate control activities to limit the spread of adverse health conditions, ultimately reducing morbidity, mortality and negative economic impact. It can be used, for example, to identify risk groups and guide the implementation of relevant intervention activities such as educational messages. Surveillance can also be used to evaluate the effectiveness of national programmes and provide a basis for shaping public health policy. As shown in the World Health Organization (WHO) framework, surveillance systems need to collect different types of data, different types of information and use different methods to achieve different objectives (Figure 1.2).

1. **Epidemic response objective:** Information is needed to monitor trends (to generate baseline rates) and for the early detection of unusual events, clusters, outbreaks, and epidemics so that timely and relevant responses can be initiated. For example, an increase in cases of fever and respiratory symptoms should trigger an investigation to determine the source, aetiology, and relevant response for control.
2. **Control activities objective:** Programme indicators are needed to monitor programme effectiveness; for example, vaccine coverage rates are important to monitor the performance of an immunization programme.
3. **Health policy objective:** Monitoring health status is necessary for developing health policy. For example, due to the high reported cases of HIV/AIDS in Jamaica, the government expanded the National AIDS Programme to include the prevention of mother-to-child transmission (PMTCT), and the provision of free care and treatment of people living with HIV/AIDS.
4. **Resource allocation objective:** Epidemiological and administrative data are needed for appropriate resource allocation. For example, Surveillance data for COVID-19 was used to inform resource allocation for national interventions. In order to generate a complete and accurate picture of a given health situation the surveillance process requires data from several sources such as:
 - Vital statistics
 - Morbidity and mortality reports
 - Case investigations
 - Disease registries
 - Outbreak reports
 - Laboratory reports
 - Sentinel/ Hospital Active Surveillance reports
 - Environmental reports
 - Agricultural (animal and plant health) reports
 - Surveys
 - Censuses
 - Class 1 notifications
 - Class 2 and Class 3 reports
 - Port health surveillance (airports, seaports)
 - Tourist Establishment Surveillance Reports

Climate Change and the Surveillance System

Jamaica's disease profile is shaped by its climate, and that climate is changing. Rising temperatures, shifting rainfall patterns, more frequent drought and more intense hurricanes are altering the distribution, seasonality and intensity of several conditions under surveillance — notably the Aedes-borne arboviruses, malaria, leptospirosis, water-borne and food-borne illness, and heat-related conditions. Higher temperatures shorten the extrinsic incubation period of arboviruses and lengthen the periods in which transmission is efficient. Disrupted water supply drives household water storage, which creates vector breeding sites. Flooding and storm damage produce the post-disaster conditions described in [Chapter 10](#).

The practical implication for surveillance is that historical baselines and expected seasonal patterns can no longer be assumed to hold. The system must therefore be adaptive so alert thresholds and expected case ranges should be reviewed periodically rather than treated as fixed. Surveillance data should also be interpreted alongside meteorological and environmental data; and risk assessment ([Chapter 5](#)) should treat climate as a standing driver of vulnerability rather than an occasional consideration.

Types of Surveillance Systems

Passive: A passive surveillance system is one in which the data provider is responsible for sending surveillance data to the next level in the system. Healthcare providers send reports to a designated public health agency in compliance with a known set of rules or regulations. The most commonly used passive systems are disease notification systems, requiring disease reports from healthcare providers to a central public health agency/ department. Reports of deaths and disease registries are other examples of passive surveillance.

Active: An active surveillance system is one in which data is routinely requested or collected from the data provider. Such systems seek out data from selected groups or networks put together for specific purposes. Examples of active systems include Class 1 disease hospital active surveillance (HAS) and repeated or serial health surveys. The essential feature of active surveillance is that staff of the health agency calls and/or visits the sites to gather information on a set group of suspected and confirmed diseases/events, conditions and clinical syndromes.

Passive surveillance requires less human and financial resources than active surveillance. Jamaica's national surveillance system can be classified as both active and passive. While the health department (HD) of each parish is expected to send data to the National Surveillance Unit (NSU) in specified timeframes, NSU contacts all reporting sites that have not met the reporting deadline and solicit the relevant data from them.

Key Message: Passive surveillance is good for establishing baseline rates and monitoring trends. This is especially useful for monitoring diseases of moderate to high prevalence. However, passive surveillance is not useful for accurate estimates of disease burden or for outbreak monitoring and response. In this instance, active surveillance is used.

Active surveillance is most often used for diseases of special interest, for example, those with high case fatality rate, subject to elimination and/or eradication, and with emerging or re-emerging diseases such as measles, polio, malaria, yellow fever, severe dengue, SARS, tuberculosis and Highly Pathogenic Avian Influenza. However, active surveillance requires more resources and commitment.

Surveillance systems can also be classified as **syndromic or disease-based systems**. As the names suggest, syndromic systems conduct surveillance of syndromes while disease systems conduct surveillance of diseases with specific aetiologies. Syndromic surveillance is particularly useful as an early alert system for changing disease situations. Surveillance of diseases is more useful for monitoring trends of a particular disease.

Key Message: Syndromic surveillance is useful for the early detection of and response to public health threats (e.g., identifying influenza season, alert and outbreak levels in a population). Disease surveillance is good for monitoring disease trends and is important for evaluating programmes and planning interventions. Timeliness is of the essence when it comes to disease outbreaks, whether natural or intentional.

Laboratory surveillance is necessary for case detection and disease confirmation and for monitoring specific trends. It is also important for evaluating programs and planning mid to long-term interventions.

Syndromic, disease and laboratory surveillance are complementary to each other.

The surveillance system can also be structured in terms of sources of information: exhaustive or based on sentinel sites. An exhaustive system is one which collates information from all health care facilities whereas a sentinel-based system collects information from selected sites based on their representation of the geographical area in terms of communities served and conditions seen.

Ideally, an effective surveillance system should combine components of the different types of surveillance: active, passive, laboratory, syndromic and disease-based.

Attributes of Surveillance Systems

When planning or evaluating a surveillance system, the following attributes can be used to gauge the overall usefulness of the system:

- Simplicity
- Flexibility
- Acceptability
- Representativeness
- Timeliness
- Sensitivity
- Positive Predictive Value (PPV)
- Negative Predictive Value (NPV)

Chapter 2: National Policy

In Jamaica, surveillance activities must be conducted within a framework that balances public health imperatives with ethical considerations and legal obligations. This chapter explores the intersection of ethical principles and the legislative environment governing disease surveillance activities.

History of the National Surveillance System

The surveillance system has evolved over time in Jamaica. Prior to 1976, disease reporting in Jamaica was mainly the responsibility of the Statistics Unit (now the Health Information Unit). Specific surveillance activities were initiated in 1976 following an outbreak of food poisoning due to the accidental contamination of counter flour with Parathion, an organophosphate insecticide. The Epidemiology Unit became operational in 1977 during an outbreak of Dengue Fever. Its mandate included communicable disease surveillance and control, outbreak investigation and quarantine. In the 1980s, HIV/AIDS surveillance and control, disaster preparedness and research were added to the Unit's responsibilities.

The goal of the surveillance system in Jamaica is to prevent and control the occurrence of communicable and non-communicable diseases and Class 1 health events.

The policies that govern surveillance in Jamaica serve to improve the health status of all its citizens by reducing the morbidity and mortality and associated costs of communicable and non-

communicable diseases and Class 1 health events. The Ministry of Health and Wellness' (MOHW) vision speaks to a society of healthy people. It envisages a health system that is client-centred, responsive and guarantees access to quality health care for every person in Jamaica and which considers the specific needs of the vulnerable. It also seeks to provide information and to educate the populace, to facilitate individuals taking responsibility for their own health, making informed decisions and adopting healthy lifestyle practices. Overall, the MOHW wants to ensure

that the highest quality of services for health promotion, protection and care is accessible to all persons in Jamaica to achieve optimal health, at an affordable cost to the government.

Regulations Governing Surveillance in Jamaica

Local Regulations

Jamaica's legislative and regulatory landscape provides the legal authority and guidelines for conducting surveillance activities ethically and effectively. These include:

- **The Public Health Act (1985)**

The Public Health Act [Public Health (Class 1 Notifiable Diseases) Order 2003] serves as the cornerstone for public health activities, including disease surveillance. It authorises the Ministry of Health and Wellness to collect, analyse, and disseminate health data to prevent and control communicable diseases.

Section 8: The Minister may at any time call upon a Local Board to investigate any disease of human beings present in their respective parishes, and to do whatever is necessary for arresting the spread of that disease.

Section 14:

(1) The Minister may make regulations generally for carrying out the provisions and purposes of this Act, and in particular, subject to section 7, but without prejudice to the generality of the foregoing, may make regulations in relation to:

- a) notifiable and communicable disease, the treatment and prevention thereof and the isolation of patients suffering therefrom;
- b) the prevention, mitigation and suppression of disease, including the disinfection, closing, or destruction of buildings in which infected persons have lodged or resided, and the restriction of movement of persons into and out of infected areas;
- c) vaccinations and inoculations;
- d) air and soil pollution;

- e) the collection and publication of epidemiological and other data pertaining to public health;
- f) occupational diseases and employment health hazards
- g) the importation, preparation, and distribution of food or drink intended for human consumption, in so far as it concerns public health;
- h) the inspection and prevention from contamination of food and drink intended for human consumption, the analyzing and testing of samples of such food and drink by an official analyst, the issuing of certificates in relation thereto, and the condemnation, seizure and disposal of such articles as are unfit for human consumption;
- i) the control and destruction of rodents, mosquitoes and other insects, termites, and other vermin;
- j) prescribing any fees in respect of any examination, certificate, license or other matter under this Act;
- k) prescribing any forms for the purposes of this Act;
- l) prescribing any other matter or anything which may be, or is required by this Act to be prescribed by the Minister.

(2) Regulations made under Subsection 1 may be applicable to the entire Island or to such part thereof may be specified therein.

Section 16:

(1) Where the Central Health Committee or a Local Board reports to the Minister that any part of the Island appears to be threatened with, or affected by, any communicable disease in epidemic proportions, and that measures apart from, or in addition to, those specifically provided for in the Act should be taken promptly, the Minister may by order direct the enforcement of any measures recommended by the Central Health Committee or Local Board, or any other measures he thinks expedient for preventing or mitigating that disease.

Section 18:

(1) Where a communicable disease is prevalent in epidemic proportions in the Island or any part of it, the Minister may, by order published in the Gazette, prohibit the assembly of persons and direct the closure of public spaces, schools and other premises.

Quarantine and isolation powers under the Act

Read together, these provisions give the Ministry a graduated set of powers rather than a single one. Section 8 allows the Minister to direct a Local Board to investigate a disease and to do whatever is necessary to arrest its spread. Section 14(1)(a) and (b) provide for regulations governing the isolation of patients suffering from a notifiable or communicable disease, and the restriction of movement of persons into and out of affected areas. Sections 16 and 18 come into play once a disease reaches epidemic proportions, and allow the Minister to direct enforcement measures by Order — including measures not otherwise provided for in the Act — and to prohibit gatherings and close premises.

The practical distinction for surveillance staff is that sections 8 and 14 support action in respect of an individual case or a defined infected area, and are available at any time. Sections 16 and 18 support population-level measures, and are available only when the epidemic level is reached and the Central Health Committee or a Local Board has so reported to the Minister.

- **Quarantine Act (1951 with latest amendment in 1991)**

The Quarantine Act provides the legal framework for controlling the spread of infectious diseases through quarantine measures.

- **Tourist Establishment Regulations (2000)**

Given Jamaica's vibrant tourism sector, the Tourist Establishments Regulations under the Public Health Act play a critical role in surveillance within hospitality settings to protect residents and visitors alike.

- **Disaster Risk Management Act (2015)**

The Disaster Risk Management Act (DRMA) provides the legal framework through which the State responds to hazards — including biological hazards and disease outbreaks — that rise to the level of a disaster or a threatened disaster.

Under Section 26, where the Minister responsible for disaster preparedness and emergency management notifies the Prime Minister that Jamaica is threatened with, or affected by, a hazard, the Prime Minister may declare all or part of the island a disaster area. Once such a declaration is made, the Prime Minister may by Order direct the enforcement measures considered necessary to prevent or minimise loss of life or harm to the health of persons.

The Act was used in this way during the COVID-19 pandemic. The entire island was declared a disaster area in March 2020, and successive Disaster Risk Management (Enforcement Measures) Orders provided for, among other things, the quarantine of specified communities, restrictions on movement, self-quarantine requirements for persons landing in Jamaica, isolation requirements for confirmed cases, and restrictions on gatherings. A subsequent amendment introduced fixed penalties for breaches of those measures.

- **Data Protection Act (Effective December 2023)**

The Data Protection Act regulates the processing of personal data, ensuring individuals' privacy rights are protected.

Public Health Act	Quarantine Act	Tourist Establishment Regulations	Disaster Risk Management Act	Data Protection Act
• Reporting of notifiable diseases by healthcare providers. • Authority for health officials to investigate outbreaks. • Powers to implement control measures to protect public health.	• Enforce quarantine and isolation for individuals or groups exposed to communicable disease. • Implement health checks at points of entry to prevent disease importation. • Balance individual rights with the need to protect public health.	• Health and safety standards for tourist establishments • Obligations for reporting outbreaks or unusual health events. • Collaboration between health authorities and tourist establishment operators to manage public health risks.	• Declaration of a disaster area where a hazard threatens or affects Jamaica. • Enforcement measures by Order, including community quarantine, movement restrictions and isolation. • Penalties for non-compliance with public health measures in force.	• Requirements for lawful and fair data collection. • Obligations to obtain consent where applicable, with exceptions for public health emergencies. • Measures to secure personal data against unauthorised access, loss, or damage.

Figure 2.1. Key stipulations and mandates of the regulations governing disease surveillance activities in Jamaica

International Health Regulations (IHR)

Jamaica, as a signatory to the International Health Regulations (IHR) (2005), is obligated to strengthen its surveillance and response systems to detect, assess, report, and respond to public health events that may constitute a public health emergency of international concern (PHEIC). These legally binding regulations, adopted by the World Health Organization (WHO), require countries to notify the WHO of any events that may pose a risk to global public health security within 24 hours of assessment.

Ethical Considerations for Public Health Surveillance

A key component of public health surveillance is to gather data that supports public health objectives while ensuring the protection of individuals and groups. However, the collection of personal data for public health purposes brings forward significant ethical concerns, especially regarding the balance between privacy and confidentiality, governance, informed consent, and other related issues. Issues about privacy, autonomy, equity, and the common good need to be considered and balanced and knowing how to do so can be challenging in practice.

Ethical considerations in public health surveillance therefore mainly center on safeguarding individual privacy, preventing stigmatisation, ensuring data security, and upholding transparency, all while effectively monitoring and addressing public health risks. Public health professionals can foresee, manage, and potentially prevent ethical issues by incorporating ethical principles throughout the planning and execution of a public health surveillance system.

The WHO Guidelines for Ethics in Public Health Surveillance, 2017¹, provide a global framework for establishing ethical national surveillance systems. The aim of the guideline development project was to highlight key ethical issues to help resolve potential controversies in surveillance, which is an ethical responsibility of governments. These guidelines should be carefully evaluated when making decisions related to the collection, analysis, sharing, communication, and use of surveillance data.

¹ World Health Organization. *WHO guidelines on ethical issues in public health surveillance* [Internet]. (June 2017). Available from: <https://www.who.int/publications/i/item/who-guidelines-on-ethical-issues-in-public-health-surveillance>

Chapter 3: Description of the Surveillance System

Mission Statement for the National Surveillance Unit

To detect, assess, and guide the neutralization of direct public health threats to the well-being of Jamaicans.

Purpose and Objectives

The purpose of the surveillance system is to collect, collate, analyse and interpret data for action on:

- A. reportable diseases or health events as stipulated in the Public Health Act and
- B. syndromes under surveillance at sentinel sites.

The objectives of the surveillance system are:

- Outbreak detection and response.
- Estimate the magnitude of the health problem and monitor trends in disease and selected health events.
- Provide data to support systematic programme planning, monitoring and evaluation.
- Provide data and information to support priority-setting, guidelines, policies and legislation towards disease prevention and control.
- Assist with the validation and evaluation of regional and national surveillance systems.
- Strengthen the regional and national surveillance systems.
- Information communication to inform those who need to know and enable appropriate action to protect and improve the health of the population.
- Test hypotheses about aetiologies, determinants of health and sources of outbreaks.
- Detect changes in health practice.
- Assess quality of health care.
- Identify research needs.

Information Flow

Figure 3.1 below demonstrates the flow of surveillance information between the various levels, both nationally and internationally.

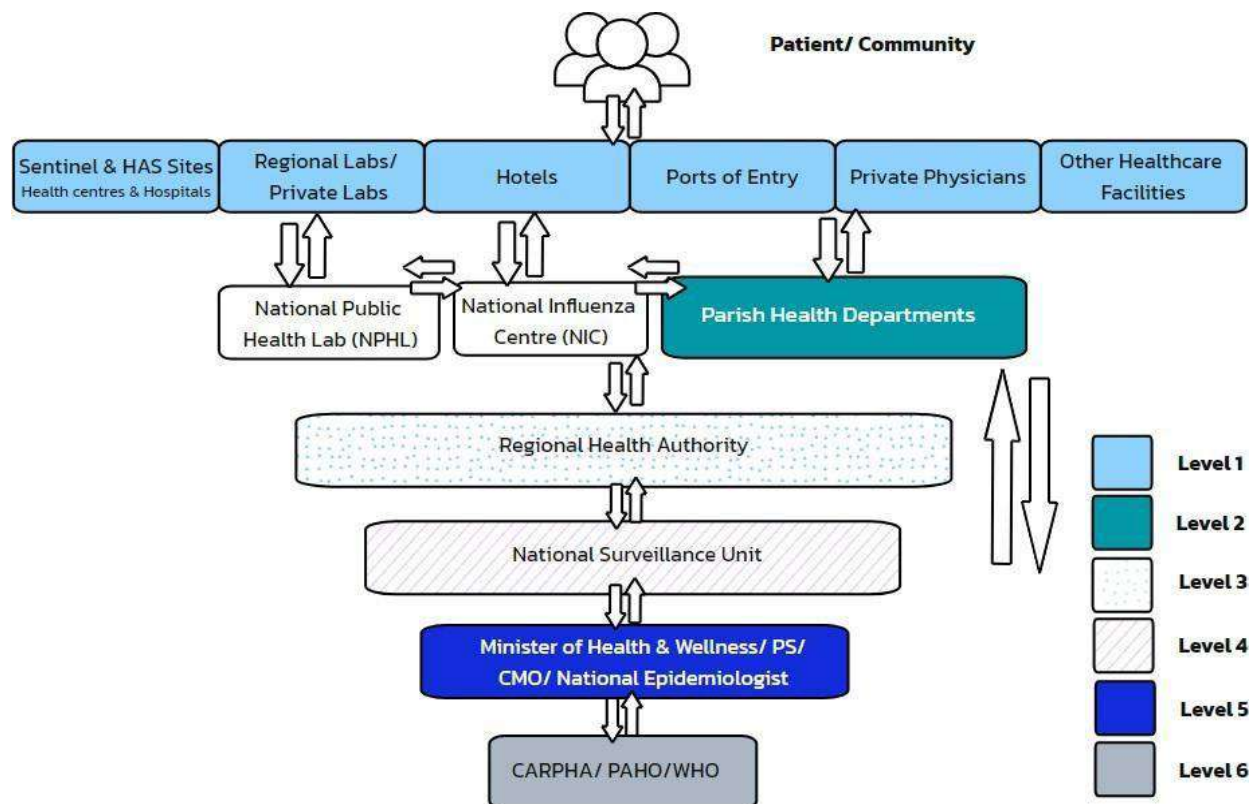


Figure 3.1. Functional Flow Chart of Surveillance Information

A contact list for different levels of the surveillance system is in [Appendix 1](#).

Level 1 (Sentinel/HAS Sites, Hotels, Points of Entry, Laboratories, Private Physicians, Other Health Care Facilities)

Facilities where clients are most likely to initiate first contact have specific roles and responsibilities.

Functions:

- Institute patient care and management.
- Make appropriate referrals.
- Collect pertinent client surveillance data, e.g., symptoms, date of onset, immunization history, travel history, etc.
- Sample collection and transportation.
- Daily/ Weekly Reporting to Level 2.

Level 2 (Parish Health Departments)

By virtue of their surveillance expertise, they will liaise with Level 1 to get appropriate client data for processing and referral to Level 3.

Functions:

- Collate aggregate client information from Level 1 or directly from clients.
- Verify the information collected.
- Assist with surveillance activities of case(s) detected at Level 1.
- Maintain regular contact with Health Personnel at Levels 3 and 4.
- Transfer collated information to Levels 3 and 4 in a timely manner.
- Conduct investigation on notified cases.
- Perform active surveillance for Class 1 diseases/ health events.
- Perform data analysis and feedback to Level 1.
- Implementation of policy decisions.

Level 3 (Regional Health Authority)

The RHAs functions include, but are not limited to the following:

- Collation of notification data from Levels 1 and 2.
- Review of notification data.
- Assist with the investigations of notified cases.
- Perform data analysis.
- Transform data into information.
- Develop prevention action and strategies.
- Disseminate data/information.
- Monitoring and evaluation of Parish Health Departments.
- Provide support and resources for surveillance at Levels 1 and 2.
- Interact with Level 5 in terms of policy formulation.

Level 4 (National Surveillance Unit)

The NSU's functions include, but are not limited to the following:

- Collation of notification data from Levels 1, 2 and 3.
- Review of notification data.
- Assist with investigations as needed.
- Perform active surveillance for some selected diseases.
- Validation of data from Levels 1, 2 and 3.
- Transform data into information.
- Develop prevention and control action and strategies.
- Disseminate data/information.
- Interact with Level 5 regarding policy formulation, development of guidelines, protocols, etc.
- Perform data analysis.
- Monitoring and evaluation of RHAs' and HDs' surveillance programmes and interventions.
- Support implementation of policy decisions at Levels 2 and 3.

Level 5 (Minister of Health and Wellness, Permanent Secretary, Chief Medical Officer, National Epidemiologist)

Health personnel at this level are involved with policy formulation and dissemination of information, e.g. through the media. They include the Minister of Health and Wellness, Permanent Secretary, Chief Medical Officer and the National Epidemiologist. At this level, decisions are made regarding advisories, declaration of warnings, alerts, or amendments of legislation to address specific public health threats.

Level 6 (CARPHA, PAHO/WHO)

Global and regional reports from Levels 4 and 5 are disseminated to the Caribbean Public Health Agency (CARPHA) and Pan American Health Organization/ World Health Organization (PAHO/WHO) at regular intervals and otherwise as necessary.

Data Collection and Continuous Data Quality Assurance

High-quality data are fundamental to effective surveillance. Continuous efforts to improve data collection processes and ensure data integrity are essential. Syndromes and diseases are to be reported according to epidemiological (epi) week. Epidemiological weeks start on a Sunday and end on a Saturday. Each year, the number of weeks is divided into 52 or 53 epi weeks, starting from the first week with at least four days in January. The first epidemiological week of the year ends, by definition, on the first Saturday of January if it falls at least four days into the month, even if it means that this first week starts in December of the previous year. Using epi weeks provides a standardised method to track disease occurrence over time ¹.

The epidemiological week for each case may be based on:

1. the date of onset of illness - for management of confirmed diseases/ outbreak cases/ suspected cases.
2. the date the patient presents to the health facility (for reporting purposes).

The date of onset for a syndrome or illness refers to the first date of symptom onset (which may differ from when a patient presents and is seen at a health facility). It is vital for understanding disease progression, identifying the period of infectiousness, and correlating with exposure events.

Other elements to focus on during data collection are ensuring information completeness and accuracy. For example, accurately recording the case's home address helps map disease distribution, identify clusters, and facilitate targeted public health interventions. In aggregate reporting, zero reporting ensures that the absence of cases is actively reported, preventing gaps in surveillance data. This practice helps distinguish between no cases and a lack of reporting.

Continuous data quality improvement strategies include regular training, standardised data collection tools, routine data audits, and feedback mechanisms for data providers.

¹ Caribbean Public Health Agency. *Regional Surveillance Policy Guidelines* [Internet]. (June 2011). Available from: [https://carpha.org/Portals/0/Publications/Regional-Surveillance-Guidelines-Interim-Policy%20\(1\)%20\(1\).pdf](https://carpha.org/Portals/0/Publications/Regional-Surveillance-Guidelines-Interim-Policy%20(1)%20(1).pdf)

Sentinel Sites

Sentinel sites are strategically selected healthcare facilities or communities that serve as representative reporting units for monitoring specific diseases and/or health events. They are crucial in enhancing disease surveillance systems by providing consistent and comparable data over time, allowing for tracking disease incidence, prevalence, and seasonal patterns that inform public health decision-making.

Sentinel sites are chosen based on the following criteria

- representative of the geographic area served
- easily accessible by all the population served
- perform all the relevant surveillance functions
- should have information about the population served
- must have information on the total number of patients seen daily and their presenting signs and symptoms
- should have adequately trained staff.

All sentinel sites must report their sentinel data weekly to the next level ([Appendix 2](#)). Health centres that are sentinel sites will only report syndromic data, while hospital sentinel sites are required to report on additional syndromes and diseases. A list of sentinel sites is included in [Appendix 3](#).

Syndromic Surveillance

Syndromic surveillance focuses on weekly collection, analysis, and interpretation of patient's presenting complaints to detect potential outbreaks early (see Figure 3.2). It relies on symptom patterns (syndromes) rather than confirmed diagnoses and involves data and sample collection.

SYNDROMIC SURVEILLANCE

- ☐ UNDIFFERENTIATED FEVER
- ☐ FEVER + RASH
- ☐ FEVER + JAUNDICE
- ☐ FEVER + HAEMORRHAGIC SYMPTOMS
- ☐ FEVER + RESPIRATORY SYMPTOMS
- ☐ FEVER + NEUROLOGICAL SYMPTOMS
- ☐ GASTROENTERITIS
- ☐ ACCIDENT
- ☐ VIOLENCE
- ☐ NONE OF THE ABOVE

To improve the accuracy and completeness of syndromic surveillance reporting, Jamaica utilises a syndromic stamp placed in medical records at the sentinel sites that report on syndromic surveillance. The clinicians then indicate by checking the appropriate box corresponding to the patient's history and examination as part of their notes. Not all patients seen will meet the definition for a syndrome; in such cases the clinician selects the option of 'None of the Above'. Medical records staff tally the number of cases seen daily and log the data for the weekly report to be submitted to the parish health department.

Samples should also be collected for a representative portion of syndromes seen each week, collecting sequentially starting from the first case seen. Undifferentiated fever, gastroenteritis, fever and respiratory symptoms should have 1 in 10 samples taken, and samples should be taken for all cases of fever and rash, fever and neurological, fever and jaundice and fever and haemorrhagic symptoms.

Hospital Active Surveillance

Hospital Active Surveillance involves proactive case finding by regularly reviewing medical records, laboratory reports, and patient admissions in healthcare facilities (see Figure 3.3). Hospital-based surveillance helps identify severe or hospitalised cases of communicable diseases or health events, providing insights into disease severity, burden, and healthcare impact. Active surveillance ensures more comprehensive data capture compared to passive systems.

Both active and passive surveillance systems work together and form the core basis for surveillance activities. They complement more specialised surveillance systems by enhancing timeliness and sensitivity in detecting public health threats and supporting rapid response and effective disease control measures.

SYNDROMIC SURVEILLANCE

Syndromic surveillance relies on symptom patterns (syndromes) rather than confirmed diagnoses. It involves both data and sample collection and is valuable as an early warning system for changing conditions.



STEP 1 – PREPARATION

Medical records staff ensure the blank sheet where the clinician makes notes has the syndromic stamp ahead of seeing patients.



STEP 2 – PATIENT CARE

During the patient exam, the clinician checks the syndrome(s) that match the symptoms of the patient's presenting complaint. If there is no syndromic match, they check **"NONE OF THE ABOVE"**.



STEP 3 – SAMPLE COLLECTION

Clinicians take off and/or request lab samples for a representative portion* of cases seen for each syndromic condition.

Work with the Parish Health Department & Regional Health Authority to establish weekly targets based off of past trends for your site.



STEP 4 - SAMPLE TRANSPORTATION

Laboratory staff should coordinate with the regional labs and/or parish health department for the timely transportation of samples to NPHL. A line listing of all samples submitted should be made and a copy sent to the parish health department.



STEP 5 – REPORTING

Medical records review patient records daily and tally each syndrome seen by relevant age group. This data is collated & reported to the health department weekly.

Figure 3.2. Syndromic Surveillance

HOSPITAL ACTIVE SURVEILLANCE

HAS is the active review of patient logs to ensure that the public health authorities have received notifications of all patients with class 1 diseases that presented since the date of last visit. These cases should be notified within 24 hours.

Preparation Phase

- Review class 1 disease list.
- Collect necessary forms including class 1 notification booklet and case investigation forms (as needed).
- Make a list of notifications already received from hospital since date of last visit.



At the Hospital

- Check in with relevant health professionals on site to confirm access to records.
- Check counterfoils of class 1 notification booklet at each location and cross-reference with your list of notifications received.
- Review daily logs for additional cases that match diseases and conditions on the class 1 list.
 - A&E/ Casualty triage book/ log
 - Inpatient wards admission/ discharge logs
 - Laboratory
 - Mortuary
- As much as is possible, validate suspected cases through patient charts, lab results, imaging reports or discussions with health personnel on site and document findings.



Data Submission

- Keep a note of cases or copies of forms for weekly reporting, follow-up and investigation.
- Submit completed forms to the parish health department and the MOHW National Surveillance Unit within 24 hours.



National Surveillance Unit:

Parish Health Department:

Figure 3.3. Hospital Active Surveillance

Community Surveillance

Community surveillance is the systematic detection, reporting and verification of unusual health events by people living and working in the community, before those events present at a health facility. It is a very important layer of Jamaica's early warning capability.

Many outbreaks — arboviral, diarrhoeal, food-borne and vaccine-preventable — begin with illness that never reaches a clinic or reaches it only once transmission is already established. Detection at community level shortens the interval between the onset of an event and the point at which the health system becomes aware of it, and that interval is an important determinant of how easily an outbreak can be contained. Community surveillance facilitates early warnings, the earliest available signal for events that have not yet reached the formal system, it captures illness clusters, unusual deaths and unusual events occurring in homes, schools, workplaces, places of worship and informal settlements and it extends surveillance to groups with limited access to, or low utilisation of health services.

Roles and responsibilities:

Community surveillance works through a chain of people rather than a single designated officer. The responsibilities set out below establish what each is expected to contribute to detecting, verifying and reporting events in the community.

The Community Health Aide (CHAs) is the principal routine point of contact between the health system and the household and is therefore central to community surveillance. In this role the CHA is expected to:

- Observe and report illness clusters, unusual deaths and disease rumours encountered during routine household and community visits;
- Report any suspected Class 1 notifiable condition immediately to the Medical Officer (Health) or designate, rather than holding it for a routine reporting cycle;
- Support case finding, contact listing and contact follow-up during case investigations and outbreaks;

- Deliver health education messages and support community-level control measures; and
- Maintain a simple record of events reported and the action taken on each.

The Public Health Inspector (PHI) provides the environmental and investigative arm of community surveillance. In this role the PHI is expected to:

- Conduct routine and event-driven inspection of premises, water supplies, food-handling establishments and vector breeding sites;
- Receive, verify and investigate reports and rumours arising from the community;
- Collect samples and assist with the completion of case and outbreak investigation forms;
- Implement and, where necessary, enforce control measures; and
- Report verified events through the parish surveillance structure.

The Public Health Nurse (PHN) supervise the community-level workforce and also carries direct case investigation responsibilities. In this role the PHN is expected to:

- Supervise the CHAs assigned to the community, and verify the reports they submit;
- Ensure that events detected in the community enter the formal reporting stream rather than remaining local knowledge, and escalate to the Medical Officer (Health);
- Conduct case investigation, including home visits, interviews with relatives and review of available clinical documentation;
- Visit homes, day-care centres, nurseries, schools and workplaces to review the immunisation status of close contacts, search for additional cases, and arrange immediate vaccination where indicated;
- Initiate community and hospital surveillance for additional cases once a case is identified;
- Maintain, jointly with the PHI, the parish record of diseases notified and the officer responsible for each notification.

However, detection capacity extends well beyond the health staff. Parish teams should establish and maintain working relationships with:

- Community and civic leaders,
- Councillors and Community Development Committees;
- Faith leaders and places of worship;
- School principals, guidance counsellors and school nurses;
- Pharmacists, private practitioners and traditional birth attendants operating in the community;
- Community-based and non-governmental organisations, and trained community volunteers;
- Funeral directors and burial societies, who are often the first to observe unusual clusters of death;
- Farmers, veterinarians and animal health officers, for zoonotic events and unusual animal deaths, consistent with a One Health approach.

These stakeholders function as community informants. Each should know what to report, to whom, and by what means. It is the responsibility of the parish team to make that pathway explicit and to provide feedback each time a report is made.

Signals arising from the community are a part of event-based surveillance signals and should thus be verified by the parish team ordinarily within 24 hours, reported onward to the region and the National Surveillance Unit, noted in the Weekly Parish Surveillance Report, and fed back to the informant — all in accordance with the procedures set out in [Chapter 9](#).

CHAs, PHIs and identified community informants should be oriented on the Class 1 notifiable conditions and syndromes under surveillance, on what constitutes a reportable event or rumour, on how and to whom to report and within what timeframe, and on confidentiality and the avoidance of stigmatisation (see [Chapter 2](#)), with regular refresher sessions.

Analysis, Utilisation and Dissemination of Surveillance Information

Effective data management is critical for safeguarding data integrity and ensuring timely analysis and utilisation at national and subnational levels. The following best practices support robust data management in surveillance:



Standardised Data Collection Tools: Ensure health workers across the levels of the surveillance system have access to the official investigation and reporting forms. These should be used as the basis for any electronic system (including spreadsheets) to ensure consistency across the levels of information flow.



Data Security and Confidentiality: Whether data is physically or electronically stored, implement strong data protection protocols to safeguard sensitive health information. This includes secure access controls, regular security audits, and data encryption (where possible or relevant) to prevent unauthorised access and data breaches.

★★★ **Data Quality Assurance:** Establish continuous quality control processes, including regular data validation, accuracy checks, and audits. Training data collectors and supervisors on data quality principles is essential.



Timely Data Reporting: Implement internal mechanisms to enable strict adherence to clear timelines for data submission from subnational to national levels and for feedback to the prior levels.



Capacity Building: Regularly train facility health workers, data managers, and surveillance officers on data management practices, including data entry, analysis, interpretation, and reporting.



Data Analysis and Utilisation: Institutionalise routine data analysis through designated persons and times to collate and analyse surveillance data to identify trends, detect outbreaks, and inform public health interventions. These analyses should be routinely shared, discussed at designated meeting spaces, and disseminated widely.

Feedback Mechanisms: Provide timely and regular feedback to data providers at all levels to improve data quality and maintain motivation. Feedback loops also help to identify gaps and areas for improvement.

The dissemination of surveillance reports is crucial for meeting reporting obligations to internal and external stakeholders. These reports provide timely and accurate data supporting decision-making, regulatory compliance, and transparency. Internally, they help organisations assess risks, allocate resources effectively, and refine strategies. Externally, they fulfill legal and regulatory requirements, foster public trust, and facilitate collaboration with industry partners, government agencies, and other entities. Clear and consistent reporting strengthens accountability and ensures surveillance efforts contribute to informed policy and operational decisions.

A weekly report summarising the cases of diseases, syndromes and conditions under surveillance produced at Health Departments and submitted to the RHA and NSU is compiled by the NSU into a weekly bulletin (National Epidemiology Bulletin). Monthly, quarterly and annual reports are also prepared at Levels 2-4 and disseminated to the relevant stakeholders and agencies. The method of dissemination may change based on evolving technology and resources. All outgoing reports from the NSU are reviewed internally by the Head of the NSU before dissemination.

Table 3.1. Dissemination of Reports

Report	Recipient	Method of Dissemination
Weekly Syndromic Surveillance Report (Appendix 4)	Caribbean Public Health Agency (CARPHA)	Email
Weekly EPI (Expanded Programme on Immunization) report		
Monthly (4-Weekly) Surveillance Report (Syndromic & Disease-Specific)		
Annual Disease Report/ EPI reports		
Annual Hansen's Disease Report	CARPHA, PAHO/WHO	Email
Annual Malaria Report	PAHO/ WHO	Website
Annual TB Report	PAHO/ WHO, CARPHA	Website
Weekly Epidemiology Bulletin	Internal and external stakeholders, and the general public	Website
Alerts	*Intra-Ministry personnel and other in-country stakeholders	Email
Monthly Disease-Specific Programme Area Report		
Quarterly HIV/AIDS Report		
Annual Surveillance Report		Hard copy distribution

*Intra-ministry personnel include the Minister of Health and Wellness, the Chief Medical Officer (CMO), the Permanent Secretary (PS), and other select technical personnel. In-country stakeholders include the Regional Health Authorities, Parish Health Departments, the Statistical Institute of Jamaica and the Planning Institute of Jamaica.

Specific Uses of Surveillance Information

Data will be used nationally to inform policies and direct action for preventing, controlling, and eliminating diseases, such as vaccination, prophylaxis, outbreak control, and therapy. It may also be used at all levels for:

- Monitoring and evaluation
- Estimation and projections
- Assessing programme needs
- Allocating resources
- Advocacy
- Identifying and setting priorities
- Developing guidelines, policies and legislation
- Identifying training needs
- Research purposes
- Determine endemic levels for communicable diseases.

Monitoring and Evaluation

When developing monitoring and evaluation indicators for a public health surveillance system, it is essential to focus on the system's performance in terms of data quality and public health impact. These indicators must be simple, easy to measure, and capable of providing insights into the system's timeliness, completeness, data quality, sensitivity, and specificity. They may also assess the status of diseases targeted for elimination or eradication, such as measles, poliomyelitis, and rubella.

The performance indicators should be assessed monthly and reported during parish and regional surveillance meetings. Additionally, the RHAs and NSU will conduct internal surveillance audits (see audit guidelines and sample audit tool in [Appendix 5](#)) annually, while external assessments will occur periodically. The guidelines for evaluating a surveillance system produced by Jamaica, CARPHA, and PAHO/ WHO will be utilised for external evaluations.

Monitoring indicators will include: -

- The proportion of sentinel sites submitting weekly surveillance reports on time to the Health Department (HD).
- The proportion of the HDs submitting weekly surveillance reports on time to the Regional Health Authority (RHA) and the National Surveillance Unit (NSU).
- The proportion of Class 1 notifications submitted within 24 hours of suspicion.
- The proportion of Class 1 notifications with investigations completed within specified timeframes.
- The average time between the reporting date of Class 1 notification and the date the investigation was initiated.

Evaluation indicators will include:

- The proportion of fever and rash and acute flaccid paralysis cases reported to the HD, RHA, or NSU using the appropriate reporting form.
- The proportion of suspected outbreaks notified to the RHA and NSU within 48 hours of determination.
- The proportion of confirmed outbreaks receiving a public health response at the regional or national level.
- The proportion of outbreaks detected at regional and national levels through analysis of surveillance data from parishes, including those missed at the parish level.

Chapter 4: Notifiable Diseases and Syndromes

Class 1, 2 and 3 Notifiable Diseases and Health Events

The diseases/health events listed in this manual are divided into three classes according to their surveillance and reporting requirements. Three classes of notifiable diseases/health events have been defined for Jamaica: Classes 1, 2, and 3.

Class 1 diseases/ health events are of the highest priority because of their potential to cause high morbidity and/or mortality. They are mainly infectious diseases and account for just over three-quarters of all notifiable conditions. Healthcare providers must report Class 1 diseases/health events immediately on suspicion to the Medical Officer of Health at their Parish Health Departments or the National Surveillance Unit within 24 hours of contact.

Class 1 diseases/health events include:

- Diseases subjected to International Health Regulations
- Diseases under international surveillance
- Diseases/Events of national and international importance
- Vaccine-preventable diseases
- Diseases/Events of national interest
- Any exotic or unusual communicable disease
- Case report universally required by International Health Regulations or as a disease under surveillance by WHO.

An outbreak of a communicable disease not normally classified as Class 1 but which has the potential to cause high morbidity or mortality may result in its reporting status being upgraded to Class 1.

A list of Class 1 conditions, case definitions and sample collection requirements is displayed in all public health facilities on a Class 1 Poster ([Appendix 6](#)).

Class 2: This class includes diseases based either on the relative urgency of investigating contacts and the source of infection or for initiating control measures in the Jamaican context. Healthcare providers must report Class 2 diseases weekly on a line listing submitted to the Medical Officer of Health at their Parish Health Departments.

Class 3: Healthcare providers must report monthly totals for Class 3 diseases to the Medical Officer of Health at their Parish Health Departments. A line listing may also be used to report the cases to the next level.

Diseases may also be classified based on features related to their transmission:

Foodborne Diseases

Foodborne diseases (food poisoning) refer to illnesses that occur due to the consumption of contaminated food or water by harmful pathogens. These contaminants could be faecal matter, toxins, metals or due to uncooked/ undercooked meat.

Sexually Transmitted Infections (STI)

A sexually transmitted infection refers to an infection that is transferred from one human to another human through sexual activity. It can also be acquired congenitally from mother to foetus.

Syndromic

Syndromes (syndromic) refer to conditions, signs, and symptoms that are correlated and tend to occur simultaneously, signalling a particular illness or the likelihood of the illness occurring.

Vaccine-Preventable Diseases

Vaccine-preventable diseases refer to infectious diseases that vaccines can prevent.

Vector-borne Diseases

Vector-borne diseases are human illnesses caused by parasites, viruses and bacteria that are transmitted by vectors. Vectors include arthropods (such as mosquitoes, ticks, fleas, lice, and flies), which act as carriers of the infectious agent.

Water-borne Diseases

Water-borne diseases refer to illnesses caused by drinking contaminated water that contain harmful microbes or pathogens.

Zoonotic Disease

Zoonotic diseases refer to infectious diseases that are transferred between animals and humans. These infections can be caused by harmful pathogens such as viruses, bacteria, fungi, parasites, and prions. They can be transmitted through food, water or direct contact.

Table 4.1. Diseases, health events and syndromes under surveillance in Jamaica

CLASS 1 DISEASES/ HEALTH EVENTS	
HEALTH EVENT	CATEGORY
Accidental Poisoning	-
Maternal Death	-
DISEASE	CATEGORY
Acute Flaccid Paralysis (Poliomyelitis)	Vaccine-Preventable Disease
Anthrax	Zoonotic Disease
Chikungunya	Vector-borne Disease
Cholera	Water-borne Disease
Congenital Rubella Syndrome	Vaccine-Preventable Disease
Congenital Syphilis	Sexually Transmitted Infection (mother to foetus)
COVID-19	Zoonotic Disease
Dengue (Severe)	Vector-borne Disease
Diphtheria	Vaccine-Preventable Disease
Encephalitis	
Fever and Rash (Measles, Rubella)	Vaccine-Preventable Disease
Haemophilus Influenzae Meningitis	Vaccine-Preventable Disease
Hepatitis B and C	Vaccine-Preventable Disease
HIV/AIDS	Sexually Transmitted Infection
Legionnaire's Disease	
Leprosy (Hansen's Disease)	
Malaria	Vector-borne Disease
Measles	Vaccine-Preventable Disease
Meningitis	
Meningococcal Meningitis	Vaccine-Preventable Disease
Mpox	Zoonotic Disease
Neonatal Tetanus	Vaccine-Preventable Disease
Ophthalmia Neonatorum	Sexually Transmitted Infection (mother to child)

CLASS 1 DISEASES/ HEALTH EVENTS	
<i>HEALTH EVENT</i>	<i>CATEGORY</i>
Pertussis-Like Syndrome (Whooping Cough)	Vaccine-Preventable Disease
Plague	Zoonotic Disease
Rabies (in humans)	Zoonotic Disease
Rheumatic Fever	
Rubella	Vaccine-Preventable Disease
Tetanus	Vaccine-Preventable Disease
Tuberculosis	Vaccine-Preventable Disease
Typhoid and Paratyphoid Fevers	Water-borne Disease
West Nile Virus	Vector-borne Disease
Yellow Fever (Urban or Sylvatic)	Vector-borne Disease
Zika Fever	Vector-borne Disease

CLASS 2 DISEASES/ HEALTH EVENTS	
<i>DISEASE</i>	<i>CATEGORY</i>
Dengue Fever	Vector-borne Disease
Influenza	Zoonotic Disease
Viral Hepatitis A	Water-borne Disease

CLASS 3 DISEASES/ HEALTH EVENTS	
<i>DISEASE</i>	<i>CATEGORY</i>
Bacterial Vaginosis	Sexually Transmitted Infection
Campylobacter	Food-borne Disease
Chancroid	Sexually Transmitted Infection
Chicken Pox	Vaccine-Preventable Disease
Chlamydia	Sexually Transmitted Infection
E. coli (EHEC)	Food-borne Disease

CLASS 3 DISEASES/ HEALTH EVENTS	
<i>DISEASE</i>	<i>CATEGORY</i>
Genital discharge syndrome	Sexually Transmitted Infection
Genital ulcer syndrome	Sexually Transmitted Infection
Gonorrhoea	Sexually Transmitted Infection
Herpes Simplex Virus (HSV)	Sexually Transmitted Infection
Lymphogranuloma Venereum (LGV)	Sexually Transmitted Infection
Mumps	Vaccine-Preventable Disease
Non-Specific Urethritis	Sexually Transmitted Infection
Salmonellosis	Food-borne Disease
Shigellosis	Food-borne Disease
Syphilis	Sexually Transmitted Infection
Trichomonas	Sexually Transmitted Infection
Unspecified STI	Sexually Transmitted Infection

SYNDROMES UNDER SURVEILLANCE	
<i>SYNDROME</i>	<i>CATEGORY</i>
Fever and Rash	Syndromic
Fever (< 5 years/ ≥5 years)	
Gastroenteritis (< 5 years/ ≥5 years)	Syndromic
Accidents (< 5 years/ ≥5 years)	Syndromic
Violence (< 5 years/ ≥5 years)	Syndromic
Fever and Respiratory Symptoms/ Influenza-like Illness (ILI) (< 5 years/ ≥5-59 years/ ≥60 years)	Syndromic
Fever and Haemorrhagic Symptoms	Syndromic
Fever and Neurological Symptoms (Hospital sentinel only)	Syndromic
Fever with Jaundice (Hospital sentinel only)	Syndromic

National/Regional Requirements

All case investigations should be initiated as soon as possible after contact. Below are general guidelines for completion of investigations based on national and international guidelines. Under outbreak conditions, the timelines for investigation are subject to change.

Table 4.2. Case definitions and investigation requirements for Class 1 diseases and health events in Jamaica

	Disease	Case Definition	Specimen	Laboratory	Timeline for Investigation
1.	Accidental Poisoning	Any case of suspected poisoning regardless of route of exposure (e.g. bleach or kerosene ingestion) considered to have occurred by accident.	For suspected Lead poisoning- Blood in purple top tube	Government Chemist, Hope Road, Kingston 6, Jamaica, (876-927-1829). For other agents, seek advice from the National Surveillance Unit (876-633-7937)	2 weeks
2.	Maternal Death	Any death of a woman while pregnant or within 42 days of termination of pregnancy. For surveillance purposes, all maternal deaths within 1 year post termination of pregnancy should be reported.	N/A	N/A	6 weeks
3.	Acute Flaccid Paralysis / Poliomyelitis	Acute onset of flaccid paralysis in the absence of trauma.	Two stool specimens (at least 24 hours apart) within 14 days of onset of paralysis. Place each in a sterile universal container. Transport on ice immediately to NPHL.	National Public Health Lab	First contact strategy, but within 48 hours. Follow up at 60 days.
4.	Chikungunya	Fever and severe joint pains	Serum specimen in red top tube on ice to NPHL.	National Public Health Lab	48 hours

	Disease	Case Definition	Specimen	Laboratory	Timeline for Investigation
5.	Cholera	Profuse, watery diarrhoea rapidly leads to dehydration or death, especially in persons 5 years of age and older.	Stool specimen in Cary-Blair Media to NPHL. DO NOT REFRIGERATE AND DO NOT SEND ON ICE.	National Public Health Lab (Bacteriology Section)	24 hours
6.	Congenital Rubella Syndrome	An infant with one or more of the following: Congenital cataracts, glaucoma, deafness, microphthalmia, microcephaly, congenital heart defects, meningoencephalitis, purpura at birth; or a history of maternal rubella.	1. Acute and convalescent serum specimen in red top tube <u>on ice.</u> 2. Nasopharyngeal swab (in viral transport medium) <u>and</u> urine sample, <u>both on ice.</u>	National Public Health Lab (Immunology Section)	48 hours
7.	Congenital Syphilis	Failure to thrive, snuffles, skin peeling, dactylitis, fever, anemia, jaundice, hepatosplenomegaly.	Serum specimen in red top tube. CSF as indicated.	National Public Health Lab (Serology Section)	48 hours
8.	COVID-19	Acute onset of fever AND cough (ILI) OR acute onset of three or more of the following signs or symptoms: fever, cough, generalized weakness/fatigue, headache, myalgia, sore throat, coryza, dyspnea, nausea/diarrhea/anorexia.	Nasopharyngeal swab OR combined nasopharyngeal and oropharyngeal swabs (nylon, Dacron, or polyester flocked swabs) in viral transport medium <u>on ice.</u>	National Public Health Lab	48 hours
9.	(Dengue Fever)	Acute onset of fever plus at least two of the following: Vomiting or nausea, rash, headache or retro-orbital pain, arthralgia and/or myalgia, hemorrhagic tendencies (petechiae, positive tourniquet test), leukopenia.	Serum specimen in red top tube <u>on ice.</u>	National Public Health Lab	48 hours

	Disease	Case Definition	Specimen	Laboratory	Timeline for Investigation
	Severe Dengue	Dengue case with any of the following: shock or respiratory distress due to severe plasma leakage; severe bleeding; severe organ involvement	Serum specimen in red top tube <u>on ice</u> . Specimens such as heart, kidney, liver and spleen in a sterile container suspended in normal saline, if the patient dies.	National Public Health Lab	48 hours
10.	Diphtheria	Acute pharyngitis or laryngitis with a tonsillar or laryngeal pseudomembrane	Throat swab	National Public Health Lab (Bacteriology Section)	48 hours
11	Encephalitis	Acute onset of headache, high fever, meningeal signs, stupor, disorientation, coma, spasticity, tremors, occasional convulsion (especially in infants) and spastic, but rarely flaccid, paralysis.	CSF sample <u>on ice</u> Acute and convalescent serum specimens in red top tubes <u>on ice</u> Stool sample in sterile universal container <u>on ice</u>	National Public Health Lab (Bacteriology Section)	48 hours
12	Fever and Rash	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) and a generalised maculopapular rash.	Serum specimen (collected less than 7 days after onset of rash) in a red top tube <u>on ice</u> .	National Public Health Lab (Immunology Section)	First contact strategy but within 48 hours
13	Haemophilus Influenzae Meningitis	Sudden onset of fever, vomiting, convulsions, listlessness, neck rigidity and bulging fontanelle.	1-2 sets of blood cultures and CSF	Hospital Bacteriology Lab or National Public Health Lab for culture	48 hours
14	Hepatitis B and C	Anorexia, abdominal pain, nausea, vomiting, jaundice.	Serum specimen in red top tube <u>on ice</u> .	National Public Health Lab (Immunology Section)	2 weeks
15	HIV/ AIDS	Severe weight loss ($\geq 10\%$ of body weight), fever, diarrhoea (>1 month duration), lymphadenopathy, oral thrush or other signs of immunosuppression.	Serum specimen in red top tube <u>on ice</u> .	National Public Health Lab (Immunology Section)	2 months

	Disease	Case Definition	Specimen	Laboratory	Timeline for Investigation
16	Leprosy (Hansen's Disease)	Sharply demarcated anesthetic skin patches.	Skin smear or Skin Biopsy	Refer to Dermatologist for specialist testing using skin smear or skin biopsy.	2 weeks
17	Malaria	Intermittent fever with chills and headache, especially in those who recently traveled to malaria endemic areas.	Thin and thick blood smears on glass slide and EDTA (purple top) tube with 1 ml blood.	National Public Health Lab (Parasitology Section)	48 hours
18	Measles	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) and a generalized maculopapular rash.	Serum specimen (collected less than 7 days after onset of rash) in a red top tube on ice .	National Public Health Lab (Immunology Section)	First contact strategy but within 48 hours
19	Meningitis	Sudden onset of fever with signs and symptoms of meningeal involvement.	1-2 sets of blood cultures and CSF For suspected viral etiology send CSF on ice	Hospital Bacteriology Lab or National Public Health Lab	1 week
20	Meningococcal Meningitis	Sudden onset of fever, intense headache, nausea, vomiting, lethargy, and frequently, petechial rash.	Blood and CSF	Hospital Bacteriology Lab or National Public Health Lab for culture	48 hours
21	Mpox	Unexplained acute rash AND one or more of the following signs or symptoms: Fever ($>38.5^{\circ}\text{C}$ or 101.3°F), headache, myalgia (muscle pain/body aches) back pain, profound weakness or fatigue.	Skin lesion swabs (nylon, Dacron or polyester flocked swabs) from more than one lesion, including exudate, roofs, lesion crusts. Place specimen in viral transport medium on ice .	National Public Health Lab	48 hours
22	Ophthalmia Neonatorum	Purulent conjunctivitis in any neonate.	Swab of eye discharge.	National Public Health Lab (Bacteriology section)	1 week
23	Pertussis-Like Syndrome	Paroxysms of coughing, characteristic high-pitched inspiratory whoop.	Nasopharyngeal aspirate or cough plate Contact the laboratory for details prior to specimen collection.	National Public Health Lab (Bacteriology section)	48 hours

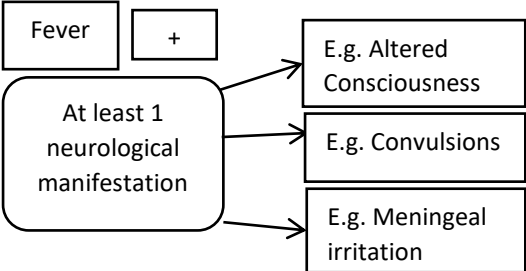
	Disease	Case Definition	Specimen	Laboratory	Timeline for Investigation
24	Plague	Swollen, tender, inflamed lymph nodes, lymphadenitis, fever and respiratory symptoms.	Blood culture or bubo aspirate. Notify laboratory prior to specimen submission.	National Public Health Lab (Bacteriology section)	24 hours
25	Rheumatic Fever	Recurrent inflammatory disease affecting large joints (arthritis), the heart (carditis), the brain (chorea), the skin (erythema marginatum) and subcutaneous tissue (subcutaneous nodules) occurring as sequelae of streptococcal upper respiratory tract infection.	Serum for ASTO titres.	National Public Health Lab (Serology Section)	6 weeks
26	Rubella	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) and a generalized maculopapular rash.	Serum specimen (collected less than 7 days after onset of rash) in a red top tube <u>on ice</u> .	National Public Health Lab (Immunology Section)	48 hours
27	Tetanus	Children -difficulty in sucking, opisthotonus; Adult -difficulty swallowing, followed by muscle spasm and rigidity	Clinical Diagnosis	N/A	48 hours
28	Tuberculosis (Extra-pulmonary and Pulmonary)	Persistent cough (>3 weeks), fever, night sweats, and weight loss.	3 consecutive early morning sputum sample or gastric washing (for children) or relevant biopsy specimens.	National Public Health Lab (Tuberculosis Section)	Initiate within 2 weeks and complete within 10 weeks
29	Typhoid and Paratyphoid Fevers	Insidious onset of sustained fever, headache, malaise, and anorexia.	Blood culture and stool specimen in a sterile container	National Public Health Lab (Bacteriology section)	48 hours
30	Yellow Fever (Urban or Sylvatic)	Sudden onset of fever, chills, headache, myalgia, vomiting, prostration, jaundice (especially in persons who recently visited Yellow Fever endemic areas).	Serum specimen in red top tube <u>on ice</u> .	National Public Health Lab	24 hours

	Disease	Case Definition	Specimen	Laboratory	Timeline for Investigation
31	Zika Fever	A patient with a rash or elevated body temperature (>37.2°C or 99.0°F) and <u>one or more</u> of the following symptoms: (i) Arthralgia or myalgia (ii) Non-purulent conjunctivitis or conjunctival hyperaemia (iii) Headache or malaise	Serum specimen in red top tube <u>on ice.</u>	National Public Health Lab	48 hours

Syndromes and Other Health Conditions

Table 4.3. Case Definitions for Syndromes and Other Health Conditions

Disease/ Syndrome	Case Definition	Special Note
Undifferentiated Fever	A person presenting with a body temperature of $\geq 38^{\circ}\text{C}$ / 100.4°F (or recent history of fever) without an obvious diagnosis or focus of infection. A person may also have headache, retro-orbital pain, myalgia, arthralgia, nausea or vomiting but without any symptoms fitting another syndromic definition.	Conditions from presumed viral infection should be included in surveillance. If no thermometer is available, a patient who was or complained of being 'hot to touch' may be considered to have a fever. These conditions for fever should also be applied to the other fever syndromes below.
Fever and Rash	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) (or recent history of fever) and a maculopapular rash.	There should be no obvious cause of one or both symptoms, indicating that the symptoms are related to each other. This case definition will capture suspected measles, rubella, dengue and roseola cases.
Gastroenteritis	A person presenting with 3 or more loose stools within 24 hours.	
Violence	Any injury for which the cause is intentional, e.g. gunshot wounds, stab wounds, etc.	
Accidents	Any injury for which the cause is unintentional, e.g. motor vehicle accident, falls, burns, etc.	
Fever and Respiratory Symptoms (Influenza- like illness)	A body temperature of $\geq 38^{\circ}\text{C}$/ 100.4°F (or recent history of fever) and cough in a previously healthy person who does not require admission. Fever + Cough	See guidelines for SARI/ ILI
Fever and Haemorrhagic Symptoms	A body temperature of $\geq 38^{\circ}\text{C}$/ 100.4°F (or recent history of fever) in a previously healthy person presenting with at least one haemorrhagic (bleeding) manifestation with or without jaundice (e.g. haematemesis or purpura or haemoptysis or melena). Fever + At least 1 bleeding manifestation E.g. Haematemesis E.g. Purpura E.g. Melaena E.g. Haemoptysis	Fever with haematuria and epistaxis should not be counted.

Disease/ Syndrome	Case Definition	Special Note
Fever and Neurological Symptoms	A body temperature of $\geq 38^{\circ}\text{C}$/ 100.4°F (or recent history of fever) in a previously healthy person with or without headache and vomiting. The person must also have meningeal irritation, convulsions, altered consciousness, altered sensory manifestations or paralysis (except AFP).  <pre> graph LR Fever[Fever] --- Plus[+] Plus --- Neuro[At least 1 neurological manifestation] Neuro --> Altered[E.g. Altered Consciousness] Neuro --> Convulsions[E.g. Convulsions] Neuro --> Meningeal[E.g. Meningeal irritation] </pre>	Any person that is a case of suspected meningitis, encephalitis or meningo-encephalitis should be counted as fever and neurological symptoms.
Fever and jaundice	A body temperature of $\geq 38^{\circ}\text{C}$ / 100.4°F (or recent history of fever) in a previously healthy person presenting with jaundice.	
Admitted Lower Respiratory Tract Infection (LRTI)	Any person admitted to hospital with a diagnosis of Lower Respiratory Tract Infection/ Pneumonia	
Deaths from LRTI/ Pneumonia-Like Illness	Any person who has died whose cause of death is listed as LRTI/ Pneumonia	
Foodborne Illness (Outbreak)	An incident in which 2 or more people experience a similar illness, after ingestion of a common food or drink.	Stool and/or vomitus, or rectal swab (if stool unavailable) or leftover food or other food should be sent to the National Public Health Lab.
Leptospirosis	A patient with fever, headache, myalgia of calves and/or thighs, and weakness and one or more of the following symptoms/signs: • Conjunctival suffusion • Anuria or oliguria • Jaundice • Cough, haemoptysis or shortness of breath • Haemorrhage (gastrointestinal bleeding, lung bleeding) • Meningeal irritation • Skin rash • Cardiac arrhythmia or failure	Whole blood with anticoagulant in a purple-top tube (collected ≤ 5 days from onset of symptoms) should be sent to the National Public Health Lab.

No definitive time period is given for *recent history of fever*. This is left to the discretion of the clinician, who will take into account the disease(s) being queried.

Relationship of Syndromes to Diseases

Table 4.4. Syndromes and Possible Associated Pathogens/ Diseases

Syndromes	Potential Pathogen/Diseases	
Acute Flaccid Paralysis	1. Poliomyelitis 2. Guillian Barré Syndrome	
(Undifferentiated) Fever	1. Dengue 2. Leptospirosis 3. Malaria 4. Influenza	5. Parvovirus B19 6. Oropouche 7. Mayaro 8. Measles
Fever and haemorrhagic symptoms	1. Severe Dengue 2. Leptospirosis 3. Yellow fever 4. Bacterial (meningococcal) meningitis	5. Malaria 6. Hantavirus 7. Adenovirus 8. Other haemorrhagic fevers e.g. Lassa fever, Ebola virus, Marburg virus)
Fever and Jaundice	1. Hepatitis A, B or C 2. Dengue fever 3. Severe Dengue 4. Leptospirosis	
Fever and neurological symptoms	1. Enteroviruses (Polio and other enteroviruses) 2. West Nile virus 3. Bacterial or viral meningitis 4. Malaria	5. St. Louis encephalitis virus 6. Herpes simplex. 7. Dengue
Fever and Rash	1. Measles 2. Rubella 3. Dengue fever 4. Scarlet fever	
Fever and Respiratory symptoms	1. Influenza 2. Respiratory syncytial virus (RSV) 3. Metapneumovirus 4. Hantavirus	5. Leptospirosis 6. SARS CoV 7. Legionellosis
Gastroenteritis	1. Salmonella 2. Shigella 3. Rotavirus 4. Norwalk	5. Enterotoxigenic E. coli 6. E. coli 0157:H7 7. Staphylococcus aureus 8. Foodborne chemical toxins

Forms for Surveillance

The following national surveillance forms can be found in the Appendix section:

- Class 1 Notification form ([Appendix 7](#))
- Weekly Parish Surveillance Report Form ([Appendix 2](#))
- General surveillance line listing ([Appendix 8](#))
- Case investigation forms ([Appendix 49-61](#)).

Table 4.5. *Guidelines for the Use of Surveillance Forms*

Type of form	When to use	Frequency of reporting
Class 1 Notification form	Whenever a Class 1 disease is suspected/identified	Within 24 hours/daily
Case investigation forms	During case/early outbreak investigation	Daily during early outbreak investigation or within 48 hours (or as specified for a particular condition) of completing a case investigation.
Weekly Parish Surveillance Report Form	To submit parish data	Weekly by 3 p.m. on Tuesday afternoons to the NSU (this may vary if the Tuesday falls on a public holiday).
General surveillance line listing	During an outbreak investigation	Daily during early outbreak or within 48hrs when the outbreak has waned.
Syndromic surveillance	Whenever a syndrome under surveillance is identified	Use daily at Health Centres. Send aggregate numbers to HD/RHA weekly.
Post disaster surveillance form	During and after a disaster	Used to capture and transfer information to the next level daily.

Chapter 5: Risk-Informed Surveillance

Risk Assessment

Risk assessment is a core feature of both generalised surveillance systems and more specialised surveillance structures. It enables public health workers to implement effective control measures. Risk assessment in the context of public health surveillance refers to the structured process of evaluating the probability and potential impact of an adverse health event, such as an outbreak, epidemic, or emergence of a novel pathogen on a population.

Purpose of Risk Assessment

Risk assessment is a critical component of epidemiological surveillance. It enables public health authorities to identify, prioritise, and respond to potential threats to population health.

Through systematic evaluation of available data, risk assessment supports evidence-based decision-making, helping to allocate resources efficiently and implement timely interventions.

Key Components

Risk assessment typically involves the following components:

1. Hazard Identification

Determining the biological agent, disease, or event that threatens public health. This includes infectious diseases, zoonotic threats, environmental hazards, or antimicrobial resistance.

Identifying aetiological agents during disease outbreaks is critical to effective management. Laboratories are key in assisting with outbreak detection, especially when the same serotype/subtype is detected from several sources or places without clinical or epidemiological information to suggest an outbreak. Additionally, they may help predict

potential epidemics (e.g., changes in dengue serotypes) and are key to antimicrobial resistance surveillance.

2. Exposure Assessment

Evaluating how, where, and to what extent populations are exposed to the hazard. This includes modes of transmission, geographic spread, and population mobility.

3. Vulnerability Assessment

Identifying populations or subgroups at increased risk due to age, immunity status, living conditions, socioeconomic status, or access to healthcare.

4. Impact Assessment

Estimating the potential consequences of the hazard on health outcomes (morbidity, mortality), health systems, and social or economic sectors.

5. Risk Characterisation

Integrating information from the above components to produce a qualitative or quantitative estimate of risk. Risks are often categorised as low, moderate, or high, with justification provided.

Types of Risk Assessment

- **Rapid Risk Assessment**

Conducted quickly in response to emerging or urgent public health threats. It often uses limited data and is updated as more information becomes available.

- **Comprehensive Risk Assessment**

A more detailed evaluation is conducted during ongoing or post-outbreak phases and is used for planning, preparedness, and policy development.

Sources of Data

Risk assessments draw from multiple data sources, including:

- Routine surveillance data (case reports, laboratory confirmations)
- Syndromic surveillance
- Event-based surveillance (e.g., media monitoring, community reports)
- Field investigations
- Epidemiological modelling
- Demographic and social data
- Environmental and meteorological data (e.g. rainfall, temperature, drought and storm forecasts from the Meteorological Service of Jamaica)
- International health alerts

Risk Communication and Decision-Making

The risk assessment results should be communicated clearly to decision-makers, stakeholders, and, where appropriate, the public (see Risk Communication Protocol). Effective communication includes:

- A summary of findings
- Assumptions and uncertainties
- Recommended actions
- Risk classification and rationale.

Risk assessments inform key decisions, such as:

- Activation of emergency response systems
- Resource mobilisation
- Public health advisories
- Travel and trade recommendations
- Coordination with regional and global health partners.

Chapter 6: Laboratory Surveillance

Laboratory surveillance is essential for detecting and controlling infectious diseases and enabling rapid identification of pathogens and their sources. Thus, laboratories play a vital role, especially in communicable disease surveillance, including trend analysis, outbreak identification, intervention assessment, and support for early warning systems.

Key responsibilities of laboratories include:

- **Diagnostic Testing:** The laboratory's primary role in public health surveillance is confirming aetiology.
- **Reference and Specialized Testing:** Utilizing advanced techniques to support national surveillance.
- **Quality Assurance:** Maintaining rigorous quality control for reliable data.
- **Routine Surveillance:** Routine, standardised reporting representing tests done nationally to the National Surveillance Unit is essential for a cohesive and responsive surveillance system. Standardisation ensures data consistency, comparability, and reliability for national disease monitoring and trend analysis. It also enables data aggregation from diverse sources, providing a comprehensive picture of disease patterns across the country.

Although many methods of laboratory confirmation are available, at the national level, only specific tests are offered based on national standards. The National Public Health Lab (NPHL) and the National Influenza Centre (Virology Department, University of the West Indies (UWI)) laboratories can provide most diagnostic tests required for disease confirmation in Jamaica. However, some sophisticated diagnostic tests that are unavailable will be sent to the appropriate reference laboratory per the existing laboratory protocol and standard operating procedures (SOPs).

Laboratory Documents

The current Jamaica laboratory investigation form can be found in [Appendix 9](#).

Chapter 7: The National Surveillance Committee and Response Teams

The surveillance response teams will be parish-based to ensure timely response to disease outbreaks. In the event of a multi-parish or multi-region outbreak, regional and/or national teams will be constituted under the leadership of the Regional Technical Director (RTD) or National Epidemiologist, as the case may be. This national or regional team will use the core of the parish response team that exists in the parishes.

Members of a Parish response team:

- Medical Officer of Health
- Medical Officer
- Senior Public Health Nurse/Public Health Nurse
- Chief Public Health Inspector/ Public Health Inspector
- Parish Surveillance Nurse/ Officer
- Epidemiological Clerk/ Data entry personnel
- Community Health Aide
- Contact Investigator
- Health Education Officer
- Laboratory Technologist
- Administrative Staff for Procurement
- Others will be co-opted as deemed necessary.

Members of a Regional response team:

- Regional Medical Epidemiologists
- Regional Epidemiologist/ Surveillance Officers
- Regional Nursing Supervisor
- Regional Environmental Health Officer
- Regional Health Education Officer
- Director of the Regional Laboratory
- SMO, Regional Hospitals
- Parish MOs(H)
- Parish Senior Public Health Nurse

- Parish Chief Public Health Inspectors
- Others will be co-opted as deemed necessary.

The Regional Technical Director will give oversight.

Members of the National Response team:

- National Epidemiologist
- Director, Emergency Disaster Management and Special Services
- Director, Health Promotion and Protection Branch
- Regional Technical Directors
- Senior Medical Officers (Health)
- Chief Nursing Officer
- Medical Epidemiologist, National Surveillance Unit
- National Surveillance Officers
- Regional Medical Epidemiologists
- Parish MOs(H) [of affected parishes]
- Director, Environmental Health Unit
- Director of National Public Health Laboratory
- Consultant Microbiologist, NPHL
- Consultant Microbiologist, UHWI
- Others will be co-opted as deemed necessary.

Terms of Reference for Response Teams:

- The parish response team will take the lead in all outbreak investigations confined to that parish.
- The regional response team will take the lead if more than one parish within that region is involved.
- The national response team will take the lead if more than one health region is involved.
- The regional response team will collaborate with the parish response team in all outbreaks by providing guidance and additional support for staff if needed.
- The national response team will collaborate with the regional and parish response teams by providing guidance and additional support if needed.
- In all outbreaks involving exotic diseases or diseases of international concern, the national response team shall take the lead irrespective of the parish or region where the outbreak is occurring.

- The response team will meet as frequently as necessary, but not less than weekly, to review communicable disease data and respond appropriately until the situation has been resolved.
- The parish response team will primarily be responsible for active case detection, investigation and implementation of control measures.
- The regional response team will be responsible for developing regional outbreak preparedness and response plans, enabling timely resource mobilisation from the central level or partners and a prompt response in the event of an outbreak declaration.
- The national response team will be responsible for reviewing and consolidating regional plans, as well as mobilising resources for the implementation of regional plans.

National Advisory Committee on Surveillance

The National Advisory Committee on Surveillance will meet monthly at the Extended Surveillance Meeting to review data and other matters pertaining to the epidemiological surveillance of diseases and health events in Jamaica and across the world. A similar Committee should be established at the regional and parish levels.

The objectives of the meetings are to:

1. Monitor trends in communicable diseases and selected health-related events
2. Monitor the performance of the national surveillance system
3. Detect outbreaks and gauge appropriateness of response
4. Estimate the magnitude of public health problems
5. Identify opportunities for strengthening the public health surveillance system, including best practices at all levels.

The recommended attendees at the Extended Surveillance Meeting include:

- National Epidemiologist
- Medical Epidemiologist, NSU, MOHW
- PAHO Advisor on Epidemiology/Surveillance
- Surveillance Officers, NSU, MOHW
- Regional Medical Epidemiologists
- Surveillance Officers, Health Region and Parish
- Medical Officers of Health

- Public health nurses, including Hospital Active Surveillance nurses
- Port Health Staff
- Food Safety Specialists
- TB Coordinator
- National Environmental Health Specialists
- Family Health Unit representatives
- Non-communicable Diseases and Injuries Prevention representative
- International Health Regulations representative
- National HIV/ STI Programme representative
- Microbiologist, NPHL
- Head of Department, Microbiology, UHWI laboratory
- Director, Veterinary Public Health, MOHW
- Veterinary public health representatives from the Ministry of Agriculture, Fisheries and Mining
- Others (including individuals from other ministries, departments, and agencies) will be co-opted as deemed necessary.

Extended Surveillance Meetings will be held on the third Thursday of each month except for the months of August and December. Information presented from all the parishes and health regions in Jamaica will be discussed. Partner Ministries, departments or agencies such as the Ministry of Agriculture, Fisheries and Mining will also present at these meetings and further collaboration advanced.

- The meeting will be chaired by the Medical Epidemiologist in the National Surveillance Unit or designate.
- The meeting minutes should clearly reflect task assignments, timelines, and any other issues discussed related to surveillance.

This meeting should include training and updates on topical and/or relevant areas in surveillance.

Epidemiology Meeting Guidelines

Monthly epidemiology meetings are essential avenues for discussing and disseminating information related to diseases in the country, region, and parish. To better standardise these meetings throughout Jamaica, guidelines have been developed.

- It is recommended that epidemiology meetings be held monthly to determine the burden of notifiable diseases and health events in the parishes, as well as for early detection of changing trends and outbreaks. These meetings should also serve as a forum for field staff to discuss investigations of diseases and the control measures implemented. The MO(H) should chair these meetings.
- Parishes/ regions are expected to have visual representations of data e.g. line graphs, bar charts, and spot maps for diseases, events and syndromes of significance to them e.g. gastroenteritis, typhoid, tuberculosis, ophthalmia neonatorum, etc. This is in order to facilitate early detection of rising numbers.
- These should be updated weekly or more frequently as the situation demands, for example, in outbreaks.
- Registers of all Class I diseases notified should be kept (by disease) to enable quick access to details on the cases, e.g. date of onset, laboratory investigations and reports, final classifications, outcomes etc.
- A list of diseases notified, date of notification and persons (PHI/PHN) responsible for the investigation should be done monthly and taken to each meeting.
- Copies of the weekly parish surveillance reports submitted to the NSU for the previous four weeks (or a summary) should be circulated and discussed at the meetings.
- At each meeting, the following should be discussed:
 - Any problems with the collection, investigation or submission of these reports
 - Line graphs and spot maps of communicable diseases/ health events and their trends
 - Review of outstanding investigation reports
 - Overview of completed investigation reports.
- A list of actions required should be prepared for dissemination to all concerned at all levels.
- This meeting also provides an ideal opportunity for training and updates on topical and/or relevant issues in surveillance.

An accurate record of all reports, presentations, and discussions shared during the meeting should be maintained and circulated to the meeting participants.

Chapter 8: Public Health Surveillance Investigations

Two types of investigations are related to public health surveillance and response to communicable diseases or health-related conditions. These are:

- A. Case Investigations
- B. Outbreak Investigations

Case Investigations

Case investigation refers to actively looking into and gathering detailed information about a reported case of a disease or health event being monitored under public health surveillance. The goals of case investigation are to:

- Ensure early detection of emerging infectious diseases or potential outbreaks
- Understand the case's characteristics
- Identify potential source(s) of infection and contributing factors for health events
- Inform preventive measures and control strategies.

Case Investigation can be triggered by:

- Notification of cases
- Alerts from health practitioners
- Public
- Rumours
- Media
- Research activity.

There are standardized procedures for public health authorities to follow when investigating cases to control the spread of disease or to respond to health events and threats.

What is a case definition?

In public health surveillance, a case definition is defined as a standard set of criteria for deciding whether an individual should be classified as having a health condition of interest.

The case definition:

- Includes information relating to person, place and time
- Includes signs and symptoms
- Should differentiate between suspected, probable or confirmed cases of a disease or health event
- Should differentiate between cases that are confirmed clinically, by laboratory, or by epidemiologic linkage.

Specifics related to the investigation of a disease or health event are adjusted based on the specific disease(s) or health event under surveillance or the public health context.

To capture all cases, the case definition should be broad, with minimal criteria for exclusion. Many investigations start with a broad case definition, which becomes more precise as the investigation proceeds.

Table 8.1. *Case Classifications*

Case Category	Description
Confirmed	Laboratory confirmation of agent, confirmed clinically, or by epidemiologic linkage
Probable	Typical clinical features of the illness AND Partial laboratory results (confirmation pending) OR Epidemiologic link to a laboratory-confirmed case
Suspected	Typical clinical features of the illness AND Missing laboratory and epidemiological information

An individual's case classification can change throughout the investigation. For example, an individual may be first classified as a suspected case but reclassified as confirmed as laboratory information becomes available.

Case Searches

It is important to actively search for similar cases, especially outside the sentinel surveillance system. Two scenarios can be considered:

- (i) If other cases are detected, an assessment of the epidemic risk or risk of exposure to the population will determine whether the investigation should then be equivalent to an outbreak investigation.
- (ii) If no other cases are detected, the investigation will then focus on ruling out or better understanding a potentially emerging infectious disease, together with a close assessment of the related epidemic risk or potential event of public health and/or international concern.

If investigated cases are found to be isolated cases for which no diagnosis has been made, further laboratory investigations should be pursued in keeping with available information, technology and resources at the time. Finally, when no epidemic risk is identified (e.g., isolated or unrelated cases of a known aetiology occurring in a low-risk environment), the Surveillance Team should still ensure that adequate and close monitoring is maintained to detect any significant changes in the epidemiological situation.

The availability of an aetiologic (laboratory) diagnosis or agent at the time of the case investigation will determine what objectives to aim for and the appropriate protocol to follow.

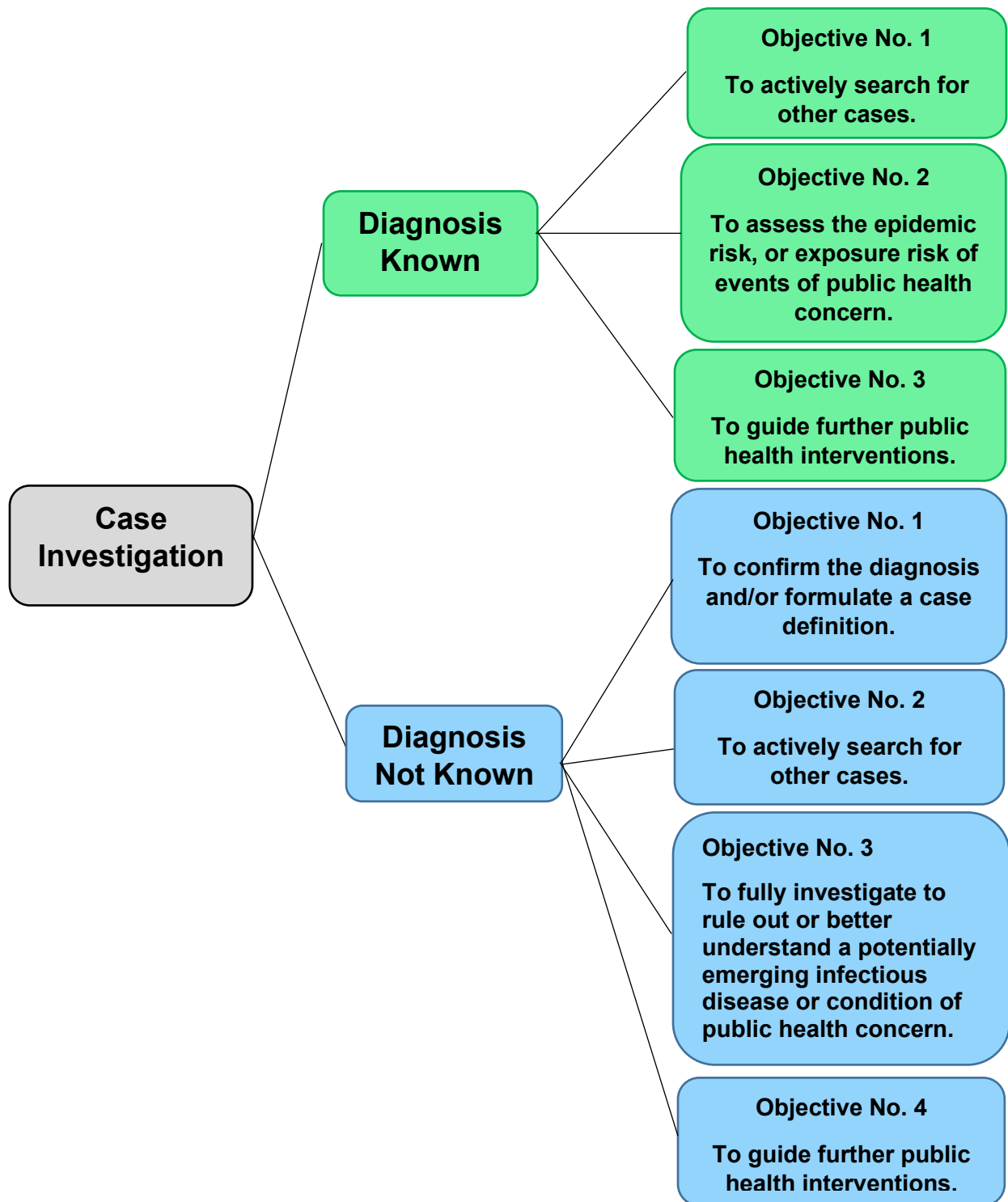


Figure 8.1. Case Investigation Activities by Availability of Aetiologic Diagnosis

Diagnosis Known:

Objective No. 1 - To actively search for other cases

Activities:

- a. Review existing recent epidemiological data from health centres.
- b. Actively collect the missing information from non-reporting health centres.
- c. Further seek reporting physicians' judgment about similar cases.

Objective No. 2 - To assess the epidemic risk, or exposure risk of events of public health concern

Activities:

- a. Investigate around the index and other existing cases
- b. Collect samples appropriately, according to the situation (i.e. clinical, environmental, animal).

Objective No. 3 - To guide further public health interventions

Activities:

- a. Develop response plans and implement control measures that would be guided by information obtained through activities under objectives 1 and 2 above.

Diagnosis NOT Known:

Objective No. 1 - Confirm the diagnosis and/or formulate a case definition

Objective No. 2 - To actively search for other cases

Activities:

- a. Review existing recent epidemiological data from health centres
- b. Actively collect the missing information from non-reporting health centres
- c. Further seek reporting physicians' judgment about similar cases.

Objective No. 3 - To fully investigate to rule out or better understand a potentially emerging infectious disease or condition of public health concern

Activities:

- a. Consult partners for assistance e.g. PAHO/WHO, CARPHA, CDC
- b. Collect samples to facilitate further testing

Objective No. 4 - To Guide further public health interventions

Activities:

- a. Develop response plans and implement control measures that would be guided by information obtained through activities under objectives 1, 2 and 3 above.

Outbreak Investigation

An effective outbreak investigation is essential for protecting public health, minimising morbidity and mortality, and preventing future outbreaks. A systematic approach to identifying the source and cause of an epidemic, implementing control measures, and communicating findings ensures a timely and coordinated response. Routine surveillance and reports from event-based surveillance data are critical in detecting outbreaks early and providing evidence to guide public health actions.

What is an outbreak?

An outbreak or epidemic refers to cases of illness, health-related behaviour, or other health events in a community or region clearly in excess of what is normally expected. While the terms outbreak and epidemic are often used interchangeably, "epidemic" typically conveys a greater sense of urgency and a wider spread. It is important to note that if a disease is not currently endemic to Jamaica, especially if it was previously eradicated or eliminated locally over five years and appears in a single case, it is considered an outbreak. Outbreaks can have serious consequences, not just for those directly affected but for the entire community. Determining an outbreak involves factors such as epidemic thresholds, seasonality, and regional spread for diseases regularly occurring in the country (endemic conditions).

Outbreak Investigation can be triggered by:

- The monitoring of trends of diseases or reported syndromes from the sentinel sites
- Notification of cases
- Alerts from health practitioners
- Media reports
- Public
- Rumours.

How are outbreaks determined?

The Chief Medical Officer (CMO) is responsible for advising the Minister of Health and Wellness on the detection of an outbreak in Jamaica, based on evidence from the surveillance system, as presented by key technical officers, including the National Epidemiologist, Regional Technical Director, and the Medical Officer of Health at the parish level. All suspected outbreaks should be reported immediately to the Medical Officer of Health and the NSU. The criteria for declaring an outbreak varies for different diseases.

Normal operations cannot continue during an outbreak. Surveillance must be intensified, and cases should be reported daily or in real-time. Continuing laboratory testing is not cost-effective once the cause of the outbreak is known unless it is needed for patient management. Once an outbreak has been declared and the aetiology is known, testing can be reduced to 10% of the cases that fit the case definition, and all severe cases, including deaths. Once cases have reduced by about 75%, general testing can resume until it is determined that the outbreak is over.

The CMO will advise the Minister of Health and Wellness on the end of an outbreak when the public health threat has been reduced. After the outbreak ends, surveillance activities can return to normal.

Outbreak Investigation

Outbreak investigation identifies the source, cause, and spread of an outbreak of illness or health-event. It involves systematically collecting and analyzing data to determine the nature and extent of the outbreak, identify risk factors, and implement control measures to stop the spread of the disease.

The Four Goals of an Outbreak Investigation are:

- I. **Case management** – this activity aims to minimize the effects of the disease causing the outbreak or to minimize the occurrence of severe morbidity and mortality.
- II. **Containment of infection** – which aims to break the chains of transmission and prevent further spread of infection in the local, national and international environment.

- III. **Active search for new cases** – this is to monitor the development of the outbreak and assess the effectiveness of control measures being implemented. This may be achieved through the dissemination of case definitions to all health workers or by visiting hospitals and/or clinics to examine medical records.
- IV. **Protection of susceptible individuals** – this is the identification of risk factors and populations in danger of contracting the disease and using methods (e.g. immunization, public education) to protect these groups from becoming infected.

The primary objectives of an outbreak investigation include:

- (i) **Confirming the outbreak:** Determine if the observed number of cases exceeds normal expectations for the disease or health event.
- (ii) **Identifying the source and cause:** Pinpoint the factors responsible for initiating or spreading the outbreak.
- (iii) **Implementing control measures:** Minimize further transmission and reduce the impact on public health. This may include vaccines, antibiotics, oral rehydration solutions or non-pharmaceutical interventions.
- (iv) **Communicating findings:** Inform stakeholders, including the public and health authorities, to support further action and awareness.

When planning the activities to be conducted during the investigation, you must find a balance between what is ideal and what is achievable, between what is needed and what you can provide and afford. This process typically includes ten (10) key steps that must be performed in a successful outbreak investigation. These steps are not necessarily conducted sequentially; often, more than one step is performed at the same time.

Ten Steps of Outbreak Investigation

1. Determine the existence of an Outbreak:

Compare number of cases to the expected number. If no past data available, knowledge and experience of local health staff is critical.

2. Assemble Team:

Identify investigation team and needed resources

3. Verify and Confirm Diagnosis:

Review clinical findings and lab results to verify diagnosis.

****Implement immediate control measures to contain outbreak*

4. Establish a case Definition:

Create clear and concise definition of a case, including symptoms, onset date, and other relevant factors.

5. Conduct Case Finding:

Identify and collect information from all possible cases, including demographic, data, exposure history, and symptoms. Develop line listing.

6. Develop Hypotheses:

Generate possible explanations for outbreak, based on the data collected and known epidemiology.

7. Test Hypotheses:

Conduct additional Investigations, such as environmental sampling, laboratory testing, or case-control studies, to confirm or refute the hypotheses.

8. Implement Control Measures:

Based on findings, recommend & implement actions to prevent further spread, such as isolation, quarantine, vaccination, or environmental interventions. Share results to inform response efforts & improve preparedness. Continue to track the outbreak and adjust control measures as needed until the outbreak is resolved.

9. Perform Descriptive Analyses:

Analyse the data to identify patterns and trends, such as time, place, and person characteristics.

10. Communicate Findings & Monitor:

Share results to inform response efforts and improve preparedness. Continue to track the outbreak and adjust control measures as needed until the outbreak is resolved.

Figure 8.2. Ten Steps of an Outbreak Investigation

Outbreak Investigation Team (Refer to [Chapter 7](#) above)

Investigating an outbreak is a team effort with each team member having specific functions. The National Response Team (see [Chapter 7](#)) has a crucial role. In Jamaica, the roles and functions of the response teams include:

Team Lead

- In parish outbreaks, MO(H) of the parish will be the lead investigator and may request help from the regional or national level.
- For multi-regional or national outbreaks, the National Epidemiologist (or designate determined by the Chief Medical Officer) will be the lead investigator.

Public Health Nurse

- Collect and collate data on cases and contacts during the time of the outbreak and past data so that disease events over time can be observed and reported on.

Environmental Health Officers / Public Health Inspectors

- Conduct site investigations and collect data and samples when appropriate. In an outbreak of food borne illness, these sites may be food establishments or hotels.

Health Educators / Promoters / Coordinators

- Planning of health promotion at national level and within the communities, and to work in collaboration with the Communications Unit.

Designated and Authorized Spokespersons

- Communicate with the media so that clear, consistent messages are delivered to the public. It is important that the public receives accurate information from the MOHW, which is not necessarily the type of information that sells newspapers, etc. (see MOHW Communication Policy and Guidelines)

Data Entry Support

- Support data entry, preliminary analysis of data as the investigation proceeds.

Statistical Analysis Support

- Analyze data, generate information and produce relevant tables, charts for reports.

GIS Support

- Analyze and display geographically referenced data as needed.

Laboratory Support

- Confirm the aetiological or causative agent responsible for the outbreak.

Clinical Specialist

- Provide advice and guidelines on diagnosis and protocols for patient care and management.

Depending on the availability of resources and the size of the outbreak, one team member may perform more than one of these roles. While each person has specific expertise, the team leader may assign other responsibilities within the context of investigating an outbreak.

Administrators, human resource and finance personnel are also critical to the process to meet with the team to discuss response strategies and to ensure that human and financial resources are available for the outbreak response.

Outbreak Report

Communicating the findings to inform response efforts and improve preparedness is crucial to any investigation. Team leads are responsible for ensuring that the report is produced in a timely manner, but the distribution of any report to the public is determined at the national level. An outbreak report form is in [Appendix 10](#). The following format can be used as an outline for an outbreak report.

Introduction

- Background
- Reason for investigation/ Rationale

Objectives of Investigation

Methods

- Dates of investigation
- Site(s) of investigation
- Case finding – indicate what was done regarding case finding
- Lab specimens collected
- Describe the response and intervention

Results

- Date and location of first known case (index case)
- Results of additional case finding
- Lab analysis and results
- Describe key features of results of time, place and person analysis
- Results of response and evidence of impact

Discussion

- Based on the result, describe the events leading to the outbreak
- Emphasize the lessons learnt from the incident
- Limitations of the investigation

Conclusion and Recommendations

- Emphasize the lessons learnt from the incident
- Make recommendations for action at each level: local/parish, intermediate/regional and national/ policy formulation level

References

Appendices

- Questionnaires
- Maps
- Investigation forms

Risk Communication Protocol

A key aspect of an investigation is risk communication. To ensure effective prevention and control while minimising panic, the key elements and risk factors must be communicated clearly and appropriately. Spokespersons are determined at the national level. Remember only one spokesperson at a time.

The three (3) paradigms to remember:

1. When people are insufficiently alarmed about a serious hazard, the task is to increase their concern and motivate them to take appropriate action.
2. When people are excessively alarmed about a small hazard, the task is to diminish their concern and deter them from unnecessary and potentially harmful actions.
3. When people are justifiably alarmed about a serious hazard, the task is to harness their concern and guide their actions.

At the Facility Level:

- The SMO or CEO should be the spokesperson.
- The target audience should be specific and confined to the facility that they serve.
- They should use a news medium that serves their target population.
- They must clearly state in their communication that the outbreak or health event is localised to their facility.
- They must not speak for the parish, region or the country.
- The message should be developed using recommendations in the MOHW communication plan.
- Generally, they must not speak on diseases or health events that have international and regional implications without first consulting with the MOHW. Some examples include Cholera, Yellow Fever, Malaria, Plague, SARS, outbreaks in a hotel, and outbreaks of vaccine-preventable diseases.
- A copy of official communication must be shared with the Parish, RHA and MOHW.

- When diseases or health events span across more than one facility, communication will be coordinated through the Parish, RHA or MOHW.

At the Parish Level:

- The MO(H) or Parish Manager should be the spokesperson. In some circumstances, this task can be delegated.
- The target audience should be specific and confined to the parish that they serve.
- They should use a news medium that serves their target population.
- They must clearly state in their communication that the outbreak or health event is localised to their parish.
- They must not speak for the region or the country.
- The message should be developed using recommendations in the MOHW communication plan.
- Generally, they must not speak on diseases or health events that have international and regional implications without first consulting with the MOHW. Some examples include Cholera, Yellow Fever, Malaria, Plague, SARS, outbreaks in a hotel, and outbreaks of vaccine-preventable diseases.
- A copy of official communication must be shared with the RHA and MOHW.
- When diseases or health events span across more than one parish, communication will be coordinated through the RHA or MOHW.

At the Regional Level:

- The RD/ RTD of the region should be the spokesperson. In some circumstances, this task can be delegated.
- The target audience should be specific and confined to the health region they serve.
- They should use a news medium that only serves their target population.
- They must clearly state in their communication that the outbreak or health event is localised to their health region.

- They must not speak for the other health regions or the country.
- The message should be developed using recommendations in the MOHW communication plan.
- Generally, they must not speak on diseases or health events that have international implications without first clearing with the MOHW. Some examples include cholera, yellow fever, Malaria, plague, SARS, and outbreaks of vaccine-preventable diseases.
- A copy of the official communication should be shared with the MOHW.
- When diseases or health events span multiple health regions, communication should be coordinated through the MOHW.
- At the conclusion of the event, an official statement should be issued to confirm its end and outline the next steps for future communications on the topic.

At the National Level:

- The Minister of Health and Wellness/ CMO/ Permanent Secretary/ National Epidemiologist should be the spokesperson. In some instances, this task can be delegated.
- The target audience is the entire Island.
- The national level should ensure that the communication is well coordinated with the regional and parish level spokespersons.
- Input from the parish and regional level should be encouraged and incorporated.
- At the conclusion of the event, an official statement should be issued to confirm its end and outline the next steps for future communications on the topic.
- The message should be formulated with input from the NSU and health personnel involved in surveillance, keeping the three communication paradigms in mind.

N.B: If you are in doubt about what to do, refer to the MOHW Communication Policy and Guidelines or call the MOHW.

Chapter 9: Event-Based Surveillance (EBS)

Event-based surveillance (EBS) refers to a system for rapidly detecting public health threats by collecting and analyzing unstructured, ad-hoc information from both formal and informal sources, such as media, community reports, and healthcare workers' observations. Event-based Surveillance is a critical enhancement to the existing indicator-based surveillance (IBS) system in Jamaica. While the National Surveillance Unit has always emphasised the need to be open to rumours and data from unofficial sources that may have public health implications, EBS builds on this and emphasises the organised collection, monitoring, and assessment of this information from diverse sources. Community-sourced signals described in [Chapter 3](#) are handled under the verification procedure in this chapter.

Objectives

The objectives of EBS integration into the national surveillance system are to:

- Enhance early detection of outbreaks, emerging diseases, and public health threats.
- Strengthen the capacity for timely and coordinated response to public health events.
- Improve the overall sensitivity and timeliness of the surveillance system.
- Facilitate rapid verification and assessment of potential health threats.

The EBS system should be integrated into the existing surveillance structure at the parish, regional, and national levels.

Roles and Responsibilities

1. Parish Team
 - a. Act as the primary EBS data collectors and first responders.
 - b. Investigate community-reported health events.
 - c. Implement initial containment and control measures as required
 - d. Report verified events to the regional and national levels.

2. Regional Team

- a. Provide technical support to parishes.
- b. Coordinate event responses.
- c. Evaluate EBS data for regional trends and threats.
- d. Report regional-level verified events to the national level.

3. National Team

- a. Monitor national EBS trends and signals.
- b. Conduct internet-based searches for public health-related signals.
- c. Lead national-level responses to major health events.
- d. Liaise with international health authorities.

Verification and Reporting

All EBS signals, regardless of source, must undergo verification before action is taken. These EBS signals include:

- All Class 1 notifiable conditions
- Any exotic or unusual health events
- Communicable diseases such as Anthrax, Legionnaire's Disease, Rabies, Severe Acute Respiratory Syndrome (SARS)
- Diseases of public health importance that are never/ rarely diagnosed in the country and/or linked to travel and trade (e.g. Rift Valley fever)
- Clusters of deaths in a health care facility, community, construction site, school or other institution over a period of two weeks
- Contamination of food or medicines
- Illness or death in reservoir animals
- Chemical contamination of air, soil, or natural water
- Excess exposure to radiation (e.g. faulty X-ray machine, inadequate radiotherapy)
- Radiation or nuclear exhaust
- Natural or man-made disasters or risk of occurrence.

The verification process involves:

- Cross-checking reports with existing health data.
- Conducting preliminary field investigations when necessary.
- Consulting with healthcare providers and community informants.

Verified events should be prioritized based on urgency and potential public health impact, which guides the level of response required. All signals detected, the source(s), status of rapid risk assessment (discounted or verified and level of urgency) along with any public health interventions taken should be documented in a register at the parish level.

The number of signals detected, and number verified should be reported on the Weekly Parish Surveillance Report. Where no signals were detected, this should also be noted. Details of verified incidents should be documented and reported.

Training and Capacity Building

Training will be integrated into existing capacity-building initiatives for surveillance teams. Areas of focus will include:

- Identifying and evaluating EBS signals
- Conducting rapid assessments
- Coordinating multi-level responses.

Refresher training sessions will be conducted to ensure continued proficiency.

Monitoring and Evaluation

The effectiveness of EBS integration will be assessed through existing surveillance monitoring and evaluation mechanisms. Key evaluation performance indicators will include:

- Timeliness of signal detection and verification
- Proportion of verified signals leading to public health actions.

The following metrics will be reviewed during monthly epidemiology meetings at all levels to ensure the effectiveness and responsiveness of EBS within the broader surveillance system.

Teams will:

- Review EBS signals detected in the previous month
- Evaluate response actions taken
- Discuss challenges and best practices.

The means of verification for documentation of EBS activities include:

- Parish registers or logs
- Weekly Parish Surveillance Report
- EBS reports
- Minutes of epidemiology/ surveillance meetings.

Indicator performance results will inform adjustments to EBS procedures, ensuring continuous improvement.

Chapter 10: Disaster Surveillance

Introduction

The United Nations Office for Disaster Risk Reduction (UNDRR) defines disaster as a severe disruption of the functioning of a community or a society at any scale due to hazardous events interacting with conditions of exposure, vulnerability and capacity, leading to one or more of the following: human, material, economic and environmental losses and impacts. Disasters can be either natural or man-made and include events such as hurricanes, tropical storms, floods, earthquakes, volcanic eruptions, industrial accidents, fires and acts of terrorism, including bioterrorism ¹. Widespread epidemics of serious diseases, pandemics and events of public health concern may also be viewed as disasters and managed as such.

During a disaster, surveillance is important to determine the extent and scope of health impacts on those affected. Many communicable diseases thrive in post-disaster climates, along with disrupted public utilities and health services, and the potential for a large-scale outbreak becomes a very real possibility. As the risk of communicable disease spread is heightened following a disaster, there is a need for intensified and enhanced surveillance. Additionally, there can be an increase in disaster-related injury and illness due to displacement, and difficulty managing chronic non-communicable diseases.

Disaster surveillance enables public health systems to identify adverse health effects in the community, and to track morbidity, mortality, disease trends and risk factors. It provides information to determine action items, and tailor interventions to prevent or reduce injury, illness, or death and guide prevention strategies for future disasters.

¹ United Nations Office for Disaster Risk Reduction (UNDRR). *The Sendai Framework Terminology on Disaster Risk Reduction - "Disaster"* [Internet]. 2017. Available from: <https://www.undrr.org/terminology/disaster>.

Disaster management is organising resources and responsibilities to reduce the impact of disasters. The disaster management cycle typically includes mitigation/ prevention, preparedness, response, and recovery phases. Disaster surveillance is a crucial component of the disaster management cycle, acting as the "eyes and ears" to monitor potential hazards and early warning signs of a disaster (preparation / pre-disaster), allowing for timely intervention (during and post-disaster response) and preventative measures during the pre-disaster phase.



Figure 10.1. Overview of Key Stages of Disaster Management

Preparation	Response	Recovery	Mitigation
• Ensure access to routine surveillance data from pre-disaster periods. • Identify contingency mechanisms to maintain two-way communication across the surveillance system.	• Coordinate surveillance activities through established emergency operation centres across parish, regional & national levels • Daily reporting from sentinel sites, including shelters	• Rapid assessments to determine the extent of damage • Continuous daily reporting, analysis and publication of data received • Maintain routine surveillance activities in unaffected areas	• Routinely review and update emergency plans. • Address deficiencies in preparedness plans. • Ensure emergency protocols are updated and disseminated as necessary

Figure 10.2. Key Surveillance Actions at Each Stage in Disaster Management

Preparation (Pre-Disaster)

In the preparation phase, ensure access to routine surveillance data from the pre-disaster period, and prepare for communication contingencies. With disaster surveillance reporting, emphasis should be placed on the system's sensitivity to detect minor changes in disease occurrence so that analysis and appropriate action can be taken immediately.

Well-defined and mutually understood relationships must be established between the health network and the national coordinating agency for disaster preparedness and response. This is usually coordinated through established Emergency Operations Centres (EOCs) at parish, regional and national levels. A surveillance coordinator should be assigned, and this individual should report directly to the EOC manager.

The location(s) of the Emergency Operations Centre(s) depends upon:

- Extent of the disaster, local or nationwide.
- Communication facilities with special emphasis on contingency telephone and/or radio links with national coordinating agencies and field reporting units. Computer links could be beneficial where these exist and are not interrupted by the disaster. Maintaining the rapid two-way flow of information between the peripheral and the central level is crucial as critical and urgent decisions will need to be made.

Additionally, routine surveillance data in pre-disaster periods are essential in assessing the risk of communicable diseases in communities and nationally. The risk of a particular disease resurging following a disaster depends on many factors, one of which is the endemic level of that disease. Routine data should thus be used as a baseline for post- disaster surveillance activities. This data should be readily assessable for scrutiny following a disaster.

Response (During and Immediately After a Disaster)

Activities in the response stage enable the authorities to take swift action during and immediately after a disaster. The public health teams should conduct surveillance activities at each shelter and report to the parish health department. Daily information on how many people reside in shelters or seek attention at a health facility is required.

Reporting

Surveillance during disaster periods should be based on existing systems with minimum essential modification. Reporting may necessitate limiting the number of diseases under surveillance in disaster zones, becoming more flexible regarding diagnostic criteria in laboratory work, and relying on reporting symptom complexes (syndromic reporting).

- Timeliness of reporting is essential. Since the situation is changing daily, daily reporting is necessary, and multiple reports, such as shelter reports, may be required during the response period. The collection of reporting forms should be organised daily (see Post-Disaster Daily Shelter Surveillance Form and Report in [Appendix 11 and 12](#)).
- Completeness of data may not be feasible in disaster situations. However, it is important that the data that is collected can be interpreted as overall indicators on which appropriate and effective public health interventions can be based.
- Where there are no cases of a disease, event or syndrome, zero (0) should be reported, to differentiate between no cases and a lack of reporting.
- Information, reports and “rumours” arising from informal channels should not be ignored. Action should be taken to verify the source and reliability of the information, confirm veracity, and institute necessary measures where indicated.
- Collaboration with other units to ensure monitoring activities extend beyond disease occurrence to include other conditions with public health implications, e.g., information on the status of water supplies and food safety.

In addition to the morbidity reporting of disease occurrence, mortality surveillance is also critical. During a disaster, it tracks deaths related to the disaster, which can help determine the disaster's impact and guide public health responses. It is an important indicator of a disaster's gravity. Identifying the leading cause(s) and circumstances of death can help guide immediate and future prevention strategies. Mortality surveillance, therefore, aims to identify the number of deaths related to the disaster, determine the causes of death, assess the impact of the disaster, inform future planning and mitigation efforts, and support public health decisions. During a disaster, deaths should be reported using the Class 1 Reporting Form (see [Appendix 7](#)) or as instructed by the National Surveillance Unit.

Data from investigations and field reports should be analysed, and the findings should be published in an official daily or weekly report.

Recovery (Post-disaster)

The objective of the recovery phase is to restore normalcy as soon as possible and rebuild after a disaster.

Assessment of Damage and Subsequent Disease Potential

Damage assessments should not wait until detailed reports on specific locales are received; a rapid assessment should be initiated as early as possible. As far as possible, information should be displayed on wall maps and updated as new information becomes available. At the local level, a rapid assessment of the extent of damage should place special emphasis on:

- Communications
- Roads, including state bridges
- Telephone links
- Health facilities
- Areas flooded
- Water supply systems
- Sewerage systems
- Solid waste disposal systems.

Epidemiologic factors which influence the potential risk of communicable disease transmission after a disaster include:

- Changes in pre-existing levels of disease
- Ecological changes resulting from the disaster
- Population displacement (persons in shelters, etc.)
- Changes in population density
- Disruption of public utilities
- Interruption of basic public health services.

Most prominent are the influences on the modes of transmission of communicable diseases. For example:

- Crowding in evacuation centres (shelters) can increase transmission of diseases caused by respiratory and person-to-person spread.
- Tropical depressions and hurricanes can cause flooding, increase contact with contaminated water and increase the risk of a leptospirosis outbreak or skin infection. Leptospirosis is an acute and often severe bacterial zoonotic disease that frequently affects the liver and other organs.
- Flooding can damage or overwhelm water treatment plants, pumping stations, and distribution mains, disrupting or contaminating supplies and increasing the risk of gastrointestinal or foodborne illnesses.
- Stagnant water following floods provides fertile breeding grounds for several vectors, such as mosquitoes, which can contribute to a dengue outbreak.

Routine surveillance in non-disaster areas should not lapse, as outbreaks in regions unaffected by the disaster may still occur. In addition, people from the disaster areas may move to other regions while incubating an infection.

Mitigation

The mitigation phase of the disaster management cycle is about reducing or eliminating the risks and impacts of disasters before they happen. It focuses on preventing future emergencies or minimising their effects when they occur. While special needs may be peculiar to certain types of disasters, both in terms of surveillance activities and public health action, there are common basic areas which must be addressed:

- The existence of a health sector plan for disaster preparedness is assumed, and it should be reviewed and updated annually ¹ (see Jamaica's Comprehensive Disaster Risk Management Policy and Strategy 2020-2040). If any deficiencies are identified, they should be addressed in the interim, during simulation exercises, or in an actual disaster situation.
- Provisions allowing ready access to baseline and other data, including reference maps.
- Clear guidelines for what to report and how. This should include reports from the Event-based Surveillance System.
- Guidelines and resources for the appropriate analysis of the collected surveillance data.
- Mechanisms for feeding field information to the command centre with provisions to cater for breakdown in standard communication systems.
- Appropriate feedback provisions.
- Backup laboratory services, the use of which should be rationalised.
- Suitable field equipment for monitoring and recording essential surveillance data and collecting and transporting clinical and environmental specimens.
- Inputs from Epidemiologists at both the planning and field operations stages.
- Suitable mechanism for disseminating information and advice to the public.

¹ Ministry of Local Government and Rural Development, Jamaica, Office of Disaster Preparedness and Emergency Management, *Jamaica's Comprehensive Disaster Risk Management Policy and Strategy 2020-2040* [Internet]. (July 2022). Available from: [chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.odpem.org.jm/wp-content/uploads/2024/08/Jamaicas-Comprehensive-Disaster-Risk-Management-Policy-and-Strategy-2020-2040.pdf](https://www.odpem.org.jm/wp-content/uploads/2024/08/Jamaicas-Comprehensive-Disaster-Risk-Management-Policy-and-Strategy-2020-2040.pdf)

Chapter 11: Tourist Establishment Surveillance

Tourism plays an important role in the economic development of Jamaica, but it also presents a risk for transmission of communicable diseases, such as emerging and re-emerging diseases and foodborne illnesses. Tourist Establishment Surveillance is therefore integral to the control of public health threats associated with tourism. Early detection, notification, and response to disease outbreaks are vital to securing and maintaining Jamaica's tourism industry. Effective collaboration among key stakeholders—such as the MOHW, RHAs, Parish Health Departments, Ministry of Tourism, hotel management, hotel nurses, laboratories, and international organizations (e.g., PAHO/WHO, CARPHA, CDC)—is necessary to ensure the prompt detection and reporting of communicable diseases of interest.

The surveillance process at tourist establishments begins with the reception personnel collecting information on guests' permanent residence and other demographic information. Upon arrival, guests should be informed of the location of the nurse's station and encouraged to visit or contact the healthcare provider onsite if they feel ill at any time during their stay. The tourist establishment nurse represents one of the first lines of contact for ill staff or guests. All staff at tourist establishments should be vigilant and report any signs of communicable disease among guests and staff.

Tourist establishments are routinely monitored to ensure compliance with relevant public health regulations including disease surveillance. In Jamaica tourist establishment surveillance has two (2) key components, the weekly reporting of syndromes and reporting on suspicion or confirmation of any outbreaks of communicable disease or foodborne illness occurring at a tourist establishment.

Routine Weekly Reporting of Syndromes and Other Health Events

Weekly surveillance reports are to be submitted by hotels to the parish health departments. The hotel manager or nurse should use a standardised form for reporting syndromes and health events under surveillance, such as fever and respiratory symptoms (ILI), fever and rash, diarrhoea/gastroenteritis, and injuries ([Appendix 13](#)). This is a requirement of the Public Health (Tourist Establishment) Regulation. Syndromic data is reported in aggregate numbers. In aggregate reporting, zero reporting ensures that the absence of cases is actively reported, preventing gaps in surveillance data and helps to distinguish between no cases and a lack of reporting.

Data reported from hotels is collated and analysed at the parish level, and a weekly parish surveillance summary report is submitted to the RHAs and NSU simultaneously. The parish summary report must be submitted on a standardized form. The data from these reports are further analysed at the regional and national levels to monitor trends and respond accordingly.

Reporting of Communicable Diseases and Foodborne Illness Outbreaks

Where there is suspicion of a communicable disease or foodborne illness outbreak at a tourist establishment, a report shall be made to the parish Medical Officer of Health or parish health department immediately or within 24 hours by the hotel manager or the nurse on duty. In the event that the outbreak is noted prior to the routine time for reporting, the nurse will notify the health department verbally and should not wait on the routine reporting time. This is important as an outbreak can spread due to delayed implementation of control measures resulting from late reporting.

Upon receiving the notification, the health department mobilises the outbreak response team to launch an investigation in collaboration with the hotel team. The health department shall submit a preliminary investigation form and report to the RHAs and NSU simultaneously ([Appendix 10: Outbreak Investigation Form](#)). The outbreak investigation steps should be applied according to

the situation to ensure that all necessary details are acquired and that timely interventions are made. All the necessary preventive and control actions should be employed to prevent spread to the susceptible population. The laboratory should be notified as their services may be necessary to identify pathogen(s).

Chapter 12: Port Health Surveillance

The points of entry are the first contact with visitors who may be travelling from countries with diseases that are not endemic to Jamaica. The International Health Regulations and the Quarantine Act provide specific requirements to decrease the risk of entry and spread of diseases. Port Health Surveillance facilitates the recognition of infectious or potentially infectious persons (especially those with diseases of international and local concern), prompting the implementation of appropriate response interventions, such as quarantine and isolation, when necessary. In order to support these processes, cooperation, participation and high levels of communication are needed from key stakeholders such as airport authorities, airline management staff, immigration officers, public health personnel and other relevant agencies or institutions.

Surveillance at Points of Entry

The surveillance system at points of entry is designed according to the International Health Regulations (IHR) and adopts an “all hazards approach” to monitor biological, chemical, and radiological threats. The primary focus for surveillance includes travelers such as crew members, tourists, business travelers, and local residents. Additionally, surveillance extends to animals, plants, and goods from various origins. This comprehensive monitoring often involves collaboration with other governmental bodies or agencies to ensure a thorough and effective response to potential hazards. Based on the kinds of events occurring at points of entry, an interdisciplinary approach is required.

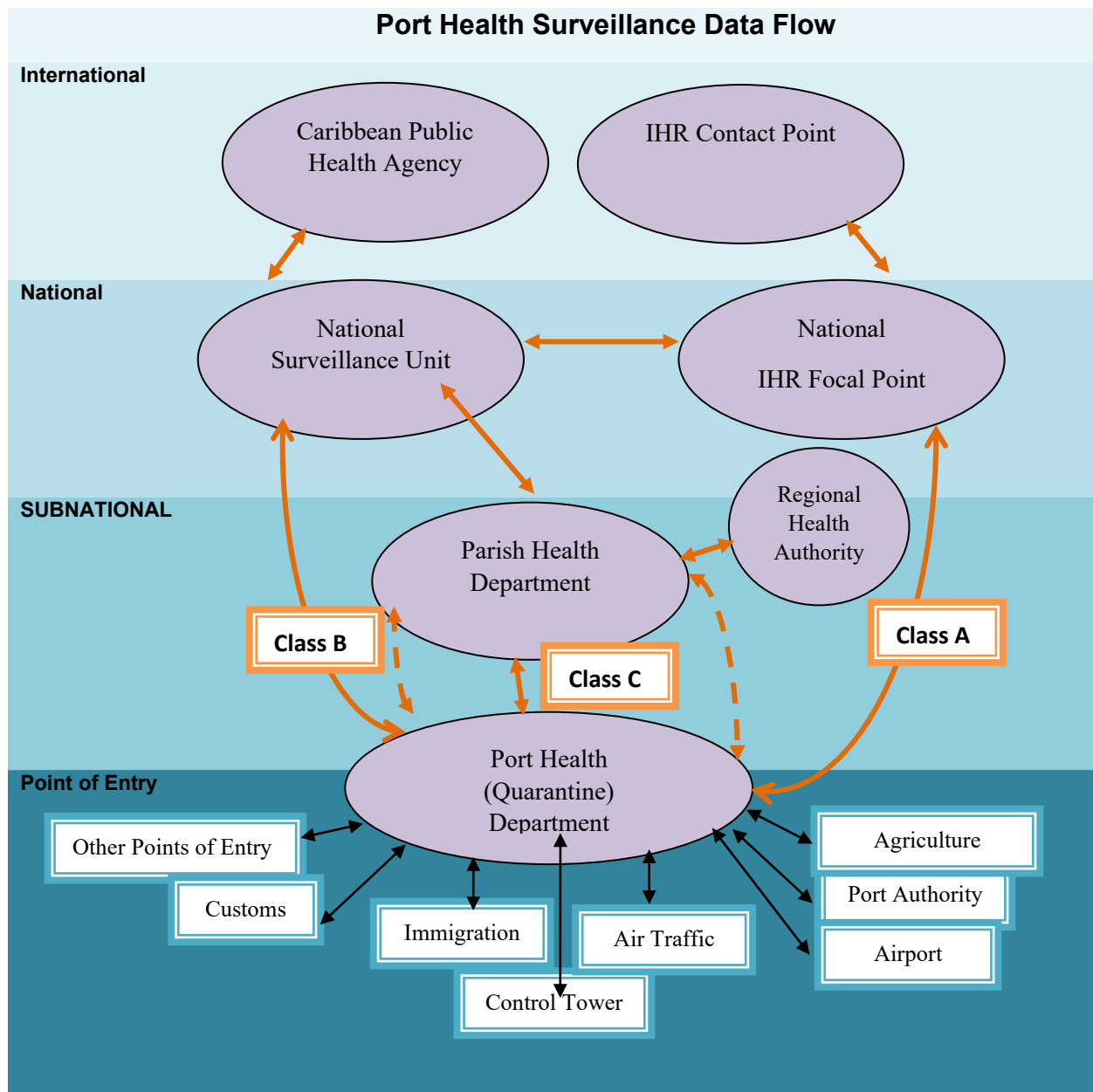


Figure 12.1. Event-Based Surveillance between Points of Entry and the National Surveillance System

Class A Event: Infectious diseases required by IHR, Biological, Chemical and Radiological Events
 Class B Event: Class 1 Notifiable Diseases including outbreaks
 Class C Event: Environmental events, stowaways/refugees and deaths (not included in classes A or B)

Roles and Responsibilities of the Public Health Team

The Public Health Team is responsible for conducting interviews with travellers who report feeling ill or display suspicious symptoms. They should acquire information about their recent travel history and collect contact information to facilitate follow-up if needed. In addition, travellers are to be advised to seek medical care if they begin to feel ill while in Jamaica (even if it is just a fever). The team is also to distribute health alert cards to travellers who visited countries that are endemic for communicable diseases of interest (see List of Malaria and Yellow Fever Endemic Countries – [Appendix 14](#)).

Immigration cards from the previous day are to be reviewed and line lists prepared of travellers arriving from endemic countries. Note the place of stay/address in Jamaica, arrival date, length of stay, and the endemic country/countries visited within the last six weeks. Submit this information promptly to the relevant Health Departments in all parishes where the travellers/guests are staying and to the nurses in the hotels (where applicable) where the guests are booked.

By law, all vessels are required to report notifiable diseases. The Maritime Declaration of Health or the Aircraft General Declaration must be examined prior to disembarkation of passengers.

Chapter 13: Surveillance of Non-communicable Diseases

Introduction

Non-communicable diseases (NCDs), also known as chronic diseases, tend to have a long duration and are the result of a combination of genetic, physiological, environmental, and behavioural factors. This group of conditions are not usually caused by acute infections but result in long-term health consequences and often create a need for long-term treatment and care. NCDs include conditions such as cancers, cardiovascular disease (CVD), diabetes, chronic lung illnesses, mental illness, arthritis, and injuries. However, some NCDs are caused by infections, such as cervical, vulvar, vaginal, penile, laryngeal, oral and anal cancers (human papillomavirus), liver cirrhosis (hepatitis B or C viruses) and stomach cancer (*H. pylori*)¹.

In 2021 NCDs accounted for 75% of non-pandemic related deaths (43 million) globally¹. The Americas had the highest increase in the total NCD deaths from 2010 – 2019 of all the WHO regions, with men having higher rates of death and disability. The Caribbean had higher disability-adjusted life year (DALY) rates for CVDs, cancers, and diabetes than any other sub-region in the Americas and was disproportionately affected². In 2022, approximately 9.74 million people died from cancer globally, with lung cancer being the leading cause, followed by breast cancer. The Caribbean recorded an estimated 66,364 cancer deaths³.

In Jamaica, over the past 50 years, NCDs have been the leading cause of death, premature mortality, morbidity, and disability. Between 2006 and 2021, on average 76% of all deaths were due to NCDs (excluding injuries), which accounted for 12% of all deaths. Cancers had the highest rates of increase, and Jamaica had 4,632 cancer-related deaths, with prostate cancer being the leading cause, followed by breast, colon, lung, and cervical cancers³.

¹ World Health Organization. *Noncommunicable diseases* [Internet]. (January 2025), Available from: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>

² Pan American Health Organization. *Leading causes of death and disease burden in the Americas: Noncommunicable diseases and external causes*. Washington, D.C.: PAHO [Internet]. (March 2025). <https://doi.org/10.37774/9789275128626>.

³ International Agency for Research on Cancer, World Health Organization. *Global cancer observatory* (2025) [Internet]. Available from: <https://gco.iarc.fr/en>

The prevalence of risk factors for NCDs continue to increase. The distribution of mortality, morbidity, disability, and risk factors is related to age and gender ¹. The NCD epidemic is also occurring in the context of emerging and re-emerging infectious diseases and climate change and worsening the trajectory of other disease outcomes ².

NCDs do not only have an impact on health systems but on development. The total economic losses for the period 2015–2030 due to all NCDs, including mental health conditions, are projected to be approximately USD 18.45 billion ³. Additionally, direct costs from violence related injuries for the Jamaican health sector was JMD 3.6 billion each year. This was approximately 22% of the recurrent hospital budget of the Ministry of Health and Wellness ⁴.

Most of the major NCDs can be prevented, onset delayed and managed using evidence- based, cost-effective public health interventions, such as the WHO Best-buys. As such, Jamaica has committed to meeting the Sustainable Development Goal (SDG) 3.4 (*to reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being*), and other associated SDGs ⁵.

Non-communicable Disease Surveillance

Non-communicable Diseases Surveillance is the ongoing systematic collection, management, analysis, interpretation, and dissemination of data to provide information on NCD disease burden and track NCD related health outcomes and risk factors over time ². The implementation of a National NCD surveillance system is an essential component of any programme and strategy for the prevention and control of NCDs and their risk factors.

¹ Ministry of Health and Wellness, *Vitals – NCD Edition* [Internet]. (April 2023), Available from: <https://www.moh.gov.jm/wp-content/uploads/2023/07/Vitals-NCDs-MOHW-April-2023-1.5.23.pdf>

² Banatvala, N., & Bovet, P. *Noncommunicable diseases: A compendium (1st ed.)*. Routledge. (Eds.). (2023). Available from: <https://doi.org/10.4324/9781003306689>.

³ Bloom, D. E., Chen, S., & McGovern, M. E. *The economic burden of noncommunicable diseases and mental health conditions: Results for Costa Rica, Jamaica, and Peru*. *Revista Panamericana de Salud Pública* (2018), 42, e18. <https://doi.org/10.26633/RPSP.2018.18>

⁴ Ward, E., Toppin, J., McCartney, T., Lyew-Ayee, P., McGaw, K., & Ashley, D. *Violence, health and development: Examining the costs of violence-related injuries to Jamaica*. *West Indian Medical Journal*. (2018), 67(Suppl. 4), 23.

⁵ World Health Organization. *Noncommunicable diseases* [Internet]. (January 2025), Available from: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>

Injury surveillance is a sub-component of NCD surveillance. ‘An injury is caused by acute exposure to physical agents such as mechanical energy, heat, electricity, chemicals, and ionizing radiation interacting with the body in amounts or at rates that exceed the threshold of human tolerance’¹. Injuries may be unintentional, intentional or violence related.

An unintentional injury is defined as: physical damage to the body; damage resulting from excessive energy applied to the body (physical, radiant, etc.); or from exposure to external agents (e.g., poisons); or from the absence of essentials (warmth, oxygen); the application, exposure, or deprivation not done deliberately to oneself or by another person. These injuries may occur by the following mechanisms: falls, road traffic crashes, water hazards, fire and hot liquids, and poisonings².

Violence is the intentional use of physical force or power, threatened or actual, against oneself, another person, or against a group or community, that either results in or has a high likelihood of resulting in injury, death, psychological harm, maldevelopment, or deprivation ³. Violence related injuries can be self-directed, interpersonal violence inflicted by another individual or by a small group of individuals; or violence inflicted by larger groups such as states, organised political groups, militia groups, and terrorist organisations. The four types of violent acts are physical violence, sexual violence, psychological violence, and violence involving deprivation or neglect.

These definitions inform the current injury surveillance system known as the Jamaica Injury Surveillance System (JISS). Jamaica has been monitoring the burden of injuries since 1996 using data from the accident and emergency departments at public hospitals.

¹ Espitia-Hardeman V, Paulozzi L. *Injury Surveillance Training Manual* [Internet]. Atlanta (GA): Centers for Disease Control and Prevention, National Center for Injury Prevention and Control (2005).

² Nolan, M. B., Wegner, M. V., & Remington, P. L. *Chronic disease prevention and control (5th ed.)*. Washington, DC: American Public Health Association Press (2023).

³ World Health Organization. *Noncommunicable diseases* [Internet]. (January 2025), Available from: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>

Use of the NCD Surveillance Information for Action

The NCD (including injuries) surveillance data have been used to inform national policies, laws, plans, programmes, resource allocation, and research within and outside of the health sector at national, regional, and global levels. This occurs through a complex network of data systems and collaborating partnerships.

Purpose and objectives of the National NCDs surveillance system

The purpose of the National NCDs surveillance system (NNCDSS) is to track the distribution of NCDs and their risk factors, and socio-economic determinants, including injuries and data on programme and policy performance in order to inform and prioritise prevention and control efforts and resource allocation. The COVID-19 pandemic was a reminder of the need for a robust, responsive, and integrated surveillance system that can generate information to aid in the development of an effective response to NCD public health threats.

The objectives of the NNCDSS are to:

Primary objectives:

- Track the prevalence, incidence, morbidity, mortality and the national rates of NCDs with focus on the priority diseases.
- Identify and track the behavioural, biological, environmental, socio-economic determinants and risk factors for NCDs.
- Establish, update, and maintain, as required, a standardised system for collecting, analysing, and reporting NCD-related data from multiple sources.
- Provide evidence-based data to inform national health policies, resource allocation and programmes.

Secondary objectives:

- Align the National NCD surveillance efforts with agreed international targets, goals, and commitments.

- Provide data for NCD-related epidemiological and public health research, and clinical studies.
- Evaluate the impact of the national NCD prevention and control programmes.
- Utilise surveillance data to develop and guide public education campaigns.
- Ensure standards are in place for data security and ethical use throughout the surveillance process in keeping with the data protection laws.

National Non-communicable Disease Framework

The Ministry of Health and Wellness (MOHW) collects data on NCDs, guided by the framework in Figure 13.1, through a complex network of several surveillance systems using multiple data sources that are within the network of the public health sector, private health sector and outside of the health.

These are the main disease/conditions and risk factors under surveillance:

- Priority diseases/conditions: diabetes, cardiovascular disease, cancers, chronic respiratory diseases, sickle cell disease, injuries, mental health conditions, chronic kidney diseases, eye diseases, and oral diseases.
- Priority risk factors: obesity, unhealthy diet, air pollution, physical inactivity, harmful use of alcohol, human papilloma virus, rheumatic fever, and hepatitis B.

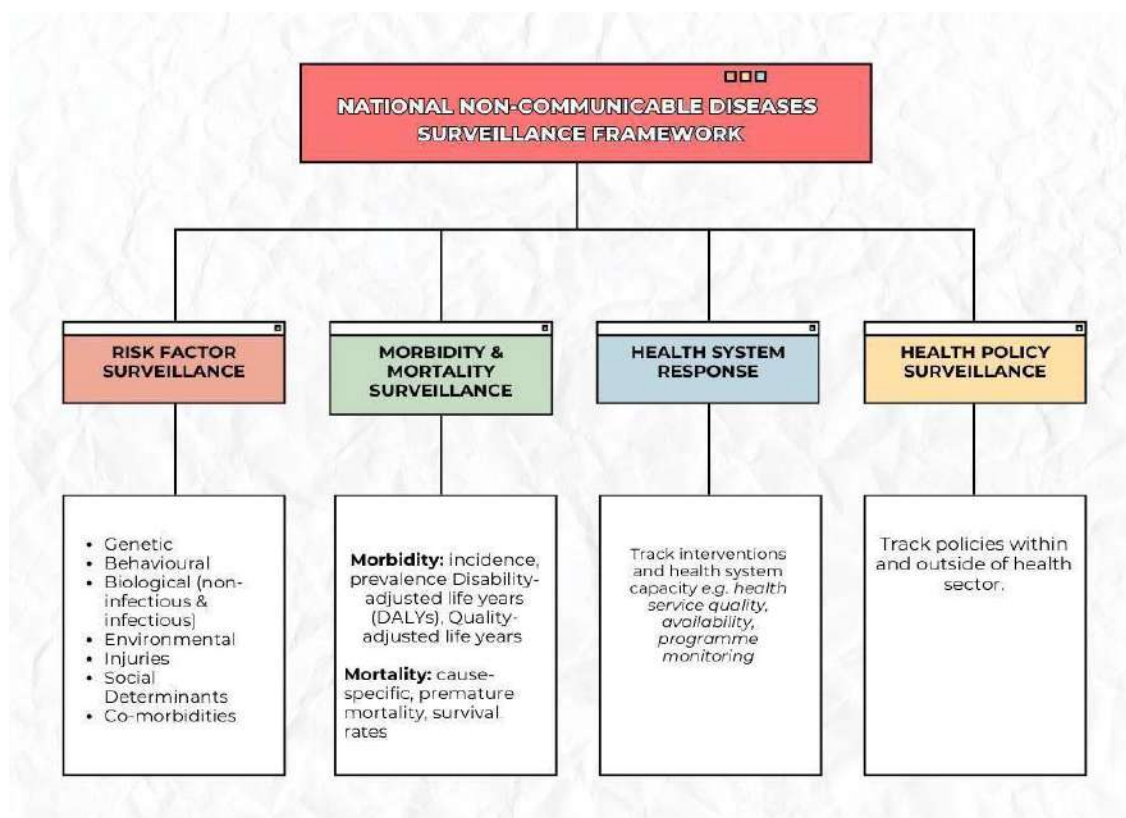


Figure 13.1. National Non-communicable Disease Framework

Both passive and active surveillance methods are employed, and the following are the data systems and sources that are currently utilised for NCD surveillance by the MOHW (Table 13.1):

- Sentinel surveillance – e.g., Jamaica Injuries Surveillance System.
- Hospital active surveillance – e.g., cancer registration.
- Non-communicable disease registries – e.g., population-based cancer registry.
- NCDs registers – e.g., pap smear registers I and II.
- Population-based health surveys – e.g., Global School Health Survey, Global Youth Tobacco Survey, and Jamaica Health and Lifestyle Survey.
- Health system surveillance – e.g., hospital discharges, and primary care Monthly Clinic Summary Report (MCSR).
- Health policy surveillance – e.g., compliance with the tobacco law.
- Mortality surveillance – e.g., vital statistics or records.

There is a set of national indicators that are routinely reported. This is guided by the MOHW national policies and plans, Service Level Agreements for NCDs, international commitments and reporting needs. The indicators should be reviewed at least every two (2) years to check on their relevance or need for updating. The case definitions, indicators, frequency, and level of reporting are unique to each data system. The existing surveillance tools, plans, protocols should therefore be referred to for guidance.

Table 13.1. *Non-communicable Disease Data Systems and Sources*

Data System	Data Source	Data collected on NCDs
Health Surveys (major)	Global School Health Survey Global Youth Tobacco Survey	NCD risk factors and Tobacco use in youth 13 - 17 years old
	Jamaica Health and Lifestyle Survey	NCD prevalence, risk factors, health system response, social determinants.
	Reproductive Health Survey	Programme monitoring/ health system response data on cancer, and violence-related injuries
	Jamaica Survey of Living Conditions – Health section	Priority NCDs prevalence, health system response, social determinants.
	Census	Population demographics
	Other Ministries Departments and Agencies (MDAs) led surveys	Select priority NCDs prevalence, injuries, health system response, policy/ programme monitoring, socio-economic data
	World Health Organization Capacity Surveys	Status on implementation of international commitments for NCDs & prevalence estimates are typically every 2 years.

Data System	Data Source	Data collected on NCDs
Sentinel surveillance system	Weekly Sentinel Surveillance System information is collected along with communicable diseases data	Information collected weekly from sites on Intentional and Unintentional injuries: <5 and ≥5 years age groups
	Jamaica Injuries Surveillance System (JISS) 3 - hospital- based, web-based system that gathers data from twelve public hospital sentinel sites: Western Region • Cornwall Regional • Savanna-la-mar Southern Region • Mandeville • May Pen • Lionel Town South-East Region • Spanish Town • Kingston Public • Bustamante Hospital for Children • Princess Margaret North-East Region • Port Antonio • Annotto Bay • St. Ann's Bay	Four categories of injuries: • Unintentional Injuries (UIs) • Violence- Related Injuries (VRIs) • Suicide Attempts (SAs) • Road Traffic Crash injuries (RTCs)
NCD Registries	Diabetes and Hypertension Registry – primary care	Patient demographics, clinical monitoring information for patients with Diabetes and Hypertension seen in health centres

Data System	Data Source	Data collected on NCDs
	Sickle Cell Disease Registry primary care and outpatient clinics.	Patient demographics, clinical monitoring information on patients with SCD presenting for routine health maintenance in health centres and outpatient clinics.
	National Cancer Registry – population-based coverage island-wide with four (4) sub-Registries and UWI Cancer Registry covering Kingston and St. Andrew	Cancer incidence and mortality
Health System surveillance	Monthly Clinic Summary Report (MCSR)	Collects aggregate data on select NCD patient and programme monitoring indicators e.g. screening, visits and control status for diabetes and hypertension.
	Hospital Monthly Statistical Report (HMSR) database	Collects aggregate data on patients seen for asthma, Lower Respiratory Tract Infection and Injuries at public hospitals.
	Medical Records Case Abstract (MRCA)	Hospital discharges for NCDs as specified on the ICD-10 codes.
	National Health Fund	Patient and medication information for 22 chronic diseases
	Pap Smear Register I and II	Patient demographics, screening, referrals, quality of smear, and pap smear results
	Outreach/mobile services	Aggregate data, age, and sex at birth, service utilisation.
Mortality surveillance	Vital statistics	Cause- and age-specific mortality, premature mortality, survival rates

Governance Structure and Roles and Responsibilities

The NNCDSS is implemented in accordance with the existing governance structure and surveillance system, and its implementation varies depending on the disease, condition, or risk factor programme as outlined in Figure 13.2.

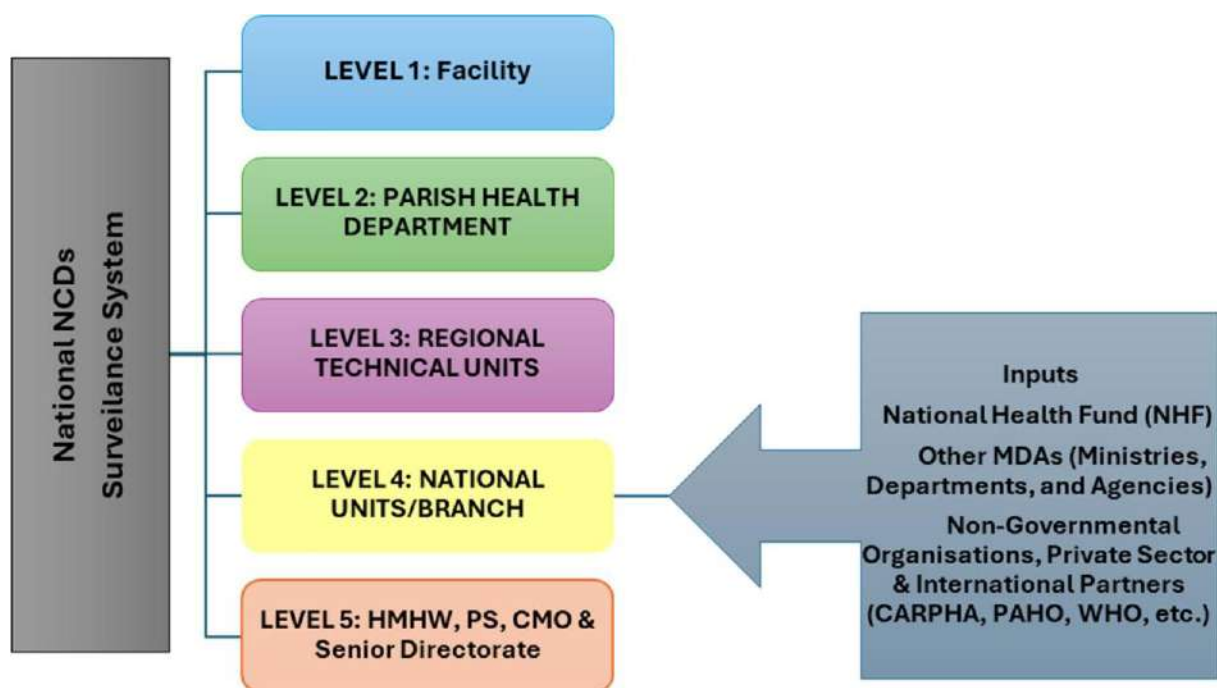


Figure 13.2. Governance Structure of National Non-communicable Diseases

Data coordination and data management should take place at Levels 1-4, with varying levels of data analysis. Depending on where the primary data is generated and the data systems used at Levels 2-3 (parish/regional), reporting and policy recommendations should be done and submitted to Levels 4-5. Level 4 responsibilities include comprehensive national analysis, reporting, and policy/programme recommendations. At Level 5, health personnel are involved in policy formulation and the dissemination of information. Dissemination of reports should be done to and at all levels and as required/relevant to external parties.

What to report

Regional Health Authorities should submit both quarterly and annual surveillance reports for the priority NCDs to the Ministry of Health and Wellness, using the standard template. Other reports will be generated depending on the system, specific programme requirements, Service Level Agreement, and National NCD Plan. Reports are mainly descriptive and can be generated weekly, monthly, quarterly and annually for some systems, e.g. NCD Registries and JISS 3, while health surveys typically are reported every 5 years. Refer to specific programmes and protocols for guidance.

Integration of National with PAHO, CARPHA and WHO Systems

NCD Surveillance linkages with other national health programmes, such as reproductive health, and oral health is ongoing.

There are also continuous efforts to ensure data systems have been aligned with best practices for NCD surveillance and for reporting on the Caribbean Public Health Agency's and WHO's NCD Monitoring Framework, Sustainable Development Goals, and other international commitments.

Chapter 14: Mortality Surveillance

General surveillance of mortality in a population and mortality associated with specific outbreaks allow for characterisation of the demographic impact of diseases, outbreaks and health events on a country. Patterns and trends in mortality may also be identified, which allows for the implementation of strategies to address specific public health concerns.

Identifying and validating deaths associated with outbreaks is a global challenge. In some jurisdictions there are limitations in testing capacity that may lead to under-reporting of deaths due to the disease causing the outbreak. Disruptions in the health system and the society in general may contribute to increased deaths from other diseases or events during outbreaks, and individuals may avoid visiting health facilities due to fear of contracting infections. To characterize the effect of outbreaks on mortality, excess mortality calculation would give a better understanding of the demographic impact of the outbreak on the population.

Sources of Mortality Data

- **National Surveillance System:** All deaths associated with outbreaks of Class 1 notifiable conditions are to be reported to the National Surveillance Unit (NSU) within 24 hours of suspicion. This includes suspected deaths due to COVID-19, severe dengue and maternal deaths. Databases containing information on these deaths are housed in the NSU and reports on these health events are prepared and disseminated on schedule.
- **Hospital Monthly Statistical Reports (HMSR):** monthly number of deaths occurring in public hospitals.
- **Civil registration and vital statistics system (CRVS):** The National Identification and Registration Authority (NIRA), formerly the Registrar General's Department (RGD), is responsible for registering all vital events occurring in Jamaica, including births, deaths and marriages. Mortality data from the CRVS account for community deaths and deaths occurring outside of public hospitals but tend to lag behind routine surveillance data as it

is dependent on the registration of deaths. Annual number of deaths in preceding years are obtained from NIRA. This includes disaggregation by age, sex and cause of death.

- **Statistical Institute of Jamaica (STATIN):** Data from the RGD are transmitted to STATIN which is responsible for the collection, compilation and publication of Jamaica's statistical information. STATIN reports on births, deaths and population data (including mid-year population data). These are used in the calculation of mortality rates.

Notification and Investigation of Outbreak-associated Deaths

All outbreak-associated deaths are Class 1 notifiable health events and should be notified within 24 hours of suspicion to the Parish Health Departments and the NSU. Case investigations must be initiated within 24 hours of notification, and a case investigation form must be completed for each death and submitted within 48 hours of notification.

Deaths associated with outbreaks (e.g., Dengue and COVID-19) are investigated by the parish teams. Investigation forms are completed and submitted to the NSU via the Parish Health Departments. It is recommended that specific laboratory tests be performed for all suspected deaths associated with outbreaks. A panel of medical doctors/ epidemiologists review the case reports and classify the deaths based on specific case definitions.

Minimum Data Required for Reporting and Classification of Deaths

Detailed information on each case is required to accurately determine the cause of death and appropriate classification. The following data must be provided to allow for the classification of notified deaths associated with outbreaks:

Demographic data:

- a. Name
- b. Age
- c. Sex
- d. Address
- e. Health facility/ institution where patient was seen or treated

Clinical Data:

- f. Date of admission to hospital
- g. Date of onset of symptoms
- h. Symptoms – all symptoms should be stated
- i. Comorbid conditions
- j. Source of infection – imported, contact of a confirmed case, local transmission not epidemiologically linked
- k. Summary of clinical course in hospital (if admitted) – narrative report or a copy of the medical record with the last admission
- l. Date of admission to ICU (if applicable)
- m. Date of recovery or end of isolation period (if applicable)
- n. Date of death

Data Analysis

Case fatality rate/ratio (CFR) for the specific epidemic (e.g., COVID-19, dengue) is calculated by dividing the number of deaths (due to a specific condition) by the number of cases (of the specific condition).

$$\text{Case fatality ratio (CFR) (\%)} = \frac{\text{Number of deaths from a disease}}{\text{Number of cases of a disease}} \times 100$$

Excess Deaths - The monthly or annual number of deaths in Jamaica in a given year (from the HMSR or NIRA) are compared to at least five years prior to the event. The deaths are examined against the threshold of the five-year mean number of deaths (monthly or annually) plus two standard deviations, for possible excess deaths. There are excess deaths if the threshold is exceeded.

Crude mortality rate (CMR) refers to the total number of deaths in a population in a specific period of time (usually one year).

$$\text{Crude mortality rate (CMR)} = \frac{\text{Number of deaths}}{\text{Mid-year population}} \times 100,000$$

Premature mortality rate for the four leading non-communicable diseases (NCDs) (cardiovascular diseases, cancer, diabetes, and chronic respiratory diseases) is calculated and reported annually. This is the unconditional probability of death between ages 30 and 70 years from cardiovascular diseases, cancer, diabetes, and chronic respiratory diseases. Specific methodology using direct standardization is used to ensure comparability across jurisdictions.

Child mortality rates are calculated using mortality and births data from the NIRA or STATIN (data from both public and private sector) or the HMSR (only public sector hospitals):

Table 14.1. Child Mortality Rate Calculations

Variables	Formulae
Stillbirth Rate	$\text{Stillbirth rate} = \frac{\text{number of stillbirths}}{\text{Total births (alive or dead)}} \times 1,000$
Early neonatal mortality rate	$\text{Early neonatal mortality rate} = \frac{\text{No. of deaths during the first 7 days of life}}{\text{No. of live births}} \times 1,000$
Neonatal mortality rate	$\text{Neonatal mortality rate} = \frac{\text{No. of deaths under 28 days of life}}{\text{No. of live births}} \times 1,000$
Perinatal mortality rate	$\text{Perinatal mortality rate} = \frac{\text{No. of early neonatal deaths} + \text{No. of stillbirths}}{\text{Total births (alive or dead)}} \times 1,000$
Infant mortality rate	$\text{Infant mortality rate} = \frac{\text{No. of deaths during the first 365 days of life}}{\text{No. of live births}} \times 1,000$
Under 5 mortality rate	$\text{Under 5 mortality rate} = \frac{\text{No. of under 5 deaths}}{\text{No. of live births}} \times 1,000$

Cause-specific mortality rates are calculated based on specific ICD-10 codes for diseases and conditions that are recorded as the underlying cause of death in NIRA mortality tables or on line lists. This information is explored based on conditions of interest or of concern in the population e.g. NCDs and injuries. Trends are identified and the findings reported periodically.

Chapter 15: International Health Regulations

Introduction

Under Articles 21(a) and (22), the World Health Organization (WHO) constitution confers upon the World Health Assembly the authority to adopt regulations designed to prevent the international spread of disease.

The International Health Regulations (IHR) 2005, are a legally binding international agreement involving 196 countries, including all WHO Member States. The IHR aim to prevent, detect, and respond to public health threats that have the potential to cross borders and affect global populations ¹.

The IHR (2005) require countries to develop and maintain core public health capacities for surveillance, assessment, reporting, and response. This chapter outlines the relevance of IHR to national surveillance systems and describes the obligations for information sharing and coordination.

The IHR (2005) is central to global health security. Effective implementation relies on strong surveillance systems, timely risk assessment, and transparent communication. National and subnational surveillance teams play a critical role in identifying, reporting, and responding to public health events following IHR requirements.

Role of Surveillance Systems in IHR Implementation

National surveillance systems are the backbone of IHR compliance, providing the data and analysis needed to:

- Detect unusual or severe public health events
- Conduct risk assessments

¹ World Health Organization. *International Health Regulations (2005) Third Edition* [Internet]. Geneva 2016. Available from: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/1632f577-e897-4da2-8d4b-d0d37751b65c/content>

- Support early warning and rapid response
- Enable timely reporting to WHO

Surveillance systems should be designed to:

- Monitor notifiable diseases and syndromes
- Incorporate event-based surveillance (EBS)
- Include intersectoral coordination (e.g., animal health, environmental monitoring)

Purpose and Scope of IHR (2005)

The IHR (2005) provide a framework for:

- Detecting public health risks at an early stage.
- Assessing and reporting events that may constitute a Public Health Emergency of International Concern (PHEIC).
- Strengthening national capacity for surveillance and response.
- Ensuring rapid information sharing between countries and the WHO.
- Avoiding unnecessary interference with international traffic and trade.

As part of IHR's (2005) implementation, all States Parties must conduct an annual self- assessment of their core capacities using the State Party Self-Assessment Annual Reporting (SPAR) tool. This tool allows countries to evaluate their progress in developing and maintaining the IHR core capacities (including legislation, surveillance, response, and laboratory systems). The SPAR results are submitted to WHO and contribute to global health security monitoring. They also help identify capacity gaps, inform national planning and investment, and prioritise areas for technical support and partnership. Regular and transparent reporting through SPAR is essential for improving preparedness and fostering international accountability.

Core Capacity Requirements for Surveillance and Response

Each country must establish and maintain a set of core capacities at national, intermediate (regional health authority), and local (parish) levels to detect and respond to public health threats.

1. States Parties shall utilise existing national structures and resources to meet their core capacity requirements under these Regulations, including with regard to:
 - (a) their surveillance, reporting, notification, verification, response and collaboration activities;and
 - (b) their activities concerning designated airports, ports and ground crossings.
2. Each State Party shall assess, within two years following the entry into force of these Regulations for that State Party, the ability of existing national structures and resources to meet the minimum requirements described in this Annex. As a result of such assessment, States Parties shall develop and implement plans of action to ensure that these core capacities are present and functioning throughout their territories as set out in paragraph 1 of Article 5 and paragraph 1 of Article 13.
3. States Parties and WHO shall support assessments, planning and implementation processes under this Annex.
4. At the local community level and/or primary public health response (parish) level
The capacities:
 - (a) to detect events involving disease or death above expected levels for the particular time and place in all areas within the territory of the State Party; and

¹ World Health Organization. *International Health Regulations (2005) Third Edition* [Internet]. Geneva 2016. Available from: [chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/1632f577-e897-4da2-8d4b-d0d37751b65c/content](https://iris.who.int/server/api/core/bitstreams/1632f577-e897-4da2-8d4b-d0d37751b65c/content)

- (b) to report all available essential information immediately to the appropriate level of healthcare response. At the community level, reporting shall be to local community healthcare institutions or the appropriate health personnel. At the primary public health response level, reporting shall be to the intermediate or national response level, depending on the organizational structures. For this Annex, essential information includes the following: clinical descriptions, laboratory results, sources and type of risk, numbers of human cases and deaths, conditions affecting the spread of the disease and the health measures employed;
- (c) to implement preliminary control measures immediately.

5. At the intermediate public health (regional health authority) response levels

The capacities:

- (a) to confirm the status of reported events and to support or implement additional control measures;
- and
- (b) to assess reported events immediately and, if found urgent, to report all essential information to the to the national level. For this Annex, the criteria for urgent events include serious public health impact and/or unusual or unexpected nature with high potential for spread.

6. At the national level

Assessment and notification. The capacities:

- (a) to assess all reports of urgent events within 48 hours and
- (b) to notify WHO immediately through the National IHR Focal Point when the assessment indicates the event is notifiable pursuant to paragraph 1 of Article 6 and Annex 2 and to inform WHO as required pursuant to Article 7 and paragraph 2 of Article 9.

Public health response. The capacities:

- (a) to determine rapidly the control measures required to prevent domestic and international spread;
- (b) to provide support through specialised staff, laboratory analysis of samples (domestically or through collaborating centres) and logistical assistance (e.g. equipment, supplies and transport);
- (c) to provide on-site assistance as required to supplement local investigations;
- (d) to provide a direct operational link with senior health and other officials to approve rapidly and implement containment and control measures;
- (e) to provide direct liaison with other relevant government ministries;
- (f) to provide, by the most efficient means of communication available, links with hospitals, clinics, airports, ports, ground crossings, laboratories and other key operational areas for the dissemination of information and recommendations received from WHO regarding events in the State Party's own territory and in the territories of other States Parties;
- (g) to establish, operate and maintain a national public health emergency response plan, including the creation of multidisciplinary/multisectoral teams to respond to events that may constitute a public health emergency of international concern; and
- (h) to provide the foregoing on a 24-hour basis.

Event Notification and Information Sharing

What Must Be Reported?

Under Article 6 of the IHR, countries must notify WHO through the National IHR Focal Point of:

- Any event that may constitute a PHEIC within 24 hours of assessment
- Public health risks that spread internationally or require a coordinated international response.

The Decision Instrument

A four-question decision tool helps determine whether an event should be reported (See Figure 15.1 – IHR Annex 2) ¹:

1. Is the public health impact of the event serious?
2. Is the event unusual or unexpected?
3. Is there a significant risk of international spread?
4. Is there a significant risk of international travel or trade restrictions?

The event must be reported to WHO if two or more answers are yes.

Follow-up Reporting

Countries must continue to share:

- Timely updates
- Case counts and distribution
- Laboratory and epidemiological findings
- Control measures taken.

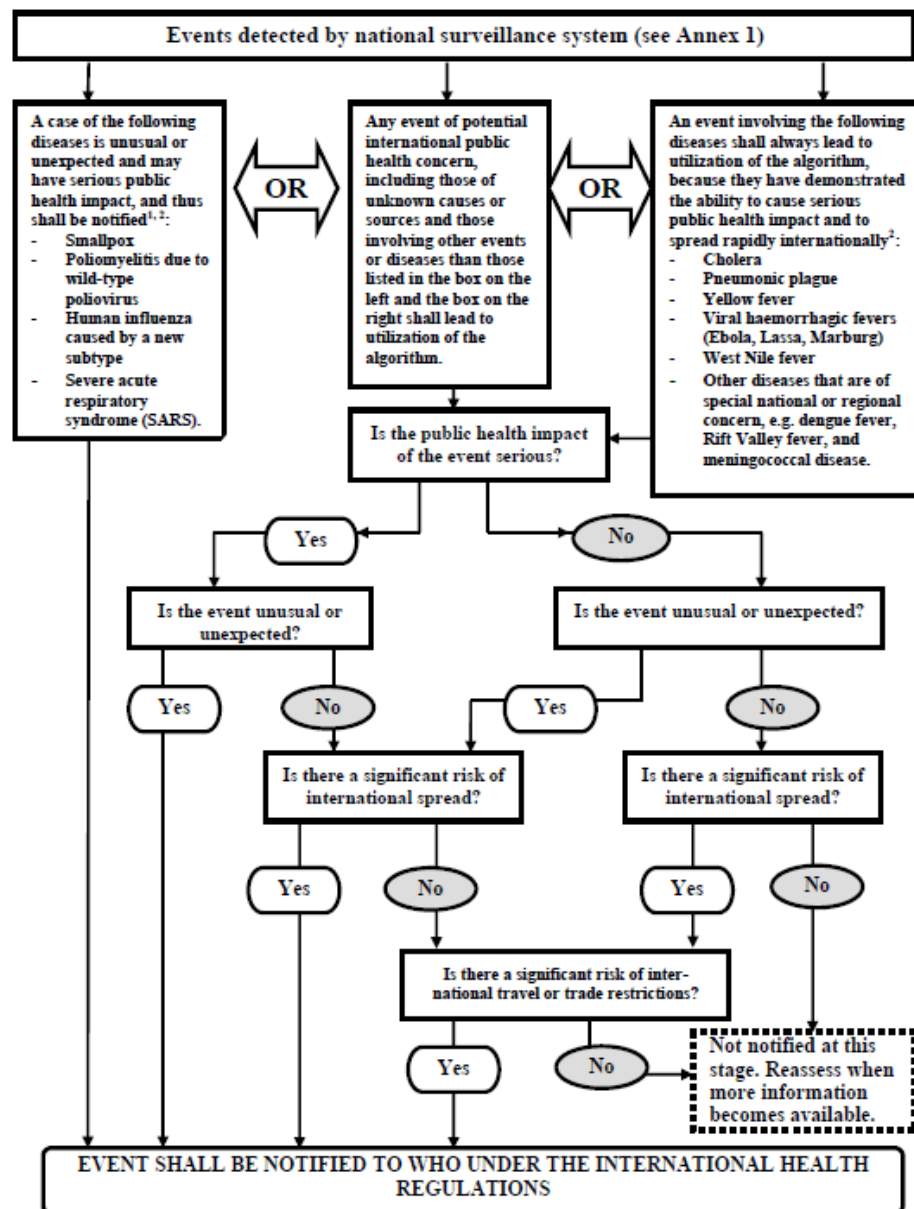
Cross-Border and Regional Coordination

Surveillance systems must also be equipped to:

- Share information across borders
- Coordinate investigations and responses with neighbouring countries
- Participate in regional networks (e.g., Integrated Disease Surveillance and Response (IDSR), the Global Outbreak Alert and Response Network (GOARN)).

¹ World Health Organization. *International Health Regulations (2005) Third Edition* [Internet]. Geneva 2016. Available from: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/1632f577-e897-4da2-8d4b-d0d37751b65c/content>

ANNEX 2
DECISION INSTRUMENT FOR THE ASSESSMENT AND NOTIFICATION
OF EVENTS THAT MAY CONSTITUTE A PUBLIC HEALTH EMERGENCY
OF INTERNATIONAL CONCERN

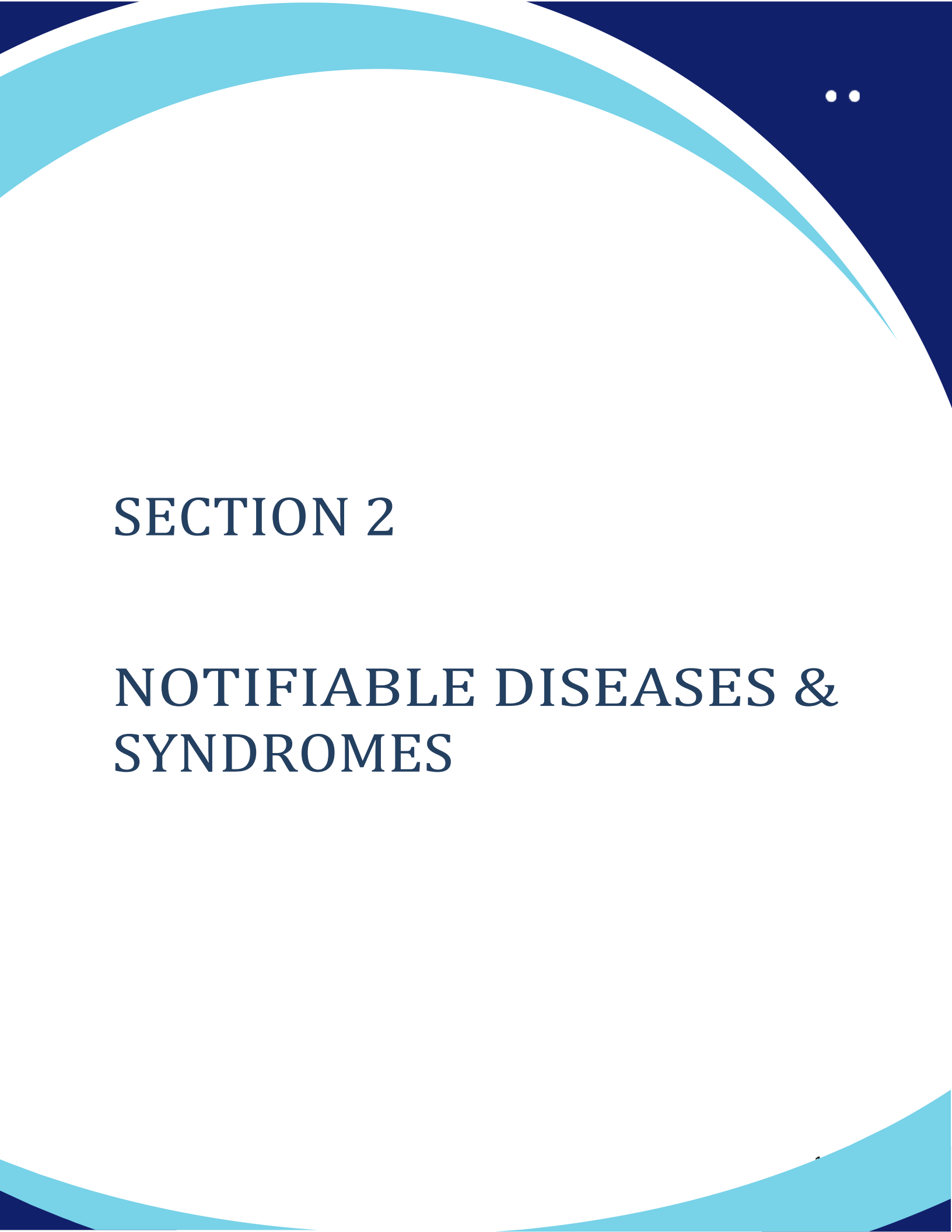


¹ As per WHO case definitions.

² The disease list shall be used only for the purposes of these Regulations.

Figure 15.1. Annex 2 Decision Instrument for the Assessment and Notification of Events that may Constitute a Public Health Emergency of International Concern¹

¹ World Health Organization. *International Health Regulations (2005) Third Edition* [Internet]. Geneva 2016. Available from: [chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/1632f577-e897-4da2-8d4b-d0d37751b65c/content](https://iris.who.int/server/api/core/bitstreams/1632f577-e897-4da2-8d4b-d0d37751b65c/content)



SECTION 2

NOTIFIABLE DISEASES & SYNDROMES

Chapter 16: Chikungunya Epidemiological Surveillance Protocol

Background

Chikungunya is an arthropod-borne viral disease that is characterized by fever and severe joint pains (arthralgia). It is transmitted to humans by the bites of infected female *Aedes aegypti* or *Aedes albopictus* mosquitoes. The Chikungunya virus (CHIKV) that causes Chikungunya was first identified in the United Republic of Tanzania in 1952, but it was in the 1970s that urban outbreaks were first reported in Asia. Since 2004, outbreaks of CHIKV have become more frequent and widespread.

The Pan American Health Organization (PAHO) / World Health Organization (WHO) issued an Epidemiological Alert on December 9, 2013, regarding the detection of the first cases of autochthonous transmission of chikungunya fever in the Americas in Saint Martin on December 6, 2013. Large outbreaks have subsequently occurred affecting most of the countries in the region of the Americas and Chikungunya has now been reported in over 110 countries in Asia, Africa, the Americas and Europe. Jamaica recorded its first case of Chikungunya in early 2014. In that outbreak there were over 1,600 suspected cases, of which 89 were laboratory confirmed.

Based on the similarities in the transmission of chikungunya and dengue, the surveillance for chikungunya was built on the existing dengue surveillance system.

The epidemiological surveillance system of Chikungunya in Jamaica was strengthened through the following strategies:

- Training the epidemiological and clinical staff on Chikungunya management
- Standardizing case definitions
- Designation of Chikungunya as a Class 1 Notifiable disease
- Laboratory surveillance for confirmation of autochthonous transmission of Chikungunya

Case Definition

Suspected Case

Fever of acute onset of $>101.3^{\circ}\text{F}$ (38.5°C) **and** severe arthralgia (or arthritis) not explained by other medical conditions in someone who resides in or has visited epidemic or endemic areas within two weeks prior to the onset of symptoms. Other features may be present:

- Headache
- Diffuse back pain
- Rash - maculopapular (in children, may be bullous lesions)
- Myalgia
- Nausea
- Vomiting
- Conjunctivitis

Confirmed Case

A suspected case with a positive result in any of the following CHIKV specific tests: Viral isolation, detection of viral RNA by RT-PCR, detection of IgM in a single serum sample (collected during the acute or convalescent phase) or four-fold increase in CHIKV-specific antibody titers (samples collected at least two to three weeks apart).

Designation of Chikungunya as a Class 1 Notifiable Disease

Chikungunya was gazetted as a Class 1 Notifiable disease and should therefore be notified to the Parish Health Department or the National Surveillance Unit, Ministry of Health and Wellness within 24 hours of suspicion.

Laboratory Testing for Chikungunya

Samples for serology should be sent to the National Public Health Laboratory along with the completed Jamaica Laboratory Investigation Form ([Appendix 9](#)). The following are the requirements for sampling and for the conservation of the sample:

Type of sample: Serum: 4-5 mls of blood in a plain red top tube

- Acute phase: 0-8 days after symptom onset
- Convalescent phase: 10-15 days after symptom onset

Conservation of the sample:

- Keep refrigerated (2 – 8°C) if sample will be processed within 48 hours at the NPHL.
- Keep frozen (-10 to – 20°C) if sample will be processed after the first 48 hours.
- Maintain frozen (-70°C) if sample will be processed after one week.

Method for Confirmation of the Start of a Chikungunya Outbreak in Jamaica

Laboratory surveillance will be used as the methodology for the confirmation of autochthonous transmission.

Inclusion Criteria

The following are inclusion criteria for the laboratory surveillance to confirm autochthonous transmission:

1. Samples that are negative for Dengue testing AND
2. Cases consistent with clinical features of Chikungunya: fever and arthritis/arthralgia AND
3. The samples should be processed according to the day of sampling with respect to the onset of symptoms. The algorithm in [Appendix 15](#) should be strictly followed by the laboratory. At most, 25 of the samples positive for CHIKV should be sent to the collaborating laboratory, as well as 10% of the negative samples that fit the case definition.

Established Outbreak

The confirmation of an imported or autochthonous case of Chikungunya will activate the Ministry of Health and Wellness' response mechanisms. The National and International Epidemiological Focal Point, National IHR Focal Point, and the PAHO / WHO IHR Contact Point will be notified, as described by the International Health Regulations (2005).

Enhanced Surveillance

The clinical and epidemiological surveillance system will be enhanced based on the epidemiology of the confirmed case(s).

- All persons fitting the case definition should be notified immediately to the Parish Health Department and the National Surveillance Unit, Ministry of Health and Wellness.
- A detailed investigation of all suspected cases is to be conducted. A report should be submitted within 48 hours using the standard Investigation Form ([Appendix 16](#)), but not limited to this form.
- Surveillance of persons with similar exposure is to be conducted for at least 21 days post exposure. A line listing is to be completed, indicating the possible date of exposure, the date that the person was contacted and their clinical status recorded.
- Active community surveillance is to be conducted where indicated.

Characterisation of the Outbreak

The data related to the Chikungunya outbreak will be analysed regularly (weekly or daily as required) to characterise the outbreak.

- Demographic characteristics of the outbreak
- Monitoring and description of the spread of CHIKV
- Clinical features
- Clinical severity and impact on society
- Identification of risk factors for severe disease

End of the Outbreak

Based on the epidemiological assessment, a decision will be made regarding the declaration of the end of the outbreak.

Endemic Seasonal Transmissions

Chikungunya as a Class 1 Notifiable disease, will be monitored according to the current Integrated Surveillance system. The preparedness plan will be reviewed and revised in keeping with the epidemiological situation to maintain preparedness.

Framework for Epidemiological Surveillance of Chikungunya

Table 16.1. Framework for Epidemiological Surveillance of Chikungunya

Expected Result	Indicators
CHIK Epidemiological Surveillance system for timely alert and opportune response implemented	● CHIKV surveillance systems in place (e.g., Lab data, clinical data, entomological data) ● Number of sites at the country level with the information about CHIKV (Hospitals, Health Centres) ● CHIK outbreaks reported according to IHR guidelines.

Activities	Tasks	Responsible Persons
<u>Phase 1:</u> <u>Preparedness</u> 1. Strengthen the epidemiological and laboratory surveillance systems for CHIK in the countries of the Caribbean Sub-region	Include CHIK as part of Disease Surveillance System - Class 1 Notifiable disease (reported immediately on suspicion, within 24hrs). Standardize a CHIK case definition (clinical and epidemiological) based on PAHO guidelines. Set up clinical, laboratory and epidemiological surveillance. Standardize the methods used to determine the criteria (clinical, epidemiological and laboratory) to confirm the start of a CHIK outbreak. Train the epidemiological, laboratory and clinical staff on Chikungunya management	CMO, National Epidemiologist, Medical Epidemiologist - Communicable Diseases, Director - National Laboratory Services (NLS), Director - EDMSSB

Activities	Tasks	Responsible Persons
<u>Phase 2: Established outbreak</u> 1. Confirmation/ Declaration of the beginning of an outbreak	Classify cases as locally- acquired or imported cases. Notify the National and International Epidemiological Focal Points, National IHR Focal Point and PAHO / WHO IHR Contact Point, according to IHR. Enhance clinical, epidemiological and laboratory surveillance systems.	National Epidemiologist, Medical Epidemiologist - Communicable Diseases IHR NFP; Director - EDMSSB/ Officer with Primary Responsibility National Epidemiologist, Medical Epidemiologist - Communicable Diseases, Director - NLS
2. Monitor and assess the epidemic situation	Activate and maintain the National, Regional and Parish Emergency Operations Centres / Coordination mechanisms. Establish routine communication mechanisms with parish and regional health authorities and international organizations. Analyze and interpret weekly data and develop a daily and weekly outbreak report. Provide support and technical assistance to the field.	CMO, Director - EDMSS, National Epidemiologist, MOHW NEOC, Ministry of Health and Wellness' Emergency and Disaster Risk Management Committee
<u>PHASE 3: End of the outbreak</u> 1. Analyze the actions taken during the outbreak	Maintain the monitoring and evaluation activities. Conduct after-action evaluations to identify and remedy gaps in the country surveillance and response program.	CMO, Director - EDMSS, National Epidemiologist, Ministry of Health and Wellness' Emergency and Disaster Risk Management Committee, National Public Health Laboratory (NPHL)

Activities	Tasks	Responsible Persons
<u>PHASE 4: Endemic seasonal transmissions</u> 1. Maintain the surveillance activities	Establish a regular, integrated surveillance program. Review evaluation of the plan and revise the preparedness plan to ensure preparedness is maintained.	CMO, Director - EDMSS, National Epidemiologist, Ministry of Health and Wellness’ Emergency and Disaster Risk Management Committee

Chapter 17: Coronavirus Disease - 19 (COVID-19)

Background

Coronaviruses (CoV) are a large family of viruses that cause illnesses ranging from less severe disease, such as the common cold, to more severe diseases such as Middle East Respiratory Syndrome (MERS-CoV) and Severe Acute Respiratory Syndrome (SARS-CoV). A novel coronavirus (nCoV) is a new strain that has not been previously identified in humans. Globally, novel coronaviruses emerge periodically in different areas, including SARS in 2002 and MERS in 2012 ¹.

A novel (new) coronavirus, SARS-CoV-2, which causes Coronavirus Disease 2019 (COVID-19), was identified in Wuhan City, Hubei Province, China in December 2019. The virus has caused significant morbidity and mortality in China, has spread globally and was declared a pandemic and a public health emergency of international concern in early 2020.

Jamaica reported its first case of COVID-19 on March 10, 2020.

Transmission Patterns for Jamaica include:

Date of Transmission Pattern Jamaica	Category	Definition
Up to March 9, 2020	No cases	Countries/territories/areas with no cases
March 10, 2020	Sporadic cases	Countries/territories/areas with one or more cases, imported or locally detected
March 14, 2020	Clusters of cases	Countries/territories/areas experiencing cases, clustered in time, geographic location and/or by common exposures
September 2, 2020	Community transmission	Countries/area/territories experiencing larger outbreaks of local transmission defined through an assessment of factors including, but not limited to: - Large numbers of cases not linkable to transmission chains - Large numbers of cases from sentinel lab surveillance - Multiple unrelated clusters in several areas of the country/territory/area

¹ World Health Organization. *Coronavirus disease (COVID-19)* [Internet]. (2020). Available from: <https://www.who.int/health-topics/coronavirus>

Jamaica's epidemiological surveillance system will be used to detect and report on potential cases of COVID-19. The components of this surveillance system which will be used and enhanced as necessary are:

1. The Class 1 Notification System (Case-based surveillance)
2. Sentinel Surveillance
3. Hospital Active Surveillance
4. Mortality Surveillance

Version 19 of the COVID-19 Surveillance Protocol for Jamaica included the following updates, which have now been removed in version 20, and therefore enhanced surveillance in these situations is no longer required:

- Surveillance of COVID-19 in Healthcare Workers
- General Hospital Surveillance
- Community Surveillance
- Surveillance in Special Setting
- Surveillance in Special Groups
- Tourist Establishment Surveillance

Routine surveillance in these settings is to be continued. However, surveillance for workers and residents/guests will follow the same general guidelines as outlined in this protocol. While these methods of surveillance have been removed from this update, testing for COVID-19 in patients also remains at the clinician's discretion and should be based on clinical presentation.

This protocol shall be adhered to for surveillance activities related to COVID-19 in the following areas:

- A. Case-Based Surveillance
- B. Case Reporting and Investigation
- C. Contact Tracing
- D. Respiratory Surveillance
- E. Mortality Surveillance
- F. Specimen Collection and Testing
- G. Data Analysis and Interpretation
- H. Data Dissemination and Outputs

Purpose of these Guidelines

To provide guidance on implementing surveillance standards for COVID-19.

Objectives

The objectives of epidemiological surveillance of COVID-19 are:

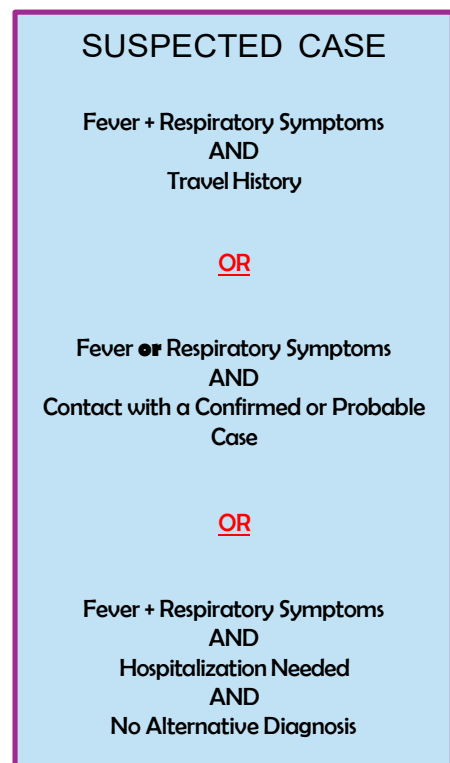
1. To monitor COVID-19 trends in Jamaica
2. To establish epidemiological characteristics of COVID-19 infection in Jamaica
3. To inform risk assessment and decision-making.

Case-Based Surveillance

COVID-19, by Public Health Order, was made a Class 1 Notifiable Disease in March 2020.

Suspected Case ¹

- ✓ A person who meets the clinical criteria:
 1. Acute onset of fever AND cough (influenza-like illness)OR
 2. Acute onset of ANY THREE OR MORE of the following signs or symptoms: fever, cough, general weakness/fatigue, headache, myalgia, sore throat, coryza, dyspnoea, nausea, diarrhoea, anorexia.OR
- ✓ A person with fever or any acute respiratory illness AND having been in contact with a confirmed or probable case of COVID-19 case or linked to a COVID-19 cluster, in the 14 days prior to the onset of illness.



¹ World Health Organization. *Public health surveillance for COVID-19 Interim guidance* [Internet]. (July 2022). Available from: [chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/5586e0ff-87fc-4bd5-966c-60494975dc00/content](https://iris.who.int/server/api/core/bitstreams/5586e0ff-87fc-4bd5-966c-60494975dc00/content)

- ✓ A patient with severe acute respiratory illness (SARI: acute respiratory infection with history of fever or measured fever of $\geq 38^{\circ}\text{C}$; and cough; with onset within the last 10 days; and who requires hospitalization).

Probable Case ¹

- ✓ A patient who meets clinical criteria above AND is a contact of a probable or confirmed case or is linked to a COVID-19 cluster OR
- ✓ A suspected case for whom testing for COVID-19 virus is inconclusive.

ACTION: MAINTAIN ISOLATION

Confirmed Case ¹

- ✓ A person with laboratory confirmation of COVID-19 infection, irrespective of clinical signs and symptoms.

ACTION: MAINTAIN ISOLATION

COVID-19 Death ¹

- ✓ A death resulting from a clinically compatible illness in a probable or confirmed COVID-19 case, unless there is a clear alternative cause of death that cannot be related to COVID-19 disease (e.g. trauma). There should be no period of complete recovery between the illness and death.

ACTION: NOTIFY PARISH HEALTH DEPARTMENT, TAKE A SAMPLE AND COMPLETE CASE INVESTIGATION

¹ World Health Organization. *Public health surveillance for COVID-19 Interim guidance* [Internet]. (July 2022). Available from: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/5586e0ff-87fc-4bd5-966c-60494975dc00/content>

Case Reporting and Investigation

Notification

COVID-19 is a Class 1 notifiable condition. This means that all cases thought to be COVID- 19 must be notified by the medical practitioner (public and private) within 24 hours of suspicion. Specifically, for COVID-19, the local Parish Health Department and National Surveillance Unit must be notified. A **Class 1 Notification Form** must be submitted within 24 hours of initial notification ([Appendix 7](#)).

Clusters of respiratory infections or undifferentiated fever must be notified by the medical practitioner (public and private) within 24 hours of suspicion. The local Parish Health Department or National Surveillance Unit must be notified immediately. Clusters should be investigated and cases in a cluster line listed.

Investigation

The Parish Medical Officer (Health) leads the case investigation team and must:

- Initiate case investigation within 24 hours of notification. A preliminary case or cluster investigation report must be submitted to the National Surveillance Unit within 24 hours of this notification ([Appendix 17](#)).
- Immediately initiate community outbreak control measures, including contact tracing, searching for other cases, and line listing all contacts using the **Contact Tracing Intake and Daily Tracking Line Listing Forms** ([Appendix 18 and 19](#)).

Contact Tracing

A contact is a person who experienced any one of the following exposures (face-to- face, direct physical contact, or direct care for a patient) during the 2 days before and the 14 days after the onset of symptoms or date of sample collection (if asymptomatic) of a probable or confirmed case.

Contact Tracing Requirements:

1. Identify all familial, social, social, work, and healthcare worker contacts who have had contact with a confirmed case from 2 days before symptom onset of the case and up to 14 days after their symptom onset.
2. Create a line list, including demographic information, date of first and last common exposure or date of contact with the confirmed or probable case, and date of onset if fever or respiratory symptoms develop.
3. The common exposures and type of contact with the confirmed or probable case should be thoroughly investigated.

All contacts must be line listed. The recommended minimum dataset for contacts is in [Appendix 18](#) and [Appendix 19](#).

Risk assessment must be completed for all contacts. Contacts are to be classified as follows:

4. Face-to-face contact with a probable or confirmed case within 1 meter and for more than 15 minutes
 - (a) Bedroom
 - (b) Household / Family
 - (c) Conveyance - travelling in close proximity with (that is, having less than 1m separation from) a COVID-19 patient in any kind of conveyance
5. Direct physical contact with a probable or confirmed case
6. Direct care for a patient with probable or confirmed COVID-19 disease without using proper personal protective equipment
7. Other situations as indicated by local risk assessments

Management of Contacts:

- Persons who fall in category 1, 2 or 3 (face-to-face, direct physical contact or direct care for a patient) will be placed in quarantine.

- Contacts will be sampled if they are symptomatic when identified or if they develop symptoms at any time within the 10-day quarantine period.
- The period of quarantine and observation will end on Day 11 after most recent contact.
- All contacts will be given explicit instructions (verbal and written) regarding the steps to be taken if symptoms develop.

The Parish Health Department must keep records of the initial contact assessment, any significant events during the quarantine period (e.g., the development of symptoms), and the end-of-quarantine assessment.

Respiratory Surveillance During an Outbreak

ACTION: NOTIFY THE PARISH HEALTH DEPARTMENT AND TAKE A SAMPLE

Severe Acute Respiratory Illness (SARI) Surveillance

- ✓ A person who presents with symptoms of a measured fever or history of fever and a cough with onset within the last 10 days or less AND who requires hospitalization for their illness.

ALL HOSPITALS TO REPORT SARI CASES

All specimens must be sent to the National Public Health Laboratory, accompanied by a completed Jamaica Laboratory Investigation Form ([Appendix 9](#)). Submit a copy of the Jamaica Laboratory Investigation Form and the daily COVID-19 Laboratory Sample Line Listing ([Appendix 20](#)) to the Parish Health Department/National Surveillance Unit (NSU). Ensure the information is as complete as possible. Complete the weekly SARI / ILI Surveillance Reporting Form ([Appendix 29](#)).

*For SARI Sentinel Sites – In addition, please continue to report as per updated guidelines in the Ministry of Health and Wellness, National Surveillance Manual.

Influenza-Like Illness (ILI) Surveillance

- ✓ A person who presents with symptoms of a measured fever or history of fever and a cough with onset within the last 10 days or less.

ALL HEALTH FACILITIES TO REPORT ILI CASES

All specimens must be sent to the National Public Health Laboratory, accompanied by a completed Jamaica Laboratory Investigation Form ([Appendix 9](#)). Please submit weekly number of cases of ILI to the Parish Health Department/National Surveillance Unit (NSU) on the Parish Weekly Sentinel Reporting Form ([Appendix 2](#)). Ensure the information is as complete as possible.

*For ILI Sentinel Sites – In addition, please continue to report as per updated guidelines in the Ministry of Health and Wellness, National Surveillance Manual.

Admitted Lower Respiratory Tract Infection (LRTI) / Pneumonia Surveillance

- ✓ A Person who presents with a lower respiratory tract infection with onset within the last 14 days or less AND who requires hospitalization for their illness.

ALL HOSPITALS TO REPORT ADMITTED LRTI / PNEUMONIA CASES

**ACTION: NOTIFY THE PARISH HEALTH DEPARTMENT AND TAKE A
SAMPLE**

**ACTION: NOTIFY THE PARISH HEALTH DEPARTMENT OF TOTAL CASES SEEN
WEEKLY AND TAKE SAMPLES OF CASES BASED ON QUOTAS**

All specimens must be sent to the National Public Health Laboratory, accompanied by a completed Jamaica Laboratory Investigation Form ([Appendix 9](#)). Please submit weekly number of cases of aLRTI/ pneumonia to the Parish Health Department/National Surveillance Unit (NSU) on the Parish Weekly Sentinel Reporting Form ([Appendix 2](#)). Ensure the information is as complete as possible.

For All CONFIRMED COVID-19 CASES

Case Investigation & Contact Tracing is to be done.

A COVID-19 Case Investigation Form must be completed and submitted to the Parish Health Department/ National Surveillance Unit.

Mortality Surveillance

Objective:

Assess the demographic impact of the COVID-19 pandemic.

COVID-19 Death¹

- ✓ A death resulting from a clinically compatible illness in a probable or confirmed COVID-19 case, unless there is a clear alternative cause of death that cannot be related to COVID-19 disease (e.g. trauma). There should be no period of complete recovery between the illness and death.

Identifying COVID-19-specific mortality is a challenge across the globe. In some places there are limitations in the availability of COVID-19 tests leading to lack of comparability in the number of COVID-19 cases and deaths. Without COVID-19 testing, deaths can be misclassified as it becomes difficult to differentiate between deaths from COVID-19 and deaths from some other causes. If there are disruptions in the health system and the society in general, this may contribute to deaths from other causes. Individuals may also avoid visiting health facilities due to fear of contracting infections. It is therefore useful to examine the excess mortality because of the COVID-19 epidemic to account for these situations.

The monthly number of deaths occurring in public hospitals is obtained from the **Hospital Monthly Statistical Reports (HMSR)**. The monthly number of deaths in public hospitals is

¹ World Health Organization. *Public health surveillance for COVID-19 Interim guidance* [Internet]. (July 2022). Available from: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/5586e0ff-87fc-4bd5-966c-60494975dc00/content>

compared to the corresponding periods in the previous five (or more) years. The monthly deaths are reviewed for excess using a threshold of the five-year mean number of monthly deaths plus two standard deviations. Mortality data from the **civil registration and vital statistics system** account for community deaths and deaths occurring outside of public hospitals but tend to lag behind routine surveillance data as it is dependent on the registration of deaths. The number of deaths monthly in preceding years are obtained from the National Identification and Registration Authority (NIRA), formerly the Registrar General's Department. The monthly deaths are reviewed for excess using a threshold of the five-year mean number of monthly deaths plus two standard deviations.

In addition, the **crude case fatality rate** for the COVID-19 epidemic is calculated by dividing the number of confirmed COVID-19 deaths by the number of confirmed COVID-19 cases.

The COVID-19 deaths are required to be notified within 24 hours of suspicion to the Parish Health Departments and the National Surveillance Unit. Case investigation must be initiated within 24 hours of notification and a case investigation form ([Appendix 17](#)) completed for each death and submitted within 24 hours of notification.

A sample must be taken for all suspected COVID-19 deaths.

The following information is to be provided for all suspected COVID-19 deaths to aid in the classification:

- Age, sex, address (and other identifying data)
- Date of onset of symptoms
- Respiratory symptoms and signs - indicate if they are not present as well
- Other symptoms - if present
- Date of admission (if admitted to hospital)
- Health Facility / Institution
- Date sample was collected
- Course in hospital (if applicable)
- COVID-19 vaccination history - including type (brand) of vaccine and date(s) administered

- Comorbid conditions
- Date of Death

Specimen Collection and Testing

WHO¹ Laboratory Strategy Guideline based on Transmission Scenario

Transmission Scenario:

- Community transmission Public Health Aim:
- Stop transmission and prevent spread

Testing Strategy Guidance:

- Test all individuals meeting the suspected case definition
- Consideration in the investigation of cases and clusters of COVID-19
- Clinical management of severe acute respiratory infections when COVID-19 is suspected
- SARI/ILI surveillance for COVID-19 and reporting

Specimens must be collected from and will be tested for SARS-CoV-2 for the following persons:

- i. All suspected cases (as per the case definition above)
- ii. All symptomatic contacts of confirmed COVID-19 cases
- iii. All SARI cases from ALL Hospitals
- iv. All admitted LRTI / Pneumonia from ALL Hospitals
- v. ILI Cases as per syndromic surveillance protocol
- vi. ALL suspected COVID-19 deaths
- vii. Other individuals based on clinician's judgement and clinical presentation

¹ World Health Organization. *Laboratory testing strategy recommendations for COVID-19 Interim Guidance* [Internet]. (March 2020). Available from: https://apps.who.int/iris/bitstream/handle/10665/331509/WHO-COVID-19-lab_testing-2020.1-eng.pdf

Type of specimen

The recommended sample is a **nasopharyngeal swab OR combined nasopharyngeal and oropharyngeal swabs**. **Dacron or polyester flocked swabs** should be used for sample collection and once collected should be placed in viral transport medium. Avoid using cotton tipped swabs for specimen collection.

In cases where a nasopharyngeal or oropharyngeal swab cannot be obtained, a lower respiratory tract specimen (e.g., endotracheal aspirate, sputum or broncho-alveolar lavage) can be taken ([Appendix 30](#)). These specimens must be placed in a sterile container to be transported for testing.

Labelling of specimens

All specimens must be labeled with:

1. Patient Name
2. Referring Facility
3. Date of Birth
4. Diagnosis: Suspected COVID-19
5. Date and time of sample collection

Specimens must be placed on ice at 4-8°C and transported **immediately** to the National Public Health Laboratory.

All specimens must be accompanied by a completed Jamaica Laboratory Investigation Form (see [Appendix 9](#)). Complete the COVID-19 Laboratory Sample Line Listing ([Appendix 20](#)) and submit a copy to the Parish Health Department/National Surveillance Unit (NSU). Ensure the information is as complete as possible.

Data Analysis and Interpretation

Review and analysis of surveillance data must be completed at all levels.

The Parish Medical Officer (Health) must ensure that Class 1 Notification Forms and Case Investigation Forms are forwarded simultaneously to the Regional Health Authorities and the National Surveillance Unit, within the timelines specified above. A line listing of all reported cases should be maintained at the parish health department, along with contact listings for each case. Epidemic curves, as well as age, sex and geographic distribution of cases, must be maintained at the parish level.

The Regional Technical Director, in collaboration with the Regional Medical Epidemiologist, must ensure that the line and contact listings are maintained for each parish. The age, sex, and geographic distribution, as well as the severity of cases

should be monitored. Depending on the situation, daily or weekly reports may be required.

The National Epidemiology Branch will conduct analysis of national data, including the epidemiological profile of cases and the epidemic curves as the situation evolves. The National Epidemiology Branch will prepare appropriate reports showing information on the patterns of disease within the population.

Data Dissemination and Outputs

The National Epidemiology Branch will be responsible for forwarding the information obtained from national level analysis to the Ministry of Health and Wellness National Emergency Operations Centre (MOH-NEOC).

Chapter 18: Dengue Surveillance Protocol

Background

Dengue (*break-bone fever*) is an acute viral infection that is caused by an arbovirus, which has four related virus serotypes (DENV-1, DENV-2, DENV-3, and DENV-4). It is the most important arthropod transmitted human viral disease and constitutes an important worldwide health problem ¹. Dengue is transmitted between humans by the *Aedes aegypti* and *Aedes albopictus* mosquitoes.

It is more common in tropical and subtropical climates. Most cases are asymptomatic, but for symptomatic cases, the most common symptoms are high fever, headache, body aches, nausea and rash. Most patients will also get better in 1–2 weeks. However, some people develop severe dengue and need care in a hospital and in severe cases, dengue can be fatal. There is no specific treatment currently.

Jamaica experienced its first large dengue outbreak in 1977, although the dengue virus was identified years before. The dengue virus is endemic to Jamaica, and all four serotypes have been identified and have caused outbreaks. The recent dengue outbreaks occurred in 2007, 2010, 2012/2013, 2016, 2018/2019 and 2023/2024, with increasing frequency noted over the period. Additionally, the number of cases in the inter- epidemic periods were higher demonstrating dengue hyperendemicity in Jamaica.

Dengue is usually seasonal with increases in cases occurring September to March of each year.

Severe Dengue is a **Class 1 Notifiable Disease**, while Dengue Fever is reported as a Class 1 Notifiable Disease for notification purposes in order to assist with enhanced monitoring. There is currently enhanced surveillance for Dengue fever so cases must be reported **immediately on suspicion**. Refer to Ministry of Health and Wellness, National Surveillance Manual which outlines

¹ Pan American Health Organization. *Dengue guidelines for patient care in the region of the Americas Second edition*. Washington, D.C. (2016). Available from: <https://iris.paho.org/bitstream/handle/10665.2/31207/9789275118900-eng.pdf?sequence=1&isAllowed=y>

the reporting relationship between the Parish Medical Officer (Health) and the National Surveillance Unit (NSU).

The dengue protocol shall be adhered to for surveillance activities and reporting.

Purpose of these Guidelines

To provide guidance on how to implement surveillance standards for Dengue.

Objectives

The objectives of epidemiological surveillance of Dengue are:

- To monitor Dengue trends in Jamaica
- To inform risk assessment and decision-making.

Case-Based Surveillance

Dengue Fever and Severe Dengue are to be reported as Class 1 Notifiable Conditions. The following dengue severity classification was adopted from the PAHO/WHO Dengue guidelines for patient care in the Region of the Americas ¹:

Dengue without warning signs - DNWS

Person who presents with fever, usually of 2 to 7 days duration, and at least 2 of the following criteria:

1. Nausea/ vomiting
2. Exanthema
3. Headache/ retro-orbital pain
4. Myalgia and arthralgia

¹ Pan American Health Organization. *Dengue guidelines for patient care in the region of the Americas Second edition*. Washington, D.C. (2016). Available from: <https://iris.paho.org/bitstream/handle/10665.2/31207/9789275118900-eng.pdf?sequence=1&isAllowed=y>

5. Petechiae or positive tourniquet test
6. Leukopenia

Cases also include any child with acute febrile illness, usually of 2 to 7 days and no apparent focus.

ACTION: NOTIFY PARISH HEALTH DEPARTMENT, TAKE A SAMPLE AND COMPLETE CLASS 1 NOTIFICATION FORM

Dengue with warning signs - DWWS

Every dengue case that, near and preferably at defervescence, presents with one or more of the following signs:

1. Intense abdominal pain or tenderness
2. Persistent vomiting
3. Fluid accumulation (ascites, pleural effusion, pericardial effusion)
4. Mucosal bleed
5. Lethargy/restlessness
6. Postural hypotension (lipothymia)
7. Liver enlargement >2 cm
8. Progressive increase in haematocrit

ACTION: NOTIFY PARISH HEALTH DEPARTMENT, TAKE A SAMPLE AND COMPLETE CASE INVESTIGATION FORM

STRICT OBSERVATION AND MEDICAL INTERVENTION

Severe dengue – SD

Every dengue case that has one or more of the following manifestations:

1. **Shock or respiratory distress due to severe plasma leakage.**

Shock evidenced by:

weak or undetectable pulse, tachycardia, cold extremities, and capillary perfusion >2 seconds, pulse pressure $\leq$ 20 mmHg: hypotension in late phase.

2. Severe bleeding:

Based on evaluation by the attending physician (e.g. hematemesis, melena, ample metrorrhagia, central nervous system bleeding)

3. **Severe organ involvement**, such as liver impairment (AST or ALT ≥ 1000 IU), CNS (impaired mental state), heart (myocarditis), or other organs.

ACTION: NOTIFY PARISH HEALTH DEPARTMENT, TAKE A SAMPLE AND COMPLETE CASE INVESTIGATION FORM

STRICT OBSERVATION AND MEDICAL INTERVENTION

Dengue Death

Every patient who meets the definition for suspected, probable, or confirmed case who dies as a consequence of dengue.

Note: It is recommended that specific laboratory tests be performed on every dengue death. Similarly, an interdisciplinary committee should analyze all dengue deaths.

ACTION: NOTIFY PARISH HEALTH DEPARTMENT, TAKE A SAMPLE AND COMPLETE CASE INVESTIGATION FORM AND DETAILED NARRATIVE REPORT OR SUBMIT A COPY OF THE DOCKET

Case Definitions

The following case definitions should be used to classify Dengue cases and were adopted from the PAHO/WHO Dengue guidelines for patient care in the Region of the Americas ¹.

Classification	Characteristics of the classification
Suspected Case (Dengue No Warning Signs)	Dengue No Warning Signs (DNWS) Person who presents with acute fever, usually of 2 to 7 days duration, and two or more of the following manifestations: ▪ nausea, vomiting ▪ exanthema ▪ myalgia and arthralgia ▪ headache, retro-orbital pain ▪ petechiae or positive tourniquet test ▪ leukopenia Any child with acute febrile illness, usually of 2 to 7 days duration and no apparent etiology, can also be considered a suspected case.

¹ Pan American Health Organization. *Dengue guidelines for patient care in the region of the Americas Second edition*. Washington, D.C. (2016). Available from: <https://iris.paho.org/bitstream/handle/10665.2/31207/9789275118900-eng.pdf?sequence=1&isAllowed=y>

Classification	Characteristics of the classification
Suspected Case (Dengue With Warning Signs)	Dengue With Warning Signs (DWWS) Every dengue case that presents near or at defervescence or in the following hours one or more of the following signs: ▪ intense and continuous abdominal pain or tenderness ▪ persistent vomiting ▪ fluid accumulation (ascites, pleural effusion, pericardial effusion) ▪ mucosal bleeding ▪ lethargy, restlessness/irritability ▪ postural hypotension (lipothymia) ▪ liver enlargement >2 cm ▪ steady increase in hematocrit
Suspected Case (Severe Dengue)	Severe Dengue Every dengue case that has one or more of the following manifestations: Shock or respiratory distress due to severe plasma leakage Shock evident from tachycardia, cold extremities, and capillary refill time equal to or greater than three seconds, weak or undetectable pulse, convergent/differential blood pressure ≤ 20 mmHg: hypotension in late phase. Severe bleeding, based on evaluation by the attending physician (e.g. hematemesis, melena, ample metrorrhagia, and central nervous system bleeding). Severe organ involvement, such as major liver impairment (AST or ALT >1000 IU), central nervous system (change in mental state), heart (myocarditis), or other organs. Note: Every severe case should be confirmed by laboratory tests specific for dengue.

Classification	Characteristics of the classification
Probable Case	Every suspected dengue case that has a positive IgM or NS1 result or clinical-epidemiological link. Serological diagnosis should include an evaluation of cross-reactivity with other flaviviruses.
Confirmed Case	Every laboratory-confirmed dengue case via real-time polymerase chain reaction (RT-PCR). Note: Laboratory diagnosis should include differential diagnosis for other diseases.
Dengue Death	Every patient who meets the definition for suspected, probable, or confirmed case who dies as a consequence of dengue. Note: It is recommended that in every death from dengue, specific laboratory tests be performed. Similarly, all deaths from dengue should be analyzed by an interdisciplinary committee.
Discarded or Ruled out Case	Every suspected dengue case that meets one or more of the following criteria: ▪ has a negative laboratory diagnosis. (Confirm that samples were obtained within the adequate time frame) ▪ does not have a clinical-epidemiological link, ▪ has a laboratory diagnosis of another clinical disorder, is a case without laboratory testing whose clinical and epidemiological investigations are compatible with suspicion of other disorders.

Case Reporting and Investigation

Notification

Dengue Fever and Severe Dengue are to be reported as Class 1 Notifiable Conditions.

This means all cases thought to be Dengue must be notified by the healthcare worker (public and private) within 24 hours of suspicion. A Class 1 Notification Form ([Appendix 7](#)) must be filled out for **all suspected cases of dengue fever**, regardless of severity, and submitted to the Parish Health Department within 24 hours.

These notifications are then forwarded from the Parish Health Department to the Regional Health Authorities and the National Surveillance Unit (NSU) simultaneously within 24 hours of receipt.

Investigation

The Parish Medical Officer (Health) designates officers for case investigation. A case investigation form ([Appendix 21](#)) is to be completed based on findings of the investigation:

- Initiate case investigation within 24 hours of notification for severe cases. A case investigation report must be submitted to the National Surveillance Unit within 72 hours of this notification.
- All cases of severe dengue must be admitted to hospital, investigated and managed in accordance with current Dengue Clinical Management Protocols.
- All suspected dengue-related deaths must be reported within 24 hours of suspicion using a Class 1 Notification Form ([Appendix 7](#)). Investigations for all suspected dengue related deaths must be initiated within 24 hours and report submitted to the National Surveillance Unit within 72 hours of this notification.

For All Severe Cases and Deaths

Case Investigation is to be done

**A Dengue Case Investigation Form must be completed and submitted to the
Parish Health Department/ National Surveillance Unit.**

Mortality Surveillance

Dengue Death

Every patient who meets the definition for suspected, probable, or confirmed case who dies as a consequence of dengue.

Without Dengue testing, deaths can be misclassified as it is difficult to differentiate between deaths from dengue and deaths from some other causes.

Dengue-related deaths are required to be notified within 24 hours of suspicion to the Parish Health Departments and the National Surveillance Unit. Case investigation must be initiated within 24 hours of notification and a case investigation form ([Appendix 21](#)), including the section specific for deaths, completed for each death and submitted within 72 hours of notification. A copy of the patient's medical record or a detailed narrative report must also be submitted to the National Surveillance Unit, Ministry of Health and Wellness for final review within 72 hours of notification of death.

A sample must be taken for all suspected Dengue deaths.

The following information is to be provided for all suspected Dengue deaths to aid in the classification:

- Age, sex, address (and other identifying data)
- Date of onset of symptoms
- All symptoms and signs
- Date of admission (if admitted to hospital)
- Health Facility / Institution

- Date sample was collected
- Comorbid conditions
- Investigations done and results
- Date of Death
- Course in hospital (if applicable) in the form of a detailed narrative report.

Specimen Requirements

Type of specimen

The recommended sample is **five (5) to seven (7) milliliters of blood** collected in a **red top tube** and transported on ice to the National Public Health Laboratory (NPHL) within 48 hrs.

All specimens must be accompanied by a completed Jamaica Laboratory Investigation Form (see [Appendix 9](#)). Ensure the information is as complete as possible, including the date of onset of symptoms.

Private practitioners should contact the Parish Medical Officer (Health) at their local Parish Health Department to arrange for transportation of samples.

Samples should be taken for the following patients:

- Suspected dengue cases (based on the case definitions)
- All severe cases or those hospitalized with a differential diagnosis that includes dengue
- Patients with co-morbid conditions
- Dengue-related deaths

Testing

Refer to *Guidelines for Laboratory Detection and Diagnosis of Dengue Virus*.

Real-time polymerase chain reaction (RT-PCR) is the confirmatory test for Dengue. RT-PCR will be done for samples taken within 5 days of onset of symptoms.

Antibody testing (IgM) will be done for samples taken more than 5 days after symptom onset.

Data Analysis and Interpretation

Review and analysis of surveillance data must be completed at all levels.

The Parish Medical Officer (Health) must ensure that Class 1 Notification Forms and Case Investigation Forms are forwarded simultaneously to the Regional Health Authorities and the National Surveillance Unit, within the timelines specified above. All notifications should be collated and cases classified as suspected, probable, confirmed or discarded. A line listing of all reported cases should be maintained at the parish level. Epidemic curves (for the cases that are classified as suspected, probable, or confirmed), as well as age, sex and geographic distribution of cases must be maintained at the parish level.

The Regional Technical Director, in collaboration with the Regional Medical Epidemiologist, must ensure that the line listings are maintained for each parish. The age, sex, and geographic distribution, as well as the severity of cases should be monitored. Depending on the situation, daily or weekly reports may be required.

The National Surveillance Unit (NSU) is responsible for the collation of all notifications and classification of cases as well as data entry into and maintenance of the relevant databases. Data analysis will be conducted on national data, including the determination of the epidemiological profile of cases and updating epidemic curves as the situation evolves. The NSU will also prepare appropriate reports showing information on the patterns of disease within the population.

Data Dissemination and Outputs

The National Surveillance Unit will be responsible for forwarding the information obtained from national-level analysis to the Ministry of Health and Wellness - Health Emergency Operations Centre (MOHW-HEOC).

Chapter 19: Malaria Surveillance Guidelines

Background

Malaria is a disease resulting from the infection of red blood cells with protozoan parasites, of the genus *Plasmodium*, transmitted primarily to people through the bite of infected female *Anopheles* mosquitoes, of which there are more than 30 anopheline species that can transmit it. It predominantly occurs in regions with hot and wet climates, with the infective mosquitoes typically biting between sunset and sunrise. Though life threatening, malaria is preventable and treatable. Global efforts are focused on reducing its prevalence and ultimately its eradication.

In humans, malaria is caused by five species of the *Plasmodium* parasite: *P. falciparum*, *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi*, which are geographically distributed as shown in the table below. Among these species, *P. falciparum* and *P. vivax* are the most prevalent, with *P. falciparum* being the most dangerous ^{1, 2}.

Table 19.1. Major Distribution of Malaria Parasite that Cause Human Infection

	<i>P. falciparum</i>	<i>P. vivax</i>	<i>P. ovale</i>	<i>P. malariae</i>	<i>P. knowlesi</i>
Major Distribution	West, East & Central, Middle East Africa; Far East, South America	India, North & East Africa, South America, Far East	Tropical Africa	Tropical Africa, India, Far East	Forested regions of South East Asia
Common Name	Malignant Tertian	Benign Tertian	Ovale Tertian	Quartan	Monkey Malaria

¹ World Health Organization. *World Malaria Report: 2012* [Internet]. Geneva, Switzerland (2012). Available from: http://www.who.int/malaria/publications/world_malaria_report_2012/wmr2012_full_report.pdf

² Rawlins SC, Hinds A, Rawlins JM. *Malaria and its vectors in the Caribbean: The Continuing Challenge of the Disease Forty-five Years after Eradication from the Islands*. West Indian Medical Journal 2008; 57 (5): 462. Available from: chrome-extension://efaidnbmnnnibpcjpcglclefindmkaj/https://www.mona.uwi.edu/fms/wimj/system/files/article_pdfs/rawlins_malaria_and_its_vectors_in_the_caribbean_the_continuing_challenge_of_the_disease_forty_five_years_after.pdf

Malaria was eradicated in Jamaica after the successful Malaria Eradication Campaign of 1958-62. The last endemic case was in 1961¹ until 2006 and 2007, when the country experienced malaria outbreaks from both *P. falciparum* (in KSA and St. Catherine) and *P. vivax* (in St. Elizabeth) species respectively. These outbreaks revealed the vulnerability of Jamaica to the reintroduction of malaria, leading to the World Health Organization in 2008 classifying Jamaica as, “a country registered (November 1966) as having achieved malaria eradication but unable to maintain the absence of local transmission”². Later, having controlled the outbreaks and with no local transmission reported since 2009; in 2012 Jamaica was reinstated in the Official Register of areas where malaria eradication had been achieved¹. This achievement was due to the successful implementation of its malaria control programme.

Jamaica continues to prevent the re-introduction of malaria through public health surveillance, identifying imported cases and instituting the necessary control measures to prevent local spread. The two main factors contributing to the risk of re-introduction are:

- a) The presence of the *Anopheles* mosquito vector throughout the island.
- b) Individuals entering Jamaica either by sea or air travel, from countries where malaria is endemic.

The *Anopheles albimanus species* is considered the principal vector for transmission in Jamaica.

Epidemiology

Each year, about 300-500 million clinical cases of malaria occur worldwide. In 2024, the estimated global number of malaria cases was over 280 million. The WHO African Region bears the majority of the burden of malaria cases worldwide in 2024. The five countries with the highest estimated

¹ Pan American Health Organization. *Status of Malaria in the Americas. XVIII report* [Internet]. Washington D.C. (1970) Available from: http://hist.library.paho.org/english/gov/csp/18_7.pdf.

² World Health Organization. *World Malaria Report 2025* [Internet]. (2025) Available from: <https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2025>

burden of malaria cases in 2023 were Nigeria, the Democratic Republic of the Congo, Uganda, Ethiopia, and Mozambique. The global number of deaths was estimated at 600,000 ¹.

The Life Cycle of the Malaria Parasite

The life cycle of the malaria parasite involves two hosts: humans and the female *Anopheles* mosquitoes. During a blood meal, a malaria-infected female *Anopheles* mosquito inoculates sporozoites into the human host. Sporozoites infect liver cells and mature into schizonts, which rupture and release merozoites. In *P. vivax* and *P. ovale* a dormant stage (hypnozoites) can persist in the liver and cause relapses by invading the bloodstream weeks, or even years later.

After this initial replication in the liver (exo-erythrocytic schizogony), the merozoites then infect red blood cells where they undergo asexual multiplication (erythrocytic schizogony). In the red blood cells, the parasites form immature (ring stage) trophozoites which then mature into schizonts. Once this stage is complete the infected red blood cells rupture, releasing merozoites into the blood stream initiating another cycle of asexual development and multiplication. Blood stage parasites are responsible for the clinical manifestations of the disease (Figure 19.1).

Some parasites differentiate into sexual erythrocytic stages (gametocytes). The gametocytes are ingested by an *Anopheles* mosquito during a blood meal. The parasites' multiplication in the mosquito is known as the sporogonic cycle. While in the stomach of the mosquito, the microgametes penetrate the macrogametes generating zygotes. The zygotes in turn become motile and elongated ookinetes which invade the midgut wall of the mosquito where they develop into oocysts. The oocysts grow, rupture, and release sporozoites, which make their way to the mosquito's salivary glands. Inoculation of the sporozoites into a new human host perpetuates the malaria life cycle².

¹ World Health Organization. *World malaria report 2024: addressing inequity in the global malaria response* [Internet]. Geneva: License: CC BY-NC-SA 3.0 IGO (December 2024). Available from: <https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2024>

² Centers for Disease Control and Prevention. *Malaria – Epidemiology and Clinical Features* [Internet]. (2024) Available from: <https://www.cdc.gov/malaria/about/index.html>

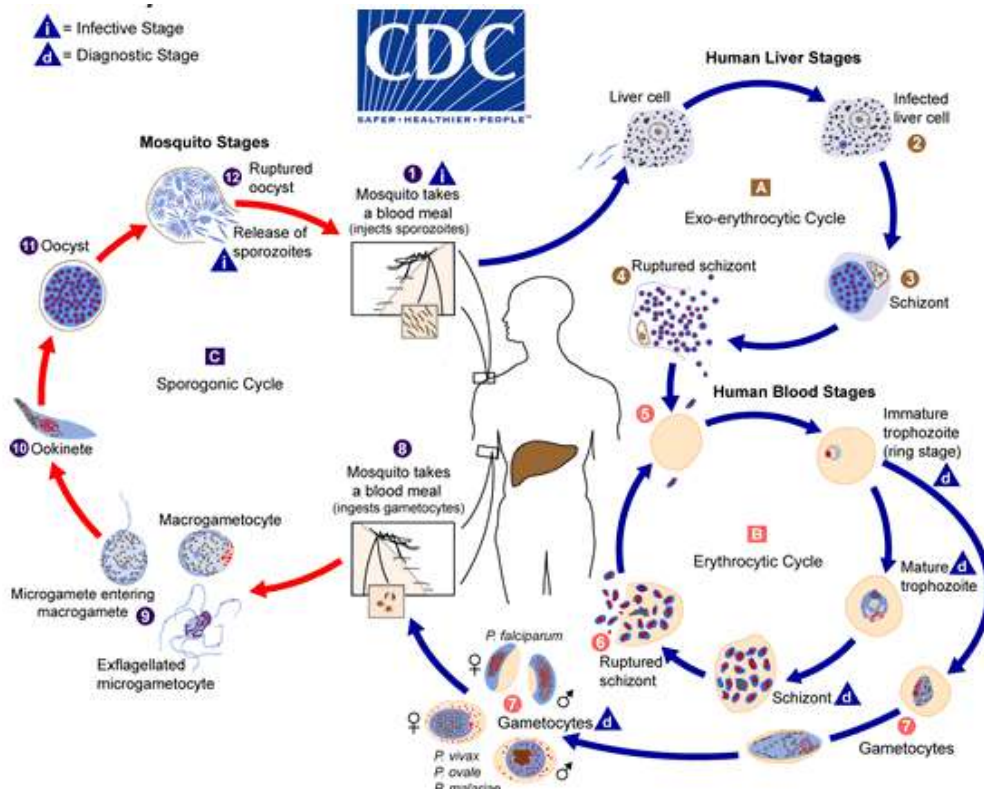


Figure 19.1. Life Cycle of the Malaria Parasite ¹

Clinical Features

Clinical features depend on age, immune status and the species of parasite. Immunity to malaria develops gradually and seems to need repeated boosting. Early symptoms are non-specific—such as malaise, fever with chills, sweats, headache, nausea, vomiting, diarrhoea, tiredness, muscle and joint pain—and resemble other infections (e.g., influenza or other common viral infections). Likewise, the physical findings are also often general including elevated temperature, perspiration, jaundice, hepatomegaly and splenomegaly. Symptoms usually appear 10–15 days after an infective mosquito bite. Delay in treatment or ineffective medicine, particularly for *P. falciparum* malaria leads to increased parasite burden and severe malaria (bleeding disorders, kidney and liver failure, acute encephalopathy, pulmonary and cerebral edema, coma and death)

¹ Centers for Disease Control and Prevention – DPDx. *Laboratory Identification of Parasites of Public Health Concern: Malaria* [Internet]. (December 2024) Available from: <https://www.cdc.gov/dpdx/malaria/index.html>

may ensue within a few hours. In severe malaria (caused by *P. falciparum*), clinical findings (confusion, coma, neurologic focal signs, severe anemia, respiratory difficulties) are more striking and may increase the suspicion index for malaria^{1,2}. The case fatality rate at this stage for persons receiving treatment is typically 10–20%.

Paroxysmal fever is the hallmark symptom of malaria with the duration lasting up to 8 or more hours at a time and varying by plasmodium species (Table 19.2). It typically begins with a flu-like illness classically described as a quartan fever period (occurs every 72 hours for paroxysms due to *P. malariae*); or tertian fever period (occurs every 48 hours for paroxysm caused by *P. vivax* and *P. ovale*). This is synchronized with the *Plasmodium* merozoites bursting from red blood cells during the parasite's asexual erythrocytic phase. These paroxysms occur with abrupt cold “chills” that transitions after about an hour, to profuse sweating, high fever, headache, malaise, and myalgia; then subsequent defervescence.

The classic presentation with predictably recurring fever and chills is highly variable and may not be present at all. This is particularly so early in the course of the illness, when the patient may have been on antipyretics or antimalarials or when partial immunity exists.

Malaria should be considered in persons presenting with the following febrile syndromes:

- Undifferentiated Fever
- Fever and Jaundice
- Fever and Haemorrhage

Characteristics of the Different Malaria Parasite Infections

The incubation period for malaria varies among plasmodium species but ranges from 7 – 40 days. *P. falciparum* infections can have a daily or irregular pattern of symptoms. Infections with *P. ovale*, *P. vivax* and *P. malariae* tend to run a benign course but may relapse many years after the initial attack. It is estimated that about 1% of patients with *P. falciparum* infection die.

¹ World Health Organization. *Malaria* [Internet]. (December 2025) Available from: <https://www.who.int/news-room/fact-sheets/detail/malaria>

² Centers for Disease Control and Prevention. *Malaria – Epidemiology and Clinical Features* [Internet]. (2024) Available from: <https://www.cdc.gov/malaria/about/index.html>

Table 19.2. Characteristics of the Different Malaria Parasite Infections

Characteristic	<i>P. Falciparum</i>	<i>P. Vivax</i>	<i>P. Ovale</i>	<i>P. Malariae</i>	<i>P. Knowlesi</i> ¹
Pre-erythrocytic Stage - Days	5 ½ - 7	6 - 8	9	14 - 16	8 - 9
Incubation Period (days)	12 (9 -14)	15 (12 -17)	17 (16 -18) to > 50	28 (18 - 40) to >72	12
Erythrocytic (Fever) Cycle (hours)	48	48	50	72	24
Primary Attack (Severity)	Most severe; potentially fatal	Mild to severe	Mildest; rarely fatal	Generally mild; chronic infection possible	Potentially severe and rapidly fatal
Febrile Paroxysm (Duration of Fever) (hours)	16-36 or longer	8 -12	8 -12	8 -10	8 - 12
Relapses (Dormant Liver stage)	None	Yes	Yes	None	None
Duration of Untreated Infection (years)	1 - 2	1.5 - 5	Probably same as vivax	3 - 50	
Major Complications	Cerebral malaria, anaemia, hypoglycaemia, jaundice, pulmonary oedema, shock	Rare	Rare	Nephrotic Syndrome	

¹ Antinori S, Galimberti L, Milazzo L, Corbellino M. *Biology of human malaria plasmodia including Plasmodium knowlesi*. *Mediterr J Hematol Infect Dis*. 2012;4(1):e2012013. doi: 10.4084/MJHID.2012.013. Epub 2012 Mar 10. PMID: 22550559; PMCID: PMC3340990. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3340990/>

Characteristic	<i>P. Falciparum</i>	<i>P. Vivax</i>	<i>P. Ovale</i>	<i>P. Malariae</i>	<i>P. Knowlesi</i> ¹
Parasitaemia/mm ³	20,000 - 500,000	20,000	9,000	6,000	600 – 10,000
Recurrences	Yes (treatment failure)	Yes (treatment failure)	No	Yes (as long as 30-50 years after primary attack)	Yes

Mode of Transmission and Risk Factors

Malaria is primarily transmitted in tropical and sub-tropical regions, including sub-Saharan Africa, Central and South America, Hispaniola, the Middle East, the Indian subcontinent, Southeast Asia, and Oceania ([Appendix 14](#)). Its transmission is influenced by climatic factors such as temperature, humidity, and rainfall, which affect mosquito breeding and survival.

Mode of transmission

- *Animal to Human*

Only female *Anopheles* mosquitoes can spread malaria from one person to another. For the *Anopheles* mosquito to become infective, it must bite, or take a blood meal, from a person already infected with the malaria parasites. Most species feed at dusk or at night or very early in the morning.

- *Human to Human*

Rarely, malaria is spread from person-to-person through infected blood - during blood transfusions, organ transplantation, sharing needles or syringes contaminated with malaria-infected blood; or congenitally (from a mother to the foetus before or during delivery).

Communicability

In humans, the parasite takes 10 – 14 days to form gametocytes. When the female mosquito takes a blood meal, she becomes infected by the gametocyte (the sexual stages of the parasite). The gametocyte undergoes further development in the mosquito for a period of 6 – 12 (8-35) days. Humans may infect mosquitoes as long as infective gametocytes are present in the blood.

Risk factors

Human activities, including agricultural practices, deforestation, and urbanization, can also impact transmission dynamics. While susceptibility to malaria is universal it may be more pronounced and likely fatal in certain population groups at higher risk. These include infants, young children, individuals with concurrent disease or immunodeficiency (sickle cell disease and HIV/AIDS infection), non-immune migrants, mobile populations, travelers and pregnant women.

Malaria in pregnancy can be very severe and life-threatening to both mother-to-be and foetus. The risk of maternal death, miscarriage, stillbirth and low birth weight with associated neonatal death are increased. Hence, if malaria is suspected in a pregnant woman, immediate medical attention should be given.

Purpose of Guidelines

To promptly detect cases of malaria infection and facilitate the implementation of public health actions to mitigate impact of the infection on individuals and the population and to prevent local spread.

Objectives

The objectives of epidemiological surveillance of malaria are:

- Detecting malaria cases in Jamaica.
- Isolate cases to prevent local transmission.

- Quickly identify cases, clusters and sources of infection to provide optimal clinical care to cure the infection and prevent severe disease.
- Conduct epidemiological characterization of malaria cases in terms of time, place, and person.
- Monitor the spread of malaria if reintroduced.
- To prevent the emergence and spread of drug-resistant malaria.
- Inform public health action and policy for a national response.

The Class 1 Notification System of Jamaica will be used to detect and report cases of Malaria infection. This protocol should be adhered to for surveillance activities and reporting.

Case Definitions

Suspected Case

A person with intermittent fever (or a history of fever) with chills and headache and no other obvious cause, with a history of living in or travelling to a malaria endemic area.

Note: Other non-specific symptoms may include malaise, fatigue, abdominal discomfort, joint and muscle aches.

Confirmed case

A suspected case with laboratory confirmation - identification of *Plasmodium* species on peripheral blood smear. Confirmed cases are classified as follows:

- **Imported:** Malaria acquired outside the country.
- **Autochthonous:** cases acquired locally in an area where malaria is not endemic, without the patient travelling abroad.
- **Indigenous:** Malaria acquired by mosquito transmission in an area where malaria is a regular occurrence.

- **Introduced:** Malaria acquired by mosquito transmission from an imported case in an area where malaria is not a regular occurrence.
- **Induced:** Malaria acquired through artificial means (e.g., blood transfusion, sharing of syringes or needles)
- **Congenital:** Malaria acquired through transplacental transmission.
- **Cryptic:** An isolated case of malaria not associated with secondary cases, as determined by appropriate epidemiologic investigations.
- **Relapsing:** Recurrence of disease after it has been apparently cured. Relapses are caused by reactivation of dormant liver-stage parasites (hypnozoites) of *P. vivax* and *P. ovale* ¹.

Case Reporting and Investigation

ACTION: NOTIFY PARISH HEALTH DEPARTMENT, TAKE A SAMPLE AND COMPLETE CLASS 1 NOTIFICATION AND CASE INVESTIGATION FORMS

Notification

Malaria is a Class 1 notifiable zoonotic disease. Due to the public health risks, it must be reported immediately on suspicion within 24 hours to the Medical Officer (Health) at the parish health department and the National Surveillance Unit, even if other diagnoses are considered. A Class 1 Notification Form ([Appendix 7](#)) shall be completed when reporting each case, regardless of severity. The receiving entity is to share the notifications received with the other levels of the surveillance system.

Case reports should include at a minimum the following information:

- Date of report
- Reporting location
- Name, age, sex and residence of the case
- Date of onset of first symptoms

¹ Caribbean Public Health Agency. *CARPHA's Regional Public Health Surveillance Manual*. Port of Spain, Trinidad and Tobago, (June 2021)

- Recent travel history
- Presence of fever
- Presence of other clinical signs or symptoms as per case definition
- Laboratory findings

Investigation

The Parish Medical Officer (Health) leads the case investigation team and must initiate a preliminary case investigation and submit an investigation form ([Appendix 22](#)) and report within 24 hours of notification.

Special attention must be paid to travel history including local and international travel.

Contact Tracing

All contacts should be promptly investigated and attempts made to identify the source of infection.

- Record addresses and travel history of recent visitors to the household.
- Determine whether there are persons in the household or immediate neighbours with similar symptoms who will need to be sampled for testing and treated as necessary.

Specimen Requirements for Confirmatory Testing

The diagnosis of malaria is based on the detection of parasites in the blood in a person that is clinically suspected of having malaria. Microscopic examination of blood remains the reference method for laboratory confirmation of malaria, particularly for parasite detection, quantification, species identification, and monitoring of the response to therapy¹. Thick blood films are more sensitive for detecting malaria parasites. Thin blood films are important for species identification and assessment of parasite density.

¹ World Health Organization. *World Malaria Report 2009* [Internet]. (November 2009) Available from: http://whqlibdoc.who.int/publications/2009/9789241563901_eng.pdf

The National Public Health Laboratory (NPHL) is the reference laboratory for malaria diagnosis. All persons suspected of having malaria should have **blood taken for a thick and thin smear on glass slides and 1 ml of blood in a tube with EDTA (purple top tube)**. This should be sent to the **NPHL within 6 hours** accompanied by a completed Jamaica Laboratory Investigation Form ([Appendix 9](#)). Ensure the information on the form is as complete as possible, including the date of onset of symptoms and that slides and blood tubes are properly labelled, including sample dates and time. Dates are essential since serial specimens are taken from each case and the results are important to case management.

Private practitioners should contact the Parish Medical Officer (Health) at their local Parish Health Department to arrange for transportation of samples. Specimens should be addressed to the Malaria Laboratory, in the Parasitology Division, NPHL and the laboratory should be notified prior to sending samples. The Medical Officer (Health) should be advised of laboratory results for malaria tests as soon as possible. The positive identification of *Plasmodium* parasite and species should be reported.

All laboratory forms must be completed with the following information:

- Patient Name
- Patient Address
- Date of Birth
- Referring Facility
- Diagnosis: Suspected Malaria
- Date of onset of symptoms
- Date and time of sample collection
- Type of sample collected
- Hospital ID/Registration number

Microscopic examination of a blood sample remains the ‘field standard’ for confirmation of a malaria diagnosis. A single negative blood film does not exclude malaria, particularly in non-immune individuals who may be symptomatic at low parasite densities. For suspected malaria cases with a negative initial smear, blood films should be repeated at 12–24 hour intervals, with a total of three sets examined before the diagnosis of malaria can be ruled out.

Rapid Diagnostic Tests (RDTs) offer a useful alternative to microscopy where reliable microscopic diagnosis is not available. It is recommended that all RDTs are followed up with microscopy to confirm the results.

Community Fever Surveillance for Malaria

Conducting community fever surveillance for malaria helps in the early detection of secondary cases. Fever surveillance usually begins from about day 21 to day 28 post symptom onset in the index case (dependent on the *Plasmodium* species).

Information needed to determine when to start fever surveillance:

- the time (date) of onset of symptoms in the index patient.
- the extrinsic incubation period (days taken for the mosquito to become infective after biting the index case (9-40 days).
- the intrinsic incubation period (days taken for symptoms to develop in the secondary case, once bitten by the infected mosquito (7-14 days).

Persons in the community who are identified with fever or history of fever should be sampled and tested for malaria.

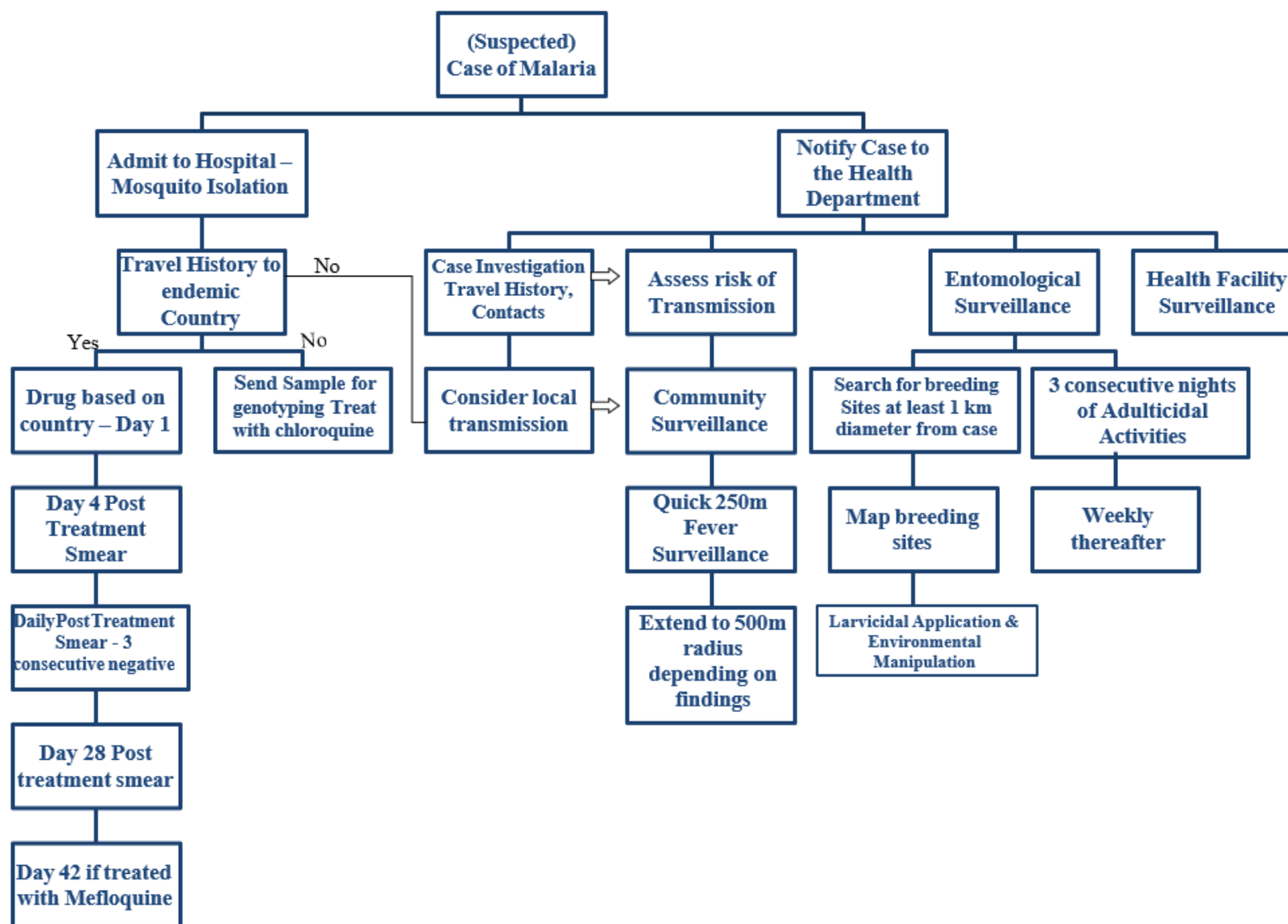


Figure 19.2. Algorithm for the management of a case of malaria in Jamaica

Case Management

A suspected case **MUST BE ADMITTED TO HOSPITAL** and isolated in a mosquito-proof area, or under a mosquito net while awaiting further investigations. Anti-malarial drugs should commence (after blood sample is taken).

Value of hospitalization:

- Allows for appropriate management under a mosquito-proof net.
- Allows better supervision of anti-malarial therapy.
- Allows on-the-spot recognition of complications of malaria or side effects of drug therapy.

Data Analysis and Interpretation

Review and analysis of surveillance data must be completed at all levels.

The Parish Medical Officer (Health) must ensure that Class 1 Notification Forms and Case Investigation Forms are forwarded simultaneously to the Regional Health Authorities and the National Surveillance Unit. All notifications should be collated and cases classified as suspected, confirmed or discarded. A line listing of all reported cases should be maintained at the parish level. Epidemic curves (for the cases that are classified as suspected or confirmed), as well as age, sex and geographic distribution of cases must be maintained at the parish level.

The Regional Technical Director, in collaboration with the Regional Medical Epidemiologist, must ensure that the line listings are maintained for each parish. The age, sex, and geographic distribution, as well as the severity of cases should be monitored. Depending on the situation, daily or weekly reports may be required.

The National Surveillance Unit (NSU) is responsible for the collation of all notifications and classification of cases as well as data entry into and maintenance of the relevant databases. Data analysis will be conducted on national data, including the determination of the epidemiological profile of cases and updating epidemic curves as the situation evolves. The NSU will also prepare appropriate reports showing characterization of the disease within the population.

Data Dissemination and Outputs

The National Surveillance Unit will be responsible for forwarding the information obtained from national level analysis to the Ministry of Health and Wellness – Health Emergency Operations Centre (MOHW-HEOC).

International Reporting

Given that malaria has been eradicated in Jamaica, and there is an existing risk for malaria re-introduction, any incidence of local malaria infection spread, is reported through the appropriate regulatory mechanisms. The situation is to be evaluated using the "Decision Instrument for the Assessment and Notification of Events That May Constitute a Public Health Emergency of International Concern, as outlined in Annex 2 of the International Health Regulations (2005).

Chapter 20: Maternal Mortality Surveillance

Maternal Death is a Class 1 notifiable health event and requires notification within 24 hours of suspicion to the parish health department or the National Surveillance Unit. For surveillance purposes all deaths in women of reproductive age (10-50 years) in whom there is evidence of pregnancy within a year preceding death are to be notified.

Classification of Maternal Deaths

Maternal Deaths

Case Definition:

The death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of pregnancy, from any causes related to or aggravated by the pregnancy or its management but not from accidental or incidental causes ¹.

- This definition includes:
 - Direct: Obstetric complications of the pregnant state from interventions, omissions, incorrect treatment or from a chain of events resulting from any of the above.
 - Indirect: Previous existing disease or disease that develops during pregnancy and which was not due to obstetric causes but was aggravated by the physiological effects of pregnancy.
- This definition excludes (Late and Coincidental deaths)

Late Maternal Deaths

Case Definition

The death of a woman from direct or indirect obstetric causes more than 42 days but less than one-year after termination of pregnancy (included in surveillance but excluded from maternal mortality statistics).

¹ World Health Organization. *The WHO Application of ICD-10 to deaths during pregnancy, childbirth and the puerperium: ICD-MM* [Internet]. Geneva (2012). Available from: [chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/1bbeb2a8-e4c0-46c8-aa0d-0056627a17b5/content](https://iris.who.int/server/api/core/bitstreams/1bbeb2a8-e4c0-46c8-aa0d-0056627a17b5/content)

Coincidental Deaths

Case Definition

The death of woman while pregnant but from causes unrelated to the pregnancy i.e., accidental or incidental causes (included in surveillance but excluded from maternal mortality statistics).

Maternal Mortality Case Definition for Surveillance

Death in any woman of reproductive age (10-50 years) in whom there is evidence of pregnancy or termination of pregnancy within the past one year.

Surveillance Criteria:

Gender:	Female
Age:	10-50 years
Evidence of Pregnancy:	Pregnant and died undelivered, history of abortion, miscarriage, stillbirth or live birth
Period of observation for pregnancy:	From date of death to one-year preceding death

Cases are to be notified on suspicion and investigated using the forms in the Appendices 36-40 (Maternal Mortality Clinical Summary, Maternal Mortality Home Visit and Antenatal Report, Maternal Mortality Post-Mortem Summary and Maternal Mortality Case Review Summary). The information documented should be reviewed by the parish surveillance teams and all reports submitted to the Regional Health Authority and the National Surveillance Unit, Ministry of Health and Wellness. The purpose of the investigation is to determine:

- a. Cause of death
- b. Underlying risk factors/ conditions
- c. Delays associated with accessing/receiving appropriate care
- d. Classification as either direct, indirect, coincidental or late

The National Maternal Mortality Committee will make a final decision on which cases are to be included in determining the national maternal mortality ratio for Jamaica and the final classification of the cases.

Surveillance Process

Step 1: Identification Of Cases

- Notification of a suspected maternal death immediately on occurrence by each medical practitioner/clinician
- Monthly contacts/ visits with the local registrar of deaths, the police and the funeral homes
- Monthly contacts/visits to private maternity facilities of Obstetricians
- Weekly visits to hospitals

Responsible Person: Hospital Active Surveillance (HAS) Officer

Activities:

- Review all death and ward registers (obstetrics, medicine, surgery, ICU, accident & emergency department) to identify deaths in females 10-50 years.
- Review each docket for evidence of pregnancy or delivery within the last year. If there is evidence of pregnancy, complete a Class I Notification Form ([Appendix 7](#)) as well as a preliminary Maternal Mortality Clinical Summary Form (Form 1 – [Appendix 23](#))
- Leave another Maternal Mortality Clinical Summary Form with the consultant clinician to be completed. Ensure that the patient is adequately identified on the form by completing the demographic information.

Timeframe: As case is identified or notified by institution or weekly

Step 2: Logging and Routing of Cases for Investigation

Review Maternal Death Notifications

Responsible Person: Parish Medical Officer of Health [MO(H)]

Activities:

- Log suspected case in a surveillance register
- Inform by forwarding the notification and the preliminary Maternal Mortality Clinical Summary Form to the following persons:

- Senior Public Health Nurse or Public Health Nurse responsible for community of residence to the suspected case if case is resident in the parish.
- Regional Medical Epidemiologist/ Regional Epidemiologist/ Regional Surveillance Officer
- Ministry of Health and Wellness, National Surveillance Unit
- MO(H) of the parish of residence if the suspected case lives in another parish

Time Frame: within 24 hours of receipt of notification

Step 3: Conduct Case Investigation

Complete Clinical Review

Responsible Person: Consultant obstetrician (Type A or B Hospital/Medical Officer in Charge (Type C or Community Hospital)

Activities:

- Complete and review the Maternal Mortality Clinical Summary (Form 1) and ensure that all available information has been documented.
- Submit completed Maternal Mortality Clinical Summary Form (Form 1) to the relevant Parish Health Department (parish of residence of the case through the Hospital Active Surveillance Officer if necessary)

Time Frame: Within 72 hours of the death or notification of the death

Complete Home and Antenatal Investigation

Responsible Person: Senior Public Health Nurse/Public Health Nurse

Activities:

- Conduct home visit and interview relatives
- Visit antenatal clinic(s) attended by the suspected case including
 - Community Health Centre
 - Private Practitioner
 - High Risk Clinic
 - Hospital

- Review all available documentation including maternal record book and docket notes
- Complete the Maternal Mortality Home Visit and Antenatal Report (Form 2 – [Appendix 24](#)) using data collected from the home visit and antenatal records
- Submit the completed Maternal Mortality Home Visit and Antenatal Report (Form 2) to the Medical Officer of Health in the parish of residence of the patient.

Time Frame: Within three (3) Weeks of the notification

Complete Postmortem Summary Report

Responsible Person: Regional Medical Epidemiologist/ Regional Epidemiologist/ Regional Surveillance Officer

Activities:

- Liaise with the Pathologist doing the post-mortem to receive the final report
- Complete Maternal Mortality Postmortem Summary (Form 3 – [Appendix 25](#))
- Submit a copy of the Maternal Mortality Postmortem Summary (Form 3) to the MO(H) in the parish of residence of the case.

Time Frame: Within four (4) weeks of receipt of notification.

Step 4: Complete Case Investigation Reports

Compile Case Investigation Reports

Responsible Person: Regional Medical Epidemiologist/ Regional Surveillance Officer/ Regional Epidemiologist

Activities:

- Compile the Maternal Mortality Clinical Summary, Home Visit and Antenatal Report and the Post-Mortem Summary (Forms 1-3)
- Forward all completed Maternal Mortality Summary Forms (Forms 1-4) to the National Surveillance Unit at the Ministry of Health and Wellness
- Inform Regional Technical Director when compiled

Time Frame: Within one (1) week of receipt of the reports

Step 5: Review Maternal Mortality Cases

Convene Regional Case Review Meeting

Responsible Person: Regional Technical Director

Activities:

- Set date, time and venue of meeting and advise all participants
- Facility and parish case reviews are to be held and information garnered used to inform discussions at the Regional Case Reviews.
- Ensure the following categories of staff attend the meeting
 - Regional Obstetrician
 - Regional Nursing Supervisor
 - Epidemiologist
 - MO(H) parish of residence
 - MO(H) parish of occurrence
 - Clinical service providers (Hospital, Community)
 - Pathologist
 - Medical Officer (Health)-Surveillance
 - Nurse Manager for Maternity Ward or designate
 - National Surveillance Officer responsible for maternal mortality
 - Director, Family Health Services

At the conclusion of the Case Review Meeting:

- Finalize the Case Review Summary (Form 4 – [Appendix 26](#))
- Ensure that all completed case reports (Forms 1-4) have been forwarded to following parties:
 - MO(H) parish of residence
 - Hospital of occurrence (SMO)
 - Ministry of Health and Wellness, National Surveillance Unit
- Update electronic database on a quarterly basis

Time Frame: Within 6 weeks of notification of the case

Step 6: Monitor Maternal Mortality Surveillance System

Monitor Maternal Mortality Surveillance System

Responsible Persons: Medical Epidemiologist / National Surveillance Officer

Activities:

- Update and maintain national maternal mortality database
- Analyze maternal mortality database
- Validate maternal deaths with registered maternal deaths
- Active search of the list of deaths in women of reproductive age from the National Identification and Registration Authority (NIRA)
- Prepare national data for stakeholders
- Monitor the surveillance process

Time Frame: Quarterly

Guidelines for Regional Case Reviews and Annual Regional Meetings

Regional Case Reviews

The objectives of the individual case review are to:

1. Identify the medical cause of death
2. Identify the preventable social and health services determinants.

The case review meetings should be conducted by a small group of persons including:

- Obstetrician or other medical officer who attended the mother during her latest illness
- Midwife or public health nurse who conducted the home visit
- Regional Medical Epidemiologist and/or Surveillance Officer
- Medical Officers of Health (parish of death occurrence and residence of mother)

The team should have available:

- Completed Forms 1-4 (Clinical Summary, Home Visit and Antenatal Report, Postmortem Summary, Case Review Summary)
- Medical record summaries from the most recent health facilities the patient attended (hospital, health center)

At the end of the review process the review team should be able to:

- Identify the underlying cause of death (the medical or obstetric complications which initiated the chain of events that led to the patient's death)

- Identify the immediate cause of death (the final complications leading to the patient's death)
- Classify the death as direct (obstetric), indirect (medical) or coincidental (pregnancy not related to patient's death e.g. accident, violence) ¹.
- Determine the interval between delivery and death to determine whether the death meets the 6-week criteria defining a maternal death.
- Identify whether there were any delays using the Three Delays Model:
 - Delay 1 - delays in the patient seeking care
 - Delay 2 - delays in reaching or accessing a health facility
 - Delay 3 - delays in the provision of adequate and appropriate care at a health facility

At the end of the review process, the decisions of the review team should be forwarded to the Ministry of Health and Wellness; with copies to (1) the hospital(s) providing care and (2) the parish health department in the mother's parish of residence to be shared with the primary care team at the primary care facility where the mother had her antenatal care.

Deliverables (monthly or quarterly at minimum):

- Copies of all forms (1-4) related to each case of maternal death

Annual Regional Meetings

Objective: Comprehensive annual review of all pregnancy related deaths (maternal, late maternal, coincidental) over the previous calendar year with a view to identifying system failures in quality of care that need to be addressed at the regional level by making policy changes to reduce the recurrence of avoidable features.

¹ World Health Organization. *The WHO Application of ICD-10 to deaths during pregnancy, childbirth and the puerperium: ICD-MM* [Internet]. Geneva (2012). Available from: [chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://iris.who.int/server/api/core/bitstreams/1bbeb2a8-e4c0-46c8-aa0d-0056627a17b5/content](https://iris.who.int/server/api/core/bitstreams/1bbeb2a8-e4c0-46c8-aa0d-0056627a17b5/content)

In preparing for the annual review, cases due to similar underlying causes should be presented as a group. The process should avoid presenting cases on a parish by parish or hospital by hospital basis; instead similar cases should be presented together (e.g. all gestational hypertension; all diabetes cases).

The purpose of the annual review is to:

- Share information on the underlying causes of death.
- Compare the recent experience with past trends.
- Identify what delays if any, occurred in accessing care (Delay 1, 2 or 3)
- Identify whether there were limitations in quality of care which need to be addressed by the health team.
- Identify what policy changes are needed to improve the quality of care and prevent similar deaths from occurring in the future.
- Identify who will be responsible for spearheading these changes.
- If external help is needed to address the problem, identify who will be responsible for identifying where this assistance will come from including potential cost of these interventions.

Indicators For Monitoring

The following are indicators to be used for monitoring maternal mortality surveillance:

- % of cases notified within 24 hours of suspicion
- % of cases investigated within 6 weeks of notifications
- % of cases with all forms (Form 1-4) completed and submitted to the Ministry of Health and Wellness
- % of cases with post-mortems done
- # of regional maternal mortality reviews held
- Maternal mortality ratio

Chapter 21: MPox Surveillance

MPox Summary

Human MPox (hMPX) is a zoonotic viral disease caused by the MPox virus (MPXV). Human MPox infection, if occurring in Jamaica, would be considered an exotic or unusual communicable disease and should, therefore, be reported as a Class 1 notifiable disease.

The incubation period for MPox is 5 to 21 days.

The ideal test sample is a swab of skin lesion material, including swabs of lesion surface and/ or roof, or vesicular fluid/exudate. All specimens must be placed on ice at 4-8°C and transported immediately to the National Public Health Laboratory. A case is confirmed by real-time polymerase chain reaction (RT-PCR).

MPox must be notified by the healthcare worker (public and private) immediately on suspicion to the local Parish Health Department and the National Surveillance Unit.

Clusters of visits for unusual rash must be notified by the healthcare worker (public and private) immediately and investigated.

Healthcare workers in all settings should be vigilant in identifying patients presenting with unusual rash associated with lymphadenopathy or fever. They should also record a detailed travel and vaccination history.

The Class 1 Notification System of Jamaica will be used to detect and report cases of MPox infection

Background

MPox virus (MPXV) is an enveloped double-stranded DNA virus with a genome size of approximately 190 kb. It belongs to the *Orthopoxvirus* genus of the *Poxviridae* family. The *Orthopoxvirus* genus also includes *Vaccinia virus*, *Cowpox virus*, *Variola virus* and several other animal-related poxviruses ¹. Human MPox (MPX) was recognized as a human disease in 1970.

There are two (2) phylogenetically distinct clades of MPXV:

1. Clade I (subclades Ia and Ib) - Central African (Congo Basin) clade - more severe with case fatality ratio of 10% and more frequent human-to-human transmission
2. Clade II (subclades IIa and IIb) - West African clade - self-limiting with a case fatality ratio of about 1%.

Mortality is higher among children and young adults; and pregnant women and immunocompromised individuals are especially at risk of severe disease. Complications such as respiratory distress, secondary bacterial infections, encephalitis, and sequelae were found to be less common in patients vaccinated against smallpox ².

Epidemiology

Human MPox virus was first isolated in 1958 from pox lesions during an outbreak of vesicular disease among captive cynomolgus macaques (monkeys) imported into Denmark for polio-vaccine-related research. Although the disease name suggests that monkeys are the primary host, the specific animal reservoir of MPXV remains unknown ². African rodents such as the Gambian giant rats (*Cricetomys gambianus*) and squirrels might be natural reservoirs of the virus³.

¹ World Health Organization. *Alert: Multi-country monkeypox outbreak in non-endemic countries* [Internet]. (May 2022) Available from: <https://www.who.int/emergencies/disease-outbreak-news/item/2022-DON385>.

² World Health Organization. *Mpox* [Internet]. (August 2024) Available from: <https://www.who.int/news-room/fact-sheets/detail/mpox>

³ Centers for Disease Control and Prevention. *Multistate outbreak of monkeypox - Illinois, Indiana, and Wisconsin* [Internet]. (2003). *JAMA*. 2003 Jul 2;290(1):30-1

In 1970, the first human isolate of MPXV was reported in a child in the equatorial region of the Democratic Republic of Congo (DRC), nine months after the eradication of smallpox in that country. Subsequently, sporadic cases were reported from the rainforest areas of central and western Africa, and large outbreaks were identified mainly in the DRC where the disease is currently considered endemic ¹. The first outbreak of MPX reported outside of Africa was an outbreak linked to importation of infected mammals in 2003 in the United States. In 2017, mpox re-emerged in Nigeria and spread between people across the country and in travellers to other destinations ¹. In 2022–2023 a global outbreak of mpox was caused by the clade IIb strain.

Transmission

MPox does not spread easily between people. Spread of MPox may occur when a person comes into close contact with an infected animal (rodents are believed to be the primary animal reservoir for transmission to humans), human, or materials contaminated with the virus. The virus enters the body through broken skin (even if not visible), the respiratory tract, or the mucous membranes (eyes, nose, or mouth).

Person-to-person spread is uncommon, but may occur through:

- contact with clothing or linen (such as bedding or towels) used by an infected person
- direct contact with MPox skin lesions or scabs
- coughing or sneezing of an individual with an MPox rash

Clinical Features

The clinical presentation of MPX includes symptoms and lesions that may be difficult to distinguish from smallpox, other orthopoxvirus, parapoxvirus infections, and chickenpox. The

¹ Di Giulio DB, Eckburg PB. *Human monkeypox: an emerging zoonosis. Lancet Infect Dis.* 2004 Jan;4(1):15-25.

main difference between smallpox and MPX is that MPXV causes lymphadenopathy (e.g., in the cervical or inguinal region) while smallpox virus and chickenpox virus usually do not.

The incubation period is the duration/time between contact with the infected person and the time that the first symptoms appear. The **incubation period** for MPox is between **5 and 21 days**.

The infection is usually mild-to-moderate in nature and can be divided into two periods.

- **Invasion/prodromal period (0-5 days)** with clinical manifestations of fever, intense headache, lymphadenopathy (swelling of the lymph node), back pain, myalgia (muscle ache) and intense asthenia (lack of energy)
- **Skin eruption period** (within 1-3 days after appearance of fever) where rashes appear in various stages often beginning on the face and then spreading elsewhere on the body. The face (in 95% of cases), and palms of the hands and soles of the feet (in 75% of cases) are most affected.

The evolution of the rash which occurs over a period of 10 days, progresses through the following stages:

- Maculopapular (lesions with a flat base)
- Vesicles (small fluid filled blisters)
- Pustules (pus-containing lesions)
- Crust (dried blisters)

MPox infection is usually a self-limiting illness, and most people recover within several weeks. However, severe illness can occur in some individuals (children & immuno-compromised).

An individual is **contagious until all the scabs have fallen off and there is intact skin underneath**. The scabs may also contain infectious virus material.



Figure 21.1. Images of MPox lesions

Notes:

Areas of erythema and/or skin hyperpigmentation are often seen around discrete lesions. Lesions can vary in size and may be larger than those shown.

Lesions of different appearances and stages may be seen at the same point in time. Detached scabs may be considerably smaller than the original lesion.

Table 21.1. Differential Diagnosis of MPox

DISEASE		CLINICAL DESCRIPTION
1	MPox	- Illness is usually mild to moderate in severity but can be fatal. Illness presenting with fever, headache, lymphadenopathy, back pain, myalgia (soreness in muscle) and asthenia (decrease in muscle strength) - Rash which follows fever starts from face, then spreads usually in a centrifugal pattern to other parts of the body especially extremities - Rash progresses from maculopapular to vesicles, pustules (rash with pus) and crusts (dried blisters) - Rashes in a particular area are usually at the same stage of development

	DISEASE	CLINICAL DESCRIPTION
2	Chicken pox	- Mild/moderate childhood infection which can also affect adults in whom it tends to be more severe - Fever, tiredness, loss of appetite and headaches - Rash that turns into itchy, fluid-filled blisters that eventually turn into scabs - The rash may first show up on the face, chest, and back then spread to the rest of the body, including inside the mouth, eyelids, or genital area - Rash is usually not pustular - Rashes are usually at different stages of development - Lymphadenopathy is not a common feature
3	Measles	- High fever, cough, watery nose (coryza), and conjunctivitis (red, watery eyes). Tiny white (Koplik) spots may appear inside mouth 2-3 days after symptoms - Flat red (maculo-papular) rashes appear on face around hairline, and spread downward to the neck, trunk, arms, legs, and feet - Small, raised bumps may also appear on top of the flat red spots
4	Syphilis	- Fever, swollen lymph glands, sore throat, patchy hair loss, headaches, weight loss, muscle aches, and fatigue - Painless chancre in primary stage of the disease - Skin rashes and/or mucous membrane lesions (sores in the mouth, vagina, or anus) mark the second stage

Surveillance Components

The protocol shall be adhered to for surveillance activities. **MPox has been designated as a Class 1 notifiable condition.** There are five (5) components to be considered in surveillance for MPox:

1. Case identification
2. Case reporting and investigation (including contact tracing)
3. Specimen collection and testing
4. Data analysis and interpretation
5. Data dissemination and outputs

Purpose of the Guidelines

To establish standards for the surveillance of MPox.

Objectives

The objectives of epidemiological surveillance of MPox are:

1. To establish epidemiological characteristics of the MPox infection
2. To inform risk assessment and decision-making.

Surveillance and Reporting

Surveillance

The key objectives of surveillance and case investigation for MPox in the current context are to: rapidly identify cases, clusters, and the sources of infection as soon as possible in order to provide optimal clinical care, isolate cases to prevent further transmission, identify and manage contacts, and tailor effective control and prevention methods based on the most commonly identified routes of transmission.

In non-endemic countries, such as Jamaica, one case is considered an outbreak. Because of the public health risks associated with a single case of MPox, clinicians should report suspected cases immediately to the **Parish Health Department** and the **National Surveillance Unit** regardless of whether they are also exploring other potential diagnoses.

Reporting

Case reports should include at a minimum the following information: (See [Appendix 27](#): MPox Case Investigation Form)

- date of report
- reporting location
- name, age, sex and residence of the case
- date of onset of first symptoms
- recent travel history, recent exposure to a probable or confirmed case

- relationship and nature of contact with probable or confirmed cases (where relevant)
- recent history of multiple or anonymous sexual partners
- smallpox vaccination status
- presence of fever occurring before rash/lesion identified
- presence of rash
- presence of other clinical signs or symptoms as per case definition
- other clinical or laboratory findings, particularly to exclude common causes of rash (see case definitions)
- hospitalisation status
- date of hospitalisation (if applicable)
- outcome at time of reporting.

Case Definitions

Suspected Case

A person of any age presenting with an unexplained acute rash

AND

One or more of the following signs or symptoms:

headache, acute onset of fever ($>38.5^{\circ}\text{C}$); lymphadenopathy (swollen lymph nodes), myalgia (muscle and body aches), back pain, and asthenia (profound weakness).

AND

for which the following common causes of acute rash do not explain the clinical picture: varicella zoster, herpes zoster, measles, Zika, dengue, chikungunya, herpes simplex, bacterial skin infections, disseminated gonococcus infection, primary or secondary syphilis, chancroid, lymphogranuloma venereum, granuloma inguinale, molluscum contagiosum, allergic reaction (e.g., to plants)

Probable case

A person meeting the case definition for a suspected case

AND

One or more of the following:

has an epidemiological link (face-to-face exposure, including health workers without eye and respiratory protection); direct physical contact with skin or skin lesions, including sexual contact.

OR

Contact with contaminated materials such as clothing, bedding, or utensils from a probable or confirmed case of MPox in the 21 days before symptom onset.

OR

Reported travel history to a MPox endemic country or to any country with an active MPox outbreak, in the 21 days before symptom onset.

Confirmed case

A case meeting the definition of either a suspected or probable case and is laboratory confirmed for MPox virus by detection of unique sequences of viral DNA by real-time polymerase chain reaction (RT-PCR).

Note: It is not necessary to obtain negative laboratory results for listed common causes of rash illness in order to classify a case as suspected.

MPox endemic countries:

Benin, Cameroon, the Central African Republic, the Democratic Republic of the Congo, Gabon, Ghana (identified in animals only), Cote d'Ivoire, Liberia, Nigeria, the Republic of Congo, and Sierra Leone. Benin and South Sudan.

Actions to be Taken

- Reporting/Notification
- Isolation of case
- Case investigation
- Contact tracing
- Quarantine of contacts (as necessary)

Case Reporting and Investigation

Notification

MPox is a Class 1 notifiable condition which means it must be notified by the healthcare worker (public and private) immediately on suspicion to the local Parish Health Department and/ or National Surveillance Unit.

The Parish Medical Officer (Health) and National Surveillance Unit, upon receiving said notification must immediately activate the call-out cascade for health emergencies. The Ministry of Health and Wellness National Emergency Operations Centre (MOHW NEOC) should be alerted immediately, via the National Epidemiological Surveillance System to all notifications for MPox.

Clusters of visits for unusual rash must be notified by the healthcare worker (public and private) immediately. Clusters should be investigated and reported.

Investigation

The Parish Medical Officer (Health) leads the case investigation team and must:

- Initiate case investigation within 24 hours of notification. A preliminary case or cluster investigation report must be submitted within 24 hours of this notification
- Immediately conduct contact tracing activities and initiate isolation and or quarantine where appropriate.

Specimen Collection and Testing

Samples must be taken from all suspected and probable cases of MPox.

- **The recommended specimen type for laboratory confirmation of MPox is skin lesion material, including swabs of lesion surface and/or exudate, or roofs from more than one lesion, or lesion crusts.**
- Swabs should be collected with Dacron or polyester flocked swabs and placed in sterile containers on ice. **Avoid using cotton tipped swabs for specimen collection.**
- The confirmatory test is the Polymerase Chain Reaction (PCR)

All specimens must be labelled with:

1. Patient Name
2. Patient Address
3. Date of Birth
4. Referring Facility
5. Diagnosis: Suspected MPox
6. Date and time of sample collection
7. Type of Sample Collected

All specimens must be placed on ice at 4-8°C and transported immediately to the National Public Health Laboratory.

All specimens must be accompanied by a completed Jamaica Laboratory Investigation Form (see [Appendix 9](#)). Contact the Consultant Microbiologist immediately to inform them of the sample.

Public Health Control Measures

Public health control measures are aimed at reducing human-to-human transmission through:

- Early recognition by specialist assessment and laboratory investigation
- Isolation of infected patients

- Implementation of appropriate infection prevention and control measures in healthcare settings (standard, contact, and droplet precautions)
- Early detection of possible new cases by contact tracing in outbreak settings.

Contact Tracing

A contact is a person who, in the period beginning with the onset of the source case's first symptoms, and ending when all scabs have fallen off, has had one or more of the following exposures with a probable or confirmed case of MPox:

- face-to-face exposure (including healthcare workers without appropriate PPE)
 - direct physical contact, including sexual contact
 - contact with contaminated materials such as clothing or bedding
- All close contacts of Probable and Suspected cases of MPox will be placed on home quarantine after most recent contact and observed daily for the development of symptoms. The period of quarantine and observation will end when a negative result for MPox is received for the Probable or Suspected case.
 - All close contacts will be given explicit instructions (verbal and written) regarding the steps to be taken if symptoms develop.
 - A record of the daily observation checks for contacts should be maintained at the Parish Health Department and daily reports submitted to the MOHW NEOC.

Considerations related to contact tracing

As soon as a suspected case is identified, contact identification and contact tracing should be initiated. Case patients should be interviewed to elicit the names and contact information of all such persons.

Contacts should be notified within 24 hours of identification.

- If the contact develops a rash, they need to be isolated and evaluated as a suspected case, and a specimen should be collected for laboratory analysis to test for MPox.

- A contact who develops initial signs/symptoms other than rash should be isolated and closely watched for signs of rash for the next seven days. If no rash develops, the contact can return to monitoring under quarantine conditions.

Epidemiological Assessment, Data Analysis and Interpretation

Epidemiological assessment and analysis of surveillance data must be completed at all levels.

The Parish Medical Officer (Health) must ensure that Class 1 Notification Forms and Case Investigation Forms are forwarded simultaneously to the Regional Health Authorities and the National Surveillance Unit, within the timelines specified above. A line listing of all reported cases should be maintained at the parish health department along with contact listings for each case. Epidemic curves as well as age, sex and geographic distribution of cases must be maintained at the parish level.

The Regional Technical Director, in collaboration with the Regional Medical Epidemiologist, must ensure that the line and contact listings are maintained for each parish. The age, sex, and geographic distribution, as well as the severity of cases should be monitored. Depending on the situation, daily or weekly reports may be required.

The National Surveillance Unit will conduct analysis of national data, including the epidemiological profile of cases and the epidemic curves as the situation evolves. The National Surveillance Unit will prepare appropriate reports showing information on the patterns of disease within the population.

Dissemination and Outputs

The National Surveillance Unit will be responsible for forwarding the information obtained from national level analysis to the Ministry of Health and Wellness National Emergency Operations Centre (MOHW NEOC).

The National Surveillance Unit will inform the Call-Out Cascade/MOHW NEOC of any notified, suspected, probable and confirmed cases of MPox virus infection immediately upon identification of a suspected, probable, or confirmed case.

Chapter 22: Surveillance of Influenza-Like Illness (ILI) and Severe Acute Respiratory Infections (SARI)

Introduction

Influenza, commonly referred to as ‘flu’ is a respiratory illness that is highly contagious. It is caused by any of several influenza viruses and can cause mild to severe illness ¹. There have been four (4) influenza pandemics in recent history: the Spanish Flu of 1918 (H1N1); the Asian Flu of 1957 (H2N2); the Hong Kong Flu of 1968 (H3N2); and the 2009 Pandemic Flu with H1N1 as the etiologic agent. The emergence of Avian Influenza A (H5N1) or ‘bird flu’ in 1997 raised concerns that a new human pandemic virus might develop. These concerns increased during outbreaks of human H5N1 influenza in 2006.

Under the International Health Regulations of 2005, PAHO/WHO should be notified immediately of all conditions that constitute a public health emergency of international significance. In 2007, WHO and Member States came together to start negotiating and interacting with industry, civil society organizations and other stakeholders to develop and implement a global approach to pandemic influenza preparedness and response.

The pandemic of 2009 uncovered several gaps in the global influenza surveillance capacity. There was a lack of any established surveillance system for severe respiratory disease in most countries. This resulted in the absence of historical epidemiological data which limited the ability of investigators to compare the severity of the pandemic with previous outbreaks or observe changes in the behavior of the virus. There was also no standardized approach to data collection and no pre-existing international mechanism for sharing epidemiological data.

On May 24, 2011, the Pandemic Influenza Preparedness (PIP) Framework came into effect when it was unanimously adopted by 194-member states during the Sixty- Fourth World Health Assembly. Its key goals were to improve and strengthen the sharing of information related to

¹ Centers for Disease Control and Prevention. *Influenza (Flu)* [Internet]. (Aug 2024) Available from: <http://cdc.gov/flu/about/index.html>

influenza viruses with human pandemic potential and to increase the access of developing countries to vaccines and other pandemic related supplies.

The National Surveillance Unit captures and analyzes data on influenza symptoms weekly and monitors the trends to allow for early detection of any outbreak and facilitate an appropriate response. Influenza-like illness (ILI) and severe acute respiratory infections (SARI) are syndromes that are used as indicators/proxies for influenza virus infection. The surveillance of influenza in Jamaica is a part to our broader surveillance system for communicable diseases.

Jamaica is part of the World Health Organization's Global Influenza Network, an international network of sentinel stations for influenza surveillance. The surveillance of respiratory infections has been ongoing since 1952 when the Global Influenza Surveillance Network (GISN) was established. In 2011, the World Health Assembly also formally recognized the need to expand influenza surveillance to include more epidemiological information to complement the virologic data and the GISN was renamed the Global Influenza Surveillance and Response System (GISRS). GISRS aims to serve as a global mechanism for monitoring influenza epidemiology and genetic profile of influenza viruses occurring globally, in human and animal populations. Its primary aim was to monitor the antigenic changes of the influenza viruses and to guide the selection of viruses for the production of influenza vaccines. The National Influenza Centre (NIC) Jamaica, established in 1952 and located within the Department of Microbiology of the University Hospital of the West Indies is a member of the GISRS. The NIC provides virological surveillance data for Jamaica and engages internationally with the World Health Organization (WHO) for participating in the work of the WHO Global Influenza Programme and Centers for Disease Control and Prevention (CDC).

The National Surveillance Manual for Jamaica guides the collection of data, analysis and reporting of information. This document reflects the revisions for Influenza-like Illness (ILI) and Severe Acute Respiratory Infections (SARI) surveillance.

Epidemiology of Influenza

The influenza virus is a highly infectious RNA virus of the orthomyxovirus family. Respiratory infections cause significant morbidity and mortality each year, and influenza presents with a

clinical syndrome that is not easy to distinguish from other respiratory infections. There are more defined seasonal trends in temperate countries, with the majority of cases occurring in the winter months. In tropical regions, the trend is less obvious, and epidemics can occur throughout the year. For Jamaica, influenza cases occur throughout the year with a major peak in cases from December through to March and a less defined peak from August to October.

There are four types of seasonal influenza viruses, types A, B, C and D. Influenza A and B viruses circulate worldwide and cause seasonal flu epidemics. Only influenza type A viruses are known to have caused pandemics. Influenza C virus usually causes mild infections while Influenza D viruses primarily affect cattle and are not known to cause illness in humans ¹. There are 3 types of influenza viruses that cause disease in humans; Influenza type A, type B and type C. Influenza A has 198 sub-types, and these are named based on the hemagglutinin (H) (18) and neuraminidase (N) (11) antigens that are found on the surface of the virus particle. Influenza virus A causes seasonal influenza, epidemics and pandemics while Influenza B viruses cause only seasonal influenza and some epidemics. The influenza type C viruses cause a mild respiratory illness and are not usually associated with epidemics.

The features that give influenza A major pandemic potential are:

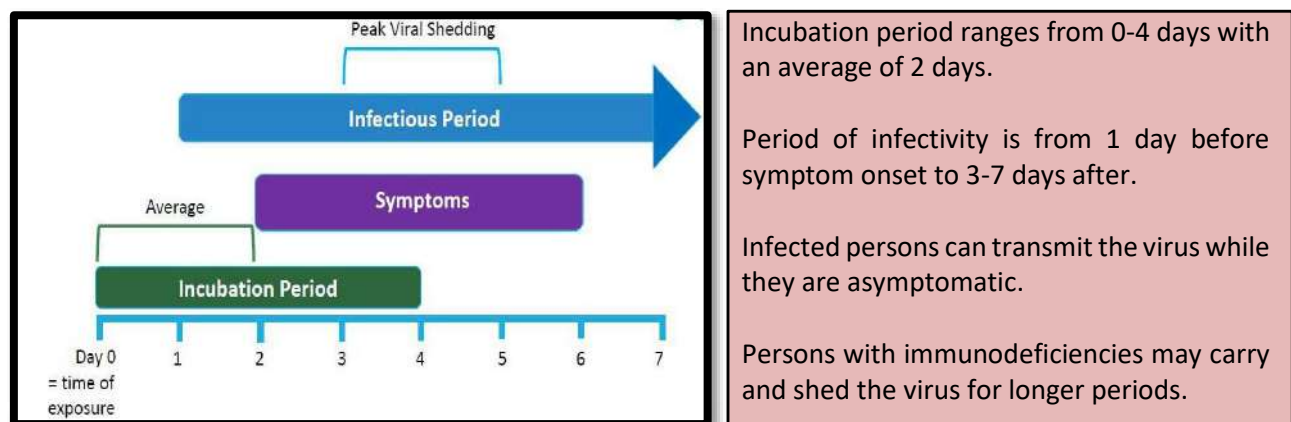
1. Its antigenic variability - The Influenza A virus undergoes antigenic changes that are known as:
 - a. antigenic drift, which are minor genetic changes that alter the antigens on the surface of the virus
 - b. antigenic shift, more radical genetic changes in the virus, including re-assortment of genetic material and adaptive mutation
2. An extensive animal reservoir - Influenza A is found in many animal species. The principal reservoir for Influenza A is the wild aquatic birds but it may be transmitted to other birds and mammals e.g. humans, pigs, horses, domestic and wild felines ².

¹ World Health Organization. *Influenza* [Internet]. (2025) Available from: <https://www.who.int/teams/health-product-policy-and-standards/standards-and-specifications/norms-and-standards/vaccine-standardization/influenza>

² Pan American Health Organization. *Operational Guidelines for Sentinel Severe Acute Respiratory Infection (SARI) Surveillance* [Internet]. (2014) Available from: http://www.paho.org/hq/index.php?option=com_docman&task=doc_view&Itemid=270&gid=31162&lang=en

Influenza Transmission

The influenza virus can be transmitted directly by coming in contact with respiratory droplets from coughs or sneezes from an infected person or by touching the contaminated hands of an infected person. Droplets can travel for up to one meter. Indirect transmission occurs by coming into contact with contaminated objects (fomites). These include hands, clothing, etc. The virus can survive in the environment for 5 minutes on hands, 8-12 minutes on paper, cloth and other fibers and 24-48 hours on hard surfaces.



Source: PAHO/WHO Influenza Sentinel Surveillance training

Figure 22.1. Graphic representation of incubation and infectious periods of influenza viruses.

Prevention and Control of Influenza

- Hand washing is the most efficient way to decrease the person to person spread of influenza
- Practicing "respiratory etiquette" also helps prevent transmission.
- Influenza vaccination prior to the start of the seasonal virus circulation. It takes about 2 weeks for protection to develop after vaccination, and protection lasts through the flu season.

Influenza Vaccine

There are two types of influenza vaccines; the inactivated influenza vaccine (IIV) which is administered by the intramuscular or intradermal route and the live attenuated influenza vaccine (LAIV). This LAIV is administered intra-nasally and is not offered in Jamaica, currently. In general, seasonal influenza vaccines are trivalent, containing a mixture of influenza A and B strains thought most likely to circulate in the coming season. The vaccine may not prevent one from getting influenza but may reduce the severity of the disease and the number of days with influenza.

Persons at risk for complications from Influenza and who should be given priority for vaccination are:

- The elderly > 65 years and children under 2 years of age
- Institutionalised persons
- Persons with chronic illnesses including metabolic diseases, diabetes, asthma, heart disease, HIV/AIDS
- Pregnant women have shown a greater tendency to develop severe forms of the disease.

Healthcare professionals and those who work with certain animals should also be given priority when distributing the influenza vaccine.

Influenza Surveillance

Overall Goal of Influenza Surveillance

To minimize the impact of influenza on the population by providing useful information to public health authorities to:

1. Plan appropriate control and intervention measures
2. Allocate health resources
3. Make case management recommendations

Specific Goals of Influenza Surveillance

To provide timely and high-quality epidemiological data and viral isolates to perform the following functions:

1. Describe the seasonality of influenza.
2. Establish baseline levels of activity for influenza and severe influenza- related disease.
3. Signal the start and end of the influenza season.
4. Identify and monitor groups at high risk for severe disease and mortality.
5. Evaluate the impact and severity of each season and predict future pandemic events.
6. Estimate the burden of influenza and help decision-makers prioritize resources and plan public health interventions.
7. Describe the antigenic characteristics and genetic makeup of circulating viruses.
8. Provide candidate viruses for vaccine production.
9. Identify locally circulating virus types and subtypes and their relationship to global and regional patterns.
10. Monitor antiviral sensitivity.
11. Assist in developing an understanding of the relationship between virus strains and disease severity.
12. Detect unusual and unexpected events such as outbreaks of influenza outside the typical season, severe influenza among healthcare workers, or clusters of vaccine failures that may herald a novel influenza virus.
13. Provide a platform for the evaluation of the effectiveness of vaccines and other interventions.

Surveillance of Influenza-Like Illness (ILI) and Severe Acute Respiratory Infections (SARI)

There are five components involved in ILI/SARI surveillance. These are:

1. Case identification
2. Case reporting and investigation
3. Specimen collection and testing
4. Data analysis and interpretation
5. Data dissemination and outputs

Case Identification

Any case that presents with fever and respiratory symptoms that fits the ILI or the SARI definition as stated below, must be identified as such. The use of a common case definition globally helps with the interpretation of data in an international context. The WHO Global 2014 definitions which are currently in use are as follows:

ILI CASE DEFINITION

An acute respiratory infection with:

- ***history of fever or measured fever of $\geq 38^{\circ}\text{C}$;***
- ***and cough;***
- ***with onset within the last 10 days***

SARI CASE DEFINITION

An acute respiratory infection with:

- ***history of fever or measured fever of $\geq 38^{\circ}\text{C}$;***
- ***and cough;***
- ***with onset within the last 10 days;***
- ***and requires hospitalization***

ILI cases may be identified in the emergency room or health centres. An ILI becomes SARI when the patient is admitted to hospital. All SARI cases are to be reported, case investigation completed and sample taken. Active case finding using various sources of hospital data is recommended to identify all possible SARI cases. This should be done during the weekly active surveillance visit by the assigned Public Health Nurse. All conditions that are coded according to the ICD10 codes - J 09 to J 18 and J 20 to J 22 - are potential SARI cases and should be reviewed. Coding is usually done after the patients are discharged and a review of codes can be used as a data quality measure.

Surveillance Indicator: Reported SARI cases in the period/SARI cases identified in the period by active case finding X 100. Goal: 80%

Case Reporting and Investigation

Influenza-like Illness (ILI):

ILI data are collected through seventy-eight (78) Sentinel Sites across Jamaica. The number of patients fitting the ILI case definition at each sentinel site must be recorded in the Weekly Parish Surveillance Report (See [Appendix 2](#)).

Severe Acute Respiratory Infection (SARI):

SARI cases are reported from the six (6) named SARI sentinel surveillance sites. These sites are:

1. Kingston Public Hospital (KPH)
2. Bustamante Hospital for Children (BHC)
3. Cornwall Regional Hospital (CRH)
4. Mandeville Regional Hospital (MRH)
5. St. Ann's Bay Hospital (SABH)
6. University Hospital of the West Indies (UHWI)

An individual SARI case investigation form must be completed for each reported case of SARI. The case investigation form must be completed by

1. the physician reporting the case OR
2. the infection control nurse OR
3. the hospital active surveillance nurse/ public health nurse.

Each SARI sentinel site surveillance team must assign a member to complete the SARI Investigation Form ([Appendix 28](#)). The case investigation form should include the documentation of demographic, clinical and epidemiological information about the patient including date of onset of symptoms, a travel history and the presence of any co-morbidities that may increase the risk of complications. If the patient has received a flu shot or is on any antiviral, antibiotic or other medications, this should also be documented.

The Weekly Data Collection Form: SARI/ILI Hospitalizations and Deaths ([Appendix 29](#)) must be completed by the Hospital Active Surveillance Nurse and should include data on total admissions as provided by the Medical Records team. The weekly data collection form and the individual case investigation forms must be submitted to the Parish Health Department by end of day on Mondays.

Surveillance Indicator: The median interval in days between hospitalization and notification of SARI case. Goal: 1 day.

Specimen Collection

Influenza-like Illness (ILI):

For ILI cases only one in ten cases (10%) of ILI should have a nasopharyngeal specimen collected. During an outbreak the requirements, including sites, reporting and sampling may change.

Severe Acute Respiratory Infection:

All cases that are identified as SARI must have a nasopharyngeal specimen taken. The Ministry of Health and Wellness requires that 100% of reported SARI cases should have a nasopharyngeal specimen collected for laboratory testing.

ALL SARI CASES SHOULD BE REPORTED & SAMPLE TAKEN

A nasopharyngeal (NP) swab is the optimal upper respiratory tract specimen collection method for influenza testing. Where there is difficulty obtaining a NP specimen e.g. from infants and some older patients, an oropharyngeal specimen may be taken. For children under five years of age, a nasopharyngeal aspirate is recommended (see [Appendix 30](#) for Additional Sample Taking Techniques where one is unable to obtain a Nasopharyngeal Specimen).

While all SARI cases should have a respiratory tract specimen taken up to ten days after onset of symptoms, the highest yield is obtained from a specimen taken in the first 5 days after onset of symptoms. The molecular or nucleic acid amplification tests (such as RT-PCR) are the most sensitive and accurate tests for influenza virus detection ¹.

In Jamaica, specimens should be transported to the National Public Health Laboratory for processing.

The successful isolation of the influenza virus depends on the prompt collection of high-quality specimens, appropriate transport and storage conditions of the specimen prior to testing and the rapid transport of the specimens to the testing laboratory.

¹ Pan American Health Organization. *Operational Guidelines for Sentinel Severe Acute Respiratory Infection (SARI) Surveillance* [Internet]. (2014) Available from: http://www.paho.org/hq/index.php?option=com_docman&task=doc_view&Itemid=270&gid=31162&lang=en

TAKING A NASOPHARYNGEAL SAMPLE

- Assemble equipment and forms

RESOURCES NEEDED

Universal Transport Media (UTM)

or

Viral Transport Media + Synthetic (Sterile Dacron / Nylon) Swab

Lab Investigation Form

Gloves

Face Mask

- Explain the procedure to the Patient
- Gain the patient's permission to perform the procedure
- Complete the Jamaica Laboratory Investigation Form
- Label Universal Transport Medium or Viral Transport Medium (VTM) tube
- Wash hands; put on gloves & mask on
- Have the patient evacuate mucous (if present) from both nostrils
- Tilt patient's head back 70 degrees
- Insert swab into nostril (to a depth equal to distance from nostrils to outer opening of the ears)
- Leave swab in place for several seconds to absorb secretions
- Slowly remove swab while rotating it – swab both nostrils with the same swab
- Place tip of swab into the sterile UTM/VTM tube below the level of the liquid media.
- Break/Cut off the applicator stick at the scored point or to a length that allows it to fit the tube
- Seal the cap tightly on the UTM/VTM tube
- Place UTM/VTM tube on frozen cold pack
- Wash hands

!!! DO NOT USE !!!

A WOODEN APPLICATOR SWAB OR A COTTON TIPPED SWAB TO COLLECT A NASOPHARYNGEAL SAMPLE FOR INFLUENZA TESTING

How to take respiratory tract sample:

1. The sample should be collected by a physician or health professional who is adequately trained to do so.
2. Personal protective equipment must be used:
 - a) gloves
 - b) masks - N95 Respirator may be used when available. Surgical masks can also be used.
3. Hands should be thoroughly washed before and after specimen collection.
4. Specimens should be transported in suitable Universal Transport Medium or Viral Transport Medium (VTM) at 4° C and in an upright position.
5. The VTM should be stored at 4° C prior to inoculation.
6. The UTM/VTM should be stored at 4° C after inoculation.
7. The specimen should be clearly and completely labeled (patient name, registration number date and time of collection).
8. The standard Jamaica laboratory investigation form must be completed and sent with the sample to the lab (See Jamaica Laboratory Investigation Form, [Appendix 9](#)).
9. The sample must be taken immediately to the hospital laboratory in an igloo with ice packs. It must be appropriately packaged for onward transportation to the National Public Health Laboratory (NPHL). The NPHL will send the samples for testing to the National Influenza Center (NIC) at the UHWI. Samples taken from patients admitted to the UHWI shall be sent directly to the NIC.
10. Ideally all respiratory swabs should be transported to the NPHL within 24 to 48 hours of collection and the maximum storage time at 4°C is 48 hours. If the specimens cannot be processed within 48 - 72 hours, they should be kept frozen at or below -70°C. Care should be taken to prevent repeated freeze/thaw cycles that can result in the loss of virus viability.

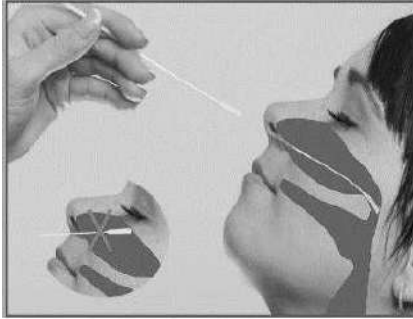


Figure 22.2. *Taking Respiratory Tract Sample*

Serum Samples for Influenza Diagnosis

Serum samples are only useful for diagnosis if an acute sample, taken within 7 days of onset of symptoms, is paired with a convalescent sample taken 14 days after the onset of symptoms. As often only the acute sample is collected, the **use of sera to make a diagnosis of influenza is not recommended.**

Surveillance Indicator: Percent of quality samples received (good quality means specimens properly taken, preserved and transported until arrival at the laboratory). Goal: 90%.

Data Analysis and Interpretation

Surveillance nurses conducting hospital active surveillance (Hospital Active Surveillance or HAS Nurses) should compare SARI cases identified by active surveillance with the total number of SARI cases for which completed investigation forms have been submitted. The HAS Nurse should assess forms for complete documentation of demographic, epidemiologic, clinical data and laboratory results. Electronic and paper databases containing all SARI cases should be maintained at each Parish Health Department.

Analysis of the data should be completed at all levels and action taken to improve the quality of surveillance at each level. Surveillance and infection control personnel at the hospital should carry out some level of analysis and report to the senior management of the hospital (See [Appendix 31](#) – SARI Data Analysis Guide). For example, the time from hospitalization to when a case is

reported and the sample is taken can be addressed promptly if this analysis is done at the institution. Any changes in numbers of patients seen at each site can first be identified by the local team. Analysis done at the level of the Parish Health Department allows for the endemic levels of influenza by SARI site and institution to be established.

At the National Level analyses of the data are conducted, appropriate graphs and charts are produced that allow for the determination of the endemic levels of the disease and identification of changes in the patterns of disease within the population. Average epidemic curves are produced based on the number of cases that have been reported over a 4 to 5-year period. The alert threshold which is a line graph plotting values that are 1.645 standard deviations above the mean for each epidemiological week and the epidemic threshold indicating values that are 2 standard deviations above the mean for each epidemiological week can be determined after the analysis of the data is done.

The analysis and interpretation of the data allows for the comparison of current influenza activity to that of previous years. It also permits the surveillance teams to detect periods of increased activity signaling the start of an epidemic.

Based on the five components involved in ILI/SARI surveillance, there are indicators that are used to measure the performance of the surveillance system for influenza at each stage:

1. Case identification
2. Case reporting and investigation
3. Specimen collection and testing
4. Data analysis and interpretation
5. Data dissemination and outputs

The surveillance system performance indicators with the specific goals as listed in [Appendix 32](#) provide guidance as to the performance of the influenza surveillance system.

Data Dissemination and Outputs

The National Surveillance Unit will be responsible for dissemination of the information obtained from national level analysis. Information is to be forwarded to the heads of the National Authorities including the Minister of Health and Wellness, and the Permanent Secretary, to international counterparts such as the Pan American Health Organization/World Health Organization (PAHO/WHO), the Caribbean Public Health Agency (CARPHA) and the Centers for Disease Control and Prevention (CDC). This information must also be fed back to the SARI sites and the Parish Health Departments.

For dissemination to PAHO/WHO, the epidemiological data is imputed into the **Fluid** which is a WHO web-based application for entry, collection and dissemination of epidemiological data on influenza. It allows countries to enter epidemiological data online and access processed information instantly. Fluid complements the **FluNet** which is a virological data collection tool into which the National Influenza Center enters the data regarding laboratory testing and results. Together these systems allow for the production of information and outputs in graphical and other forms that are then disseminated through surveillance bulletins. The National Surveillance Unit produces a Weekly Epidemiological Bulletin that is sent to surveillance teams across the island. It is also posted on the Ministry of Health and Wellness' website. The PAHO/WHO also produces a surveillance bulletin that is circulated and posted on the WHO website.

[Appendix 33 and 34](#) show a process flow chart describing the processing of data from SARI and ILI sites, respectively to the dissemination of the information.

Chapter 23: Tuberculosis

Goal and Objectives of Tuberculosis Surveillance

Tuberculosis (TB) surveillance aims to detect and measure the number of TB cases and related deaths that occur each year and to assess trends in these parameters over time.

Objectives of TB surveillance

1. Describe the number of new TB cases by category (e.g., Multidrug- Resistant Tuberculosis, HIV-positive, age/sex distribution, geographic distribution).
2. Measure the number of TB deaths.
3. Track new cases and deaths over time.
4. Measure treatment outcomes by age/sex, classification (pulmonary vs. extra-pulmonary), time (trends), and geographic distribution.

Case Detection

Medical practitioners must notify all patients presenting with a chronic cough (>3 weeks duration) as suspected TB cases within 24 hours to the Parish Health Department or National Surveillance Unit [Public Health (Class 1 Notifiable Diseases) Order 2003, Government of Jamaica, 2003].

Case Definitions

- a) **Suspected case** –A person who presents with a persistent cough for greater than 3 weeks, fever, night sweats and weight loss.
- b) **Confirmed case**
 - Bacteriologically confirmed - Person from whom a biological specimen is positive by smear microscopy, culture or PCR (Such as Xpert MTB/RIF).

- Clinically diagnosed case – An individual who does not fulfil the criteria for bacteriological confirmation but has been diagnosed with active TB by a clinician or other medical practitioner who has decided to give the patient a full course of TB treatment.
- c) **Pulmonary TB** - A bacteriologically confirmed or clinically diagnosed case of TB, with disease involving the lung parenchyma or the trachea-bronchial tree. Tuberculous intra-thoracic lymphadenopathy or tuberculous pleural effusion without radiographic evidence of lung lesions is classified as extra-pulmonary TB. A case with pulmonary and extra-pulmonary TB should be classified as a case of pulmonary TB. Miliary TB is classified as pulmonary TB.
- d) **Extra-pulmonary TB** - A bacteriologically confirmed or clinically diagnosed case of TB with disease involving organs other than the lungs and no pulmonary involvement (see pulmonary TB above).
- e) **New patient** - A patient with bacteriologically confirmed or clinically diagnosed Tuberculosis who has never been treated for TB or who has taken anti-TB drugs for less than one month.
- f) **A relapsed patient** - A patient who has previously been treated for TB and declared cured or treatment completed and is now diagnosed with TB (Bacteriologically Confirmed or Clinically Diagnosed) ¹.

Notification of TB Cases by Healthcare Workers

All suspected cases of Tuberculosis shall be notified using a Class 1 notification form ([Appendix 7](#)).

The notification form must be completed with all available information about the suspected case and transmitted to the Parish Health Department or the National Surveillance Unit within 24 hours

¹ World Health Organization. *Definitions and reporting framework for Tuberculosis - 2013 revision* [Internet]. (April 2013) Available from: <https://www.who.int/publications/i/item/9789241505345>

of suspicion. The Parish Health Department must forward notifications to the Regional Health Authority and the National Surveillance Unit as detailed in the National Surveillance Manual for Jamaica. Hospital Active Surveillance Officers are responsible for searching the wards and laboratory of each hospital for cases of tuberculosis that were not reported as above.

Case Investigation

The responsibility for case investigation lies with the Parish Medical Officer (Health) who will delegate as appropriate. Contact investigation should be carried out for all confirmed cases of TB (bacteriologically and clinically confirmed). The schedule for completing the Case Investigation Forms ([Appendix 35](#)) is as follows:

- Case Investigation Form 1 should be submitted to the Parish Health Department (Parish Medical Officer of Health) within ten (10) weeks of the initial Class 1 Notification of a suspected case of tuberculosis.
- Case Investigation Form 2 should be submitted to the Parish Health Department (Parish Medical Officer of Health) within 6 to 8 months of starting anti-TB therapy. Submit this form at the end of anti-TB therapy and/or after follow-up.

A review of the suspected case's medical records will be required with special attention paid to the following:

- Previous History of TB
- Chest X-Ray findings
- Mantoux Test result
- Sputum sample collection
- Sputum results
- HIV testing
- HIV status
- Comorbid conditions (E.g. diabetes, renal disease, etc.)
- Treatment History

Contact Tracing

Contact tracing is a critical component of the case investigation and aims to limit the spread of disease ¹. The investigating officer must:

- Identify all household and bedroom contacts (initiated once notification received)
- Screen all close contacts as follows:
 - Take a thorough history from each contact focused on HIV status, other comorbidities, and recent history of cough, weight loss, and night sweats
 - Conduct a Mantoux test on each household, bedroom, and other close contact
 - Refer as appropriate for further management

Contacts are prioritized for investigation based on:

1. **Closeness of contact.** For example, persons sharing a bedroom and/or a bed with a clinically diagnosed or bacteriologically confirmed individual with Tuberculosis are more likely to become infected. This is not limited to the household but may include homeless shelters and other settings with multi-person 'wards' such as an infirmary or areas of confinement such as prisons or lock ups.
2. **Duration of contact.** The longer a person spends in close contact with a person suspected or proven to have Tuberculosis (as above), the greater the risk of transmission and infection.
3. **Immune status of the contact.** A person living with HIV is more likely to progress rapidly to active TB disease once infected.
4. **Presence of pulmonary Tuberculosis in the index case** (the case around whom the investigation is centered). Transmission of disease is more likely in a person suspected or proven to have Pulmonary Tuberculosis than from a person with Extra-Pulmonary Tuberculosis.
5. **Age of the Contact.** Children < 5 years old are more likely to develop severe disease.

¹ World Health Organization. *Recommendations for investigating contacts of persons with infectious tuberculosis in Low- and Middle-Income Countries* [Internet]. (2012) Available from: http://apps.who.int/iris/bitstream/handle/10665/77741/9789241504492_eng.pdf;jsessionid=37CD6EAA7BA86AF8D02B248837E14EEB?sequence=1.

Management of Contacts

The management of contacts will depend on the results of initial screening and on the immune status of the contact (Figure 23.1).

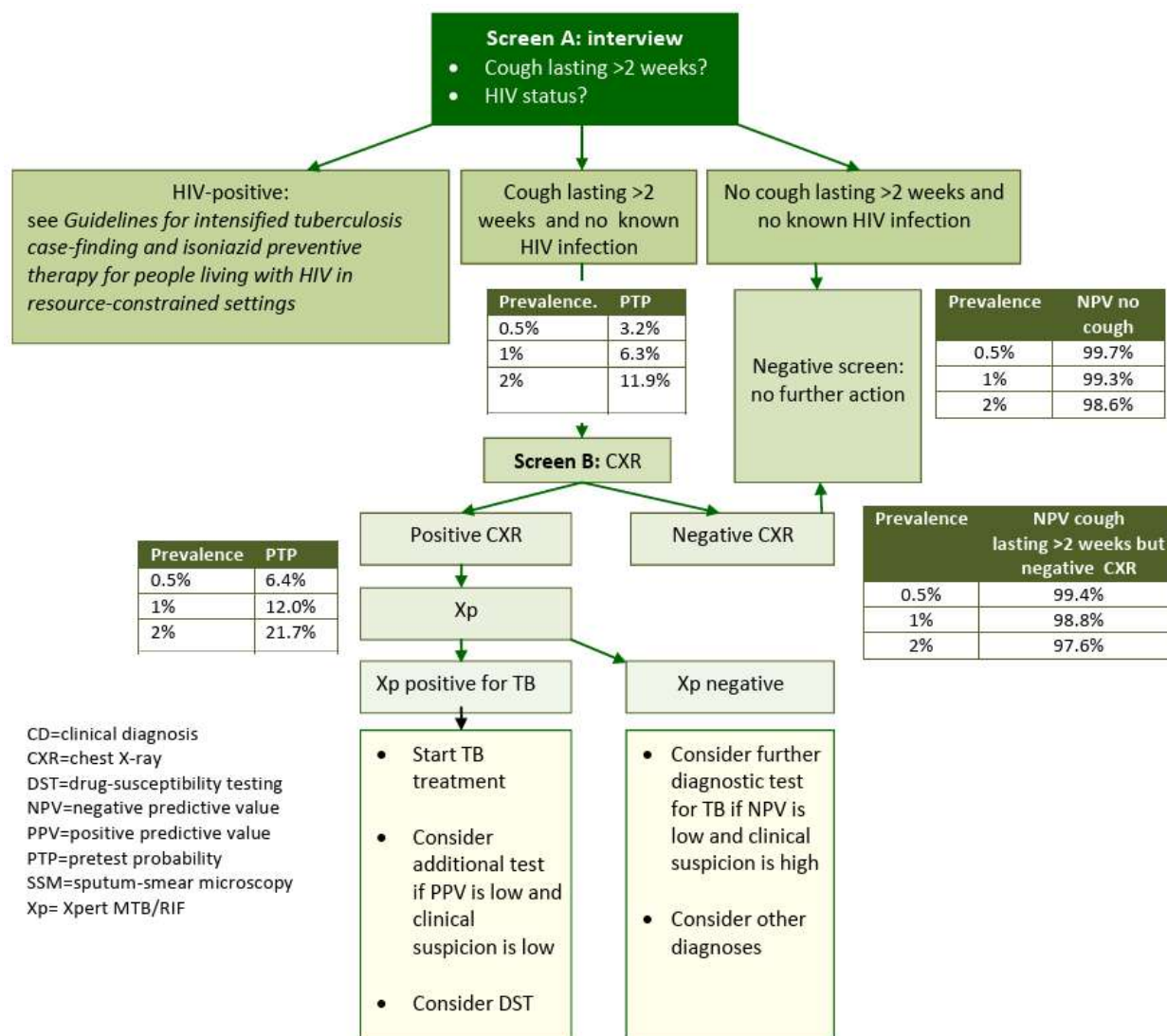


Figure 23.1. Algorithm for screening of contacts for active Tuberculosis¹

¹ World Health Organization. *Systematic screening for active tuberculosis: Principles and recommendations* [Internet]. Geneva (2013) Available from: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.ncbi.nlm.nih.gov/books/NBK294083/pdf/Bookshelf_NBK294083.pdf

Data Management

Collection

- A Class 1 form is to be submitted by the medical practitioner to the Parish Health Department or to the National Surveillance Unit within 24 hours of suspicion of a diagnosis of Tuberculosis.
- Sections 1 – 4 of the Tuberculosis Case Investigation Form ([Appendix 35](#)) are to be completed and submitted within 2 weeks of notification.
- Section 5 will be completed at the central level (MOHW).
- Section 6 of the Tuberculosis Case Investigation Form should be completed within 6 weeks of the initial notification.
- Section 7 should be completed when the case has completed treatment.
- Section 8 to be completed by the MOHW within 6 weeks of completion of treatment.

Collation

- Parishes, Regional Health Authority and the National Surveillance Unit will maintain databases of all notified, suspected, and confirmed Tuberculosis cases.

Reporting

- Parishes will submit to the regional and national levels, case investigation updates ([Appendix 36](#)) as additional data become available.

Analysis

- An annual Tuberculosis report is to be completed by the National Surveillance Unit, in keeping with the epidemiological requirements of the country and the reporting framework of the WHO.

Data Quality

- Biannual TB reports to be obtained from each Regional Health Authority containing the number of notified, confirmed, and co-infected TB cases by age-group and sex.

- Biannual data validation checks are to be carried out by the National Surveillance Unit in keeping with part B of the Standards and benchmarks for tuberculosis surveillance and vital registration systems ¹.

Monitoring and Evaluation

- Ensure TB notifications meet programme requirements, maintain records, complete field investigations, compile and submit reports, monitor drug and supply availability, and conduct audits.
- Monitor frequency of report submissions, analyze parish reports, audit parish TB surveillance system, and prepare budget.
- Determine implementation of policy decisions, status of surveillance strategies, achievement of targets, conduct biennial audits, and establish a multi-stakeholder TB Diagnostic Committee.

Contents of Monthly Reports

- Number of notifications in each health district.
- Percentage of notifications confirmed, discarded, and under investigation.
- Number of extra-pulmonary culture-positive and negative cases on treatment.
- Number of new sputum-positive and smear-negative cases.
- Number of new smear-positive cases grouped by age and sex.
- Number of relapses, failures, defaulters, and deaths.

¹ World Health Organization. *Standards and benchmarks for tuberculosis surveillance and vital registration systems* [Internet]. (May 2014) Available from: <https://www.who.int/publications/i/item/9789241506724>

Chapter 24: Vaccine-Preventable Disease (Class 1) Surveillance

This section discusses surveillance definitions, processes and procedures for the following Class I notifiable Vaccine-Preventable Diseases (VPDs):

- Acute Flaccid Paralysis (AFP)/Polio
- Fever and Rash - Measles and Rubella
- Congenital Rubella Syndrome (CRS)
- Diphtheria
- Hepatitis B
- Pertussis
- Tetanus
- Tuberculosis

Areas discussed under each disease/condition include:

- Case definitions and classification
- Case investigation
- Specimen collection and laboratory investigation (including handling, storage and transport of specimens)
- Prevention and control measures

The required reporting procedure is the same for all Class I notifiable diseases/conditions:

- Healthcare providers should report suspected cases to the Parish Health Department within 24 hours using the Class I Diseases Notification Form ([Appendix 7](#)).
- The healthcare worker who is the first point of contact with the case has the responsibility to notify the case and initiate the case investigation.
- Health Departments should notify the Health Region and the NSU immediately by telephone or other electronic medium. Hard copies of the notification form must follow.
- NSU reports cases to the Family Health Unit and the EPI Regional Advisor as required.

Note that patient information should not be transmitted over unsecured internet channels.

Table 24.1. Summary of Local Reporting Requirements for Class I Conditions

Entity	Reporting Content	To	Using	Timeline
Health Provider	Suspected case of Class I condition or health event. Indicate: - demographic and contact information for the patient - symptoms - date of onset of symptoms - travel history - immunization history - laboratory information (date of sample collection) - other pertinent information to assist with case identification, investigation and data analysis	Medical Officer of Health at the Parish Health Department or the National Surveillance Unit	Class I Notification Form (fill out form at first contact with the patient)	Report on suspicion within 24 hours of contact
Parish Health Department	Class I notification form	Regional Epidemiologist / Surveillance Officer; National Surveillance Unit Parish Health Department in which patient resides, if it differs from the one in which the case sought healthcare.	Telephone, or other electronic medium followed by hard copies	Immediately on receiving Class I notification

Acute Flaccid Paralysis (AFP)/Poliomyelitis

Polio has been targeted for worldwide eradication by the World Health Organization. In order to verify eradication, a robust surveillance system that is capable of detecting possible cases of polio should be in place. AFP is the syndrome monitored for the detection of possible cases of poliomyelitis. The reportable surveillance indicator to international health organisations such as PAHO/WHO for the independent verification of and maintenance of Jamaica's certification status as having eliminated polio, is detection of at least 1 case of AFP per 100,000 population under age 15 years.

Case Definition and Classification

Suspected Case

AFP is the acute onset of a flaccid paralysis in the absence of trauma.

This includes cases diagnosed with Guillain-Barré syndrome or Transverse Myelitis. Cases of AFP in the under 15 years age group are of particular interest, however, AFP in any patient of any age should be notified and investigated.

Probable Case

A suspected case in which the paralysis is flaccid and no other cause for the paralysis can be immediately identified.

Confirmed Case

A suspected or probable case, with or without residual paralysis, from which poliovirus is isolated from stool culture or detected through molecular methods such as polymerase chain reaction (PCR).

Case Investigation

The healthcare worker should (see [Appendix 37](#) for flow chart):

- Initiate a case investigation as soon as the case is identified:
 - Obtain clinical history

- Obtain travel and immunization history.
 - Ensure appropriate specimen is collected and transported to the appropriate laboratory.
 - Visit the home, school, nursery or workplace as needed to search for other cases and obtain immunization history of close contacts.
 - Initiate community and hospital surveillance for additional cases.
 - Follow-up with attending physician/specialists for additional test results and clinical history, in preparation for completing the 60-day follow-up.
-
- Complete and submit an Acute Flaccid Paralysis Case Investigation Form to the parish MO(H) within 48 hours of notification ([Appendix 38](#)).
-
- Complete a 60-day Follow-up Form ([Appendix 39](#)) based on a complete neurological examination to assess residual paralysis; information on this form should be based on consultation with the patient, physician, or physiotherapist.
-
- Ensure that the patient is assessed by a physician at or after 60 days post onset of paralysis.

Specimen Collection and Laboratory Investigation

Confirmation of poliomyelitis is based on identification of the poliovirus in an adequate stool specimen.

Specimen Collection

- Collect two stool samples of approximately 12 grams (3 adult thumbnails) each at least 24 hours apart and within 14 days of the onset of paralysis.
- Rectal swabs are not recommended because the amount of stool collected is inadequate for testing.

Handling, Storage and Transportation of Specimens

- Place stool in a clean, dry, screw-capped container without any preservative.
- The container should be labelled with the patient's name and date of collection.
- Place specimen(s) immediately in a cold box or refrigerator at 4°C pending transport.

- Transport on ice.
- Specimens should be sent to the Enteric Section of the National Public Health Laboratory (NPHL), which forwards them to the Caribbean Public Health Agency (CARPHA) in Trinidad.
- The Jamaica Laboratory Investigation Form ([Appendix 9](#)) must accompany the specimen. The NPHL should be informed by phone that a specimen is en route.

Note: If the patient dies before stool samples can be collected, a stool sample can be extracted prior to sending the body to the mortuary.

Alternatively, an autopsy should be performed within 24 hours of death to determine cause of death and collect appropriate samples. These should be sent to the NPHL for preparation and shipment to CARPHA.

Prevention and Control Measures

- Containment activities are required as part of the outbreak response, following the identification of a suspected case. These include:
 - intensified surveillance
 - immunization of close contacts
 - review of vaccination coverage and targeted immunization
 - public education.

Global Surveillance Indicators ¹

Table 24.2. Overall indicators on timeliness

Indicator	Calculation (expressed as a percentage)		Target
Overall detection of WPV/VDPV	For AFP	# of AFP cases* with WPV/VDPV final lab results ≤35 days of onset / # of AFP cases* with WPV/VDPV final lab results	≥80%
	System capacity†	# of WPVs and VDPVs with final lab results ≤35 days of onset for AFP cases / # of WPVs and VDPVs	≥80%
AFP detection – system		# of AFP cases* with final lab results ≤35 days of onset / # of AFP cases*	≥80%

AFP = acute flaccid paralysis; VDPV = vaccine-derived poliovirus; WPV = wild poliovirus

* Aggregated results: all lab results (AFP + contacts) used to classify AFP case as confirmed/discarded.

† Specimen-based calculation

Table 24.3. Indicators on timeliness for field activities

Indicator	Calculation (expressed as a percentage)	Target
Timeliness of notification	# of AFP cases reported ≤7 days of onset / # of AFP cases	≥80%
Timeliness of investigation	# of AFP cases investigated ≤48 hours of notification / # of AFP cases	≥80%
Timeliness of field activities	# of AFP cases with 2 stool specimens collected ≥24 hrs apart AND ≤11 days of onset / # of AFP cases	≥80%
Timeliness of field and shipment activities	# of AFP cases with 2 stool specimens collected ≥24 hours apart AND received in good condition* at a WHO-accredited laboratory AND ≤14 days of onset / # of reported AFP cases	≥80%
Timeliness of stool specimen shipment	# of stool specimens that arrive in good condition* at a WHO-accredited lab AND ≤3 days of specimen collection / # of stool specimens collected	≥80%

AFP = acute flaccid paralysis; WHO = World Health Organization

*For calculations: missing stool condition = poor condition

¹ World Health Organization. *Global guidance for conducting acute flaccid paralysis (AFP) surveillance in the context of poliovirus eradication*. Geneva: Licence: CC BY-NC SA 3.0 IGO (2024). Available from: <https://iris.who.int/server/api/core/bitstreams/eb517fee-4a13-4d21-852c-0b6e1c21eb93/content>

Fever and Rash (Measles and Rubella)

Surveillance for Measles and Rubella is integrated into fever and rash surveillance. Although both diseases have been declared eliminated from Jamaica, the country remains on high alert for imported cases.

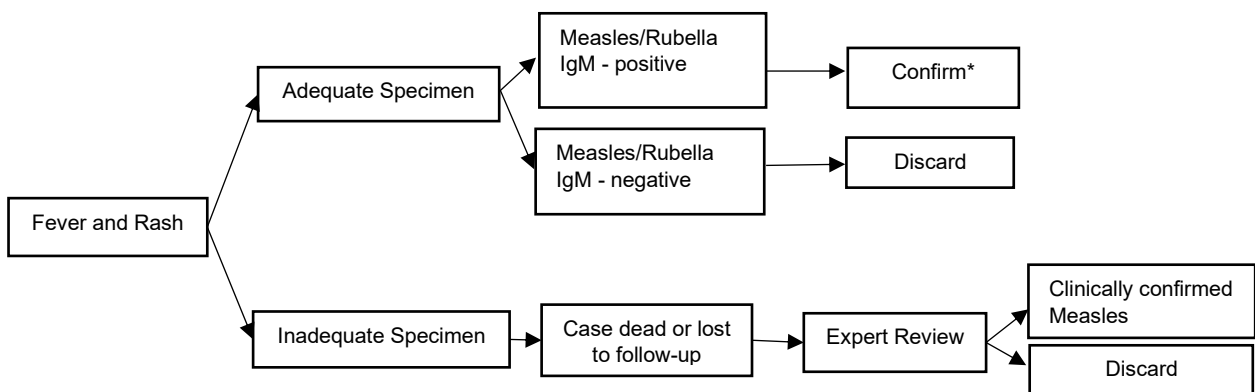
Case Definition and Classification

Suspected Case

Any person presenting with fever ($\geq 38^{\circ}\text{C}$ / 100.4°F) or giving a history of fever AND a generalized maculopapular rash.

Case Reclassification

- Suspected cases are re-classified as confirmed or discarded based on laboratory investigations for measles and rubella.
- For cases lost to follow-up, a review of the medical records by a physician should be done to classify the case.



* If a case has documentation of immunization with MMR vaccine 2 to 4 weeks prior to specimen collection, then discard as per advice from the CARPHA laboratory.

Figure 24.1. Scheme used in the PAHO Region for Case Classification

Case Investigation

Fever and rash case investigation requires the ‘first contact strategy’, that is, the case investigation is conducted by the first healthcare worker to see the case. Complete and submit a Fever and Rash Investigation Form ([Appendix 40](#)) to the parish MO(H):

Initiate and submit a case investigation immediately (see [Appendix 41](#) for flow chart):

- Obtain clinical history and laboratory test results from physician.
- Obtain travel and immunization history.
- Ensure specimen collection and referral to the appropriate laboratory.
- The PHN assigned to the case should: visit the home, daycare, nursery, school or workplace to review immunization history of close contacts and search for other cases. All persons found not to be fully immunized, or not have documentation of immunization, should be vaccinated immediately with MMR.
- Initiate community and hospital surveillance for additional cases.
- Follow-up including review of the case with the attending physician/specialists to obtain the final diagnosis, and follow-up vaccinations at the home of the patient.

Specimen Collection and Laboratory Investigation

Laboratory testing for measles (and confirmation of rubella) must be done in a WHO accredited laboratory which uses standard methods and approved reagents, and passed a recent proficiency test. At the writing of this document, the only WHO accredited laboratory within the Caribbean sub-region is located at the Caribbean Public Health Agency (CARPHA) in Trinidad.

The presence of IgM antibodies in serum indicates recent exposure to the virus, either through infection or immunization. Highly suspicious cases or confirmed cases should also have urine cultures processed for isolation of measles or rubella virus. This will assist in identification of the source of the virus.

Specimen Collection

Blood:

- Approximately 5 to 10 mL of blood should be collected in a red-top tube, i.e. one without any preservatives or anticoagulants.
- The specimen should be collected at first contact (within 30 days of onset of rash).
- Depending on the type of test done, a convalescent specimen may be requested which should be taken 14 to 21 days after the onset of rash.

Urine:

- Urine specimen should be taken from high-risk individuals. The purpose of collecting a urine specimen is for virus isolation. This allows the laboratory to study the virus and obtain epidemiological information necessary to determine the original source.
- High-risk individuals are any of the following persons:
 - Visitors to Jamaica (especially from a country where measles is circulating).
 - Locals who have been in contact with a visitor or recently visited another country where measles is endemic.
 - Persons who form a cluster of suspected measles cases (random urine samples should be taken).
- Urine should be collected within 10 days of rash onset (1-7 days is best).
- First-morning voided urine is ideal, but any urine specimen is adequate.
- A volume of up to 100 mL of urine should be collected in an empty sterile specimen container.

Nasopharyngeal swab:

- Nasopharyngeal swabs should be taken as soon as possible after the onset of the rash. Ideally, swabs should be collected within 5 days (but up to 10 days is acceptable).

Handling, Storage and Transportation of Specimens

Blood:

- Blood specimens should be kept in an upright position for at least one hour to allow the blood to coagulate.
- If possible, the serum should be separated from the clot and placed into a clean receptacle i.e. a different red-top tube, a cryovial, or other suitable vial for safe transport of serum.
- The serum should be stored at 4°C or less until transportation to the laboratory.
- Storage of separated serum at -20°C is optimal.
- If serum separation is not possible at the collection site, the tube containing the clotted blood should be kept cool (but not frozen) and shipped to the laboratory within 24 hours after collection.
- The specimen should be transported to the Immunology Section of the NPHL for shipment to CARPHA. The laboratory should be informed by phone that a specimen is en route.

Urine:

- The urine specimen must be kept cool at 4°C (ice packs or wet ice) and transported to the immunology section of the NPHL immediately or within 24 hours.
- The laboratory and the NSU should be notified that a specimen is en route as these specimens require special handling.
- At the NPHL:
 - 50 mL of urine specimen should be transferred to a sterile plastic centrifuge tube and centrifuged at 1500rpm for 5-10 minutes at 4°C to pellet the sediment.
 - If no refrigerated centrifuge is available, an ordinary centrifuge will do.
 - The sediment should be re-suspended in 2 mL of viral transport media and stored at -70°C (if available) or -20°C pending shipment to CARPHA.
 - It is very important that the samples arrive at CARPHA within 5 days of the date of collection. This ensures optimal virus recovery.

Nasopharyngeal swab:

- Dacron or polyester flocked swabs should be used for sample collection and once collected should be placed in viral transport medium. Avoid using cotton tipped swabs for specimen collection.
- Specimens must be placed on ice at 4-8°C and transported immediately to the National Public Health Laboratory.
- All specimens must be accompanied by a completed Jamaica Laboratory Investigation Form ([Appendix 9](#)).
- The specimen should be transported to the Immunology Section of the National Public Health Laboratory for shipment to CARPHA. The laboratory should be informed by phone that a specimen is en route.

Prevention and Control Measures

- Public education and routine immunization are key to measles and rubella prevention.
- Timely investigation of cases is critical for control of an outbreak and includes:
 - home visit within 48 hours of notification
 - adequate specimen collection
- Other control measures required following the identification of a suspected case are outlined below:
 - Suspected cases should be separated from young children and infants, until they have been classified as discarded.
 - Highly suspicious (symptomatic high-risk) cases should be placed in respiratory isolation and treated until laboratory results are obtained.
 - Confirmed measles cases should be treated appropriately and placed in respiratory isolation until day 5 after the onset of rash.
 - A contact is a person that lives in the same household of the case or has close contact with the case 5 days before to 5 days after the onset of rash in the case:

- Symptomatic close contacts of a confirmed or highly suspicious case should be placed in respiratory isolation and investigated as suspected cases.
- Asymptomatic close contacts of a confirmed or highly suspicious case should be quarantined at home until 21 days after last exposure when they can no longer infect others. They require vaccination against measles and rubella if not fully immunized or if evidence of immunization is not available.

Data Management & Analysis

The following indicators¹ for fever and rash surveillance must be monitored at the parish, regional and national levels:

- Percentage of sites (Sentinel and HAS) reporting every week (target is 90% or greater).
- Percentage of sites reporting on time (target is 80% or greater).
- Fever and rash cases with investigations completed in less than 48 hours after notification (target is 80% or greater).
- Cases with adequate blood specimens or epidemiological linkage to confirmed measles/rubella case (target is 80% or greater).
- Specimens arriving at CARPHA within 5 days of collection.
- Proportion of total laboratory confirmed cases with source of infection identified (target is 80% or greater).
- Number of cases with immunization history recorded and number with adequate immunization with MMR (i.e. 2 doses).
- Number of cases confirmed and number discarded.
- Proportion of total cases with adequately completed investigation forms, which must include the following data points:

¹ World Health Organization. *Vaccine-Preventable Diseases Surveillance Standards* [Internet]. (October 2018). Available from: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://cdn.who.int/media/docs/default-source/immunization/vpd_surveillance/vpd-surveillance-standards-publication/who-surveillancevaccinepreventable-11-measles-r2.pdf

1. Cases identification (name),
2. Date of birth/ age,
3. Sex,
4. Place of residence,
5. Rash onset date,
6. Vaccination status,
7. Travel history,
8. Sample collection status,
9. Date of notification,
10. Date of investigation.

Congenital Rubella Syndrome (CRS)

The Region of the Americas was certified to have eliminated rubella and congenital rubella syndrome in 2015.

Case Definition and Classification

Suspected Case

- An infant less than one year of age presenting with one or more of the following: congenital cataracts, glaucoma, deafness, microphthalmia, microcephaly, congenital heart defects or meningoencephalitis.
OR
- An infant born to a woman who had confirmed rubella infection during pregnancy or who received measles, mumps and rubella (MMR) vaccine during pregnancy.
OR
- An infant less than one year of age with a positive result for Rubella IgM antibodies at a non-WHO accredited laboratory.

Confirmed Case

A suspected case of CRS with supportive laboratory evidence from a WHO accredited laboratory.

Case Investigation

The healthcare worker should:

1. Initiate a case investigation within 48 hours of case identification:
 - Obtain clinical history and laboratory test results from physician.
 - Obtain the mother's immunization and travel history during pregnancy and history of illness/contact with rubella case during pregnancy.
 - Ensure appropriate specimen is collected and transported to the appropriate laboratory.
 - Visit the home to review immunization history of close contacts and search for other cases. All persons found not to be fully immunized or not have documentation of immunization should be vaccinated immediately.

- Initiate community and hospital surveillance for additional cases.
 - Follow-up includes review of the case with the attending physician/specialists to obtain the final diagnosis.
2. The CRS investigation should be completed, with the report submitted to the parish MO(H) within 48 hours of notification.

Specimen Collection and Laboratory Investigation

Diagnosis of CRS can be based on any one of the following:

- Detection of rubella-specific IgM antibodies in infant serum specimen(s).
- Maintenance or increase in IgG antibody titre between acute and convalescent samples.
- Isolation of rubella virus from nasopharyngeal swab, urine, CSF or blood.
- Detection of rubella virus in tissues by PCR.

Specimen Collection

Blood:

- At first contact, collect 5 to 10 mL of blood in a red-top tube.
- Additional convalescent blood specimens may be requested and should be collected in the same way.

Urine:

- Urine specimens should only be taken from highly suspected cases and from confirmed cases to isolate the rubella virus and determine its source.
- Collect up to 100 mL of urine in an empty sterile specimen container.
- Specimens should be collected as close to birth as possible but ideally in the first 3 months of age since about 50% of affected infants may not shed virus beyond 3 months of age¹.

¹ World Health Organization. *Manual for the Laboratory-based Surveillance of Measles, Rubella, and Congenital Rubella Syndrome Third Edition* [Internet]. (June 2018). Available from: https://www.who.int/docs/default-source/immunization/vpd_surveillance/lab_networks/measles_rubella/manual/chapter-3.pdf

Handling, Storage and Transportation of Specimens

Blood:

- Keep blood specimens in an upright position for at least one hour to allow the blood to coagulate.
- The serum should be separated and stored at 4°C or less and transported to NPHL within 24 hours.

Urine:

- Keep cool at 4°C (ice packs or wet ice) until transported to the laboratory.
- Transport specimens to the Immunology Section of NPHL for shipment to the Caribbean Public Health Agency (CARPHA).
- Notify the NSU and NPHL that the specimen is en route.

Prevention and Control Measures

- Public education and routine immunization are key to CRS prevention.
- Containment activities are required following the identification of a suspected case.
- Babies suspected of having rubella should be managed under contact isolation precautions and placed in a private room.
- Only healthcare workers who are not pregnant should be involved in their medical care.
- Medical providers should be notified immediately if the suspected case develops a rash.
- Close contacts should be immunized if they are not already and should remain at home while they are still infectious.
- Intensified surveillance, review of vaccination coverage, targeted immunization, and public education are all important components of the response.

Diphtheria

Diphtheria is a severe, acute and highly contagious disease caused by strains of the bacterium *Corynebacterium diphtheriae*. This disease has significant epidemic potential and therefore remains a global public health concern.

Case Definition and Classification

Suspected Case

A person presenting with acute pharyngitis or laryngitis AND a tonsillar or laryngeal pseudomembrane.

Confirmed Case

Cases may be confirmed in one of two ways:

- A laboratory confirmed case is a suspected case from whom *Corynebacterium diphtheriae* was isolated from bacterial culture.
- An epidemiologically confirmed case is a suspected case that is epidemiologically linked to a laboratory-confirmed case.

Case Investigation

The healthcare worker should:

1. Initiate a case investigation within 48 hours of case notification:
 - Obtain clinical history and laboratory test results from physician.
 - Obtain travel and immunization history.
 - Ensure specimen collection and transportation to the appropriate laboratory.
 - Visit the home, day care, nursery or workplace to review immunization history of close contacts and search for other cases. All persons found not to be fully immunized, or not have documentation of immunization, should be vaccinated immediately.

- Initiate community and hospital surveillance for additional cases.
 - Follow-up includes: review of the case to obtain the final diagnosis and outcome; home visit; check for additional cases; follow-up vaccinations.
2. Complete and submit an investigation report to the parish MO(H) within 7 days of notification.

Specimen Collection and Laboratory Investigation

Laboratory confirmation of Diphtheria is based on:

- Culture of *C. diphtheriae* (throat, nose, nasopharyngeal, skin)
- Gram stain may aid in diagnosis

Caution: Ensure that personal protective equipment is used, and that respiratory support is available during specimen collection.

Specimen Collection

Throat, nose, nasopharyngeal, and skin swabs:

- At first contact, swab the lesion with a cotton-tipped swab.
- Place swab in transport (Amies) medium and transport at room temperature.
- Parish Health Department should contact the Microbiologist at NPHL upon collection of the sample and verify the collection of the sample.

Handling, Storage and Transportation of Specimens

- Transport specimens to the bacteriology section of NPHL accompanied by a copy of the completed Jamaica Laboratory Investigation Form ([Appendix 9](#)).
- Age of the patient, clinical presentation, and site of the swab are mandatory fields.

Prevention and Control Measures

- Public education and routine immunization are key to diphtheria prevention.

- Control of diphtheria is based on:
 - primary prevention of disease by ensuring high population immunity (target > 90% coverage).
 - secondary prevention of spread by conducting rapid investigation of close contacts, as well as isolation/quarantine.
 - tertiary prevention of complications and death through early diagnosis and proper management.
- Control measures are required following the identification of a suspected case.
- Suspected cases should be placed in strict isolation and treated.
- All articles in contact with the patient and all articles soiled by discharges from the patient should be disinfected by terminal cleaning.
- Close contacts should remain at home until they can no longer infect others and should be vaccinated against diphtheria if they are not fully immunized.

Prophylactic Treatment of Carriers

- All close contacts should have cultures taken for testing and should be kept under surveillance for 7 days.
- A single dose of penicillin (IM) or a 7-10-day course of erythromycin (PO) is recommended for all persons exposed to diphtheria, regardless of their immunization status.
- A single dose of benzathine penicillin G (IM) (600,000 units for persons <6 years of age and 1.2 million units for persons ≥ 6 years of age) or a 7-10 day course of erythromycin (PO) (40 mg/kg/day for children and 1 g/day for adults) has been recommended.
- If cultures are positive, patients should be treated with antibiotics.
- Contacts who handle food or work with school children should be excused from work or school until bacteriologic examination proves them not to be carriers.
- Previously immunized contacts should receive a booster dose of diphtheria toxoid, and a primary series should be initiated in un-immunized contacts, using Td, DT or DPT vaccine depending on age.

- The search for carriers by use of nose and throat cultures is not ordinarily useful or indicated.

Epidemic measures

- Immunize the largest possible proportion of the population group involved, with emphasis on protection of infants and preschool children.
- In an epidemic involving adults, immunize groups that are most affected and at high risk.
- Repeat immunization procedures 1 month later to provide at least 2 doses to the recipients.
- Identify close contacts and define population groups at special risk. In areas with appropriate facilities, carry out a prompt field investigation of reported cases to verify diagnosis, determine biotype and toxigenicity of *C. diphtheriae*.

Hepatitis B

Hepatitis B is a potentially life-threatening liver infection caused by the Hepatitis B virus (HBV). It is a major global public health problem as it can cause chronic infection and puts people at risk of death from cirrhosis and liver cancer. Hepatitis B is also an important occupational hazard for healthcare workers.

Case Definition and Classification

Suspected case

A person presenting with: anorexia, abdominal pain, nausea, vomiting, jaundice.

Confirmed case

A person with laboratory confirmation of Hepatitis B infection using one of the following:

- detection of Hepatitis B surface antigen in the serum
- detection of serum antibodies to the Hepatitis B core antigen (HBcAg)
- detection of serum antibodies to the Hepatitis B envelope antigen (HBeAg)

Contacts

A person, with or without symptoms, who has been in close contact with a confirmed case. Close contact includes sexual contacts, household contacts, occupational contacts (e.g. healthcare workers) and children in nurseries/day-care centres.

Case Investigation

Case investigation is the responsibility of the public health nurse and the contact investigator as directed by the parish MO(H). In general, investigation involves:

- Taking of case history to find the source of the infection e.g. history of sexual contacts, occupational risk, or other risk (history of obtaining a tattoo, injection drug use, recipient of donated blood or blood product, etc.).

- Counselling of sexual and household contacts regarding risk of exposure, symptoms, sequelae and disease spread.
- Testing of contacts to determine if they have also been infected.

Cases should be investigated using the Hepatitis B Case Investigation Form ([Appendix 42](#)). The timeframe for submitting case investigation reports to the parish MO(H) is 2 weeks from the identification of the case.

Further details on case investigation and counseling are provided in the Ministry of Health and Wellness' Contact Investigation Field Guide Manual.

Specimen Collection and Laboratory Investigation

Three clinically useful antigen-antibody systems have been identified for Hepatitis B:

1. *Hepatitis B surface antigen (HBsAg) and antibody to HBsAg (anti-HBs)*

HBsAg can be detected in the serum from several weeks before onset of symptoms to days, weeks or months after onset. HBsAg is present in serum during acute infections and persists in chronic infections. The presence of HBsAg indicates that the person is infectious.

Persons who test positive for HBsAg should be retested 6 months after the initial test to determine if they have cleared the virus or have a chronic infection.

2. *Hepatitis B core antigen (HBcAg) and antibody to HBcAg (anti-HBc)*

Anti-HBc appears at the onset of illness and persists indefinitely. Presence of anti-HBc in serum indicates HBV infection, current or past. IgM anti-HBc is present in high titres during the acute infection and usually disappears within 6 months, although it can persist in some cases of chronic hepatitis. This test may therefore reliably diagnose acute HBV infection.

3. Hepatitis B envelope antigen (HBeAg) and antibody to HBeAg (anti-HBe)

The presence of HBeAg is associated with relatively high infectivity.

Specimen Collection

Approximately 5 to 10 mL of blood should be collected in a red-top tube i.e. one without any preservatives or anticoagulant.

Handling, Storage and Transportation of Specimens

- Keep blood samples upright for approximately 1 hour (or until blood has coagulated).
- If possible, the serum should be separated from the clot and placed into a clean receptacle i.e. a different red-top tube, a cryovial or other suitable vial for safe transport of serum.
- The serum should be stored at 4°C or less until transportation to the laboratory.
- Storage of separated serum at -20°C is optimal.
- If serum separation is not possible at the collection site, the tube containing the clotted blood should be kept cool (but not frozen) and shipped to the laboratory within 24 hours after collection.
- The specimen should be transported to the Immunology Section of the National Public Health Laboratory.

Prevention and Control Measures

There is no cure for Hepatitis B infection. The majority of cases will develop antibodies to the HBsAg antigen and clear their infection in a few weeks or months. Some persons, however, will become chronically infected. The infection is considered chronic when HBsAg persists for longer than 6 months after the resolution of symptoms.

Preventive measures

Spread of this virus can be prevented using the following strategies:

- All infants should be immunized with the Hepatitis B vaccine as per Government of Jamaica schedule.
- Universal Hepatitis B vaccination of neonates within 24 hours of birth. This is highly effective in preventing perinatal transmission of Hepatitis B.
- The general public should be educated on the nature of the infection, risk factors for contracting the virus, the symptoms, and the value of immunization against it.
- All healthcare workers (e.g. surgeons, dentists, oral surgeons, pathologists, operating room and emergency room staff, and clinical laboratory workers who handle blood are at highest risk) should receive the Hepatitis B vaccination series.
- Healthcare workers should always employ standard precautions when handling blood, blood products, organs, tissue, semen or other body fluids.
- Healthcare workers should practice the recommended procedures for safe disposal of used needles.
- Health facilities should ensure that the proper containers are used for discarding needles to decrease needle stick injuries.
- Blood donors should be tested for Hepatitis B infection.
- Infected persons should be educated on the nature of the disease and how it is spread. They should be educated on strict condom use for every sex act.
- Infected persons should not donate blood, plasma, body organs or tissues, or semen until it can be demonstrated that the HBsAg test is negative.
- Women who are infected should be made aware that if they become pregnant during the period that they test positive for the HBsAg, there is a high likelihood that they will pass the virus to their unborn child.
- Infants born to women who are actively infected should be given Hepatitis B immune globulin along with a Hepatitis B vaccination series as soon as possible after birth.

- Infected persons should be educated on methods of preventing spread to household contacts i.e. cuts or skin lesions should be covered; toothbrushes, utensils, razors etc. should not be shared with others.
- Infected persons should be educated to always inform healthcare practitioners (e.g. dentists and physicians) of their infection status.

Control measures

The main methods of control are:

- Vaccination to prevent infection.
- Contact tracing of confirmed cases to identify new cases and prevent spread.
- Treatment of recent contacts of confirmed cases with immune globulin (e.g. infants born to infected mothers).

Pertussis

Pertussis, or whooping cough, is a disease of the respiratory tract caused by the *Bordetella pertussis* bacteria that lives in the mouth, nose and throat. Pertussis is a highly communicable disease that affects unimmunized infants in particular, and therefore remains a public health concern globally, including in countries where vaccination coverage is high.

Case Definition and Classification

Suspected Case

Paroxysms of coughing with characteristic high pitched inspiratory whoop.

Confirmed Case

- A laboratory confirmed case is a suspected case with positive laboratory findings.
- An epidemiologically confirmed case is a suspected case that is epidemiologically linked to a laboratory confirmed case.

Case Investigation

The healthcare worker who is the first point of contact with the case has the responsibility to notify the case to the Parish Health Department or the National Surveillance Unit and initiate the case investigation:

Complete case investigation within 48 hours of case identification:

- Obtain clinical history and laboratory test results.
- Obtain travel and immunization history.
- Ensure specimen collection and referral to the appropriate laboratory.
- Visit the home, day care, nursery, school or workplace to review immunization history of close contacts and search for other cases. All persons under age 7 years found not to be

fully immunized, or not to have documentation of immunization, should be vaccinated immediately.

- Initiate community and hospital surveillance for additional cases.
- Follow-up includes review of the case with the attending physician/specialists to obtain the final diagnosis, and follow-up vaccinations at the home of the patient.

Specimen Collection and Laboratory Investigation

Laboratory confirmation of pertussis is based on:

- Isolation of *Bordetella pertussis* from culture of the posterior nasopharynx.
- Direct fluorescent antibody testing.
- Polymerase Chain Reaction (PCR).
- Detection of pertussis-specific serum IgM may aid in diagnosis.

Specimen Collection

Nasopharyngeal swab:

- Prior to sample collection, contact the Consultant Microbiologist at NPHL and the National Surveillance Unit.
- At first contact and as early in the illness (cough) as possible, swab the posterior nasopharynx with polyester (Dacron), rayon or nylon-tipped swab with a plastic or flexible wire shaft. Place in Universal Transport Medium (UTM), Viral Transport Medium (VTM) or Reagan-Lowe transport medium ¹ or as instructed by the NPHL. Do not use Calcium alginate or wooden-shafted swabs as they may inhibit PCR and reduce organism recovery.

¹ World Health Organization. *Pertussis - Vaccine-Preventable Diseases Surveillance Standards* [Internet]. September 2018. Available from: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://cdn.who.int/media/docs/default-source/immunization/vpd_surveillance/vpd-surveillance-standards-publication/who-surveillancevaccinepreventable-16-pertussis-r2.pdf?sfvrsn=a0157ae7_10

Blood:

- In the event of a verified or suspected outbreak, collect blood in a red top tube.

Handling, Storage and Transportation of Specimens

Nasopharyngeal swab:

- Maintain specimen at 2°C to 8°C after collection.
- Transport to the laboratory under refrigerated conditions e.g., using cold packs.
- Rapidly transport the swab to NPHL within 24 - 72 hours.
- If longer storage or shipment is anticipated, specimens should be frozen at -70°C and transported on dry ice.
- Avoid repeated freeze-thaw cycles.

Blood:

- Store blood samples upright for approximately 1 hour (or until blood has coagulated).
- Separate the serum and place it in a sterile tube for transport to the laboratory.
- Serum samples should be sent to the Immunology Section of NPHL for shipment to CARPHA.
- NPHL should be informed by phone that the serum sample is en route.

Prevention and Control Measures

- Public education and routine immunization are key to pertussis prevention.
- Control measures are required following the identification of a suspected case.
- Suspected cases should be separated from young children and infants, until they have received at least 5 days of a minimum 14-day course of antibiotics.
- Confirmed cases should be placed in respiratory isolation and treated.

- Symptomatic close contacts should remain at home until they can no longer infect others. They should also receive a 14-day course of Erythromycin or other macrolide antibiotic regardless of age and immunization status.
- Asymptomatic contacts should be vaccinated against pertussis if they are below age 7 years and not fully immunized. Whole cell pertussis-containing vaccine is not given to persons older than 7 years of age. The asymptomatic contacts should also receive a 14-day course of Erythromycin or other macrolide antibiotic regardless of age and immunization status.

Tetanus

Tetanus, also known as “lockjaw”, is a serious wound infection caused by the bacterium *Clostridium tetani*, which is found in soil, dust, and animal faeces. The germ produces toxins that affect the brain and nervous system, resulting in muscle stiffness.

Case Definition and Classification

Suspected Case

Neonatal Tetanus – A neonate between the ages of 3 and 28 days who has difficulty in sucking, or opisthotonus.

Death of a neonate between the ages of 3 and 28 days should be reviewed by a competent medical professional, given the possibility that tetanus is a differential diagnosis.

Non-neonatal Tetanus – Any person experiencing difficulty swallowing, followed by muscle spasm and rigidity.

History of an injury is supporting evidence of possible exposure but is not necessary.

Confirmed Case

Neonatal Tetanus – A suspected case with one or more of the following:

- facial grimace
- stiffness of the body and arching of back
- generalized spasms or convulsions



Figure 24.2. Neonatal Tetanus

Non-neonatal Tetanus – A suspected case exhibiting one or more of the following: trismus (lockjaw), risus sardonicus (sustained involuntary spasm of the facial muscles causing a grimace) or generalised muscle spasms or contractions.

Cases are clinically confirmed by a physician.



Figure 24.3. Risus Sardonicus facial expression in Non-neonatal Tetanus

Case Investigation

1. The healthcare worker who is the first point of contact with the case has the responsibility to notify the case to the Parish Health Department or the National Surveillance Unit and initiate the case investigation:
 - Case review to obtain the clinical history, date and site of injury, date of onset of illness, signs and symptoms, and current presentation.
 - Obtain details of immunization history.
 - Ensure specimen collection and referral to the appropriate laboratory (optional depending on the case's history and presentation).
 - Conduct a home visit to evaluate living conditions (this goes towards education for injury prevention, proper cleaning of wounds, etc.) and to review immunization status of household members, especially children and pregnant women.
 - Follow-up of case review with physician to obtain final diagnosis.
 - Follow-up home visit to review the case, ensure implementation of previous recommendations, and follow-up vaccinations as necessary.
2. Every suspected case of tetanus must be investigated within 48 hours using the appropriate investigation form. Completion of the form is the responsibility of a public health nurse assigned to the case by the parish MO(H).

Specimen Collection and Laboratory Investigation

An appropriate specimen from debridement or wound swab should be collected and sent to NPHL along with a copy of the laboratory investigation form. Note however that the diagnosis may be made without laboratory tests. The isolation of the organism from the wound is of little value since it may be recovered from the wounds of persons who show no signs of tetanus.

Prevention and Control Measures

There are three main lines of prevention:

- Active immunization against tetanus.
- Antibacterial measures for wounds and deliveries of babies.
- Passive immunization of possible cases with immune globulin.

a) Active Immunizations:

Active immunization with tetanus toxoid is the most effective method of preventing tetanus. Ideally everyone should be given a course of active immunization, according to the national immunization schedule. Ensure that the tetanus booster is given to pregnant women who have not received it in the past 10 years.

b) Antibacterial Measures:

- The general public should be educated on the hazards of puncture wounds, proper cleansing of wounds with soap and water, and seeking medical intervention when appropriate.
- Medical institutions must employ appropriate measures for adequate cleaning of wounds and careful debridement as well as protection of wounds to avoid contamination.
- Maternity personnel and institutions should institute/maintain clean delivery protocols to prevent cases of neonatal tetanus.
- Antibiotics, especially long-acting penicillin, can also be given to suppress bacterial multiplication. If the wound is old, i.e. more than 12 hours, tetanus may occur despite adequate doses of penicillin.

c) Passive Immunization:

- Tetanus antitoxin (ATS) or Tetanus Immune Globulin (TIG) are used for treatment.
- Fifteen hundred (1,500) units of ATS are given in the average case, although a larger dose may be required for patients who present with heavily contaminated wounds.

- Note that a test dose of ATS should be given to patients in order to test for adverse reactions. Repeated doses are not recommended, as it will lead to sensitization of the patient thereby creating the hazard of allergic reaction. These later doses are also rapidly eliminated from the body and are therefore less effective for prevention and treatment.
- TIG is administered intramuscularly at a dose of 3,000 to 6,000 IU.

Tuberculosis

The overall objective of tuberculosis control is to reduce morbidity, mortality and transmission of the disease until it no longer poses a threat to public health.

Refer to the Tuberculosis chapter of the *National Surveillance Manual for Jamaica* for details on case investigation, specimen collection, laboratory investigation, and prevention and control measures.

Chapter 25: Zika Surveillance

Background

The Pan American Health Organization (PAHO) / World Health Organization (WHO) issued an Epidemiological Alert on 7 May 2015 recommending that Member States establish and maintain the capacity for Zika virus infection detection, clinical management, and an effective public communication strategy to reduce the presence of the mosquito that transmits this disease, particularly in areas where the vector is present.

The PAHO/WHO issued an Epidemiological Alert on 1 December 2015, which reported that there were increases in congenital anomalies, Guillain- Barré syndrome, and other neurological and autoimmune syndromes in areas where Zika virus is circulating and their possible relation to the virus. Given this, PAHO/WHO additionally recommended that Member States establish and maintain the capacity to detect and confirm Zika virus cases, prepare healthcare facilities for the possible increase in demand at all healthcare levels and specialised care for neurological syndromes, and strengthen antenatal care.

Phase 1: Preparedness

During the preparedness phase, the epidemiological surveillance system of Zika virus infection in Jamaica will be strengthened through the following strategies:

1. Training the epidemiological and clinical staff on Zika management
2. Standardizing a Case Definition
3. Designation of Zika virus infection as a Class 1 Notifiable disease
4. Laboratory surveillance to confirm the autochthonous transmission of Zika Infection.
5. Surveillance for congenital anomalies in all hospitals.
6. Develop a research agenda for Zika virus infection in Jamaica.

Clinical Features

Common Clinical Features

- Fever
- Maculopapular rash
- Non-purulent conjunctivitis
- Headache
- Myalgia
- Arthralgia
- Asthenia
- Oedema in the lower limbs

Less Common Clinical Features

- Retro-orbital pain
- Anorexia
- Vomiting
- Diarrhoea
- Abdominal pain
- Mild Thrombocytopenia (laboratory finding)
- Mild Leucopenia (laboratory finding)

Severe Clinical Features

- Neurological syndromes, including Guillain Barré
- Autoimmune syndromes
- Thrombocytopenia Purpura
- Meningoencephalitis
- Congenital Anomalies

Case Definitions

Suspected Case

Patient with rash or elevated body temperature ($>37.2^{\circ}\text{C}$ or 99.0°F) with one or more of the following symptoms:

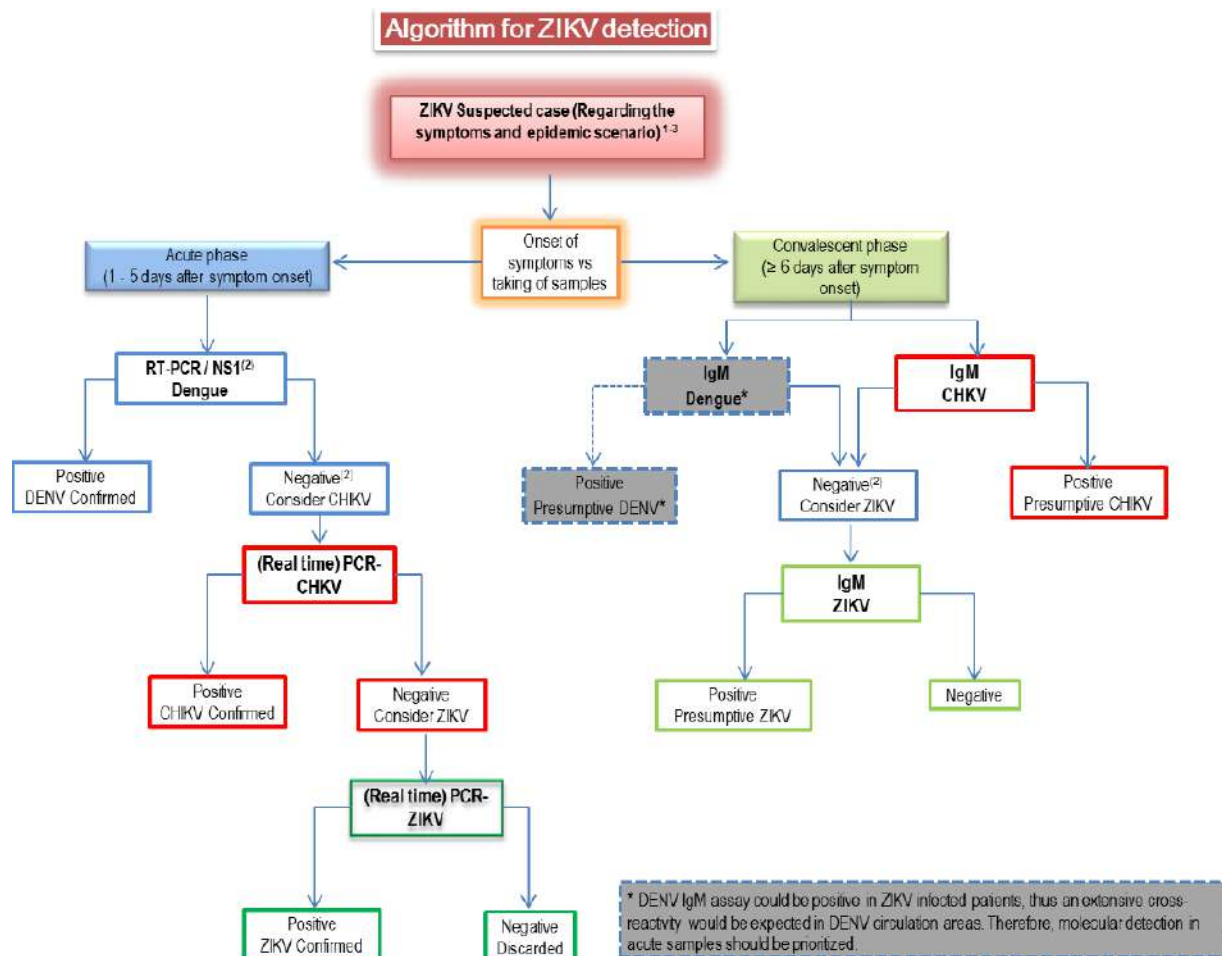
- Arthralgia or myalgia
- Non-purulent conjunctivitis or conjunctival hyperaemia
- Headache or malaise

In addition, persons presenting with neurological syndromes and autoimmune syndromes post a viral illness are to be considered a suspected case of Zika virus infection.

Mother and child: Consider Zika in a birth (including stillbirths) with congenital anomalies. Clusters of rash and fever will also be considered as suspected Zika virus infection.

Confirmed Case

A suspected case with laboratory positive result for the specific detection of Zika virus (see Figure 25.1).



CHIKV = chikungunya virus; DENV = dengue virus; ZIKV = Zika virus

¹ According to the epidemiological profile of each country and taking into account the clinical manifestations of the infection, other Arboviruses should be included in the differential algorithm for the Zika virus.

² This algorithm is not exhaustive, and dengue infection should be discarded according to the guidelines of clinical management and laboratory specific algorithm

³ These recommendations are subject to later modifications that take into account advances in knowledge of the disease and the etiologic agent.

Figure 25.1. Algorithm for detecting Zika, dengue and chikungunya viruses ¹

¹ Pan American Health Organization/ World Health Organization. *Zika virus (ZIKV) Surveillance in the Americas: Interim guidance for laboratory detection and diagnosis* [Internet]. Washington D.C. (June 2015) Available from: chrome-

Designation of Zika Fever as a Class 1 Notifiable Disease

Zika Fever is a Class 1 Notifiable Disease and is to be notified immediately on suspicion, within 24 hours, to the Parish Health Department and the National Surveillance Unit, Ministry of Health and Wellness. Cases (confirmed by laboratory testing) should also be notified as a Class 1 Notifiable Disease to the Parish Health Department and the National Surveillance Unit, Ministry of Health and Wellness.

Laboratory Testing for Zika

Samples for serology should be sent to the National Public Health Laboratory along with the completed Laboratory Investigation Form ([Appendix 9](#)). The following are the requirements for sampling and for the conservation of the sample:

Type of sample

Serum: 4 to 5 mls of blood in a plain red top tube

- Acute phase: Until 8 days after symptom onset
- Convalescent phase: 10 to 15 days after symptom onset

Conservation of the sample

- Keep refrigerated (2 to 8°C) if sample will be processed within 48 hours at the NPHL.
- Keep frozen (-10 to -20°C) if sample will be processed after the first 48 hours.
- Maintain frozen (-70°C) if sample will be processed after one week.

Method for Confirmation of the Start of a Zika Outbreak in Jamaica

Laboratory surveillance will be used as the methodology for the confirmation of autochthonous transmission.

Inclusion Criteria

The following are inclusion criteria for the laboratory surveillance to confirm autochthonous transmission:

1. Samples that are negative for Dengue and CHIK testing **AND** consistent with ZIKA.
2. Cases consistent with clinical features of Zika: Patient with rash or elevated body temperature ($> 37.2^{\circ}\text{C}$) with one or more of the following symptoms (not explained by other medical conditions): Arthralgia or myalgia; non-purulent conjunctivitis or conjunctival hyperaemia; Headache or malaise.
3. The samples should be processed according to the day of sampling with respect to the onset of symptoms. The algorithm in Figure 25.1 should be strictly followed by the laboratory.

Congenital Anomalies Surveillance

A hospital-based surveillance system will be established to identify congenital anomalies, particularly microcephaly. The surveillance system will be limited to structural anomalies that are readily identifiable and easily recognized on physical examination at birth or shortly thereafter. All hospitals will be required to submit monthly reports with the surveillance information, the Hospital Monthly Statistical Report.

The baseline information for neurological congenital anomalies, in particular for microcephaly, will be gathered for births occurring in the previous five years to detect any change or outbreak in congenital anomalies. A Line Listing ([Appendix 43](#)) needs to be completed by each hospital and reviewed by the Parish Medical Officer (Health) and the Regional Epidemiology Team. Each Regional Health Authority will submit an electronic spreadsheet to the National Surveillance Unit for the national data analysis.

Microcephaly: A disorder in which the head circumference is two standard deviations or smaller than the average for sex and age, and in some cases is associated with the altered structure of the brain and neurodevelopmental problems. The presence of a head circumference less than two standard deviations below the mean for sex and age without evidence of structural abnormalities of the brain is not considered a major anomaly ¹.

Suspected cases of congenital syndrome associated with Zika virus infection are to be reported and the case investigation form completed ([Appendix 44](#)).

[Appendix 45](#) provides an algorithm for notification and laboratory testing.

Phase 2: Established Outbreak

The confirmation of an imported or autochthonous case of Zika Fever will activate the Ministry of Health and Wellness' response mechanisms, as well as the National and International Epidemiological Focal Point and the National IHR Focal Point. The PAHO/ WHO IHR Contact Point will be notified, as described by the International Health Regulations Chapter.

Enhanced Surveillance

The clinical and epidemiological surveillance system will be enhanced based on the epidemiology of the confirmed case(s).

Notification and Investigation

- All persons fitting the case definition should be notified immediately to the Parish Health Department and the National Surveillance Unit, Ministry of Health and Wellness.

¹ World Health Organization (WHO)/Centers for Disease Control and Prevention (CDC)/International Clearinghouse for Birth Defects Surveillance and Research (ICBDSR). *Birth defects surveillance: a manual for programme managers* [Internet]. Geneva (2014). Available from: http://www.who.int/nutrition/publications/birthdefects_manual/en/

- A detailed investigation of all suspected cases is to be conducted. A report should be submitted within 48 hours using the standard Investigation Form ([Appendix 46](#)), but not limited to this form.
- Surveillance of persons with similar exposure is to be conducted for at least 21 days post exposure. A line listing is to be completed, indicating the possible date of exposure, the date that person was contacted and their clinical status recorded.
- Active community surveillance is to be conducted where indicated ([Appendix 47](#)).

See [Appendix 48](#) for the Algorithm for Notification and Intervention of a Suspected Case.

Characterisation of the Outbreak

The data related to the Zika fever outbreak will be analysed regularly (weekly or daily as required) to characterise the outbreak.

- Demographic characteristics of the outbreak
- Monitoring and description of the spread of Zika virus
- Clinical features
- Clinical severity and impact on society
- Identification of risk factors for severe disease

Phase 3: End of the Outbreak

Based on the epidemiological assessment, a decision will be made regarding declaration of the end of the outbreak.

Phase 4: Endemic Seasonal Transmissions

Zika fever as a Class 1 Notifiable Disease will be monitored according to the current Integrated Surveillance system. The preparedness plan will be reviewed and revised in keeping with the epidemiological situation to maintain preparedness.

Framework for Epidemiological Surveillance of Zika

Table 25.1. *Epidemiological Framework for Zika Surveillance*

Expected Result	Indicators
Zika Epidemiological Surveillance system for timely alert and opportune response implemented.	▪ Zika surveillance systems in place (e.g., laboratory data, clinical data, entomological data) ▪ Number of sites (Hospitals, Health Centres, and Private Doctors) within the country that have the information about Zika fever (Epidemiological Surveillance Plan) ▪ Zika fever outbreaks reported according to IHR (2005) guidelines.

Epi Activities	Tasks	Responsible Persons
<u>Phase 1: Preparedness</u> 1. Strengthen the epidemiological and laboratory surveillance systems for Zika in the countries of the Caribbean Sub- region	Include Zika as part of the Disease Surveillance System – Class 1 Notifiable Disease (reported immediately on suspicion, within 24 hours). Standardise a Zika Case Definition (clinical and epidemiological) based on PAHO/WHO guidelines. Set up clinical, laboratory and epidemiological surveillance. Standardise the methods used to determine the criteria (clinical, epidemiological and laboratory) to confirm the start of a Zika outbreak. Train the epidemiological, laboratory and clinical staff on Zika fever management. Review and revise epidemiological surveillance plan.	CMO, National Epidemiologist, Medical Epidemiologist - Communicable Diseases, Director - National Laboratory Services (NLS), Director - EDMSSB

Epi Activities	Tasks	Responsible Persons
	Develop a research agenda for Zika virus infection. Surveillance for congenital anomalies in all hospitals. Baseline incidence of congenital anomalies.	National Epidemiologist, Medical Epidemiologist - Communicable Diseases, Director - Epidemiological Research and Analysis, Regional Technical Directors, Medical Officers (Health), Senior Medical Officers.
<u>Phase 2: Established outbreak</u> 1. Confirmation/ Declaration of the beginning of an outbreak	Declare the start of the outbreak	CMO
	Classify cases as locally- acquired or imported cases.	National Epidemiologist, Medical Epidemiologist - Communicable Diseases
	Notify the National and International Epidemiological Focal Points, National IHR Focal Point and PAHO / WHO IHR Contact Point, according to IHR (2005).	IHR NFP _Officer with Primary Responsibility
	Enhance clinical, epidemiological and laboratory surveillance systems. Review and revise epidemiological surveillance plan	National Epidemiologist, Medical Epidemiologist - Communicable Diseases, Director - NLS
Monitor and assess the epidemic situation	Activate and maintain the National, Regional and Parish Health Emergency Operations Centres / Coordination mechanisms. Establish routine communication mechanisms with parish and regional health authorities and international organizations.	CMO, Director - EDMSSB, National Epidemiologist, MOHW NEOC, Ministry of Health and Wellness Emergency and Disaster Risk Management Committee

Epi Activities	Tasks	Responsible Persons
	Analyze and interpret weekly data and develop a daily and weekly outbreak report. Provide support and technical assistance to the field.	
<u>PHASE 3: End of the outbreak</u> 1. Analyze the actions taken during the outbreak	Maintain the monitoring and evaluation activities. Conduct after-action evaluations to identify and remedy gaps in the country's surveillance and response program.	CMO, Director-EDMSSB, National Epidemiologist, Ministry of Health and Wellness' Emergency and Disaster Risk Management Committee, Director - National Laboratory Services (NLS),
	Review and revise Epidemiological Surveillance Plan	National Epidemiologist, Medical Epidemiologist – Communicable Diseases, Director - NLS
<u>PHASE 4: Endemic seasonal transmissions</u> 1. Maintain the surveillance activities	Establish a regular, integrated surveillance program. Review evaluation of the plan and revise the preparedness plan to ensure preparedness is maintained.	CMO, National Epidemiologist, Director - EDMSSB, Ministry of Health and Wellness' Emergency and Disaster Risk Management Committee



APPENDICES

Appendix 1: Contact List for Different Levels of the Surveillance System

MOHW Contact List

Position	Telephone	Postal Address
Minister of Health and Wellness	Phone: 876-633-8172	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
Permanent Secretary	Phone: 876-633-8164	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
Chief Medical Officer	Phone: 876-633-8154	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
Director of Health Promotion and Protection	Phone: 876-633-8163	Ministry of Health and Wellness 52-60 Grenada Way REIT Building
Director, Health Information Unit	Phone: 876-633-8172	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
Laboratory Director, NPHL	Phone: 876-967-2234	21 Slipe Pen Road Kingston
Director of Environmental Health Unit	Phone: 876-633-7901	Ministry of Health and Wellness 15 Knutsford Boulevard ARC Building
Director of Veterinary Public Health	Phone: 876-633-7907	Ministry of Health and Wellness 15 Knutsford Boulevard ARC Building
Director of Emergency and Disaster Management and Special Services (EDMSS)	Phone: 876-727-7427	Ministry of Health and Wellness 52-56 Knutsford Boulevard IBM Building

Position	Telephone	Postal Address
Director of Family Health Unit	Phone: 876-633-7919	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
Director of Health Services Planning and Integration	Phone: 876-633-7919	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
SMO of National HIV /STI Programme	Phone: 876-633-8172	Ministry of Health and Wellness 10-16 Grenada Way RKA Building
Tuberculosis Coordinator	Phone: 876-633-8163	Ministry of Health and Wellness 52-60 Grenada Way REIT Building
Medical Epidemiologist, National Surveillance Unit	Phone: 876-633-7937	Ministry of Health and Wellness 15 Knutsford Boulevard ARC Building
Chief Nursing Officer	Phone: 876-633-7902	Ministry of Health and Wellness 15 Knutsford Boulevard ARC Building
Director of Health Promotion and Education	Phone: 876-633-8163	Ministry of Health and Wellness 52-60 Grenada Way REIT Building
Public Relations Officer	Phone: 876-633-7919	Ministry of Health and Wellness 52-60 Grenada Way REIT Building
CARPHA Head Office	Phone: (+868) 299-0820 (+868) 299-0895 (+868) 622-4261 Email: postmaster@carpha.org	16-18 Jamaica Blvd Federation Park Port of Spain Trinidad and Tobago

Parish Health Departments/ Regional Health Authorities Contact List

Parish Health Department	Telephone and Fax	Postal Address
KSA	Phone: (876) 926-1550-2 (876) 317-9681; (876) 317-9680	1A Caledonia Crescent, Cross Roads, Kingston 5
St. Catherine	Phone: (876) 538-8594	Burke Rd, Spanish Town
St. Thomas	Phone: (876) 982-1619 (876) 703-6062; (876) 982-2304	54 Lyssons Rd., Morant Bay
SERHA	Phone: (876) 754-3439; Fax: (876) 926-4109	The Towers, 15 Dominica Drive, Kingston 5
Portland	Phone: (876) 993-2557; (876) 993-2873	Smatt Road, Port Antonio
St. Mary	Phone: (876) 613-8500 (876) 994-9979	Trinity, Port Maria
St. Ann	Phone: (876) 972-2215; (876) 972-2190; (876) 972-2227	2 Owen Sound Drive, St. Ann's Bay
NERHA	Phone: (876) 795-3107; Fax: (876) 974-8819;	Shop 34-37, Ocean Village Plaza, Ocho Rios
Trelawny	Phone: (876) 954-3255-50	Rodney Street, Falmouth
St. James	Phone: (876) 979-7820-4	Creek Street, Montego Bay
Hanover	Phone: (876) 956-2733	Fort Charlotte Drive, Lucea
Westmoreland	Phone: (876) 955-2308; (876) 955-2929	Barracks Road, Savanna-la-Mar
WRHA	Phone: (876) 630-1860	Lot 31B, Fairview Shopping Centre, Montego Bay, St. James
St. Elizabeth	Phone: (876) 965-9174-5	High Street, Black River
Manchester	Phone: (876) 625-0612	Race Course Rd., Mandeville
Clarendon	Phone: 876-986-7869, Fax: 876-986-9713	1-3 Jackson Street, Denbigh
SRHA	Phone: (876) 962-8232	3 Brumalia Road Mandeville

[illegible]

Appendix 3: List of Sentinel Sites

Sentinel Sites							
Station Code	Sentinel Site Name	Parish Code	Parish Name	Date Sentinel Site Established	Active	Region	SARI Sites
CTH	CHAPELTON HOS	CLA	CLARENDON	10/2/1992	Yes	Southern	
LTH	LIONEL TOWN HOS	CLA	CLARENDON	10/2/1992	Yes	Southern	
MPH	MAY PEN HOS	CLA	CLARENDON	1/12/2005	Yes	Southern	
MPHC	MAY PEN H/C	CLA	CLARENDON	10/2/1992	Yes	Southern	
SPHC	SPAULDING H/C	CLA	CLARENDON	1/8/2011	Yes	Southern	
GIHC	GREEN ISLAND H/C	HAN	HANOVER	10/2/1992	Yes	Western	
HWHC	HOPEWELL H/C	HAN	HANOVER	10/2/1992	Yes	Western	
LUCHC	LUCEA H/C	HAN	HANOVER	10/2/1992	Yes	Western	
NHH	NOEL HOLMES HOS	HAN	HANOVER	10/2/1992	Yes	Western	
SBHC	SANDY BAY H/C	HAN	HANOVER	1/1/2000	Yes	Western	
BCH	BUSTAMANTE HOS	KSA	KINGSTON-ST. ANDREW	10/2/1992	Yes	South-East	Yes
DPHC	DUHNANEY PARK HEALTH CENTRE	KSA	KINGSTON-ST. ANDREW	1/12/2011	Yes	South-East	
EMHC	EDNA MANLEY H/C	KSA	KINGSTON-ST. ANDREW	10/2/1992	Yes	South-East	
GVHC	GLEN VINCENT P/CLINIC	KSA	KINGSTON-ST. ANDREW	10/2/1992	Yes	South-East	
HVHC	HARBOUR VIEW H/C	KSA	KINGSTON-ST. ANDREW	10/2/1992	Yes	South-East	
MFHC	MAXFIELD PARK HEALTH CENTRE	KSA	KINGSTON-ST. ANDREW	1/12/2011	Yes	South-East	
SHHC	STONEY HILL HEALTH CENTRE	KSA	KINGSTON-ST. ANDREW	1/12/2011	Yes	South-East	
UHWI	UNIVERSITY HOSPITAL	KSA	KINGSTON-ST. ANDREW	1/14/2009	Yes	South-East	Yes
UWIHC	UWI HEALTH SERVICE	KSA	KINGSTON-ST. ANDREW	10/2/1992	Yes	South-East	
CKHC	CROSS KEYS H/C	MAN	MANCHESTER	10/2/1992	Yes	Southern	
MVH	MANDEVILLE HOS	MAN	MANCHESTER	10/2/1992	Yes	Southern	Yes
MVHC	MANDEVILLE H/C	MAN	MANCHESTER	10/2/1992	Yes	Southern	

Sentinel Sites							
Station Code	Sentinel Site Name	Parish Code	Parish Name	Date Sentinel Site Established	Active	Region	SARI Sites
NPHC	NEWPORT H/C	MAN	MANCHESTER	10/2/1992	Yes	Southern	
PJH	PERCY JUNOR HOS	MAN	MANCHESTER	1/1/2000	Yes	Southern	
POHC	PORUS H/C	MAN	MANCHESTER	10/2/1992	Yes	Southern	
MAN	HARGREAVES MEMORIAL HOSPITAL	MAN	MANCHESTER	11/1/2018	Yes		
BBH	BUFF BAY HOS	POR	PORTLAND	10/2/1992	Yes	North-East	
MCHC	MANCHIONEAL	POR	PORTLAND	1/1/2000	Yes	North-East	
PAH	PORT ANTONIO HOS	POR	PORTLAND	10/2/1992	Yes	North-East	
PAHC	PORT ANTONIO H/C	POR	PORTLAND	1/1/2000	Yes	North-East	
ALXH	ALEXANDRIA HOS	STA	ST. ANN	10/2/1992	Yes	North-East	
BTHC	BROWN'S TOWN H/C	STA	ST. ANN	10/2/1992	Yes	North-East	
CMHC	CLAREMONT HEALTH CENTRE	STA	ST. ANN	1/14/2009	Yes	North-East	
MHC	MONEAGUE HEALTH CENTRE	STA	ST. ANN	9/1/2008	Yes	North-East	
ORHC	OCHO RIOS H/C	STA	ST. ANN	10/2/1992	Yes	North-East	
SABH	ST. ANN'S BAY HOS	STA	ST. ANN	10/2/1992	Yes	North-East	Yes
SABHC	ST. ANN'S BAY H/C	STA	ST. ANN	11/1/2006	Yes	North-East	
CPHC	CHRISTIAN PEN H/C	STC	ST. CATHERINE	10/2/1992	Yes	South-East	
GPMHC	GREATER PORTMORE H/C	STC	ST. CATHERINE	1/1/2000	Yes	South-East	
LSHC	LINSTEAD H/C	STC	ST. CATHERINE	10/2/1992	Yes	South-East	
OHHC	OLD HARBOUR H/C	STC	ST. CATHERINE	10/2/1992	Yes	South-East	
SIHC	ST. JAGO PARK H/C	STC	ST. CATHERINE	10/2/1992	Yes	South-East	
STH	SPANISH TOWN HOS	STC	ST. CATHERINE	1/1/2004	Yes	South-East	
BRH	BLACK RIVER HOS	STE	ST. ELIZABETH	10/2/1992	Yes	Southern	
BRHC	BLACK RIVER H/C	STE	ST. ELIZABETH	1/12/2010	Yes	Southern	
MGHC	MAGGOTY H/C	STE	ST. ELIZABETH	10/2/1992	Yes	Southern	

Sentinel Sites							
Station Code	Sentinel Site Name	Parish Code	Parish Name	Date Sentinel Site Established	Active	Region	SARI Sites
SCHC	SANTA CRUZ H/C	STE	ST. ELIZABETH	10/2/1992	Yes	Southern	
SFHC	SOUTHFIELD H/C	STE	ST. ELIZABETH	10/2/1992	Yes	Southern	
CBHC	CAMBRIDGE H/C	STJ	ST. JAMES	1/12/2010	Yes	Western	
CHHC	CATHERINE HALL H/C	STJ	ST. JAMES	10/2/1992	Yes	Western	
CRH	CORNWALL REG. HOS	STJ	ST. JAMES	10/2/1992	Yes	Western	Yes
DRSC	DRS SURGI CLINIC	STJ	ST. JAMES	11/1/2006	Yes	Western	
MB5HC	MO-BAY TYPE 5 H/C	STJ	ST. JAMES	1/10/1996	Yes	Western	
HJL	HOSPITEN JAMAICA LTD	STJ	ST. JAMES	9/1/2008	Yes	Western	
MSHC	MOUNT SALEM H/C	STJ	ST. JAMES	1/12/2010	Yes	Western	
MTHC	MAROON TOWN H/C	STJ	ST. JAMES	1/12/2010	Yes	Western	
ABH	ANNOTTO BAY HOSPITAL	STM	ST. MARY	1/9/2002	Yes	North-East	
ABHC	ANNOTTO BAY H/C	STM	ST. MARY	1/8/1997	Yes	North-East	
GAHC	GAYLE H/C	STM	ST. MARY	1/8/1997	Yes	North-East	
HGHC	HIGHGATE H/C	STM	ST. MARY	1/8/1997	Yes	North-East	
OBHC	ORACABESSA H/C	STM	ST. MARY	1/12/2010	Yes	North-East	
PMH	PORT MARIA HOS	STM	ST. MARY	10/2/1992	Yes	North-East	
PMHC	PORT MARIA H/C	STM	ST. MARY	11/1/2006	Yes	North-East	
IBH	ISAAC BARRANT HOS	STT	ST. THOMAS	10/2/1992	Yes	South-East	
MBHC	MORANT BAY H/C	STT	ST. THOMAS	10/2/1992	Yes	South-East	
PRMH	PRINCESS MARGARET HOS	STT	ST. THOMAS	3/1/2007	Yes	South-East	
SEHC	SEAFORTH HEALTH CENTRE	STT	ST. THOMAS	1/12/2011	Yes	South-East	
YSHC	YALLAHS HEALTH CENTRE	STT	ST. THOMAS	1/12/2011	Yes	South-East	
ATHC	ALBERT TOWN H/C.	TRE	TRELAWNY	1/10/1996	Yes	Western	
DDHC	DEWAR DUNCANS H/C	TRE	TRELAWNY	10/2/1992	Yes	Western	

Sentinel Sites							
Station Code	Sentinel Site Name	Parish Code	Parish Name	Date Sentinel Site Established	Active	Region	SARI Sites
FMH	FALMOUTH HOS	TRE	TRELAWNY	10/2/1992	Yes	Western	
FMHC	FALMOUTH H/C	TRE	TRELAWNY	10/2/1992	Yes	Western	
USH	ULSTER SPRING HOS	TRE	TRELAWNY	10/2/1992	Yes	Western	
DRLHC	DARLISTON H/C	WES	WESTMORLAND	10/2/1992	Yes	Western	
NEGHC	NEGRIL H/C	WES	WESTMORLAND	10/2/1992	Yes	Western	
SLMH	SAV-LA-MAR HOS	WES	WESTMORLAND	1/10/1995	Yes	Western	
SMHC	SAV-LA-MAR HEALTH CENTRE	WES	WESTMORLAND	1/14/2009	Yes	Western	
WHHC	WHITE HOUSE H/C	WES	WESTMORLAND	10/2/1992	Yes	Western	

Hospital Active Surveillance Sites						
Station Code	Sentinel Site Name	Parish Code	Parish Name	Date Sentinel Site Established	Active	SARI
AMH	ANDREWS MEMORIAL HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	
BVH	BELLEVUE HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	
BCH	BUSTAMANTE CHILDREN'S HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	Yes
KPH	KINGSTON PUBLIC HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	Yes
MAH	MEDICAL ASSOCIATES	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	
NCH	NATIONAL CHEST HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	
NMH	NUTALL MEMORIAL HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	
SJH	ST. JOSEPH'S HOSPITAL	KSA	KINGSTON/ST. ANDREW	13/1/1998	Yes	
UWIH	UNIVERSITY HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	Yes
VJH	VICTORIA JUBILEE HOSPITAL	KSA	KINGSTON/ST. ANDREW	3/3/1997	Yes	
PM	PRINCESS MARGARET HOSPITAL	STT	ST. THOMAS	3/3/1997	Yes	

Hospital Active Surveillance Sites						
Station Code	Sentinel Site Name	Parish Code	Parish Name	Date Sentinel Site Established	Active	SARI
PAH	PORT ANTONIO HOSPITAL	POR	PORTLAND	3/3/1997	Yes	
ABH	ANNOTTO BAY HOSPITAL	STM	ST. MARY	3/3/1997	Yes	
PMH	PORT MARIA HOSPITAL	STM	ST. MARY	3/3/1997	Yes	
ALXH	ALEXANDRIA HOSPITAL	STA	ST. ANN	13/1/1998	Yes	
SABH	ST. ANN'S BAY HOSPITAL	STA	ST. ANN	3/3/1997	Yes	Yes
FH	FALMOUTH HOSPITAL	TRE	TRELAWNY	3/3/1997	Yes	
CRH	CORNWALL REGIONAL HOSPITAL	STJ	ST. JAMES	3/3/1997	Yes	Yes
DRSC	DOCTORS SURGI CLINIC	STJ	ST. JAMES	1/9/2008	Yes	
MBH	MOBAY HOPE	STJ	ST JAMES	12/1/2011	Yes	
NHH	NOEL HOLMES HOSPITAL	HAN	HANOVER	3/3/1997	Yes	
ROMD	ROYAL MEDICAL	WES	WESTMORELAND	12/1/2011	Yes	
SLMH	SAV LA MAR HOSPITAL	WES	WESTMORLAND	3/3/1997	Yes	
BRH	BLACK RIVER HOSPITAL	STE	ST. ELIZABETH	3/3/1997	Yes	
MAN	HARGREAVES MEMORIAL HOSPITAL	MAN	MANCHESTER	1/11/2018	Yes	
MH	MANDEVILLE HOSPITAL	MAN	MANCHESTER	3/3/1997	Yes	Yes
LTH	LIONEL TOWN HOSPITAL	CLA	CLARENDON	3/3/1997	Yes	
MPH	MAY PEN HOSPITAL	CLA	CLARENDON	3/3/1997	Yes	
PJH	PERCY JUNOR HOSPITAL	MAN	MANCHESTER	3/3/1997	Yes	
LH	LINSTEAD HOSPITAL	STC	ST. CATHERINE	3/3/1997	Yes	
STH	SPANISH TOWN HOSPITAL	STC	ST. CATHERINE	3/3/1997	Yes	

Appendix 4: CARPHA Weekly Country Syndromic Surveillance Report



CARIBBEAN PUBLIC HEALTH AGENCY (CARPHA)

Syndromic Surveillance and COVID-19 Reporting Form

Country: _____

Year: _____

Epi Week: _____

Syndromes	Cases
Fever and Hemorrhagic Symptoms	
Fever and Neurological Symptoms	
Fever and Respiratory Symptoms < 5 yrs	ARI* = ILI=
Fever and Respiratory Symptoms ≥ 5 yrs	ARI* = ILI=
Gastroenteritis < 5 yrs	
Gastroenteritis ≥ 5 yrs	
Undifferentiated Fever < 5 yrs	
Undifferentiated Fever ≥ 5 yrs	

*Including ILI cases.

ARI case definition: Acute (sudden) febrile illness > 38°C in a previously healthy person, presenting with cough or sore throat with and without respiratory distress.

ILI case definition: An acute respiratory infection with measured fever of ≥ 38°C, and cough, with onset within the last 10 days.

Surveillance Coverage for Syndromic Reporting			
Reporting Source	Expected # of Sources	No Received	Percent Reporting
Hospitals			
Health Centres			
Private Physicians/Sentinel Laboratories			
**Collective Reporting Sources			
TOTAL			

**Note: ONLY use the row labeled 'Collective Reporting Sources' when the breakdown for the Reporting Source is unknown.

Surveillance Coverage for COVID-19 ¹					
Reporting Source	Airport		Seaport		Overall Total ²
	No.	Cum.	No.	Cum.	No.
Travellers with exposure or travel from China					
Symptomatic travellers seen at Health facilities					
Travellers quarantined by public health authority					
Travellers self-quarantined at home					
No. of travellers completing 14-day isolation/quarantine					
No. of travellers tested during the reporting period					
Outcome of tests	Positive				
	Probable				
	Negative				

¹Please complete with details for COVID-19 (formerly 2019-Novel coronavirus or 2019-nCoV) for the current EPI week.

²Include cases for which point of entry is unknown.

See Case definitions tab for detailed information on COVID-19.

Were any outbreaks/cluster/unusual events observed this week?

☐ YES

☐ NO

**IF YES, REPORT EVENT IMMEDIATELY TO CARPHA
COMPLETE OUTBREAK REPORTING FORM**

Reminder: Fever and Rash and Acute Flaccid Paralysis will continue to be reported through the PAHO/WHO Expanded Programme on Immunization (EPI) weekly notification and reporting system.

Name of Reporting Officer: _____

Signature: _____

Designation: _____

Date: _____

Submit to carpha-epidemiology@carpha.org

OR CARIBBEAN PUBLIC HEALTH AGENCY (CARPHA)

P.O. BOX 164, Port-of-Spain, TRINIDAD and TOBAGO

p: 1-868-622-4261/2 f: 1-868-622-1008

Appendix 5: Regional Surveillance Audit Guidelines

SERHA REGIONAL SURVEILLANCE AUDIT GUIDELINES¹

Preamble

This tool was designed to assist health facilities in meeting the requirements for compliance with the various standards for Epidemiology and Surveillance as outlined by the National Surveillance Manual, the Infection Control Manual and the various Operational and Service Level requirements.

The tool aims to assess the following parameters:

- 1. Management and Governance:** To assess the management structure and the levels of supervision of the programme
- 2. Knowledge:** To identify the level of knowledge of surveillance, infection prevention and control principles amongst clinical staff
- 3. Compliance:** To assess compliance with the standard operating procedures as outlined in the previously stated manuals
- 4. Equipment and Material:** To assess access to and maintenance of essential equipment and material relevant to the Surveillance and Infection Control Programme

Rationale

This audit process ensures the validity and reliability of surveillance data collected from Health Departments, Sentinel Sites, and Hospital Active Surveillance (HAS), while also evaluating the availability of necessary supplies and the staff's knowledge to effectively conduct surveillance

¹ South East Regional Health Authority, Ministry of Health and Wellness. *Regional Surveillance Audit Guidelines V1.10*. Kingston (November 2024).

activities. By doing so, it strengthens surveillance mechanisms for the timely identification and management of potential health threats. The findings are then shared with key stakeholders to guide the implementation of strategies that address and eliminate data inaccuracies.

Responsibility

It is the responsibility of the parish health departments, with the assistance of the region, to carry out audits of HAS and Sentinel Sites at least once per year. Audits can also be randomly done at any site throughout the year by either the health department or regional team.

Audit Team

The audit team should be composed of at least three persons. This should ideally include but is not limited to the:

- Medical Officer (Health)
- Public Health Nurse (Surveillance / Surveillance / another representative)
- Surveillance Officer
- Public Health Inspector
- Regional Representative

Period of Review

Hospital Active Surveillance

- One Epidemiological Week

Sentinel Surveillance

- One Epidemiological Week or
- One Day in larger facilities

Basic Preparation

1. Discuss and assemble a suitable audit team
2. Note last date of previous audit and the audit findings and recommendations that were documented.
3. Select type of audit and period to be reviewed
4. Select a date for the audit
 - a. *Give institutions enough time to share the notice of the intention to audit, and pull the

dockets. Larger institutions may need up to two months' notice.
5. Write formal letters of notice to the CEO/Parish Manager and SMO/MO(H) In Charge of the facility
 - a. CC the Regional Health Records Administrator (applicable to regional audits)
 - b. Be prepared to have to do reminders as the date nears
6. Print the necessary documents
7. Agree on meeting place and start time with audit team

Methodology

a. Hospital Surveillance Audit (HAS)

- Audits are to be conducted at HAS Sites at least once per year.
- One epidemiological week should be randomly selected for review, based on the number of weeks, year to date.
- Members of the audit team will review the case definitions for all class 1 notifiable diseases in preparation for the audit (See [Appendix 5](#) of National Surveillance Manual)
- Before each audit the team should meet and have an internal briefing session (Appendix A)
- On the day of the audit, the team should announce themselves and conduct a pre-audit briefing with the SMO/Medical Officer in-charge of the facility
- HAS Audits should include
 - Wards (Medical & Paediatric)
 - Accident and Emergency/Casualty
 - Medical Records

- These records will be reviewed to identify any notifiable disease conditions (Class 1), or any other condition or event deemed a priority of the MOHW seen in the facility for the period under review.
 - Laboratory Department and Death Registries
 - Death records are also to be surveyed. This can be at the mortuary or at the death registry.
- The number of cases for each condition identified will be compared with the reports to the Health Department or Regional office for the same period.
- Each site will also be assessed based on staff awareness, presence of class 1 notification books and critical surveillance supplies as stated in the Surveillance Unit HAS Monitoring Tool for Parish Health Departments, Health Centres and Hospitals (Appendix B)
- Upon conclusion of the audit, the audit team will meet internally to discuss findings and recommendations as per Appendix E
- At the end of this meeting, the audit team lead should conduct a debriefing session with site managers to give preliminary feedback on the rating of the site and recommendations for areas of non-conformance.
- A formal report is to be prepared and submitted to senior management of the region/parish/site being audited, to include recommendations and follow-up actions/work plans.

b. Sentinel Site Audit

- Audits are to be conducted at Sentinel Sites (Appendix G) at least once per year.
- One epidemiological week or one day should be randomly selected for review, based on the size and workload of the facility
- Members of the audit team will review the syndromic case definitions as stated in National Surveillance Manual

- Before each audit the team should meet and have an internal briefing session (Appendix A)
- On the day of the audit, the team should announce themselves and conduct a pre-audit briefing with the SMO/Medical Officer in-charge of the facility
- The audit team will review the patient log or register for the period being audited and dockets (pre-pulled prior to the arrival of the audit team) for the date(s) randomly selected. The cases will be classified based on the syndromic case definitions.
- The number of each syndrome will be tallied and then compared to:
 1. The official Sentinel Report submitted to the Health Department for the corresponding week/date
 2. Communicable Disease Surveillance Log for relevant period under review
- Each site will also be assessed based on staff awareness, presence of class 1 notification books, posters ([Appendix 6](#)) and critical surveillance supplies as stated in the Surveillance Unit Monitoring Tool for Parish Health Departments, Health Centres and Hospitals (Appendix D)
- Upon conclusion of the audit, the audit team will meet internally to discuss findings and recommendations as per Appendix E
- At the end of this meeting, the audit team lead should conduct a debriefing session with site managers to give preliminary feedback on the rating of the site and recommendations for areas of non-conformance.
- A formal report is to be prepared and submitted to senior management of the region/parish/site being audited, to include recommendations and follow-up actions/work plans.

c. Health Department Audit

The audit of the Parish Health Department aims to assess its capacity to effectively support the monitoring of the parish's surveillance system. Key individuals to be interviewed include the Medical Officer of Health, Programme Coordinators, Surveillance Nurse, Surveillance Officer, and Epidemiology Clerks. The audit will also involve reviewing relevant monitoring documents and

databases, assessing the availability of critical surveillance supplies and equipment, and conducting a brief knowledge survey of staff.

Completion

Once the audit is completed, the data should be collated, sorted and compared with the numbers reported for the EW or date under audit. The official report should include results, non-conformances identified and recommendations for addressing these non-conformances with timelines Included. Attention should be paid to the audience and jargon excluded from this section. Consider having an executive summary if the audit writing is a long one.

The report should be shared with:

- Regional Technical Director/Regional Medical Epidemiologist
- Chief Executive Officer/Parish Manager responsible for site audited
- Senior Medical Officer or Medical Officer (Health) responsible for site audited

Appendix A: Audit Briefing

The Briefing for the Audit

The audit team is to meet internally to discuss the recommended talking points listed below. Note that some of these are best done with live examples, making the briefing less a pre-audit talk, and more a 'rolling' one.

1. The type of audit: HAS vs Sentinel vs. Health Department
2. Total number of medical records to be audited
3. Total number of medical records currently available
4. The EW or day under review. Clarify the date(s).
5. Tasks to be accomplished
6. Delineation of tasks
7. Docket review
 - a. Diseases or events being audited for
 - b. How to look beyond the presenting complaint (PC) or assessment/diagnosis
 - c. Keeping count of the medical records
 - d. Suggest they make note of any recurrent issues identified
 - e. Presence and use of surveillance stamps
8. Interviews
 - a. Areas for interviews and staff category
 - b. Prompting the interviewee for answers (e.g. nearby class one posters)
 - c. Making sure the audit staff know the answers to the audit tools
 - d. Suggest they make note of any recurrent issues identified
9. Identify any early issues that may change the audit plan

Appendix B: Hospital Active Surveillance (HAS) Tool

MONITORING AND DATA COLLECTION TOOL

- HOSPITALS

Name of Facility	
Health Region	
Parish	
Date of Evaluation	
Signature	
Date	

This tool has 6 pages inclusive of this page.

HAS AUDIT DATA COLLECTION INSTRUMENT

Period of Review					
Date of Audit					
Name of Auditors					
Notifiable Disease (HAS)	#Reported to HD	#Reported from HAS Audit	Difference	% of cases reported	Comments
Accidental Poisoning					
AFP/Polio					
Congenital Rubella					
Congenital Syphilis					
Death					
Diphtheria					
Encephalitis					
Hansen's Disease					
Hepatitis B					
HIV/AIDS					
Malaria					
Maternal Death					
Measles					
Meningitis					
Ophthalmia Neonatorum					
Pertussis-like Syndrome					
Rheumatic Fever					
Tetanus					
Tuberculosis					

Typhoid Fever					
Admitted LRTI/SARI					
Respiratory Related Death					
Leptospirosis					
Cholera					
Total					

Scoring is narrative.

MONITORING TOOLS FOR HOSPITALS

The recommended persons to be interviewed are:

- Physician
- Registered Nurse
- Public Health Nurse/ Infection Control Nurse
- Health Records Clerk
- Laboratory Personnel
- Medical Records Staff

Key personnel interviewed or consulted during the audit

	Name	Position
1.		
2.		
3.		
4.		
5.		
6.		

7.		
8.		
9.		
10.		

	Indicator	Rating			Remarks
		Medicine	Surgery	Specialty	
	Staff awareness of surveillance				
	DOCTORS:				
1	State the case definitions for: AFP (1), ILI (1), Dengue without warning signs (1), TB (1), Cholera (1), Gastroenteritis (1). <i>Scoring: Inadequate: 1-3/Adequate: 4-6</i>				
2	List 5 diseases that should be reported immediately. <i>Scoring: Inadequate: 1-2/Adequate: 3-5</i>				
3	List 2 resources you can access for more information <i>Scoring: Inadequate: 0/Adequate: 1-2</i>				
4a	What sample should be sent for fever and maculopapular rash cases: (answer: serum sample OR a red top tube) <i>Scoring: Inadequate:<1/Adequate: 1</i>				
4b	What is the ideal timeline for testing of fever and maculopapular rash cases (answer: within 7 days of the onset of rash) <i>Scoring: Inadequate:<1/Adequate: 1</i>				

	Indicator	Rating			Remarks
4c.	What proportion of fever and maculopapular rash cases are to be sampled (answer: 100% or everyone) <i>Scoring: Inadequate: <1/Adequate: 1</i>				
5a	What proportion of cases seen should be routinely tested/sampled for the following conditions: ILI (1 in 10 or 10%) <i>Scoring: Inadequate: <1/Adequate: 1</i>				
5b	What proportion of cases seen should be routinely tested/sampled for the following condition: GE (answer is 100% of admitted patients, 1 in 10 for ambulatory patients) <i>Scoring: Inadequate: <1/Adequate: 1</i>				
6	What sample is to be taken for suspected leptospirosis cases (answer: purple top tube) <i>Scoring: Inadequate: <1/Adequate: 1</i>				
	NURSES:				
8	State the case definitions for: AFP (1), SARI (1), MDRO (1), TB (1), Cholera (1), Dengue with warning signs (1). <i>Scoring: Inadequate: 1-3/Adequate: 4-6</i>				
9	What type of cases require the following samples: <i>Scoring: Inadequate: 1-2/Adequate: 3-5</i>				
9a	Nasopharyngeal swab				
9b	Stool swab				
9c	Sputum sample				
9d	Serum sample				
9e	Blood sample				

	Indicator	Rating			Remarks
		Medicine	Surgery	Specialty	
10	List 5 diseases that should be reported immediately on suspicion <i>Scoring: Inadequate: 1-2/Adequate: 3-5</i>				
11	List 2 resources you can access for more information <i>Scoring: Inadequate: 0/Adequate: 1-2</i>				
12	What investigation forms do you use for a patient with:				
12a	Acute Flaccid paralysis in someone under 15 (Polio). <i>Scoring: Inadequate: <1/Adequate: 1</i>				
12b	Fever and maculopapular rash (MOH Fever and Rash/measles form). <i>Scoring: Inadequate: 0/Adequate: 1</i>				
	MEDICAL TECHNOLOGISTS (for some sites)				
	Ask the frequency of visits if a rotating medical technologist visits the health centres				
13	What forms are necessary for:				
13a	Reporting class one diseases <i>Scoring: Inadequate: wrong or no answer</i>				
14	List the media and storage temperature for: <i>Scoring: Inadequate: media and/or storage wrong. Adequate: media AND storage correct</i>				
14a	Blood for malaria				
14b	Swab for influenza				
14c	Serum for fever and rash				
14d	Stool for cholera				

	Indicator	Rating			Remarks
14e	Blood for AFP				
15	What diseases should the laboratory itself report?				
	ACTIVE OBSERVATION (Question staff and actively look for these items/information)				
16	Availability of Notification booklet <i>Scoring: Present - adequate, absent - inadequate</i>				
17	Class 1 Disease Chart <i>Scoring: Present - adequate, absent - inadequate</i>				
18	Infection Control Officer assigned <i>Scoring: Present - adequate, absent - inadequate</i>				
19	Frequency of staff meetings <i>Scoring: Adequate - at least 11, Inadequate <11 per y</i>				
20	Guidelines re: treatment of specific communicable diseases / job aids to help treatment of specific diseases <i>Scoring: Present - adequate, absent - inadequate</i>				
		Medicine	Surgery	Specialty	
21	Area designated for disease control/isolation <i>Scoring: Present - adequate, absent - inadequate</i>				
22	Guidelines for the collection, storage and transportation of samples <i>Scoring: Present - adequate, absent - inadequate</i>				
23	Availability of investigation forms e.g. Fever and Rash <i>Scoring: Present - adequate, absent - inadequate</i>				
24	Availability of Laboratory form <i>Scoring: Present - adequate, absent – inadequate</i>				
	SUPPLIES (on ward, on compound or in laboratory)				

	Indicator	Rating			Remarks
	<i>Scoring: Present - adequate, absent – inadequate</i>				
25a	Red top blood tubes (check expiration date)				
25b	Purple top blood tubes (check expiration date)				
26	Universal sterile container (for sputum and stool testing, etc.)				
28	Sample storage / transport Igloos				
29	Ice packs for sample storage / transport				
30	Transport media for bacterial/viral samples (note if room temp or cold storage)				
31	Sample Thermometer				
	HEALTH RECORDS OFFICERS				
32	What proportion of Health records personnel can produce the weekly sentinel report? (50% =adequate)				
33	What diagnoses may be recorded as unspecified fever (case definition)				
34	What diagnoses may be recorded as fever and respiratory? (Ans: Any case with fever and cough; 'Asthma' is not, and such an answer denotes an "Inadequate" assessment)				
35	What is the definition of fever and neurological (case definition)				
36	What diagnoses may be recorded as fever and rash (case definition)				
37	What diagnoses may be recorded as gastroenteritis (case definition)				
		Tally			

	Indicator	Rating		Remarks
3		Overall Score		
		Overall Rating		

Scoring

Adequate	1
Inadequate	0

1. Tally all the 'adequate' ratings: Ad
2. Tally all the possible indicators: PI
3. Tally all the 'not applicable': NA
4. Note the number of category staff you got a response from: x
 - a. You can also do this for just one category at a time, in which case x=1
5. Perform the following calculation

$$\frac{Ad}{(PI-NA)*x} * 100 = Y\%$$

This gives Y, the overall score.

6. Assign the overall score thus:

≥60%	Adequate
< 60%	Inadequate

to get the overall rating.

Appendix D: Sentinel Audit Tool

MONITORING AND DATA COLLECTION TOOL

— SENTINEL SITES

Name of Facility	
Health Region	
Parish	
Date of Evaluation	
Signature	
Date	

SENTINEL DATA COLLECTION INSTRUMENT

Period of Review			
Date of Audit			
Name of Auditors			

Fever and Rash	A person presenting with fever of 38°C (100.4°F) [†] or greater, or feels warm to the touch, or gives a history of feeling feverish and generalized maculopapular rash		
Fever <5	A person presenting with a body temperature of 38°C (100.4°F) or greater, or feels warm to the touch, or gives a history of feeling feverish.		
Fever ≥5			
GE <5			
GE ≥5	A person presenting with 3 or more loose stools within 24 hours.		
Accidents <5	Any injury for which the cause is unintentional, e.g. motor vehicle, falls, burns, etc.		
Accidents ≥5			
Violence <5	Any injury for which the cause is intentional, e.g. gunshot wounds, stab wounds, etc.		
Violence ≥5			
Fever and Respiratory/ ILI <5			
Fever and Respiratory / ILI 5-59	History of fever of 38°C (100.4°F) [†] or greater and cough; with onset within the last 10 days.		
Fever and Respiratory / ILI ≥60			
ILI tested	# of ILI cases sampled/tested		
Fever and Haemorrhagic	A body temperature of 38°C (100.4°F) [†] or greater (or recent history of fever) in previously healthy person presenting with at least one haemorrhagic(bleeding) manifestation with or without jaundice.		
Fever and Jaundice	A body temperature of 38°C (100.4°F) [†] or greater (or recent history of fever) in a previously healthy person presenting with jaundice.		

Fever and Neurological	A body temperature of 38°C (100.4°F)† or greater (or recent history of fever). The person must also have meningeal irritation, convulsions, altered consciousness, altered sensory manifestations or paralysis (except Acute Flaccid Paralysis (AFP)).		
Total cases seen			

Records Reviewed	Syndromic Stamps Seen	Syndromic Stamps Used

OTHER CLASS 1 CONDITIONS – write in other class 1 conditions seen

Class 1 Condition	Definition	# Reported from Sentinel Audit	Line List (name, DOB, date seen)
Accidental Poisoning	Susp poisoning, regardless of route of exposure, considered to have occurred by accident		
AFP/Polio	Acute onset of flaccid paralysis in the absence of trauma		
Cholera	Profuse, watery, diarrhea rapidly leading to dehydration or death, especially in persons 5 yo or older		
COVID-19	Acute onset fever and cough (ILI) or acute onset of three or more signs and symptoms (see table)		
Dengue	Acute onset of fever plus at least 2 of vomiting, nausea, rash, headache, retroorbital pain, arthralgia/myalgia, haemorrhagic tendencies, leukopenia		
Diphtheria	Acute pharyngitis or laryngitis with a tonsillar or laryngeal pseudomembrane		
Hansen's Disease/Leprosy	Sharply demarcated anesthetic skin patches		
Hepatitis B + C	Anorexia, abdominal pain, nausea, vomiting, jaundice		
HIV/AIDS	Severe weight loss (> 10% of body weight), fever, diarrhoea (>1 month duration), lymphadenopathy, oral thrush, other sign immunosuppression		
Malaria	Intermittent fever with chills and headache, especially in those who recently travelled to malaria endemic		
Maternal Death	Any death of woman while pregnant or within 42 days of term of pregnancy (early) or 1 year of pregnancy (late)		
Measles	Fever (>38°C or 101°F) and generalized maculopapular rash		
Meningitis	Sudden onset of fever with signs and symptoms of meningeal involvement		
Ophthalmia Neonatorum	Purulent conjunctivitis in any neonate		
Pertussis-like Syndrome	Paroxysms of coughing, characteristic high pitched inspiratory whoop		
Rheumatic Fever	Recurrent inflammatory disease affecting multiple areas (see table below) occurring as sequelae of streptococcal upper resp infection		
Tetanus	Children: difficulty in sucking, opisthotonus Adults: difficulty swallowing, followed by muscle spasm and rigidity		
Tuberculosis	Persistent cough (>3 weeks), fever, night sweats, and weight loss		

Typhoid Fever	Insidious onset of sustained fever, headache, malaise and anorexia		
Total			

SENTINEL MONITORING TOOL

The recommended persons to be interviewed are:

- Physician
- Registered Nurse
- Public Health Nurse/ Infection Control Nurse
- Health Records Clerk
- Laboratory Personnel*
- Medical Records Staff

**may not be present at some sites*

Key personnel interviewed or consulted during the audit

	Name	Position
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

	Indicator	Rating			Remarks
		Medicine	Surgery	Specialty	
	Staff awareness of surveillance				
	DOCTORS:				
1	State the case definitions for: AFP (1), ILI (1), Dengue without warning signs (1), TB (1), Cholera (1), Gastroenteritis (1). <i>Scoring: Inadequate: 1-3/Adequate: 4-6</i>				
2	List 5 diseases that should be reported immediately. <i>Scoring: Inadequate: 1-2/Adequate: 3-5</i>				
3	List 2 resources you can access for more information <i>Scoring: Inadequate: 0/Adequate: 1-2</i>				
4a	What sample should be sent for fever and maculopapular rash cases: (answer: serum sample OR a red top tube) <i>Scoring: Inadequate:<1/Adequate: 1</i>				
4b	What is the ideal timeline for testing of fever and maculopapular rash cases (answer: within 7 days of the onset of rash) <i>Scoring: Inadequate:<1/Adequate: 1</i>				
4c.	What proportion of fever and maculopapular rash cases are to be sampled (answer: 100% or everyone) <i>Scoring: Inadequate:<1/Adequate: 1</i>				
5a	What proportion of cases seen should be routinely tested/sampled for the following conditions: ILI (1 in 10 or 10%) <i>Scoring: Inadequate:<1/Adequate: 1</i>				
5b	What proportion of cases seen should be routinely tested/sampled for the following condition: GE (answer is 100% of admitted patients, 1 in 10 for ambulatory patients) <i>Scoring: Inadequate:<1/Adequate: 1</i>				

	Indicator	Rating			Remarks
6	What sample is to be taken for suspected leptospirosis cases (answer: purple top tube) <i>Scoring: Inadequate: <1/Adequate: 1</i>				
	NURSES:				
8	State the case definitions for: AFP (1), SARI (1), MDRO (1), TB (1), Cholera (1), Dengue with warning signs (1). <i>Scoring: Inadequate: 1-3/Adequate: 4-6</i>				
9	What type of cases require the following samples: <i>Scoring: Inadequate: 1-2/Adequate: 3-5</i>				
9a	Nasopharyngeal swab				
9b	Stool swab				
9c	Sputum sample				
9d	Serum sample				
		Medicine	Surgery	Specialty	
9e	Blood sample				
10	List 5 diseases that should be reported immediately on suspicion <i>Scoring: Inadequate: 1-2/Adequate: 3-5</i>				
11	List 2 resources you can access for more information <i>Scoring: Inadequate: 0/Adequate: 1-2</i>				
12	What investigation forms do you use for a patient with:				
12a	Acute Flaccid paralysis in someone under 15 (Polio). <i>Scoring: Inadequate: <1/Adequate: 1</i>				
12b	Fever and maculopapular rash (MOH Fever and Rash/measles form). <i>Scoring: Inadequate: 0/Adequate: 1</i>				
	MEDICAL TECHNOLOGISTS (for some sites)				

	Indicator	Rating			Remarks
	Ask the frequency of visits if a rotating medical technologist visits the health centres				
13	What forms are necessary for:				
13a	Reporting class one diseases <i>Scoring: Inadequate: wrong or no answer</i>				
14	List the media and storage temperature for: <i>Scoring: Inadequate: media and/or storage wrong.</i> <i>Adequate: media AND storage correct</i>				
14a	Blood for malaria				
14b	Swab for influenza				
14c	Serum for fever and rash				
14d	Stool for cholera				
14e	Blood for AFP				
15	What diseases should the laboratory itself report?				
	ACTIVE OBSERVATION (Question staff and actively look for these items/information)				
16	Availability of Notification booklet <i>Scoring: Present - adequate, absent - inadequate</i>				
17	Class 1 Disease Chart <i>Scoring: Present - adequate, absent - inadequate</i>				
18	Infection Control Officer assigned <i>Scoring: Present - adequate, absent - inadequate</i>				
19	Frequency of staff meetings <i>Scoring: Adequate - at least 11, Inadequate <11 per y</i>				
20	Guidelines re: treatment of specific communicable diseases / job aids to help treatment of specific diseases				
		Medicine	Surgery	Specialty	
	<i>Scoring: Present - adequate, absent - inadequate</i>				
21	Area designated for disease control/isolation <i>Scoring: Present - adequate, absent - inadequate</i>				

	Indicator	Rating			Remarks
22	Guidelines for the collection, storage and transportation of samples <i>Scoring: Present - adequate, absent - inadequate</i>				
23	Availability of investigation forms e.g. Fever and Rash <i>Scoring: Present - adequate, absent - inadequate</i>				
24	Availability of Laboratory form <i>Scoring: Present - adequate, absent – inadequate</i>				
	SUPPLIES (on ward, on compound or in laboratory) <i>Scoring: Present - adequate, absent – inadequate</i>				
25a	Red top blood tubes (check expiration date)				
25b	Purple top blood tubes (check expiration date)				
26	Universal sterile container (for sputum and stool testing, etc.)				
28	Sample storage / transport Igloos				
29	Ice packs for sample storage / transport				
30	Transport media for bacterial/viral samples (note if room temp or cold storage)				
31	Sample Thermometer				
	HEALTH RECORDS OFFICERS				
32	What proportion of Health records personnel can produce the weekly sentinel report? (50% =adequate)				
33	What diagnoses may be recorded as unspecified fever (case definition)				
34	What diagnoses may be recorded as fever and respiratory? (Ans: Any case with fever and cough; 'Asthma' is not, and such an answer denotes an "Inadequate" assessment)				
35	What is the definition of fever and neurological (case definition)				
36	What diagnoses may be recorded as fever and rash (case definition)				

	Indicator	Rating			Remarks
37	What diagnoses may be recorded as gastroenteritis (case definition)				
		Tally			
		Overall Score			
		Overall Rating			

Scoring

Adequate	1
Inadequate	0

1. Tally all the 'adequate' ratings: Ad
2. Tally all the possible indicators: PI
3. Tally all the 'not applicable': NA
4. Perform the following calculation

$$\frac{Ad}{(PI-NA)} * 100 = Y\%$$

This gives Y, the overall score.

5. Assign the overall score thus:

≥60%	Adequate
< 60%	Inadequate

to get the overall rating.

Appendix E: Internal Debriefing

The session is held internally with the members of the audit team and is meant to collate the numbers, scores and identify challenges of the day.

1. Get total medical record count from each member of the audit team
2. Get the total numbers of each disease/event identified and collate
3. Discuss if discrepancies are great
4. Discuss interview findings
5. Solicit recommendations for corrections of institution failings from the team
6. If necessary, plan a part 2 if the audit cannot be completed in one day
7. Identify challenges to the audit's process and propose solutions for future audits

Appendix 6: Class 1 Notifiable Diseases and Health Events Poster



NATIONAL SURVEILLANCE UNIT NOTIFIABLE DISEASE / HEALTH EVENT LIST Class 1 Reporting – Everyone's responsibility!

The following health events and diseases are classified as Class 1, and as such, should be reported on suspicion by phone or e-mail to the parish health department within 24 hours of initial contact (scan QR code for contact information). Reports should be made using a Class 1 notification form.

CLASS 1 HEALTH EVENTS

HEALTH EVENT	CASE DEFINITION	SPECIMEN / DESTINATION
Accidental Poisoning	Any case of suspected poisoning, regardless of route of exposure (e.g., bleach or kerosene ingestion), considered to have occurred by accident.	For suspected Lead poisoning - Blood in purple top tube to Government Chemist at Hope Road, Kingston 6 (876-927-1829). For other agents, seek advice from the National Surveillance Unit (876-633-7937).
Maternal Death	Any death of a woman while pregnant or within 42 days of termination of pregnancy. For surveillance purposes, all maternal deaths within 1 year post termination of pregnancy should be reported.	—

CLASS 1 COMMUNICABLE DISEASES

DISEASE	CASE DEFINITION	SPECIMEN / DESTINATION*
Acute Flaccid Paralysis/ Poliomyelitis	Acute onset of flaccid paralysis in the absence of trauma.	Two stool specimens (at least 24 hours apart), within 14 days of onset of paralysis. Place each in a sterile universal container. Transport <u>on ice</u> immediately to NPHL.
Chikungunya	Fever and severe joint pains.	Serum specimen in red top tube <u>on ice</u> to NPHL.
Cholera	Floose, watery, diarrhoea rapidly leading to dehydration, or death especially in persons 5 years of age and older.	Stool specimen in Cary-Blair Media to NPHL (Bacteriology Section). DO NOT REFRIGERATE AND DO NOT SEND ON ICE.
Congenital Rubella Syndrome	An infant with one or more of the following: congenital cataracts, glaucoma, deafness, microphthalmia, microcephaly, congenital heart defects, meningoencephalitis, purpura at birth, or history of maternal rubella.	1. Acute and convalescent serum specimen in red top tube <u>on ice</u> to NPHL (Immunology Section). 2. Nasopharyngeal swab (in viral transport medium) and urine sample, both <u>on ice</u> to NPHL.
Congenital Syphilis	Failure to thrive, snuffles, skin peeling, desquamation, fever, anaemia, jaundice, hepatosplenomegaly.	Serum specimen in red top tube to NPHL (Serology Section). CSF as indicated.
COVID-19	Acute onset of fever AND cough (ILI) OR acute onset of three or more of the following signs or symptoms: fever, cough, generalized weakness/fatigue, headache, myalgia, sore throat, coryza, dyspnoea, nausea/diarrhoea/anorexia.	Nasopharyngeal swab OR combined nasopharyngeal and oropharyngeal swabs (nylon, dacron or polyester flocked swabs) in viral transport medium <u>on ice</u> to NPHL.
Dengue Fever/	Acute onset of fever plus at least two of the following: Vomiting or nausea, rash, headache or retro-orbital pain, arthralgia and/or myalgia, haemorrhagic tendencies (petechiae, positive tourniquet test), leukopenia.	Serum specimen in red top tube <u>on ice</u> to NPHL.
Severe Dengue	Dengue case with any of the following: shock or respiratory distress due to severe plasma leakage; severe bleeding; severe organ involvement.	Serum specimen in red top tube <u>on ice</u> to NPHL. Specimens such as heart, kidney, liver and spleen in a sterile container suspended in normal saline, if the patient dies.
Diphtheria	Acute pharyngitis or laryngitis with a tonsillar or laryngeal pseudomembrane.	Throat swab to NPHL (Bacteriology Section).
Encephalitis	Acute onset of headache, high fever, meningeal signs, stupor, disorientation, coma, spasticity, tremors, occasional convulsion (especially in infants) and spastic, but rarely flaccid, paralysis.	1. CSF sample <u>on ice</u> to NPHL. 2. Acute and convalescent serum specimens in red top tubes <u>on ice</u> to NPHL. 3. Stool sample in a sterile universal container <u>on ice</u> to NPHL.
Fever and Rash	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) and a generalized maculopapular rash.	Serum specimen (collected less than 7 days after onset of rash) in red top tube <u>on ice</u> to NPHL (Immunology Section).
Haemophilus Influenzae Meningitis	Sudden onset of fever, vomiting, convulsions, listlessness, neck rigidity and bulging fontanelle.	1-2 sets of blood cultures and CSF to hospital bacteriology laboratory or NPHL for culture.
Hepatitis B & C	Anorexia, abdominal pain, nausea, vomiting, jaundice.	Serum specimen in red top tube <u>on ice</u> to NPHL (Immunology Section).
HIV/AIDS	Severe weight loss ($\geq 10\%$ of body weight), fever, diarrhoea (>1 month duration), lymphadenopathy, oral thrush or other signs of immunosuppression.	Serum specimen in red top tube <u>on ice</u> to NPHL (Immunology Section).
Leprosy (Hansen's Disease)	Sharply demarcated anesthetic skin patches.	Refer to dermatologist for specialist testing using skin smear or skin biopsy.
Malaria	Intermittent fever with chills and headache, especially in those who recently traveled to malaria endemic areas.	Thin and thick blood smears to NPHL (Parasitology Section) <u>plus</u> EDTA (purple top) tube with 1 ml blood.
Measles	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) and a generalized maculopapular rash.	Serum specimen (collected less than 7 days after onset of rash) in red top tube <u>on ice</u> to NPHL (Immunology Section).
Meningitis	Sudden onset of fever with signs and symptoms of meningeal involvement.	1-2 sets of blood cultures and CSF to hospital bacteriology laboratory or NPHL for culture. For suspected viral etiology, send CSF <u>on ice</u> to NPHL.
Meningococcal Meningitis	Sudden onset of fever, intense headache, nausea, vomiting, lethargy, and frequently, petechial rash.	Blood and CSF to hospital bacteriology laboratory or NPHL for culture.
Mpox	Unexplained acute rash AND one or more of the following signs or symptoms: fever ($\geq 38.3^{\circ}\text{C}$), headache, myalgia (muscle pain/body aches), back pain, profound weakness or fatigue.	Skin lesion swabs (nylon, dacron or polyester flocked swabs) from more than one lesion, including exudate, roofs, lesion crusts. Place specimen in viral transport medium <u>on ice</u> and send to NPHL.
Ophthalmia Neonatorum	Purulent conjunctivitis in any neonate.	Swab of eye discharge to NPHL (Bacteriology Section).
Pertussis-Like Syndrome	Paroxysms of coughing, characteristic high pitched inspiratory whoop.	Nasopharyngeal aspirate or cough plate to NPHL (Bacteriology Section). Contact laboratory for details prior to specimen collection.
Plague	Swollen, tender, inflamed lymph nodes, lymphadenitis, fever and respiratory symptoms.	Blood culture or bubo aspirate to NPHL (Bacteriology Section). Notify laboratory prior to specimen submission.
Rheumatic Fever	Recurrent inflammatory disease affecting large joints (arthritis), the heart (carditis), the brain (chorea), the skin (erythema marginatum) and subcutaneous tissue (subcutaneous nodules) occurring as sequelae of streptococcal upper respiratory tract infection.	Serum for ASTO titres to NPHL (Serology Section).
Rubella	Fever ($\geq 38^{\circ}\text{C}$ or 100.4°F) and a generalized maculopapular rash.	Serum specimen (collected less than 7 days after onset of rash) in red top tube <u>on ice</u> to NPHL (Immunology Section).
Tetanus	Children - difficulty in sucking, opisthotonus. Adult - difficulty swallowing, followed by muscle spasms and rigidity.	Clinical diagnosis.
Tuberculosis	Persistent cough (>3 weeks), fever, night sweats, and weight loss.	3 consecutive early morning sputum samples or gastric washings (for children) or relevant biopsy specimen to NPHL (Tuberculosis Section).
Typhoid	Insidious onset of sustained fever, headache, malaise, and anorexia.	Blood culture and stool specimen in a sterile container to NPHL (Bacteriology Section).
Yellow Fever	Sudden onset of fever, chills, headache, myalgia, vomiting, prostration, jaundice (especially in persons who recently visited Yellow Fever endemic areas).	Serum specimen in red top tube <u>on ice</u> to NPHL.
Zika Fever	A patient with rash or elevated body temperature ($> 37.2^{\circ}\text{C}$) and one or more of the following symptoms: (i) arthralgia or myalgia (ii) non-purulent conjunctivitis or conjunctival hyperemia (iii) headache or malaise.	Serum specimen in red top tube <u>on ice</u> to NPHL.

*Stool and sera should be transported on ice unless otherwise indicated. Samples should reach the designated lab within 24 hours of collection.

*Serum samples sent to NPHL should be submitted as follows: Acute blood: 1-7 days after onset of symptoms; Convalescent Blood: 14-21 days after onset of symptoms.

ANY EXOTIC OR UNUSUAL HEALTH EVENTS OR COMMUNICABLE DISEASE SUCH AS ANTHRAX, LEGIONNAIRES DISEASE, SEVERE ACUTE RESPIRATORY SYNDROME (SARS), NEW SUBTYPE INFLUENZA, PANDEMIC INFLUENZA, BABES, ETC., SHOULD ALSO BE REPORTED AS CLASS 1.

Revised August 2023

Appendix 7: Class 1 Reporting Form



National Surveillance Unit, National Epidemiology Branch,
ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew
Email: surveillance@moh.gov.jm, Telephone 876-633-7937
The General Class 1 – Individual Notification Form



INITIAL DIAGNOSIS					
Initial Diagnosis/Notification:					
Date of Notification: DD/MM/YYYY		Date Patient Seen: DD/MM/YYYY		Case type: ☐ New Case ☐ Follow-up	
Patient's Demographics					
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN			Patient's ID Number:		
First Name:			Middle Name(s):		
Last Name:			Pet Name(s):		
Patient's Maiden Name (if applicable):					
Sex Assigned at Birth: ☐ Male ☐ Female		Date of Birth: DD/MM/YYYY		Age:	
Name of Facility Where Patient Seen:			Medical Record Number:		
Country of Residence:		Community:		Parish (Jamaica):	
House Number, Street Name:					
Landmark or directions to address:					
Phone Number:		Email Address:		Occupation:	
Name of Workplace/School:					
Address of Workplace/School:					
Has the patient travelled overseas in the last 6 weeks? ☐ Yes ☐ No <i>If yes, fill out the table below:</i>					
Country Visited	City Visited	Date Arrived in Country	Date Departed Country		
		DD/MM/YYYY	DD/MM/YYYY		
		DD/MM/YYYY	DD/MM/YYYY		
		DD/MM/YYYY	DD/MM/YYYY		
Date returned to Jamaica DD/MM/YYYY					
Next of Kin					
First Name:		Last Name:		Pet Name(s):	
NOK ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN			NOK ID Number:		
Address: Lot/ Street/Community/Parish:					
Name of School/ Workplace:					
Address of School/Workplace:					
Phone number:			Email address:		
Relationship to the Patient:					
☐ Wife	☐ Son	☐ Uncle	☐ Cousin		
☐ Husband	☐ Daughter	☐ Aunt	☐ Friend		
☐ Mother	☐ Sister	☐ Grandmother	☐ Guardian		
☐ Father	☐ Brother	☐ Grandfather	☐ Other (specify):		
Clinical Profile of Patient					
Has the patient experienced/Is the patient experiencing any of the following symptoms?					
SYMPTOM	DATE OF ONSET	DURATION	SYMPTOM	DATE OF ONSET	DURATION
☐ No Symptoms	DD/MM/YYYY		☐ Jaundice	DD/MM/YYYY	
☐ Abdominal Pain	DD/MM/YYYY		☐ Joint Pain	DD/MM/YYYY	
☐ Conjunctivitis	DD/MM/YYYY		☐ Myalgia	DD/MM/YYYY	
☐ Cough	DD/MM/YYYY		☐ Rash	DD/MM/YYYY	
☐ Diarrhoea	DD/MM/YYYY		☐ Seizures	DD/MM/YYYY	
☐ Eye Pain	DD/MM/YYYY		☐ Vomiting	DD/MM/YYYY	
☐ Fever	DD/MM/YYYY		☐ Weakness	DD/MM/YYYY	
☐ Haemoptysis	DD/MM/YYYY		☐ Other	DD/MM/YYYY	
☐ Headache	DD/MM/YYYY		Specify Other:		

Patient's Name:

Medical Record Number:

Does the patient have any pre-existing conditions? ☐ Yes ☐ No ☐ Unknown

Please specify any pre-existing conditions

☐ Asthma	☐ Diabetes mellitus	☐ HIV/AIDS	☐ Malignancy
☐ Autoimmune	☐ Heart disease	☐ Hypertension	☐ Sickle cell disease
☐ Chronic obstructive pulmonary disease (COPD)	☐ Kidney disease	☐ Liver dysfunction	☐ Other Specify Other:

Vaccination Status:

Type of Vaccine	Number of doses	Date of last dose	Source of Vaccination Information (Vaccination card, Health service record, Verbal)
		DD/MM/YYYY	
		DD/MM/YYYY	
		DD/MM/YYYY	
		DD/MM/YYYY	
		DD/MM/YYYY	

Was patient admitted to hospital? ☐ Yes ☐ No If yes, date of admission DD/MM/YYYY

Name of hospital: Ward:

Was patient discharged from hospital? ☐ Yes ☐ No If yes, date discharged DD/MM/YYYY

Is patient deceased? ☐ Yes ☐ No If yes, date of death DD/MM/YYYY

Sample and Lab Information

Was a sample taken? ☐ Yes ☐ No

If yes, what specimen was taken?

☐ CSF	☐ Nasopharyngeal Swab	☐ Sputum	☐ Tissue
☐ Blood Smear	☐ Naso-/Oro-/ Pharyngeal Swab	☐ Stool	☐ Other (specify):
☐ EDTA Blood	☐ Serum	☐ Urine	

What date was the specimen taken? DD/MM/YYYY

Which lab is the sample being sent to?

☐ National Public Health Laboratory ☐ University of the West Indies ☐ Other

Specify Other Laboratory:

Lab Test(s) Requested:

☐ Antigen	☐ PCR	☐ Culture	☐ Microscopy
☐ ELISA	☐ Western Blot	☐ Other (specify):	

Lab result(s): ☐ Positive ☐ Negative ☐ Indeterminate

Notifier's Comments:

Notifier's Details

Date of Reporting (DD/MM/YYYY):

First Name:	Last Name:
Professional Group:	
Phone number:	Email Address:
Name of Institution:	
Office Number/Street Name:	
Community:	Parish:
Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA	
Name of Parish MO(H):	Signature of MO(H):
	Date signed by MO(H): DD/MM/YYYY

Appendix 8: General Surveillance Line Listing

LINE LISTING OF NOTIFIED (DISEASE/HEALTH EVENT) CASES FOR (MONTH(S), YEAR);

(TOWN, PARISH)

Name	Sex	Age	Address/Phone #	Date of Onset	Care Site	Sample Taken (Specify type)	Date Sample Taken	Symptoms Experienced

N.B. This is the basic format which may be adjusted on a situation-by-situation basis to be more specific for the diseases being captured (For e.g., immunization data will be added in the event of a fever & rash/measles outbreak)

Appendix 9: Jamaica Laboratory Surveillance Investigation Form

JAMAICA: Laboratory Surveillance Investigation Form

September 2017

1. Patient Information

Last Name

First Name

Patient ID

Gender ☐ M ☐ F Age years ☐ months

Date of Birth

Street # -

City/Parish

Postal Code Tel:

Travel History ☐ Y ☐ N Country Visited:

2. Referring Doctor

Consultant:

Attending Dr.:

Signature:

Reporting Address: HOSPITAL / WARD

Tel: Fax:

Date Specimen Taken:

3. Provisional Diagnosis

(eg. Malaria, Influenza, Measles)

5. Case/Specimen Status

☐ Single case ☐ Outbreak ☐ Survey ☐ Unknown

6. Date of Onset of Illness

7. Outcome

Hospitalized? ☐ Y ☐ N ☐ DK

Died? ☐ Y ☐ N ☐ DK

8. Signs and Symptoms

☐ Fever → Temp: → Onset:

☐ Rash → Location: → Onset:

☐ Pain → Location

☐ Haemorrhagic symptoms → describe

☐ Paralysis → Location: → Onset:

☐ Altered mental state

☐ Chills

☐ Circulatory collapse

☐ Conjunctivitis

☐ Convulsions

☐ Coryza

☐ Cough

☐ Diarrhoea

☐ Failure to thrive

☐ Hepatomegaly

☐ Jaundice

☐ Neck stiffness

☐ Lymphadenopathy

☐ Kernig's sign

☐ Vomiting

☐ Weakness of limbs

☐ Weight loss

☐ Other → specify

Chronic Conditions

☐ Asthma

☐ Autoimmune disease

☐ Cancer

☐ Diabetes Mellitus

☐ HIV / AIDS

9. Syndromic Classification

☐ Acute Flaccid Paralysis

☐ Gastroenteritis

☐ Fever & Hemorrhagic

☐ Fever (undifferentiated)

☐ Fever & Rash

☐ Fever & Respiratory

☐ Fever & Neurologic

10. Immunization History

EPI No:

BCG: ☐ Y ☐ N

DPT: ☐ Y ☐ N

HBV: ☐ Y ☐ N

MMR: ☐ Y ☐ N

*Specify

☐ Y ☐ N

☐ Y ☐ N

☐ Y ☐ N

☐ Y ☐ N

☐ Y ☐ N

MR: ☐ Y ☐ N

Polio: ☐ Y ☐ N

YF: ☐ Y ☐ N

Other†: ☐ Y ☐ N

☐ Y ☐ N

4. Food/Animal/Environment Sample Details (if relevant)

Specimen ID

Name of food/env sample

Where specimen(s) collected

☐ Outbreak ☐ Traceback ☐ Survey ☐ Other

*Serum; EDTA blood; Blood smear; Sputum; CSF; Swab; Urine; Stool; Tissue; Plasma (PPT); Food; Water; Animal; Environment; if other specify			
Physician / EHO Use	Specimen 1	Specimen 2	Specimen 3
*Type of Specimen			
Date Specimen Collected			
Lab Test(s) Requested			
Laboratory Use			
Date Received at Nat Lab			
Nat Lab Specimen ID			
Test(s) Performed			
Date(s) Tested			
Laboratory diagnosis			
Date Referred to CAREC			
Name of Testing Lab			

Approved by (Testing Lab): Date:

CARPHA USE: Specimen ID (1) (2) (3)

Specify Institution affected		
OUTBREAK REPORTING FORM		
A. Reporting Details		
1. Agency submitting report:		
2. Region:		
3. Parish:		
4. Name of person submitting report:		
5. Contact telephone number:		
6. Date this form was completed/...../.....		
7. Is this ☐ a first report or ☐ an updated/amended report?		
B. Type of Outbreak		
8. ☐ Food-borne ☐ Respiratory ☐ Water-borne ☐ Sexually transmitted infection ☐ Vector-borne ☐ Unknown at this stage ☐ EPI disease ☐ Other, please specify below		
9. Was a vehicle/vector/source identified? ☐ Yes ☐ No		
10. If yes, please specify:		
C. Descriptive Epidemiology (person, place)		
11. Number of cases: Suspected or Probable Confirmed Total		
12. List number of cases (suspect, probable and confirmed) by age group and gender:		
13. Was the whole country affected? ☐ Yes ☐ No		
14. If no, describe the areas affected:		
15. Exposure setting (check all that apply): ☐ General community ☐ Health institution (e.g. hospital, nursing home) ☐ Other institution (e.g. prison, boarding home) ☐ Hotel or resort complex ☐ Restaurant ☐ School or child care facility ☐ Other, please specify type ☐ Don't know		
D. Clinical Details		
16. Common Symptoms/Syndromes (check all that apply)		
☐ Nausea	☐ Vomiting	
☐ Diarrhea	☐ Abdominal cramps	
☐ Fever	☐ Rash	
☐ Respiratory symptoms	☐ Hemorrhagic symptoms	
☐ Genital ulcer	☐ Genital discharge	
☐ Neurological symptoms	☐ Headache	
☐ Other, specify: chills, weakness, dehydration, blood in stool		
17. Number of cases hospitalized: [] <i>(including cases that died)</i>		
18. Number of cases that died: [] <i>(including cases hospitalized)</i>		
19. Incubation period (circle appropriate units)		
Average [] hours / days		
Range: [] - [] hours / days		
20. Duration of illness (circle appropriate units)		
Average: [] hours / day		
Range: [] - [] hours / day		
E. Case Summary (time)		
21. Please record number of cases per unit time (attach epi curve). Record time interval as:		
- Month (i.e. Jan 04, Feb 04, Mar 04), or		
- Epidemiological week (i.e. 23, 24, 25), or		
- Day (record as exact date, i.e. 23/06/04)		
Time Interval	Number Suspect/ Probable Cases	Number of Confirmed Cases

F. Aetiology

22. Was a primary causative pathogen identified in the outbreak? ☐ Yes ☐ No
23. If yes, please specify the name and subtype (if known) of the pathogen *Shigella flexneri*

G. Clinical Specimens (*e.g. stool, blood, urine, nasal aspirate, etc)

24. Type of Specimen	Number Tested	Number Positive	Etiologic Agent	Subtype 1	Subtype 2	Antimicrobial Resistance Profile

H. Food or Environmental Specimens (*e.g. ground beef, raw chicken, water, surface swab, etc)

25. Type of Specimen	Number Tested	Number Positive	Etiologic Agent	Subtype 1	Subtype 2	Antimicrobial Resistance Profile

I. Results of an epidemiological study

26. What type of epidemiological study was conducted?

- ☐ Cohort study ☐ Other, please specify _____
☐ Case Control Study ☐ No epidemiological study was conducted

27. If a cohort study was conducted, what was the overall attack rate? _____ %
(note, attack rate = [number ill/total persons at risk] x 100)

28. If a cohort or case control study was conducted, please complete the following table

Risk Factor	Odds Ratio or Relative Risk	95% Confidence Intervals	p-value

Additional Outbreak Details/Notes

Please provide a brief summary of the outbreak, including information on the following if applicable and available:

Background/ Source of Notification
Initial Response Measures Taken
Case Definitions
Epidemiological Assessment
Results/ Findings of Investigation
Recommendations
Integrated Team Response/Further
Actions Taken Economic Impact
Hypothesis



52-56 KNUTSFORD BOULEVARD, KINGSTON 5, JAMAICA

HEALTH EMERGENCY OPERATION CENTRE

TELEPHONE NO. 876-633-8271

E-mail: eoc@moh.gov.jm/ mohneocja@gmail.com

DAILY SHELTER SURVEILLANCE FORM

Date: / / dd/ mm/ yyyy		Day/Week number:		Ending: / / dd/ mm/ yyyy	
Parish:		Telephone # of Reporting agency: +1 area code phone number			
SITE- <i>specify by checking the box beside the relevant option</i>					
Shelter ☐		School ☐		Senior Citizens home ☐	
Children’s Homes ☐					
Geographical area:					
Other (specify):					
Shelter/Site Name:					
Address:					
Population/ No. of Residents/Patients:					

A. SURVEILLANCE DATA: (Record no. cases in line provided)									
HEALTH CONDITIONS	AGE								
	UNDER 5 YRS		OVER 5 YRS						TOTAL
			5-17		18-59		60+		
	Males	Females	Males	Females	Males	Females	Males	Females	
1. Fever (100°F or 38°C)									
2. Fever with rash									
3. Gastroenteritis									
4. Accidents (unintentional)									
5. Violence (Intentional)									
6. Respiratory illness									
a. Upper									
b. Lower									
7. Asthma									
8. Skin Rashes									
9. Conjunctivitis									
10. Other (Specify)									
TOTAL									



52-56 KNUTSFORD BOULEVARD, KINGSTON 5, JAMAICA
HEALTH EMERGENCY OPERATION CENTRE

TELEPHONE NO. 876-633-8271

E-mail: eos@moh.gov.jm / mohneocja@gmail.com

B. ENVIRONMENTAL HEALTH CONDITIONS

1. WATER SUPPLY Available : Yes ☐ No ☐ Quantity: _____ gallons

Water Source (check in box the relevant option) Sufficient ☐ Insufficient ☐

Public system (pipe) ☐ Rain ☐ Surface ☐ Ground ☐

Other (specify): _____

Treatment: Chlorinated ☐ Filtered ☐ Boiled ☐

Other (specify) _____

Comments: (issues, supply status and areas affected)

Parish: _____

Date: _____

dd/mm/yyyy

*70 gals per day/person (of 3 gals. Of drinking water/day)

2. SANITARY FACILITIES

No. of toilets: (WC) _____

No. damaged _____

(Pit) _____

No. damaged _____

Other: _____

Sewage system: status _____

Comments (issues, supply status and no, and location of site affected):

3. SOLID WASTE MANAGEMENT

Solid waste collected/removed daily yes ☐ no ☐

Are plastic bags/bins available yes ☐ no ☐

Animal carcasses removed: yes ☐ no ☐

Buried: yes ☐ no ☐

Comments (issues, status, areas affected):

4. VECTORS:

Mosquitoes _____ specify type: Aedes ☐ Anopheles ☐ Culex ☐

Rodents ☐ Flies ☐ Other ☐

Control Activities

Specify: _____

Comments (issues, supply status, areas affected)

5. FOOD SAFETY

High risk food prepared (beef, chicken, shellfish, fish, milk and milk products, mayonnaise, gravy etc)

Quantity of food condemned: _____

Food Storage:

Food Preparation:

No. of handwashing facilities _____

Status _____

Comments (issue, supply status, and areas affected)



52-56 KNUTSFORD BOULEVARD, KINGSTON 5, JAMAICA
HEALTH EMERGENCY OPERATION CENTRE

TELEPHONE NO. 876-633-8271

E-mail: coc@moh.gov.jm/ mohneocja@gmail.com

ITEM LIST	LIST OF INSTITUTION /SITE NEEDS
Number of persons requiring meals	
Number of persons requiring basic medication	
Water supply needs	
Sanitary facilities needs	
Plastic bags need	
Other (specify)	

Person completing form_____ (print) for MO(H) Signature _____

Appendix 12: Post-Disaster Shelter Summary Report E.O.C

MINISTRY OF HEALTH
DAILY/WEEKLY PARISH SHELTER SUMMARY REPORT E.O.C.

Shelter Report: 2

From: _____

Page ____ of ____

Daily/Week number: _____ (ending ____ / ____ / ____)

Date: _____

Surveillance Data

Disease

Shelter

Shelter									
#	Name	Patients/ Residents		1	2	3	4	5	6
Shelter 1	_____	_____	1. Fever and Rash						
			<5 yrs.						
			5 yrs.						
Shelter 2	_____	_____	2. Fever						
			<5 yrs.						
			5 yrs.						
Shelter 3	_____	_____	3. Gastroenteritis						
			<5 yrs.						
			5 yrs.						
Shelter 4	_____	_____	4. Accidents						
			(Unintentional injuries)						
Shelter 5	_____	_____	< 5 yrs.						
			5 yrs.						
Shelter 6	_____	_____	5. Violence (Intentional)						
			< 5 yrs.						
			5 yrs.						
			6. Respiratory Illness						
			Upper						
			< 5 yrs.						
			5 yrs.						
			Lower						
			< 5 yrs.						
			5 yrs.						
			Asthma						
			< 5 yrs.						
			5 yrs.						
			7. Skin Rashes						
			8. Conjunctivitis						
			9. Other (specify)						

NB. Insert name of shelter and assigned shelter number to the left. In the table insert at the top (first row) the assigned shelter numbers. Use additional forms if there are more than 6 shelters.

Person completing report: -----(print) MO(H) signature: -----

MOH Comments: _____

Appendix 13: Hotel Surveillance Reporting Form

SECOND SCHEDULE
THE PUBLIC HEALTH ACT
Weekly Surveillance Report

(Regulation 29)

Name of Tourist Establishment _____

Week Ending _____

Occupancy (total number of guests) _____

	STAFF	GUESTS
Total number of cases seen	_____	_____
Total number of first visits	_____	_____
Total number of accidents/injuries	_____	_____
(a) Bruises	_____	_____
(b) Burns/Scalds	_____	_____
(c) Cuts	_____	_____
(d) Fractures	_____	_____
(e) Sprains/strains	_____	_____
(f) Other	_____	_____
Total number of gastrointestinal conditions	_____	_____
Total number of diarrhoea cases	_____	_____
Number of diarrhoea cases with vomiting	_____	_____
Number of diarrhoea cases with fever	_____	_____
Number of diarrhoea cases with blood in stool	_____	_____
Number of diarrhoea cases with abdominal cramps	_____	_____
Total acute respiratory infections	_____	_____

Name _____ Position Held _____

Signature _____ Date _____

Appendix 14: List of Malaria & Yellow Fever Endemic Countries

MALARIA ENDEMIC COUNTRIES

This list is a compilation of countries who have reported at least one indigenous Malaria case since 2022 ¹.

WHO AFRICAN REGION

- Angola
- Benin
- Botswana
- Burkina Faso
- Burundi
- Cameroon
- Central African Republic
- Chad
- Comoros
- Congo
- Côte d'Ivoire
- Democratic Republic of the Congo
- Equatorial Guinea
- Eritrea
- Eswatini
- Ethiopia
- Gabon
- Gambia
- Ghana
- Guinea
- Guinea-Bissau
- Kenya
- Liberia
- Madagascar
- Malawi
- Mali
- Mauritania
- Mozambique
- Namibia
- Niger
- Nigeria
- Rwanda
- Sao Tome and Principe

- Senegal
- Sierra Leone
- South Africa
- South Sudan
- Togo
- Uganda
- United Republic of Tanzania
- Zambia
- Zimbabwe

WHO REGION OF THE AMERICAS

- Bolivia (Plurinational State of)
- Brazil
- Colombia
- Costa Rica
- Dominican Republic
- Ecuador
- French Guiana
- Guatemala
- Guyana
- Haiti
- Honduras
- Mexico
- Nicaragua
- Panama
- Peru
- Venezuela (Bolivarian Republic of)

WHO EASTERN MEDITERRANEAN REGION

- Afghanistan
- Djibouti
- Iran (Islamic Republic of)
- Pakistan
- Somalia
- Sudan
- Yemen

WHO SOUTH-EAST ASIA REGION

- Bangladesh
- Democratic People's Republic of Korea
- India
- Myanmar
- Nepal
- Thailand

WHO WESTERN PACIFIC REGION

- Cambodia
- Indonesia
- Lao People's Democratic Republic
- Papua New Guinea
- Philippines
- Republic of Korea
- Solomon Islands
- Vanuatu
- Viet Nam

¹ World Health Organization. *World malaria report 2025* [Internet]. (December 2025) Available from: <https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2025>

COUNTRIES WITH RISK OF YELLOW FEVER TRANSMISSION

This list is a compilation of countries determined by the WHO to be at risk for **yellow fever** transmission. The list reflects countries currently reported, or having been reported in the past, and the presence of vectors and animal reservoirs representing a potential risk of infection and transmission ¹.

AFRICA

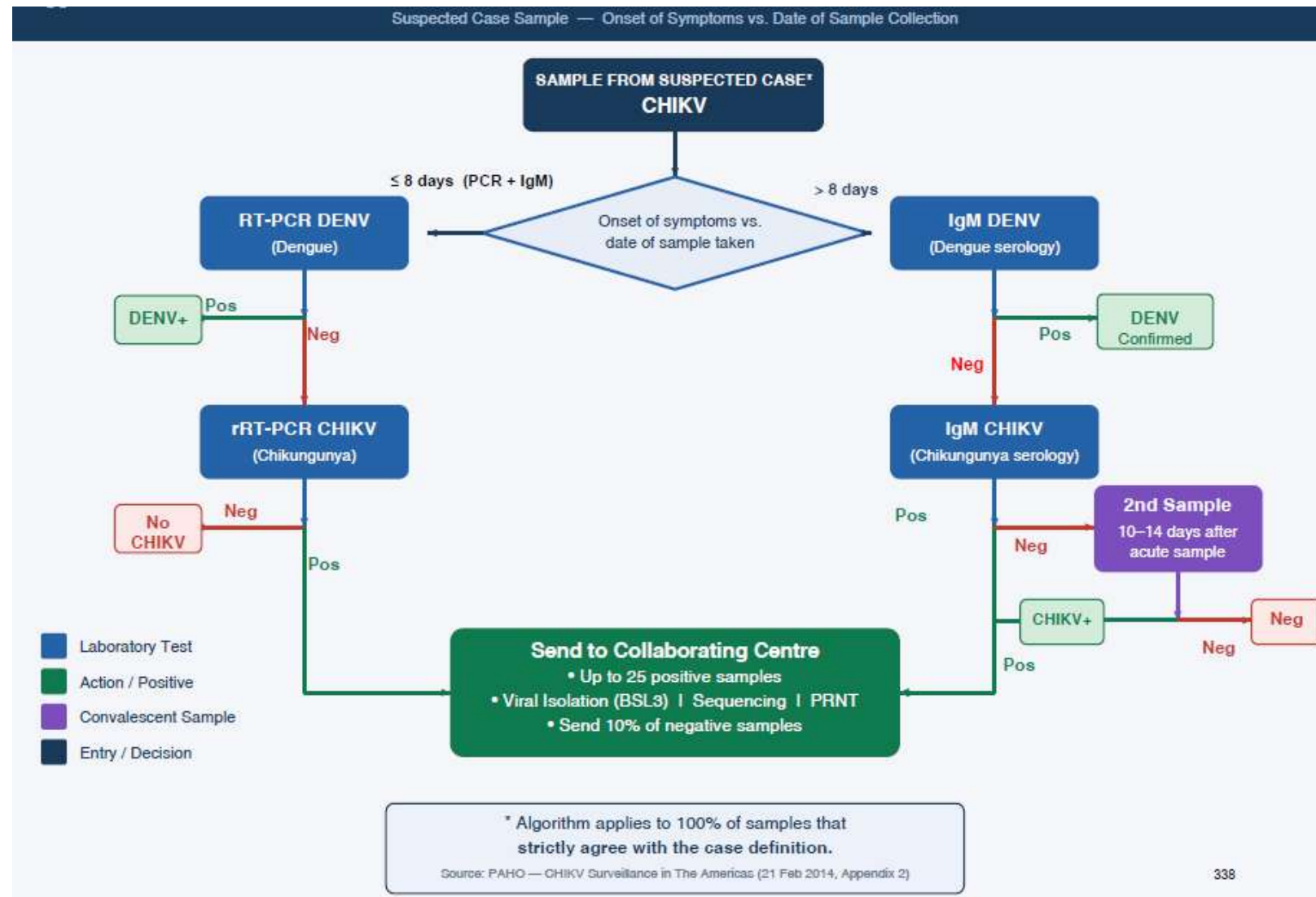
- Angola
- Benin
- Burkina Faso
- Burundi
- Cameroon
- Central African Republic
- Chad
- Congo
- Côte d'Ivoire
- Democratic Republic of the Congo (Former Zaire)
- Equatorial Guinea
- Ethiopia
- Gabon
- Gambia
- Ghana
- Guinea
- Guinea-Bissau
- Kenya
- Liberia
- Mali
- Mauritania
- Niger
- Nigeria
- Rwanda
- Sao Tome and Principe
- Senegal
- Sierra Leone
- South Sudan
- Sudan
- Togo
- Uganda

CARIBBEAN/ CENTRAL AND SOUTH AMERICA

- Argentina (Misiones and Corrientes Provinces)
- Bolivia (Plurinational State of)
- Brazil
- Colombia
- Ecuador (including Galapagos Islands)
- French Guiana
- Guyana
- Panama
- Paraguay
- Peru
- Suriname
- Trinidad and Tobago
- Venezuela (Bolivarian Republic of)

Source: Countries with risk of yellow fever transmission and countries requiring yellow fever vaccination (November 2022). (2022, November 18). World Health Organization. [https://www.who.int/publications/m/item/countries-with-risk-of-yellow-fever-transmission-and-countries-requiring-yellow-fever-vaccination-\(november-2022\)](https://www.who.int/publications/m/item/countries-with-risk-of-yellow-fever-transmission-and-countries-requiring-yellow-fever-vaccination-(november-2022))

Appendix 15: CHIKV Laboratory Testing Algorithm



Appendix 16: Chikungunya Case Investigation Form

Reporting Centre:				Date of Report / / (dd/mm/yy)	
1. Patient information					
Name			Age (yrs)	Sex: M F	
Address		Phone #	D.O.B: / / (dd/mm/yy)		
Community (STATIN)			Case #	Occupation	
Date of onset of illness! / / (dd/mm/yy)					
Clinical features	Y	N	Clinical features	Y	N
Fever	☐	☐	Skin manifestations Describe:	☐	☐
Arthritis	☐	☐	Myalgia	☐	☐
If yes, where:			Back pain	☐	☐
Hands	☐	☐	Headache	☐	☐
Feet	☐	☐	Nausea	☐	☐
Ankles	☐	☐	Mucosal bleeding	☐	☐
Other	☐	☐	Vomiting	☐	☐
Arthralgia	☐	☐	Asthenia (generalized weakness)	☐	☐
Severe	☐	☐	Other		
Periarticular oedema	☐	☐			
Risk Factors:					
Pregnancy	☐	☐	Other:	☐	☐
Diabetes	☐	☐		☐	☐
Cardiovascular diseases	☐	☐		☐	☐
Sickle Cell Disease	☐	☐		☐	☐
Clinical diagnosis:					
Is/was this patient hospitalised?	Y	N	Date (s)	Outcome of illness	
	☐	☐			

Resolution of symptoms	☐	☐					
3. Exposure and Travel History							
	Y	N	Date	Details			
Has the patient travelled to a chikungunya endemic/epidemic area within the past 3 weeks	☐	☐					
Has the patient been in contact with a chikungunya case within the past 3 weeks	☐	☐					
Places visited in the past 3 weeks – emphasis on places visited in the daylight hours							
4. LABORATORY DATA							
Specimen	Date collected	Date rec'd	Condition	Test	Result	Date sent	Comment
First blood specimen				Virus Isolation			
				IgM ELISA			
Second blood specimen				IgG antibody by HAI test			
				RT-PCR			
5. ENVIRONMENTAL SURVEY							
Community type	☐ Planned			☐ Unplanned			
Aedes population :	☐ Aegypti			☐ Albopictus			
Water supply:	☐ Piped			☐ Stored			
Solid waste collection:	☐ Frequent (at least once weekly)			☐ Infrequent			
AEDES INDICES - SURROUNDING COMMUNITY:							
	Home			Workplace			
Premises index:							
Container index:							
Breteau index:							
INVESTIGATOR:							
Name:		Signature:		Date:			

TO BE COMPLETED BY THE MEDICAL OFFICER OF HEALTH

6. Final case classification: ☐ Clinically Suspected – Not Confirmed
☐ Epidemiologically Linked - Not Confirmed
☐ Laboratory Confirmed – Imported
☐ Laboratory Confirmed – Autochthonous
☐ Discarded

Appendix 17: COVID-19 Case Investigation Form

Date of Reporting: _____		Region: _____		Parish: _____	
Doctor: _____		Hospital / Site: _____		Ward: _____	
Email: _____		Phone #: _____			

Hospital/Medical Record Number: _____		NEW CASE ☐		UPDATE ☐	
Last Name: _____		First Name: _____			
Date of Birth: _____		Age: _____		Sex: MALE ☐ FEMALE ☐	
Country of Residence: _____		Parish: _____		Community: _____	
Street #: _____		Street Name: _____			

Epidemiologic Week of Onset: _____		Date of Onset of Illness: _____		Admission Date: _____	
------------------------------------	--	---------------------------------	--	-----------------------	--

CLINICAL & EPIDEMIOLOGICAL PROFILE										
CLINICAL PROFILE										
History of Fever or Fever over 38°C (<10 days)					Yes	No	Recorded temperature _____ °C			
Cough	Yes	No	Difficulty Breathing/Wheezing		Yes	No	Abnormal Lung Auscultations		Yes	No
Rhinorrhoea	Yes	No	Dyspnoea/Tachypnoea		Yes	No	Abnormal lung x-ray findings		Yes	No
Sore Throat	Yes	No	Diarrhoea		Yes	No	Seizure		Yes	No
Shortness of Breath	Yes	No	Nausea/Vomiting		Yes	No	Other, please specify: _____			
Loss of smell (Anosmia)	Yes	No	Headache		Yes	No				
Loss of Taste (Ageusia)	Yes	No	Myalgia		Yes	No				
RISK FACTORS										
Pregnancy	Yes	No	Lung Disease including COPD		Yes	No	Immunocompromised due to disease or treatment		Yes	No
If yes, Trimester	1	2	3	Asthma	Yes	No	HIV / AIDS		Yes	No
Diabetes Mellitus	Yes	No	Neurological Disease		Yes	No	Malignancy		Yes	No
Sickle Cell Disease	Yes	No	Liver Disease		Yes	No	Hypertension		Yes	No
Heart Disease	Yes	No	Renal Disease		Yes	No	Obesity		Yes	No
EPIDEMIOLOGICAL PROFILE										
Occupation: Health Care Worker ☐ Health Laboratory Worker ☐ Working with Animals ☐ Student ☐ Other, please specify: _____										
CONTACT HISTORY										
In the 14 days before symptom onset, did the individual:										
Have close contact ³ with a person who is under investigation for COVID-19?								☐ Y ☐ N ☐ Unknown		
Have close contact ³ with a laboratory-confirmed COVID-19 case?								☐ Y ☐ N ☐ Unknown		
If yes:								☐ Y ☐ N ☐ Unknown		
Where: Home ☐ Work ☐ Health Care Setting ☐ Other, please specify: _____								☐ Y ☐ N ☐ Unknown		
Was the contact ill at the time of contact?								☐ Y ☐ N ☐ Unknown		
In which country was the contact diagnosed with COVID-19? _____								☐ Y ☐ N ☐ Unknown		
Contact's Name: _____										
Close Contact with a person with acute respiratory infection in the 14 days prior to onset of symptoms				Yes	No	Animal Contact				
If yes, where: Home ☐ Work ☐ Health Care Setting ☐ Other, please specify: _____						If yes, please specify: _____				
				Yes	No					

Fever (> 38 °C) may not be present in some patients, such as those who are very young, elderly, immunosuppressed, or taking certain medications. Clinical judgement should be used to guide testing of patients in such situations.

A contact is a person: - Providing direct care for Confirmed (Test Positive) Cases; working with health care workers infected with novel coronavirus; visiting patients or staying in the same close environment of a Positive patient; Working together in close proximity or sharing the same classroom environment with a Test Positive patient - Traveling together with a Test Positive patient in any kind of conveyance; Living in the same household as a Test Positive patient within a 14-day period after the onset of symptoms in the case under consideration.

Prepared by the National Surveillance Unit, Ministry of Health & Wellness

Revised - 2020/07/08

TRAVEL HISTORY

In the 14 days before symptom onset, did the individual:

☐ Y ☐ N ☐ Unknown

Travel to Jamaica from another Country:

If Yes:

Country Province/City Departure Date

Country Province/City Departure Date

Admitted to ICU: Yes ☐ No ☐ If Yes, Date admitted to ICU: Date Discharged from ICU:

Mechanical Ventilation: Yes ☐ No ☐ If Yes, Date started: Date stopped:

Length of Ventilation (days):

Extracorporeal Membrane Oxygenation: Yes ☐ No ☐

LABORATORY DATA

MOLECULAR TESTING

Sample Type	Collection Date	Test Type	Results	Result Date
Nasopharyngeal ☐	Date	PCR ☐ Other (specify) _____	COVID-19 +ve ☐ COVID-19 -ve ☐ Other +ve ☐	
Oropharyngeal ☐	Date	PCR ☐ Other (specify) _____	COVID-19 +ve ☐ COVID-19 -ve ☐ Other +ve ☐	
Sputum ☐	Date	PCR ☐ Other (specify) _____	COVID-19 +ve ☐ COVID-19 -ve ☐ Other +ve ☐	
Other (specify) _____				

SEROLOGICAL TESTING

Sample Type	Collection Date	Test Type	Results	Result Date
Serum ☐	Date	IgM ☐	Positive ☐ Titre _____	
		IgG ☐	Negative ☐	
Other (specify) _____		Other (specify) _____	Inconclusive ☐	

CONTACT TRACING

Contact Type	Number of Contacts Identified	Number of Contacts Sampled
Bedroom		
Household		
Conveyance		
Health Facility		
Other(specify) _____		

MO(H) Signature: Date:

Fever (≥38 °C) may not be present in some patients, such as those who are very young, elderly, immunosuppressed, or taking certain medications. Clinical judgement should be used to guide testing of patients in such situations.

A contact is a person: - Providing direct care for Confirmed (Test Positive) Cases; working with health care workers infected with novel coronavirus; visiting patients or staying in the same close environment of a Positive patient; Working together in close proximity or sharing the same classroom environment with a Test Positive patient; Traveling together with a Test Positive patient in any kind of conveyance; Living in the same household as a Test Positive patient within a 14-day period after the onset of symptoms in the case under consideration.

Prepared by the National Surveillance Unit, Ministry of Health & Wellness

Revised – 2020/07/08

Appendix 18: Fields for COVID-19 Contact Intake Line Listing



- i. Date of intake
- ii. Name of Suspected/Confirmed Case
- iii. Type of Contact
- iv. Risk level
- v. Date of most recent contact
- vi. First name
- vii. Last name
- viii. Date of Birth
- ix. Current Age
- x. Sex at Birth
- xi. Current Age
- xii. Telephone number 1
- xiii. Next of Kin name
- xiv. Next of Kin telephone number
- xv. GPS Coordinates
- xvi. Street Number
- xvii. Street Name
- xviii. District
- xix. Community
- xx. Parish
- xxi. Landmark
- xxii. Symptomatic/Asymptomatic
- xxiii. Date of onset of symptoms
- xxiv. Fever (Y/N)
- xxv. Cough (Y/N)
- xxvi. Shortness of breath (Y/N)
- xxvii. Other, specify
- xxviii. Comments

Appendix 19: Fields for COVID-19 Contact Tracing Daily Tracking Line Listing



- i. Date of assessment
- ii. Time of assessment
- iii. Parish
- iv. First Name
- v. Last Name
- vi. Day 1
- vii. Day 2
- viii. Day 3
- ix. Day 4
- x. Day 5
- xi. Day 6
- xii. Day 7
- xiii. Day 8
- xiv. Day 9
- xv. Day 10
- xvi. Day 11
- xvii. Day 12
- xviii. Day 13
- xix. Day 14
- xx. Remarks

Appendix 20: Daily COVID-19 Laboratory Sample Line Listing Report



NATIONAL LABORATORY SERVICES COVID-19 TEST LINE
LISTING REQUEST/RESULTS
FOR SURVEILLANCE PURPOSES

DATE:
PARISH:
CONTACT NUMBER:

FOR NPHL USE
ONLY

No.	Last Name	First Name	Patient Address	A G E	S E X	Care Site	Doctor	Clinical Hx	Classification (e.g. ILI, ALRTI, Day 14 sample, Pre-Op, Contact of confirmed case, etc.)	Date Collected	Date Rec'd by NPHL	Signature of NPHL Staff
<u>1</u>												
<u>2</u>												
<u>3</u>												
<u>4</u>												
<u>5</u>												
<u>6</u>												
<u>7</u>												
<u>8</u>												
<u>9</u>												
<u>10</u>												

Appendix 21: Dengue Case Investigation Form



NATIONAL SURVEILLANCE UNIT

15 Knutsford Boulevard,

Kingston 5 Jamaica, West

Indies Tel: (876) 633-7937

Date of Reporting: DD/MM/YYYY	Care Institution: _____	Parish: _____
Last Name: _____	First Name: _____	Contact #: (876) _____
Address: _____		Sex: _____
Date of Birth: DD/MM/YYYY	Date of Onset of Illness: DD/MM/YYYY	Age: _____ (months/years)
Healthcare received before admission: Health Centre ☐ Private Practitioner ☐ A&E ☐ N/A ☐		Date: DD/MM/YYYY
Admitted to Hospital? Yes ☐ No ☐	Date of Admission: DD/MM/YYYY	Ward: _____
Outcome: Survived ☐ Died ☐	Date of Discharge: DD/MM/YYYY	Date of Death: DD/MM/YYYY
Place of Death: HOME ☐ DOA ☐ A&E ☐ WARD ☐ ICU ☐	Time of Death: _____ AM/PM	
History of overseas travel in past 4-6 weeks? Yes ☐ No ☐		Specify area/Country: _____

CLINICAL & EPIDEMIOLOGICAL PROFILE					
CLINICAL PROFILE					
Dengue Without Warning Signs	Yes	No	Dengue With Warning Signs	Yes	No
Fever			Date of onset of warning signs:		
Recorded temperature: _____ °C Duration: _____			Intense Abdominal Pain or tenderness		
Nausea			Persistent vomiting		
Vomiting			Fluid Accumulation		
Exanthema			Ascites		
Headache			Pleural Effusion		
Retro-orbital Pain			Pericardial Effusion		
Myalgia			Mucosal Bleed		
Arthralgia			Lethargy/Restlessness		
Petechiae			Postural Hypotension		
Tourniquet Test Positive			Hepatomegaly (>2 cm above the normal range)		
Leukopenia			Progressive Increase in Hematocrit		
Other: _____			Other: _____		
CLINICAL PROFILE			CLINICAL PROFILE		
Severe Dengue	Yes	No	Risk Factors/Comorbid Conditions	Yes	No
Shock or respiratory Distress Due to severe plasma leakage			Pregnancy: if so, trimester?		
1. Shock evidenced by:				1	2 3
Weak or undetectable pulse			Age <1		
Tachycardia			Age >60		
Cold extremities			Diabetes Mellitus		
Capillary perfusion >2 seconds			Sickle Cell Disease		
Pulse pressure ≤ 20mm Hg/Hypotension			Hypertension		
2. Severe bleeding:			Kidney Failure		
Hematemesis			Heart Disease		
Melena			HIV/AIDS		
Metrorrhagia			Anticoagulant/ NSAID Use		
(CNS) Bleeding					
3. Severe Organ involvement:			Other: _____		
Liver impairment (AST or ALT ≥ 1000 IU) or failure					
Myocarditis					
Encephalitis (Impaired mental state)					
Kidney Failure					
Other: _____					

[illegible]

Appendix 22: Malaria Case Investigation Form

Reporting Centre: _____

Date of reporting: YYYY /MM / DD

Patient's name: _____

Age: _____ yrs Sex: M F

Present Address	Permanent Home address
Workplace Address	School Address

Clinical Data

Date of onset of THIS attack: YYYY /MM / DD Place of onset of THIS attack: _____

Name/Address/Phone No. of Reporting Physician: _____

Symptom/Signs	Y/N	Symptom/Signs	Y/N	Symptom/Signs	Y/N	Symptom/Signs
Fever		Chills		Myalgia		Headache
Sweating		Nausea		Vomiting		Other (Specify)
Complications						

LABORATORY RESULTS

Optima: Date taken YYYY /MM / DD

Positive Negative Not taken

Smear: Date taken YYYY /MM / DD

Positive Negative Not taken

SPECIES: Vivax Ovale Falciparum Not determined

Has patient been out of country? Yes No	Has patient's contact been out of country? Yes No
If yes, list all countries visited with dates	If yes, list all countries visited with dates

Malaria prophylaxis taken? Yes No Drugs: Chloroquine Primaquine Fansidar Other

Local Travel History / Overnight stay in past 3 months	(Contact info)
Blood transfusion within past 2 years? Yes No If yes, date/s	Has the patient had a past history of confirmed malaria? No/Yes

Treatment (Dose and Duration)

Chloroquine _____ Primaquine _____ Quinine _____
Pyrimethamine/sulphadoxine _____ Amodiaquine _____ Mefloquine _____

Final case classification: Malaria Y / N	CLASSIFICATION: Imported Induced Cryptic
Laboratory confirmed Y/ N	Introduced Indigenous Congenital
Additional Information: _____ _____ _____ _____	
Signature: _____	Investigator: _____ Date reported: YYYY /MM / DD

Appendix 23: Maternal Mortality Clinical Summary Form



National Surveillance Unit, National Epidemiology Branch,
15 Knutsford Boulevard, Kingston 5,
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



Maternal Mortality Form 1

Clinical Summary Form

INITIAL DIAGNOSIS		
Initial Diagnosis/Notification: Maternal Mortality		
Date of Notification: DD/MM/YYYY		Date Investigation Started: DD/MM/YYYY
Patient's Demographics		
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		Patient's ID Number:
First Name:		Middle Name(s):
Last Name:		Pet Name(s):
Medical Record Number:		Reporting Facility:
Date of Birth: DD/MM/YYYY		Age:
Country of Residence:		Parish (Jamaica):
House Number, Street Name:		
Landmark or directions to address:		
Community:		Phone Number:
Occupation:		Email Address:
Name of Workplace/School:		
Address of School/Workplace:		
Death Summary		
Date of Death: DD/MM/YYYY		Time of death: HH:MM
Age at Death: _____ years		Days post-partum: _____ days
Place of Death		
☐ Public Hospital	☐ Private Practitioner's Office	☐ Health Centre
☐ Private Hospital	☐ Private Obstetrician's Office	☐ Home
☐ En Route to Health Facility	☐ Other (specify):	
Name of Facility:		
Address Number, Street Name:		
Community:		Parish (Jamaica):
If the patient died in hospital, where was she admitted at the time of death?		
☐ Antenatal ward	☐ Accident & Emergency	☐ ICU / HDU
☐ Labour ward	☐ Operating theatre	☐ Postnatal ward
☐ Recovery Room	☐ Non-obstetrics ward	Specify:
What is the timing of the death?		
☐ Antenatal [while pregnant and before labour]	☐ In labour [during labour and/or delivery]	
☐ Postnatal [$\leq$ 42 days after the end of a pregnancy]	☐ Late postnatal [$>$ 42 days and $\leq$ 1 year after the end of a pregnancy]	
Was an autopsy requested for the patient? ☐ Yes ☐ No ☐ Unknown		

Name of Patient:

Medical Record Number:

Pregnancy History

Number of previous pregnancies (excluding current pregnancy):

Outcomes

Outcome	Number	Outcome	Number
☐ Full term live birth		☐ Induced abortions	
☐ Premature live births (<2500 g)		☐ Ectopic pregnancies	
☐ Stillbirth/foetal death		☐ Trophoblastic disease	
☐ Spontaneous abortion/miscarriage			

Any risks/complications in previous pregnancy(ies)? ☐ Yes ☐ No ☐ Unknown

Select all that diagnosis that apply:

☐ Gestational diabetes	☐ Difficult/prolonged labour	☐ Intrauterine growth retardation
☐ Gestational hypertension	☐ Forceps delivery	☐ Breech presentation
☐ Pre-eclampsia	☐ Postpartum haemorrhage	☐ Transverse lie
☐ Eclampsia	☐ Retained placenta	☐ Large for gestational age
☐ Cervical incompetence	☐ Multiple gestation	☐ Inaudible heartbeat
☐ Spontaneous abortion	☐ Preterm labour	☐ Caesarean Section
☐ Intrauterine death/stillbirth	☐ Antenatal bleeding	☐ Puerperal pyrexia/sepsis
☐ Neonatal death	☐ Excess weight gain	☐ Malignancy
☐ Persistent puerperal hypertension	☐ Foetal compromise, reduced foetal movement	☐ Other

Please specify any other problems during previous pregnancies?

Any previous surgeries? ☐ Yes ☐ No ☐ Unknown

If yes, specify

Medical HistoryDoes the patient have any pre-existing conditions? ☐ Yes ☐ No ☐ Unknown

Please specify any pre-existing conditions

☐ Asthma	☐ Diabetes mellitus	☐ HIV/AIDS	☐ Malignancy
☐ Autoimmune Disease	☐ Heart Disease	☐ Hypertension	☐ Sickle Cell Disease
☐ Other (Specify Other):			

Was the patient admitted during this pregnancy? ☐ Yes ☐ No ☐ Unknown

Name of Patient:

Medical Record Number:

If yes, when was the patient admitted?

Admitted	Discharged	Discharge Diagnosis	Discharged By	Complications
DD/MM/YYYY	DD/MM/YYYY			
DD/MM/YYYY	DD/MM/YYYY			
DD/MM/YYYY	DD/MM/YYYY			
DD/MM/YYYY	DD/MM/YYYY			
DD/MM/YYYY	DD/MM/YYYY			
DD/MM/YYYY	DD/MM/YYYY			

Was the patient transferred? ☐ Yes ☐ No ☐ Unknown

Hospital name:

Date of transfer: DD/MM/YYYY

Reason for transfer:

Current Pregnancy Summary

Pregnancy outcome

Date of Delivery: DD/MM/YYYY

- | | | | |
|--|---|---|--|
| ☐ Full term live birth | ☐ Premature live births | ☐ Stillbirth/Foetal Death | ☐ Spontaneous abortion |
| ☐ Induced abortion | ☐ Ectopic gestation | ☐ Trophoblastic disease | ☐ Died Undelivered |

Gestation at pregnancy termination/death: _____ weeks

Onset of labour

- | | | |
|---------------------------------|-----------------------------------|-------------------------------|
| ☐ No labour | ☐ Spontaneous | ☐ Induced |
|---------------------------------|-----------------------------------|-------------------------------|

Delivery Timing

- | | | |
|---|------------------------------|---------------------------------|
| ☐ Routine/ Elective | ☐ Urgent | ☐ Emergency |
|---|------------------------------|---------------------------------|

Was an anaesthetic used? ☐ No ☐ Yes, local ☐ Yes, general

Who administered the anaesthetic?

- | | | |
|--|--------------------------------|---|
| ☐ Nurse Anaesthetist | ☐ Resident | ☐ Consultant Anaesthetist |
|--|--------------------------------|---|

Attendant at delivery

- | | | |
|---|--|--|
| ☐ Nana/self | ☐ District Midwife | ☐ Registered Nurse/Midwife |
| ☐ General Practitioner | ☐ Consultant Obstetrician/ Gynaecologist | ☐ Resident |
| ☐ Other trained personnel | Specify other: | |

Medical Record Number:

[illegible]

Definitions

Labour

- Routine/Elective: The delivery, induction or Caesarean Section did not require any unexpected or urgent intervention.
- Urgent: Recognition of an increasing risk situation required the delivery to be expedited.
- Emergency: Recognition of a high-risk situation required the delivery to be expedited as an emergency.

Infant Outcome

- Livebirth: Refers to the delivery of a baby that shows any sign of life at delivery.
- Stillbirth/Foetal death: a baby is born with no signs of life after 28 or more weeks of gestation.
- Abortion/Miscarriage: Refers to the loss of the foetus before 28 weeks of gestation.

Timing of Death

- Antenatal: The woman died while pregnant and before labour.
- In labour: The woman died during labour and/or delivery.
- Postnatal: The woman died ≤ 42 days after the end of a pregnancy.
- Late postnatal: The woman died >42 days and ≤ 1 year after the end of a pregnancy.

Appendix 24: Maternal Mortality Home Visit & Antenatal Care Form



National Surveillance Unit, National Epidemiology Branch,
15 Knutsford Boulevard, Kingston 5,
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



Maternal Mortality Investigation Form

Home Visit and Antenatal Care Report

Patient's Demographics			
First Name:		Middle Name(s):	
Last Name:		Pet Name(s):	
Medical Record Number:			
Date of Birth: DD/MM/YYYY		Age:	
Marital Status:			
☐ Married	☐ Visiting union	☐ Widowed	☐ Separated
☐ Common law	☐ Single	☐ Divorced	☐ Other (specify):
Highest level of Education completed:			
☐ No formal schooling	☐ Primary/All Age /Preparatory	☐ Tertiary /Post-secondary	☐ Other
☐ Basic	☐ Secondary/High School	☐ Vocational Skills Training	
What is the patient's employment status?			
☐ Employed full time	☐ Self-employed	☐ Housewife	
☐ Employed part time (<3 days per week)	☐ Unemployed	☐ PATH programme	
Patient's Occupation:			
Does the patient fall into any of these risk categories?			
☐ Young teenager (16 years and under)	☐ Grand multipara (≥5 pregnancies)	☐ Advanced Maternal Age (>35 years)	
☐ Primigravida, 30 years or older	☐ Other (specify)		
Person Interviewed			
First Name:		Last Name:	
Relationship to the Patient:			
☐ Husband/Consort	☐ Father	☐ Sibling	☐ Friend
☐ Mother	☐ Child	☐ Other (specify):	
Were you present when patient died? ☐ Yes ☐ No ☐ Unknown			
If no, how long before death were you informed about it? ☐ Yes ☐ No ☐ Unknown			
Who told you about the death? ☐ Husband/Consort ☐ Relative ☐ Doctor ☐ Other, Specify			
Were you with her when she died? ☐ Yes ☐ No ☐ Unknown			
How long after death were you informed about it?			
Before (name) was pregnant for the last time, was she generally well? ☐ Yes ☐ No ☐ Unknown			
Please describe what happened immediately preceding death			

Name of Patient:	Medical Record Number:

ANTENATAL CARE

Did the patient receive antenatal care? ☐ Yes ☐ No ☐ Unknown

If the patient did not attend ANC what is the reason for non-attendance?

Did the patient have a maternal record booklet? ☐ Yes, seen ☐ Yes, not seen ☐ No ☐ Not known

What is the source of primary antenatal care?

Site	Number of Visits
☐ Health Centre	
☐ High Risk Clinic	
☐ General Practitioner	
☐ Obstetrician/ Gynaecologist	
☐ Other:	
TOTAL VISITS	

Date of first visit: DD/MM/YYYY	Gestational age at first visit:
Date of last visit: DD/MM/YYYY	Gestational age at last visit:
Was the patient referred to high risk clinic? ☐ Yes ☐ No ☐ Unknown	
Which high risk clinic was the patient referred to?	
Did the patient attend high risk clinic? ☐ Yes ☐ No ☐ Delayed ☐ Unknown	
If the patient did not attend high risk clinic what is the reason for non-attendance?	

Was the patient contacted by healthcare personnel? ☐ Yes ☐ No ☐ Unknown	
Date of first visit: DD/MM/YYYY	Gestational age at first visit:
Date of last visit: DD/MM/YYYY	Gestational age at last visit:
Conditions managed:	

Treatment summary:

Screening Results	
Screening results: Haemoglobin _____	
ABO Blood Group: ☐ O ☐ A ☐ B ☐ AB ☐ Unknown ☐ Not Done	
Rh(D) antigen status: ☐ Positive ☐ Negative ☐ Unknown ☐ Not Done	
Sickle Cell Screen: ☐ Positive ☐ Negative ☐ Unknown ☐ Not Done	

Name of Patient:

Medical Record Number:

If applicable, what are Sick Cell Electrophoresis results:

☐ SS	☐ S β -Thalassemia	☐ SC
☐ AS	☐ Other (Specify):	

HIV Screen: ☐ Positive ☐ Negative ☐ Unknown ☐ Not DoneSyphilis Screen (VDRL/FTA): ☐ Reactive ☐ Non-Reactive ☐ Unknown ☐ Not DoneIf reactive, was patient treated for syphilis? ☐ Yes ☐ No ☐ Unknown

Highest albuminuria result:

☐ No protein detected ☐ Trace ☐ 1+ ☐ 2+ ☐ 3+ ☐ 4+

Last blood pressure on records: _____ mm/Hg

Weight: _____ kg/lbs Height: ____-ft ____-in/ ____-cm BMI: _____ kg/m²Did the patient have oedema? ☐ Yes ☐ No ☐ Unknown

Did the patient experience any of these symptoms before delivery (select all that apply):

☐ Severe headaches	☐ Visual disturbance (seeing spots, seeing double, blindness)	☐ Epigastric pain (stomach ache)
☐ Seizures (fits)	☐ Severe abdominal pain	☐ Swelling of face or hands
☐ High fever	☐ Extremely short of breath	☐ Yellow skin or eyes
☐ Vaginal bleeding	☐ Severe chest pain	☐ Long labour (>12 hours)
☐ Coughing up blood	☐ Severe pain in calves or legs	

Postnatal

Did the patient experience any of these symptoms after delivery (select all that apply):

☐ Severe Bleeding	☐ Bad smelling discharge	☐ C-section wound re-opened
☐ Swollen C-section wound	☐ Severe abdominal pain	

Did the patient have any postnatal admissions? ☐ Yes ☐ No ☐ Unknown

Hospital name:

If yes, when was the patient admitted?

Admitted	Discharged	Discharge Diagnosis	Complications
DD/MM/YYYY	DD/MM/YYYY		
DD/MM/YYYY	DD/MM/YYYY		
DD/MM/YYYY	DD/MM/YYYY		
DD/MM/YYYY	DD/MM/YYYY		

Was the patient transferred? ☐ Yes ☐ No ☐ Unknown

Where was the patient transferred to?

Hospital name:

Date of transfer: DD/MM/YYYY

Reason for transfer:

Scheduled follow-up date post admission: DD/MM/YYYY

Medication ComplianceWas the patient prescribed medication? ☐ Yes ☐ No ☐ Unknown

What medication was prescribed?

Name of Patient:

Medical Record Number:

Was the prescription filled before patient left the health facility? ☐ Yes ☐ No ☐ Unknown

If filling the prescription was delayed what is the reason for the delay?

Did the patient take the medication? ☐ Yes ☐ No ☐ Unknown

Reason for not taking the medication:

HEALTH AND ROUTINE CHILD VISITS

Infant	Vital status	Sex	Current age
	☐ Alive ☐ Deceased	☐ Male ☐ Female ☐ Intersex ☐ Unknown	
	☐ Alive ☐ Deceased	☐ Male ☐ Female ☐ Intersex ☐ Unknown	

Health Centre/Doctor child attends:

Primary caregiver: ☐ Father ☐ Grandparent ☐ Other Relative ☐ Other

If "Other" primary caregiver, please specify:

Notifier's Details

Date Investigation Completed: DD/MM/YYYY

First Name: Last Name

Job Title:

Phone number: Email Address:

Name of Institution:

Office Number/Street Name:

Community: Parish:

Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA

Name of Parish MO(H): Signature of MO(H): Date signed by MO(H): DD/MM/YYYY

Appendix 25: Maternal Mortality Post-Mortem Form



National Surveillance Unit, National Epidemiology Branch,
15 Knutsford Boulevard, Kingston 5,
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



Maternal Mortality Investigation

Maternal Mortality Post Mortem Report

Patient's Demographics

Patient Name: _____ Medical Record Number: _____

Date of Birth: DD/MM/YYYY _____ Age: _____

Death Summary

Date of death: DD/MM/YYYY _____ Age at death: _____

Other medical problems/risk factors present: _____

Cause of Death

I.

(a) Immediate cause

(b) Intermediate cause

(c) Intermediate cause

(d) Underlying cause

II. Other significant conditions

Autopsy Completed by:

☐ DM Pathologist - Ministry of Health and Wellness/UHWI

☐ DM Pathologist - Ministry of National Security

☐ DMO

☐ Medical Officer

Date of Autopsy: DD/MM/YYYY

Notifier's Details

Date of Reporting (DD/MM/YYYY): ____ / ____ / ____

First Name: _____ Last Name: _____

Job Title: _____

Phone number: _____ Email Address: _____

Name of Institution: _____

Parish: _____ Community: _____

Office Number/Street Name: _____

Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA

Signature: _____ Date signed: DD/MM/YYYY

Appendix 26: Maternal Mortality Case Review Summary Form



National Surveillance Unit, National Epidemiology Branch,
15 Knutsford Boulevard, Kingston 5,
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



Maternal Mortality Investigation

Case Review Summary Report

INITIAL DIAGNOSIS

Date of Case Review: DD/MM/YYYY

Patient's Demographics

Patient Name:

Medical Record Number:

Date of Birth: DD/MM/YYYY

Age:

Death Summary

Date of death: DD/MM/YYYY

Time of death: HH:MM ☐ am ☐ pm

Date of delivery/termination: DD/MM/YYYY

Days to death: _____ days

Place of delivery/termination:

Gravida:

Parity:

Place of death (for hospital deaths)

☐ Antenatal ward

☐ Accident & Emergency

☐ ICU / HDU

☐ Labour ward

☐ Operating theatre

☐ Postnatal ward

☐ Non-obstetrics ward

Non-obstetrics ward:

Classification of Death

☐ Direct

☐ Coincidental

☐ Indirect

☐ Indeterminate

Timing of Death

☐ Maternal death

☐ Late maternal death

Type of Death

☐ Avoidable

☐ Unavoidable

☐ Unknown

Cause of Death:

Cause	Details
i. (a) Immediate	
(b) Intermediate	
(c) Intermediate	
(d) Underlying	
ii. Other significant conditions	

Three Delay Model

Were any delays identified? ☐ Yes ☐ No ☐ Unknown

Which delay(s) were identified?

☐ 1- Delay in seeking care

☐ 2- Delay in accessing care

☐ 3- Delay in receiving care

Name of Patient:

Medical Record Number:

DELAY 1 - Delay in seeking care

Delay Category	Delay Type	Present		
Delay in seeking care	Patient did not recognize problem	☐ Yes	☐ No	☐ Unknown
	Patient delayed seeking care	☐ Yes	☐ No	☐ Unknown
	Previous poor experience with health care	☐ Yes	☐ No	☐ Unknown
	Other	☐ Yes	☐ No	☐ Unknown

Specify details of the delay 1:

DELAY 2 - Delay in accessing care

Delay Category	Delay Type	Present		
Delay in accessing care	Delayed access due to cost	☐ Yes	☐ No	☐ Unknown
	Delayed access due to transportation	☐ Yes	☐ No	☐ Unknown
	Other	☐ Yes	☐ No	☐ Unknown
		☐ Yes	☐ No	☐ Unknown

Specify details of the delay 2:

DELAY 3 - Delays in the provision of adequate and appropriate care at a health facility

Delay Category	Delay Type	Present		
a. Providers of care at time of death	Training of the health care provider was inadequate	☐ Yes	☐ No	☐ Unknown
	Quality/seniority of the health care provider was inadequate	☐ Yes	☐ No	☐ Unknown
	Availability of the health care provider was inadequate	☐ Yes	☐ No	☐ Unknown
b. Decision-making process	Serious problem was not recognized	☐ Yes	☐ No	☐ Unknown
	Correct diagnosis was not made	☐ Yes	☐ No	☐ Unknown
	Consultation process was lacking	☐ Yes	☐ No	☐ Unknown
c. Actions taken	Referral was not done	☐ Yes	☐ No	☐ Unknown
	Emergency obstetric care was not done or inadequate	☐ Yes	☐ No	☐ Unknown
	Appropriate treatment was not given	☐ Yes	☐ No	☐ Unknown
	Referral was not done	☐ Yes	☐ No	☐ Unknown
	Emergency obstetric care was not done or inadequate	☐ Yes	☐ No	☐ Unknown
	Appropriate treatment was not given	☐ Yes	☐ No	☐ Unknown

Name of Patient:

Medical Record Number:

Delay Category	Delay Type	Present		
d. Delays in referral	Adequate transport was not available	☐ Yes	☐ No	☐ Unknown
	Financial constraints delayed execution of referral	☐ Yes	☐ No	☐ Unknown
	Permission (Refusal by sending or receiving facility)	☐ Yes	☐ No	☐ Unknown
	There was no space available to facilitate a referral	☐ Yes	☐ No	☐ Unknown
e. Facility delays	The services available at the facility were inadequate	☐ Yes	☐ No	☐ Unknown
	There were limited or no blood products	☐ Yes	☐ No	☐ Unknown
	Anaesthesia was not available	☐ Yes	☐ No	☐ Unknown
	Necessary supplies were inadequate or not available	☐ Yes	☐ No	☐ Unknown
	Necessary equipment was inadequate or not available	☐ Yes	☐ No	☐ Unknown
	Necessary medication was inadequate or not available	☐ Yes	☐ No	☐ Unknown
	Other	☐ Yes	☐ No	☐ Unknown
		☐ Yes	☐ No	☐ Unknown

Specify details of the delay 3:

Notifier's Details		
Date of Reporting (DD/MM/YYYY): /		
Comments:		
First Name:		Last Name
Professional Group:		
Phone number:		Email Address:
Name of Institution:		
Parish:		Community:
Office Number/Street Name:		
Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA		
Name of Parish MO(H):	Signature of MO(H):	Date signed by MO(H): DD/MM/YYYY

Name of Patient:

Medical Record Number:

DEFINITIONS

Classification of Death

- **Direct:** Death of a pregnant woman resulting from obstetric complications of pregnancy or its management.
- **Indirect:** Death of a pregnant woman resulting from diseases or conditions that were not due to a direct obstetric cause, but were aggravated by the physiologic effects of pregnancy.
- **Coincidental:** Death of a pregnant woman considered to be causally unrelated to pregnancy.
- **Indeterminate:** Not able to determine a classification for the death of the pregnant woman.

Timing of Death

- **Maternal death:** Death of a woman from direct or indirect causes, during pregnancy, childbirth or within 42 days of the end of the pregnancy, irrespective of the duration and site of the pregnancy.
- **Late maternal death:** Death of a woman from direct or indirect causes, more than 42 days but less than one year after the end of a pregnancy.

Appendix 27: MPox (MPX) Surveillance Case Investigation Form



NATIONAL SURVEILLANCE UNIT
24-26 Grenada Crescent, Kingston 5
Jamaica, West Indies
Tel: (876) 633-7925 / 8195
Email: surveillance@moh.gov.jm

Appendix 41: MPox (MPX) Surveillance Case Investigation Report

Date of Reporting: _____	Region: _____	Parish: _____
Doctor: _____	Hospital / Site: _____	Ward: _____
Email: _____	Phone #: _____	

Hospital Medical Record Number: _____ **NEW CASE** ☐ **UPDATE** ☐

Last Name: _____ **First Name:** _____

Date of Birth: _____ **Age:** _____ **Sex:** MALE ☐ FEMALE ☐

Country of Residence: _____ **Parish:** _____ **Community:** _____

Street #: _____ **Street Name:** _____

Date of Onset of Illness: _____ **Admitted** ☐ Y ☐ N ☐ **If Yes, Admission Date:** _____

CLINICAL PROFILE							
SYMPTOMS							
Symptoms	Yes	No	Date of Onset	Symptoms	Yes	No	Date of Onset
Rash				Asthenia (profound weakness)			
Type of rash: Maculopapular ☐ Vesicular ☐ Pustular ☐ Crusted ☐ Other, please specify: _____							
Fever over 38.5°C				Arthralgia			
Headache				Cough			
Lymphadenopathy				Other, please specify: _____			
Myalgia							
Back pain							
COMORBIDITIES							
Pregnancy			Lung Disease including COPD			Immunocompromised due to disease or treatment	
If yes, Trimester			Asthma			HIV / AIDS	
Diabetes Mellitus			Neurological Disease			Malignancy	
Sickle Cell Disease			Liver Disease			Obesity	
Heart Disease			Renal Disease			Other, please specify: _____	
Hypertension							
EPIDEMIOLOGICAL PROFILE							
Occupation Health Care Worker ☐ Health Laboratory Worker ☐ Working with Animals ☐ Student ☐ Other, please specify: _____							
Vaccination History: Smallpox (scar seen) ☐ MMR ☐ Dates _____ Varicella ☐ Dates _____ Monkeypox ☐ Dates _____							
CONTACT HISTORY							
In the 21 days before symptom onset, did the individual:							
Have close contact ² with a probable or confirmed case of monkeypox?						☐ Y ☐ N ☐ Unknown	
Have contact with contaminated materials e.g. clothing, bedding or utensils of a probable or confirmed case of monkeypox?						☐ Y ☐ N ☐ Unknown	
Have contact with animals?						☐ Y ☐ N ☐ Unknown	
If yes, please specify:							
Where: Home ☐ Work ☐ Health Care Setting ☐ Other, please specify: _____						☐ Y ☐ N ☐ Unknown	
Sexual contact						☐ Y ☐ N ☐ Unknown	
Was the contact ill at the time of contact?						☐ Y ☐ N ☐ Unknown	
In which country was the contact diagnosed with monkeypox? _____							
Contact's Name: _____							

Fever (> 38 °C) may not be present in some patients, such as those who are very young, elderly, immunosuppressed, or taking certain medications. Clinical judgement should be used to guide testing of patients in such situations
A contact is a person who experienced any one of the following exposures face-to-face, direct physical contact, or direct care for a patient during the 21 days after the onset of symptoms of a probable or confirmed case.



MINISTRY OF
**HEALTH &
WELLNESS**
NATIONAL SURVEILLANCE UNIT
24-26 Grenada Crescent, Kingston 5
Jamaica, West Indies
Tel: (876) 633-7925 / 8195
Email: surveillance@moh.gov.jm

TRAVEL HISTORY

In the 21 days before symptom onset, did the individual:

Travel to Jamaica from another Country: ☐ Y ☐ N ☐ Unknown

If Yes,

Country Province/City Departure Date

Country Province/City Departure Date

LABORATORY DATA

MOLECULAR TESTING

Sample Type	Collection Date	Test Type	Results	Result Date
Swab of lesion ☐	Date	PCR ☐ Other (specify)	Monkeypox +ve ☐ Monkeypox -ve ☐ Other+ve ☐	
Lesion/Tissue Sample ☐	Date	PCR ☐ Other (specify)	Monkeypox +ve ☐ Monkeypox -ve ☐ Other+ve ☐	
Aspirate/Exudate Swab ☐	Date	PCR ☐ Other (specify)	Monkeypox +ve ☐ Monkeypox -ve ☐ Other+ve ☐	
Other (specify)				

CONTACT TRACING

Name of Contact	Contact Type (bedroom, household, conveyance, health facility, other)	Contacts Sampled

Treatment Received (in hospital): Antiviral Yes ☐ No ☐ Type: _____

Antibiotics Yes ☐ No ☐ Type: _____

Admitted to ICU: Yes ☐ No ☐ If Yes, Date admitted to ICU _____ Date discharged from ICU _____

Recovered: Yes ☐ No ☐ Date of Recovery: ____/____/____

Died: Yes ☐ No ☐ Date of Death: ____/____/____

FINAL DIAGNOSIS: _____

MO(H) Signature: _____ Date: _____

A contact is a person who experienced any one of the following exposures face-to-face, direct physical contact, or direct care for a patient during the 21 days after the onset of symptoms of a probable or confirmed case.

National Surveillance Unit, Ministry of Health & Wellness

Revised – July 6, 2022

Appendix 28: SARI Case Investigation Form

SARI Case Investigation Form

Date of Reporting: _____ Region: _____ Parish: _____
 Doctor: _____ Hospital/Site: _____ Ward: _____
 Email: _____ Phone #: _____
 Hospital/Medical Record Number: _____ NEW CASE ☐ UPDATE ☐
 Admission Date: _____ Date of Onset of Illness: _____ Epidemiological Week of Onset: _____
 Last Name: _____ First Name: _____
 Date of Birth: _____ Age: _____ Sex: MALE ☐ FEMALE ☐
 Country of Residence: _____ Parish: _____ Community: _____
 Street #: _____ Street Name: _____

Sample taken: Yes ☐ No ☐ Sample Type: _____ Date Sample Taken: _____

Sample taken to Lab: Yes ☐ No ☐ Date sample taken to Lab: _____

Influenza Vaccine: Yes ☐ No ☐ Date of last vaccination: _____

CLINICAL AND EPIDEMIOLOGICAL PROFILE									
CLINICAL PROFILE									
History of Fever or Fever over 38°C (<10 days)				Yes	No	Recorded temperature _____ °C			
Cough	Yes	No	Otitis	Yes	No	Pneumonia	Yes	No	
Rhinorrhea	Yes	No	Conjunctivitis	Yes	No	Adenopathy	Yes	No	
Sore Throat	Yes	No	Headache	Yes	No	Vomiting	Yes	No	
Shortness of Breath	Yes	No	Myalgia	Yes	No	Diarrhea	Yes	No	
Difficulty Breathing	Yes	No	Bronchitis	Yes	No	Rash	Yes	No	
Wheezing	Yes	No	Bronchiolitis	Yes	No				
RISK FACTORS									
Pregnancy If yes, Trimester	Yes	No	Lung Disease including COPD	Yes	No	Immunocompromised due to disease or treatment	Yes	No	
	1	2	3	Asthma	Yes	No	HIV/AIDS	Yes	No
Diabetes Mellitus	Yes	No	Neurological Disease	Yes	No	Obesity	Yes	No	
Sickle Cell Disease	Yes	No	Liver Disease	Yes	No	If Yes 30 – 40 ☐ Over 40 ☐			
Heart Disease	Yes	No	Renal Disease	Yes	No	Unknown ☐			
Children under 2 years:									
Prematurity	Yes	No	Malnutrition	Yes	No		Yes	No	
Low Birthweight (LBW)	Yes	No	Lack of Breastfeeding	Yes	No		Yes	No	
EPIDEMIOLOGICAL PROFILE									
Sporadic Case	Yes	No	Animal Contact	Yes	No		Yes	No	
			If yes Home ☐ Swine ☐ Bird/Poultry ☐						
Part of an Outbreak	Yes	No	Travel abroad in the past 6 weeks	Yes	No		Yes	No	
Part of Cluster (3 or more persons geographically or socially linked with onset of disease within 10 days of each other)	Yes	No	If yes to travel, country(ies) visited: _____						
Other:									

Treatment Received (in hospital): Antiviral Yes ☐ No ☐ Type: _____ Date: _____

Antibiotics Yes ☐ No ☐ Types: _____ Date: _____

Laboratory Results: Virology Positive ☐ Negative ☐ Virus: _____

Bacteriological Positive ☐ Negative ☐ Bacteria: _____

Admitted to ICU: Yes ☐ No ☐ Date of admission to ICU: _____

Discharge Date: _____ SARI Death: Yes ☐ No ☐ Date of Death: _____

Final Diagnosis: _____ MO(H) Signature: _____

Appendix 29: Hospital SARI/ILI Surveillance Reporting Form



Data Collection Form: SARI / ILI Hospitalizations and Deaths

Parish: _____

Epidemiological Week #: _____

Reporting: Daily

Institution: _____

Reporting Date: _____

Weekly

☐
☐

Surveillance of Severe Acute Respiratory Infection (SARI) [This section is to be completed for SARI sites <u>ONLY</u>]									
	< 6 mths	6-11 mths	12-23 mths	2-4 yrs	5-14 yrs	15-49 yrs	50-64 yrs	≥ 65 yrs	Total
SARI Admissions									
SARI Deaths									
SARI ICU Admissions									
Total ICU Admissions									
SARI Samples taken									
Hospital medical admissions									
Deaths in medical admissions									
Hospital Admissions									
Deaths in hospitalized patients									

Influenza-Like Illness (ILI)				
	Total	< 5 yrs	5-59 yrs	≥ 60 yrs
ILI Cases				
ILI Samples taken				
Total Visits				

Surveillance Coordinator: _____

MO(H) Signature: _____

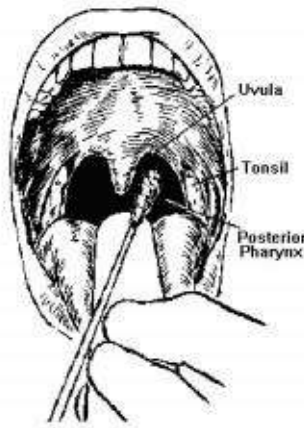
Date: _____

- Notes:
- NB. Please indicate by ticking the respective box whether the report is daily or weekly.
 - The Epidemiological Week begins on a Sunday and ends on a Saturday. The date on Sunday is recorded as the Week Start Date.
 - Hospital medical admissions constitute all admissions to the medical ward, medical admissions to the paediatric ward, and medical admissions to the intensive care unit (for each particular age group).
 - Deaths in medical admissions constitutes all deaths on the medical ward, in medical patients on the paediatric ward, in medical patients in the intensive care unit
 - Hospital admissions constitutes all admissions to hospital
 - Deaths in hospitalized patients constitute all deaths in those admitted to hospital.
 - Total visits constitute all visits to hospitals A&E
 - Form should be submitted to the surveillance unit, along with other parish weekly surveillance reports.

Appendix 30: Additional Sampling Techniques in lieu of Nasopharyngeal Specimen

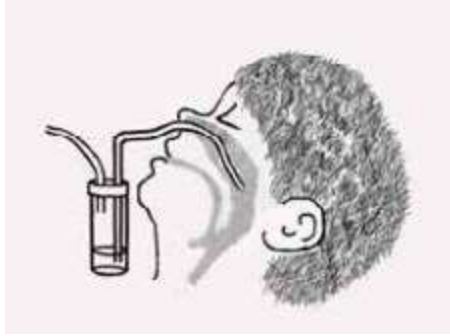
Oropharyngeal Swab:

1. Push the tongue downwards with a tongue depressor or hold it with fingers and gauze.
2. Insert swab into the posterior pharynx and tonsillar areas.
3. Rub swab over both tonsillar pillars and posterior oropharynx so that as many cells as possible are collected from the mucosal surface.
4. Avoid touching the tongue, cheek, teeth, and gums.



Nasopharyngeal Aspirate:

1. Nasopharyngeal secretions are aspirated through a catheter connected to a mucus trap and fitted to a vacuum source.
2. The catheter is inserted into the nostril parallel to the palate.
3. The vacuum is applied and the catheter is slowly withdrawn with a rotating motion.
4. Mucus from the other nostril is collected with the same catheter in a similar manner.
5. After mucus has been collected from both nostrils, the catheter is flushed with 3 ml of transport medium.



Specimen collection from a recently deceased patient:

If a patient dies before a respiratory specimen is taken collect nasopharyngeal swabs postmortem in addition to the following:

- An endotracheal aspirate
 - If the corpse has an endotracheal tube in place, collect a deep endotracheal aspirate.

OR

- A Lung Tissue Sample
 - If the circumstance allows, perform tissue sampling by incision or by needle from the affected lung(s).

Appendix 31: SARI Data Analysis Guide

SARI Sites:

The SARI Information Management System database, located at each SARI site, is to be used to synthesize the information below.

Basic information to be generated for dissemination to staff and management:

- Number of SARI cases per week for a period
- Cumulative number of SARI cases Year-to-Date
- Number of SARI Samples taken per week for a period
- Cumulative number of SARI cases with samples for whom results were received (year-to-date)
- Cumulative number of SARI cases with samples and results that have been positive for influenza

The information must be interpreted at the site and appropriate actions taken regarding aberrations. Appropriate actions include:

- Reporting of unusual cases to the parish health department
- Reporting of perceived increases in SARI cases to the parish health department and Regional Health Authority
- Requesting and implementing enhanced staff vaccination, infection prevention, and isolation measures in light of unusual or increasing SARI cases
- Increasing sensitization regarding sampling for staff in light of below normal levels of specimen collection

Appendix 32: Surveillance System Performance Indicators

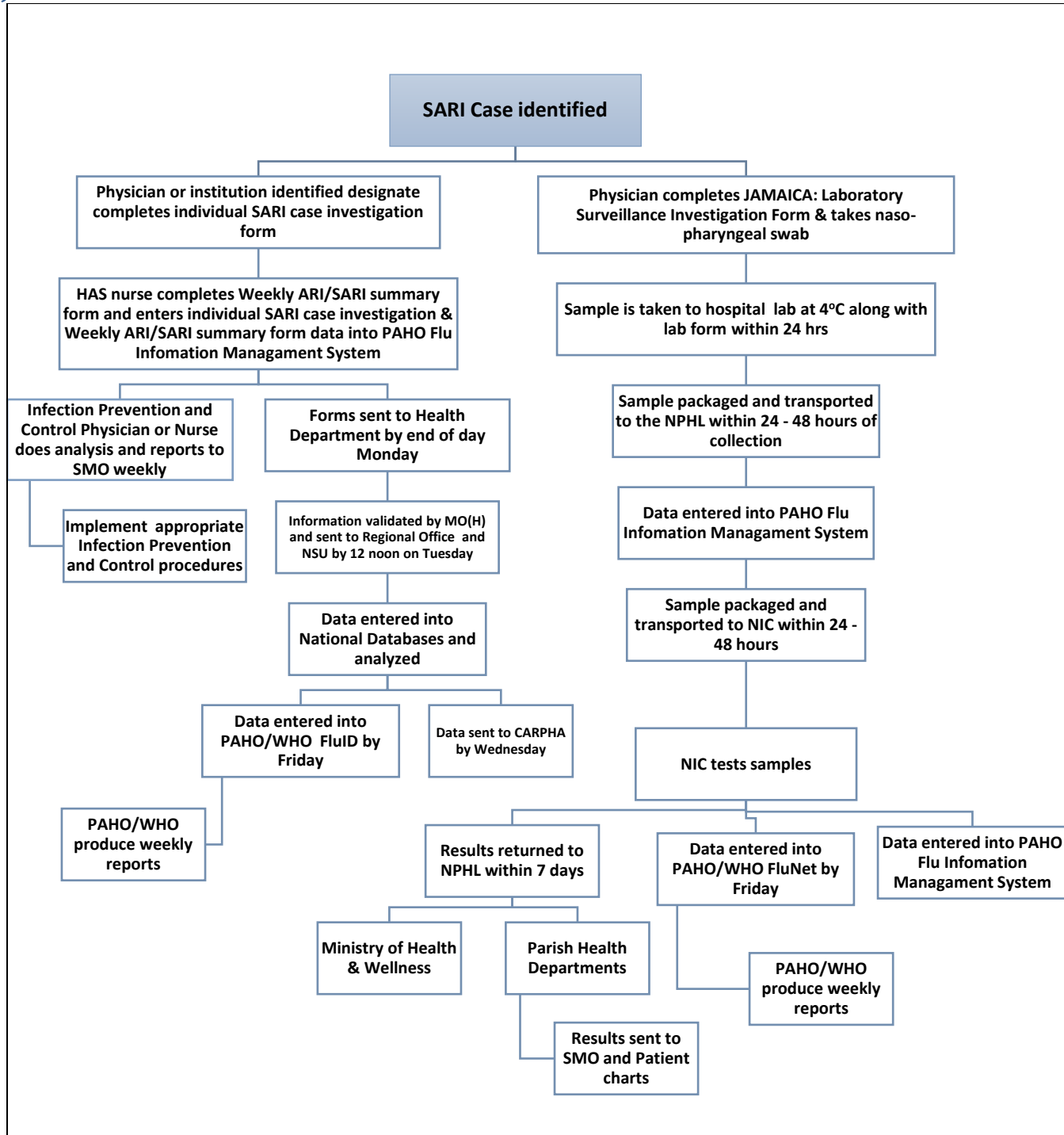
Surveillance System Performance Indicators ¹

Sentinel Surveillance Steps	Associated Monitoring and Evaluation Indicators	Indicator	Structure or Calculation	Goal
Step 1 Identify hospitalized patients that meet the SARI case definition	Percent of week's with timely notification of denominators	Timely reporting of denominators	(Number of EWs with timely reporting of denominators / total EWs reported) x 100	80%
	Percent of hospitalized SARI cases that are captured by the surveillance system	Underreporting	(Reported cases of SARI in the period / cases identified in the period by active case-finding) x 100	80%
	Median interval in days between hospitalization date and notification date	Timely case reporting	Median of the interval (number of days) between hospitalization date and reporting date	1 day
Step 2 Complete data collection form and data entry	Percent of cases investigated and closed	Case investigation coverage	(Total SARI cases completely investigated and closed / Total cases reported and discharged) x 100 * Completely investigated and closed means when test results are included (when available) for etiological diagnosis of the reported and discharged case, as well as complete clinical/epidemiological data	90%
Step 3 Respiratory sample collection and testing results	Percent of SARI cases with collected sample	Coverage of sampled SARI cases	(Number of SARI cases with collected specimen / Number of SARI cases with valid criteria for sampling) x 100 *Sampling must be within 10 days of symptom onset	90%
	Percent of quality samples received	Specimen quality	(Number of good quality specimens received / Total specimens received) x 100 * Good quality means specimens properly taken, preserved, and transported until arrival at the laboratory	90%

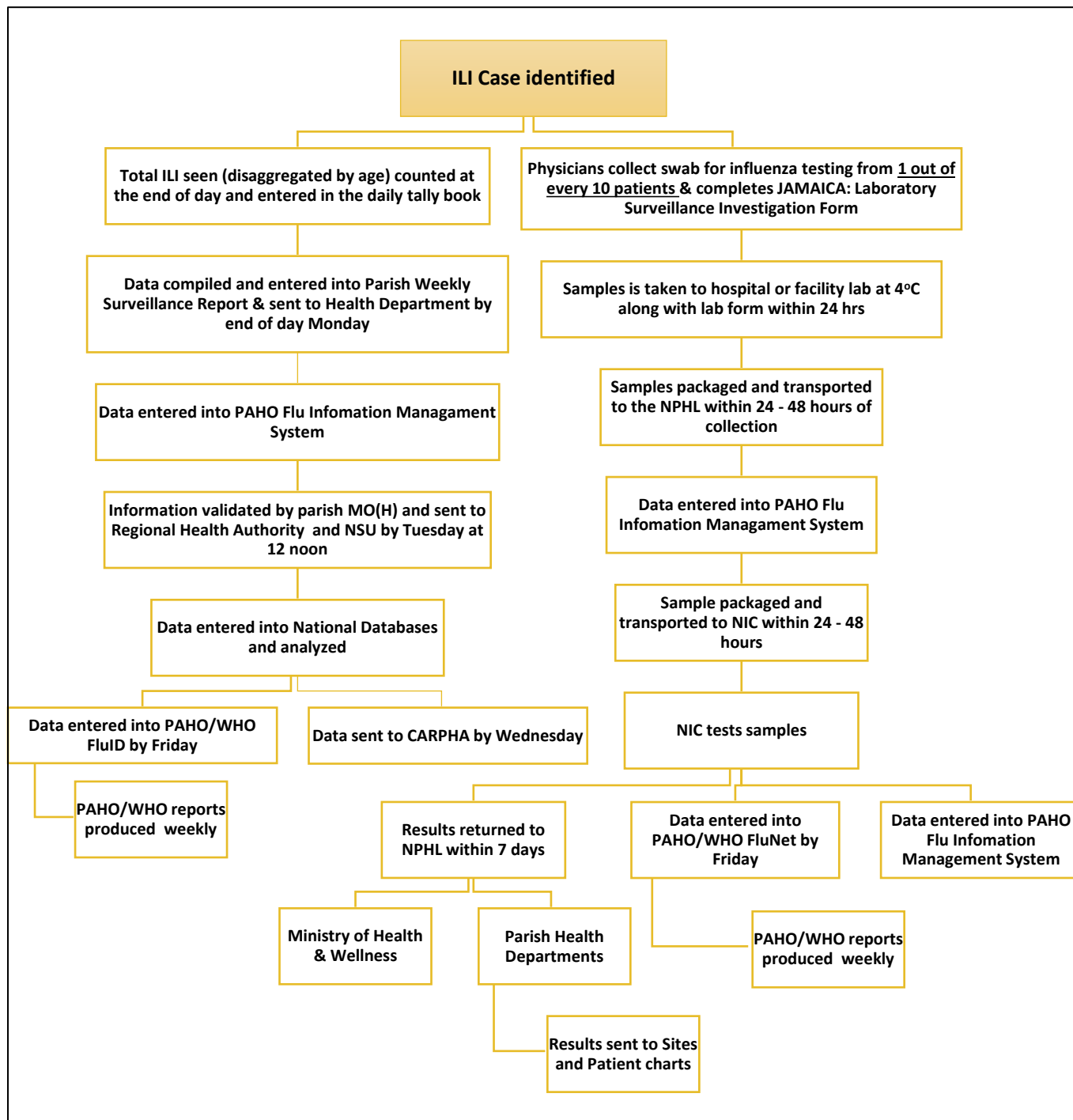
¹ Pan American Health Organization. *Operational Guidelines for Sentinel SARI Surveillance* (2014) Available from: http://www.paho.org/hq/index.php?option=com_docman&task=doc_view&Itemid=270&gid=31162&lang=en

Sentinel Surveillance Steps	Associated Monitoring and Evaluation Indicators	Indicator	Structure or Calculation	Goal
	Percent of processed quality samples received	Processing coverage	(Number of processed specimens / Total specimens correctly received) x 100	90%
	Median interval from hospitalization date to sampling date	Timely sampling	Median of the interval (number of days) between hospitalization date and sampling date	2 days
	Median interval from sample date to receipt of sample	Timely receipt of specimen	Median of the interval (number of days) between sampling date and receipt date <i>* If the date of receipt is not available, use shipment date for this indicator</i>	1 day
	Median interval from receipt date to start of processing	Timely processing	Median of the interval between date of receipt of the specimen and processing date	3 days
	Median interval from receipt date to delivery of result	Timely delivery of result	Median of the interval between date of receipt of the specimen and date of delivery of the result	3 days
	Percent of SARI ICU cases with collected sample	Coverage of sampled SARI cases in ICU	(Number of SARI cases in ICUs with collected specimen / Number of SARI cases in ICUs) x 100	100%
	Percent of SARI-associated deaths with collected sample	Coverage of sampled cases of SARI deaths	(Number of cases of SARI deaths with collected specimen / Number of cases of SARI deaths) x 100	100%
Step 4 Data analysis and interpretation	Number of cases reported each week			
	Number of specimens submitted each week			
	Percent of submitted specimens that test positive for influenza			
Step 5 Data dissemination and outputs	Percent of weeks that data are sent to the national/regional level	Timely reporting		80%
	Timeliness of data presented in weekly influenza surveillance report	Timely reporting	How many weeks are there between the current epidemiological week and the week for which data are reported	< 2 weeks

Appendix 33: Flow Chart of Data & Sample Collection Process for SARI Cases in Jamaica



Appendix 34: Flow Chart of Data & Sample Collection Processes for ILI Surveillance in Jamaica



Appendix 35: Tuberculosis (TB) Case Investigation Form 1



National Surveillance Unit, National Epidemiology Branch,
Ministry of Health and Wellness,
ARC Building, 15 Knutsford Boulevard, Kingston 5, St. Andrew, Jamaica
Telephone 876-633-7937
Email: surveillance@moh.gov.jm



TUBERCULOSIS CASE INVESTIGATION FORM 1

Form 1 shall be submitted to the Parish Medical Officer of Health, MO(H), within ten (10) weeks of the initial Class 1 Notification to the Health Department and/or Ministry of Health and Wellness of a suspected case of tuberculosis.

SECTION 1 – DEMOGRAPHIC INFORMATION

INITIAL DIAGNOSIS			
☐ Tuberculosis			
Date of Notification (DD/MM/YYYY):		Date Investigation Started (DD/MM/YYYY):	
PATIENT'S DEMOGRAPHICS			
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		Patient's ID Number:	
First Name:		Middle Name(s):	
Last Name:		Pet Name(s):	
Patient's Maiden Name (If Applicable):			
Sex Assigned at Birth: ☐ Male ☐ Female			
Name of Facility Where Patient Seen:		Medical Record Number:	
Date of Birth (DD/MM/YYYY):		Age:	
Country of Residence:		Parish (Jamaica):	
House Number, Street Name:			
Community:			
Landmark or directions to address:			
Phone Number:		Email Address:	
Occupation:			
Name of Workplace/School:			
Address of Workplace/ School:			
Workplace/School Phone Number:		Email Address:	
Marital Status:			
☐ Married	☐ Visiting union	☐ Widowed	☐ Separated
☐ Common Law	☐ Single	☐ Divorced	☐ Other (specify):
Next of Kin			
First Name:		Last Name:	
Pet Name(s):			
Address: Lot/ Street/Community/Parish:			
Name of Workplace/School:			
Address of School/Workplace:			
Phone number:		Email address:	

TUBERCULOSIS CASE INVESTIGATION FORM 1

Relationship to the Patient:			
☐ Wife	☐ Son	☐ Uncle	☐ Cousin
☐ Husband	☐ Daughter	☐ Aunt	☐ Friend
☐ Mother	☐ Sister	☐ Grandmother	☐ Other
☐ Father	☐ Brother	☐ Grandfather	

CLINICAL PROFILE					
Has the patient experienced/Is the patient experiencing any of the following symptoms?					
SYMPTOM	DATE OF ONSET (DD/MM/YYYY)	DURATION	SYMPTOM	DATE OF ONSET (DD/MM/YYYY)	DURATION
☐ Cough			☐ Haemoptysis		
☐ Fever			☐ Chest pain		
☐ Night sweats			☐ Other		
☐ Weight loss			Specify Other:		

Medical History			
Pre-existing Conditions			
☐ Asthma	☐ Diabetes mellitus	☐ HIV/AIDS	☐ Malignancy
☐ Autoimmune	☐ Heart disease	☐ Hypertension	☐ Sickle cell disease
☐ Chronic Obstructive Pulmonary Disease (COPD)	☐ Kidney disease	☐ Liver dysfunction	☐ Other Specify Other:

Pregnancy	
Is Patient Pregnant: ☐ Yes ☐ No ☐ Unsure ☐ NA	Date of LMP (DD/MM/YYYY):
Gestational Age:	Trimester: ☐ First ☐ Second ☐ Third

HIV Testing	
HIV Screening Test Type:	HIV Confirmatory Test Type:
Date of HIV Screening Test (DD/MM/YYYY):	Date of HIV Confirmatory Test (DD/MM/YYYY):
HIV Screening Test Results: ☐ Positive ☐ Negative ☐ Indeterminate	HIV Confirmatory Test Results: ☐ Positive ☐ Negative ☐ Indeterminate

Viral load and CD4 count for persons living with HIV	
Viral Load result: _____ copies per millilitre	Date viral load sample taken (DD/MM/YYYY):
CD4 Count result: _____ cells per microlitre	Date CD4 Count sample taken (DD/MM/YYYY):

ANTIRETROVIRAL (ARV) MEDICATION HISTORY?				
Past or Current Medication	Dosage	Frequency	Actual Duration Received (weeks)	Comments
☐ Tenofovir				
☐ Lamivudine				
☐ Emtricitabine				
☐ Dolutegravir				
☐ Abacavir				
☐ Zidovudine				
☐ Nevirapine				
☐ Lopinavir/ritonavir				
☐ Other (specify):				

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
NSU, MOHW, Jamaica

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TUBERCULOSIS CASE INVESTIGATION FORM 1

OTHER MEDICATION HISTORY:				
☐ Trimethoprim-Sulfamethoxazole/ Co-trimoxazole (Bactrim)				
☐ Other (specify):				
☐ Other (specify):				
☐ Other (specify):				
VACCINATION STATUS				
Type of Vaccine	Source of vaccination Information (Vaccination card, Health service record, Verbal)		BCG Scar seen	
BCG			☐ Yes ☐ No	
Substance Use				
☐ Tobacco	Frequency		Duration	
☐ Ganja/marijuana/cannabis				
☐ Alcohol				
☐ Other substance (please specify): -----				
TRAVEL PROFILE				
Are there other cases present in the case's community of residence? ☐ Yes ☐ No				
Travelled in the last 6 weeks: ☐ Yes ☐ No ☐ Unknown				
If travelled: ☐ Domestic Travel ☐ International Travel ☐ Domestic and International Travel				
If international travel, fill out the table below:				
Country Visited	City Visited	Accommodation	Date Arrived in Country (DD/MM/YYYY)	Date Departed Country (DD/MM/YYYY)
Date returned to Jamaica (DD/MM/YYYY):				
If local travel, fill out the table below:				
Address	Parish	Date Arrived (DD/MM/YYYY)	Date Departed (DD/MM/YYYY)	
Date returned to home address (DD/MM/YYYY):				
Has the patient been in contact with someone who travelled in the past 6 weeks? ☐ Yes ☐ No ☐ Unknown				

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
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TUBERCULOSIS CASE INVESTIGATION FORM 1

SECTION 2 – DIAGNOSTIC SCREENING

MANTOUX TEST			
Mantoux test done: ☐ Yes ☐ No		Mantoux test date (DD/MM/YYYY):	
Mantoux test reading ----- mm		Mantoux test interpretation: ☐ Positive ☐ Negative	
<i>Reference: Immunocompetent 10 mm or more (positive); Immunocompromised 5 mm or more (positive)</i>			
CHEST X-RAY			
Chest X-ray done: ☐ Yes ☐ No		Chest X-ray date (DD/MM/YYYY):	
Chest X-ray description/ findings/result:			
Chest X-ray interpretation: ☐ Positive ☐ Negative ☐ Indeterminate			
OTHER INVESTIGATIONS			
Other Investigation Done: ☐ Yes ☐ No		Type of Investigation:	
Date Investigation Done (DD/MM/YYYY):			
Result of Investigation:			
LABORATORY TESTING FOR MYCOBACTERIUM TUBERCULOSIS			
Sputum specimen collected	☐ Yes ☐ No		
	Sputum Sample 1	Sputum Sample 2	Sputum Sample 3
Sputum specimen collection date(s), (DD/MM/YYYY):			
Sputum microscopy for acid fast bacilli (AFB) done	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Sputum microscopy result (e.g., AFB 1+)			
Sputum microscopy interpretation	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected
	Sputum Culture 1	Sputum Culture 2	Sputum Culture 3
Sputum culture for Mycobacterium tuberculosis done	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Sputum culture result			
Sputum culture interpretation	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected

Patient's Name: _____

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Medical Record Number: _____

NSU, MOHW, Jamaica

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TUBERCULOSIS CASE INVESTIGATION FORM 1

Rifampicin resistance detected on Mycobacterium tuberculosis culture	☐ Resistant ☐ Not resistant ☐ Indeterminate		
Other drug resistance detected on Mycobacterium tuberculosis culture	☐ Yes ☐ No ☐ Not Done		
If yes, specify other drug resistance detected:			
	Sputum Sample 1	Sputum Sample 2	Sputum Sample 3
Gene Xpert or polymerase chain reaction (PCR) done	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
Gene Xpert or polymerase chain reaction (PCR) results interpretation	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected
Rifampicin resistance detected on Gene Xpert or polymerase chain reaction (PCR)	☐ Resistant ☐ Not resistant ☐ Indeterminate	☐ Resistant ☐ Not resistant ☐ Indeterminate	☐ Resistant ☐ Not resistant ☐ Indeterminate

SECTION 3 – ANTI-TUBERCULOSIS TREATMENT INITIATION: INTENSIVE PHASE

Date started Intensive Phase of anti-TB treatment (DD/MM/YYYY):	Date completed Intensive Phase of anti-TB treatment (DD/MM/YYYY):		
Name of hospital:			
Date of admission (DD/MM/YYYY):	Date of discharge (DD/MM/YYYY):		
Directly Observed Therapy (DOT): ☐ Yes ☐ No			
ANTI-TUBERCULOSIS TREATMENT			
Past or Current Medication	Dosage	Actual Duration Received (weeks)	Comments
☐ Isoniazid			
☐ Rifampicin			
☐ Pyrazinamide			
☐ Ethambutol			
☐ Streptomycin			
☐ Other			
☐ Other			

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
NSU, MOHW, Jamaica

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TUBERCULOSIS CASE INVESTIGATION FORM 1

SECTION 4 – SOCIAL ASSESSMENT

LIVING ARRANGEMENT		
☐ Lives alone	☐ Lives with partner	☐ Lives with family
☐ Lives with friends	☐ Other (Specify Other):	
Number of household members: -----		
DWELLING TYPE		
☐ Separate House-Detached	☐ Apartment Building	☐ Townhouse
☐ Other Attached	☐ Part of Commercial Building	☐ Improvised Housing Unit
☐ Homeless	☐ Non-response	☐ Other (Specify):
Number of rooms:	Number of bedrooms:	Number of persons per bedroom:
Water Supply: ☐ Good ☐ Fair ☐ Poor		
Ventilation: ☐ Good ☐ Fair ☐ Poor		
Cleanliness: ☐ Good ☐ Fair ☐ Poor		
FOOD AND NUTRITION SUMMARY		
Milk supply: ☐ Pasteurised ☐ Unpasteurised		
WORKPLACE ARRANGEMENT		
Risk Level (Crowding): ☐ Low ☐ Medium ☐ High		
Notes (e.g., about contacts; industry):		

Patient's Name: _____

Updated April 12, 2024

Medical Record Number: -----

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TUBERCULOSIS CASE INVESTIGATION FORM 1

SECTION 5 – CONTACT TRACING

First Name	Last Name	D.O. B (DD/MM/YYYY)	Sex	Relation	Address	Phone	Symptoms	Previous BCG	Mantoux		Further Evaluation	Evaluation Date (D/M/YYYY)	Findings	Classification	Comments
									Date (D/M/Y)	Result (mm)					
			☐ M ☐ F				☐ Cough ☐ Fever ☐ Night Sweats ☐ Weight Loss ☐ Asymptomatic	☐ Yes ☐ No			☐ Chest X-ray ☐ Sputum ☐ Other			☐ Suspected ☐ Previously Treated ☐ Latent TB ☐ Discarded	
			☐ M ☐ F				☐ Cough ☐ Fever ☐ Night Sweats ☐ Weight Loss ☐ Asymptomatic	☐ Yes ☐ No			☐ Chest X-ray ☐ Sputum ☐ Other			☐ Suspected ☐ Previously Treated ☐ Latent TB ☐ Discarded	
			☐ M ☐ F				☐ Cough ☐ Fever ☐ Night Sweats ☐ Weight Loss ☐ Asymptomatic	☐ Yes ☐ No			☐ Chest X-ray ☐ Sputum ☐ Other			☐ Suspected ☐ Previously Treated ☐ Latent TB ☐ Discarded	
			☐ M ☐ F				☐ Cough ☐ Fever ☐ Night Sweats ☐ Weight Loss ☐ Asymptomatic	☐ Yes ☐ No			☐ Chest X-ray ☐ Sputum ☐ Other			☐ Suspected ☐ Previously Treated ☐ Latent TB ☐ Discarded	
			☐ M ☐ F				☐ Cough ☐ Fever ☐ Night Sweats ☐ Weight Loss ☐ Asymptomatic	☐ Yes ☐ No			☐ Chest X-ray ☐ Sputum ☐ Other			☐ Suspected ☐ Previously Treated ☐ Latent TB ☐ Discarded	

Patient's Name: _____

Medical Record Number: -----

Updated April 12, 2024

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Tuberculosis Case Investigation Form 1

SECTION 6 (TB FORM 1) - CLASSIFICATION

Classification of Case		
☐ Confirmed	☐ Pending	☐ Inconclusive
☐ Discarded		
Confirmation Category		
☐ Bacteriologically confirmed	☐ Clinically diagnosed	
Anatomical site		
☐ Pulmonary	☐ Extra-pulmonary	
Drug resistance		
☐ None	☐ Rifampicin resistant	☐ Multi-drug resistant
☐ Extremely drug-resistant		
Previous TB Treatment		
Previously treated for tuberculosis?		Relapse case?
☐ Yes	☐ No	☐ Yes
		☐ No
Treatment after lost to follow-up?		Treatment after failure?
☐ Yes	☐ No	☐ Yes
		☐ No
Treatment status for current diagnosis		
☐ Completed		☐ Not completed
Status for current diagnosis		
☐ Admitted for Intensive Phase	☐ Discharged from Intensive Phase	☐ Started Continuation Phase
☐ Lost to follow-up	☐ Died	
Cause of Death		
(i)	(ii)	(iii)
Immediate	Intermediate	Underlying
Date of death (DD/MM/YYYY):		
REVIEW AND SIGNATURE		
Name of Investigator:	Signature of Investigator	Date Signed by Investigator (DD/MM/YYYY)
Comments:		
Name of Parish Medical Officer (Health):	Signature of MO(H)	Date signed by MO(H) (DD/MM/YYYY)

Patient's Name: _____

Updated April 12, 2024

Medical Record Number: _____

NSU, MOHW, Jamaica

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Appendix 36: Tuberculosis (TB) Case Investigation Form 2



National Surveillance Unit, National Epidemiology Branch,
ARC Building, 15 Knutsford Boulevard, Kingston 5, St. Andrew, Jamaica
Telephone 876-633-7937
Email: surveillance@moh.gov.jm



TUBERCULOSIS CASE INVESTIGATION FORM 2

Form 2 shall be submitted to the Parish Medical Officer of Health within 6 to 8 months after starting anti-TB therapy. Submit this form at the end of anti-TB therapy and/or after follow-up.

SECTION 1 (TB FORM 2) – DEMOGRAPHIC INFORMATION

INITIAL DIAGNOSIS			
☐ Tuberculosis			
Date of Notification (DD/MM/YYYY):		Date Investigation Started (DD/MM/YYYY):	
PATIENT'S DEMOGRAPHICS			
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		Patient's ID Number:	
First Name:		Middle Name(s):	
Last Name:		Pet Name(s):	
Patient's Maiden Name (If Applicable):			
Sex Assigned at Birth: ☐ Male ☐ Female		Date of Birth (DD/MM/YYYY):	Age:
Name of Facility Where Patient Seen:		Medical Record Number:	
Country of Residence:		Parish (Jamaica):	
House Number, Street Name:			
Landmark or directions to address:		Community:	
Phone Number:		Email Address:	
Occupation:			
Name of Workplace/School:			
Address of Workplace/School:			
Phone Number:		Email Address:	

SECTION 2 (TB FORM 2) TREATMENT – CONTINUATION PHASE

Date started continuation phase (DD/MM/YYYY):	Date completed continuation phase (DD/MM/YYYY):
Name of facility (if hospitalised or institutionalised):	
Directly Observed Therapy (DOT): ☐ Yes ☐ No	
DOTS Supervisor:	
Telephone number:	Email address:

ANTI-TUBERCULOSIS TREATMENT

Past or Current Medication	Dosage	Actual Duration Received (weeks)	Comments
☐ Rifampicin			
☐ Isoniazid			
☐ Second-line drugs			
☐ Third-line drugs/injectables			
☐ Other			
☐ Other			

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
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TUBERCULOSIS CASE INVESTIGATION FORM 2

SECTION 3 (of TB FORM 2) - FOLLOW-UP LABORATORY INVESTIGATIONS

FOLLOW-UP LABORATORY INVESTIGATIONS	
2-Month Evaluation for Mycobacterium Tuberculosis	
Sputum specimen collected? ☐ Yes ☐ No	Sputum specimen date (DD/MM/YYYY):
Sputum microscopy for acid fast bacilli (AFB) done? ☐ Yes ☐ No	Sputum microscopy result (e.g., AFB 1+):
Sputum microscopy interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
Sputum culture for Mycobacterium tuberculosis	
Sputum culture result:	
Sputum culture interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
Chest X-ray done? ☐ Yes ☐ No	Chest X-ray date (DD/MM/YYYY):
Chest X-ray description/ findings/ result:	
Chest X-ray interpretation: ☐ Positive ☐ Negative	
4 to 5-Month Evaluation Mycobacterium Tuberculosis	
Sputum specimen collected? ☐ Yes ☐ No	Sputum specimen date (DD/MM/YYYY):
Sputum microscopy for acid fast bacilli (AFB) done? ☐ Yes ☐ No	Sputum microscopy result (e.g., AFB 1+):
Sputum culture for Mycobacterium tuberculosis	
Sputum culture result:	
Sputum culture interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
Chest X-ray done? ☐ Yes ☐ No	Chest X-ray date (DD/MM/YYYY):
Chest X-ray description/ findings/ result:	
Chest X-ray interpretation: ☐ Positive ☐ Negative	
Sputum microscopy interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
6 to 8-Month Evaluation Mycobacterium Tuberculosis	
Sputum specimen collected? ☐ Yes ☐ No	Sputum specimen date (DD/MM/YYYY):
Sputum microscopy for acid fast bacilli (AFB) done? ☐ Yes ☐ No	Sputum microscopy result (e.g., AFB 1+):
Sputum microscopy interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
Sputum culture for Mycobacterium tuberculosis	
Sputum culture result:	
Sputum culture interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
Chest X-ray done? ☐ Yes ☐ No	Chest X-ray date (DD/MM/YYYY):
Chest X-ray description/ findings/ result:	
Chest X-ray interpretation: ☐ Positive ☐ Negative	
Other test	
Has any other specimen been collected? ☐ Yes ☐ No	Date Specimen collected (DD/MM/YYYY):

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
NSU, MOHW, Jamaica

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TUBERCULOSIS CASE INVESTIGATION FORM 2

Test Requested:	Test result:
Test interpretation: ☐ Positive ☐ Negative ☐ Indeterminate ☐ Sample rejected	
REPEAT CHEST X-RAY	
Chest X-ray done? ☐ Yes ☐ No	Chest X-ray date (DD/MM/YYYY):
Chest X-ray description/ findings/ result:	Chest X-ray interpretation: ☐ Positive ☐ Negative
Additional Clinical Notes:	

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: -----
NSU, MOHW, Jamaica Page 3 of 5

Tuberculosis Case Investigation Form 2

SECTION 4 (TB FORM 2) - CLASSIFICATION

Final Classification of Case			
☐ Confirmed		☐ Pending	
☐ Discarded		☐ Inconclusive	
Confirmation Category			
☐ Bacteriologically confirmed		☐ Clinically diagnosed	
Anatomical site			
☐ Pulmonary		☐ Extra-pulmonary	
Drug resistance			
☐ None	☐ Rifampicin resistant	☐ Multi-drug resistant	☐ Extremely drug-resistant
Previous Treatment for tuberculosis			
Previously treated for tuberculosis?		Relapse case?	
☐ Yes ☐ No		☐ Yes ☐ No	
Treatment after lost to follow-up?		Treatment after failure?	
☐ Yes ☐ No		☐ Yes ☐ No	
Treatment status for current diagnosis			
☐ Completed		☐ Not completed	
Case outcome for current diagnosis			
☐ Cured		☐ Treatment failure	☐ Lost to follow-up
☐ Died		☐ Not evaluated	☐ Anti-TB Treatment completed
Cause of death			
(i)		(ii)	(iii)
Immediate		Intermediate	Underlying
Date of death (DD/MM/YYYY):			
REVIEW AND SIGNATURE			
Name of Investigator:		----- Signature of Investigator	Date Signed by Investigator (DD/MM/YYYY)
Comments:			
Name of Parish Medical Officer (Health):		----- Signature of MO(H)	Date signed by MO(H) (DD/MM/YYYY)

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
NSU, MOHW, Jamaica

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Treatment History

New case: A patient who has never been treated for TB or who has taken anti-TB drugs for less than one month.

Previously treated patients: Patients who have received 1 month or more of anti-Tb drugs in the past.

- i. **Relapse patients:** A patient previously treated for TB who was declared cured or treatment completed at the end of their most recent course of treatment, and is now diagnosed with a recurrent episode of TB.
- ii. **Treatment after failure patients:** A patient who has been previously treated for TB and whose treatment failed at the end of their most recent course of treatment.
- iii. **Treatment after loss to follow up patients:** A patient who has been previously treated for TB and were declared loss to follow-up at the end of their most recent course of treatment (previously known as treatment after default patients).
- iv. **Other previously treated patients:** Patients who have been previously treated for TB but whose outcome after their most recent course of treatment is unknown or undocumented

Definitions of treatment outcomes for drug-susceptible TB

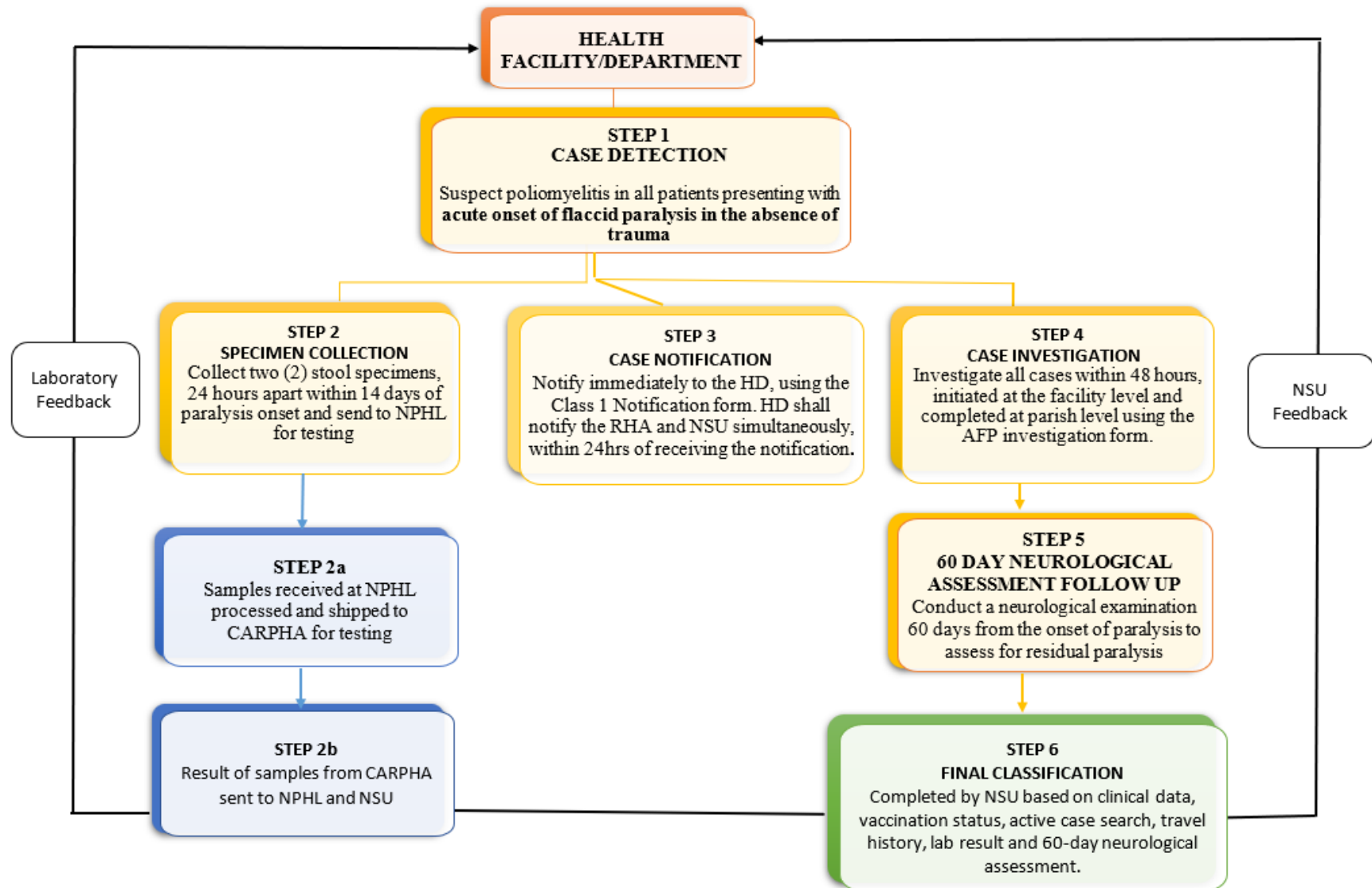
- i. **Cured:** A pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who was smear or culture negative in the last month of treatment and on at least one previous occasion.
- ii. **Treatment completed:** A TB patient who completed treatment without evidence of failure, but with no record to show that sputum smear or culture results in the last month of treatment and on at least one previous occasion were negative, either because tests were not done or because results are unavailable.
- iii. **Died:** A TB patient who dies for any reason before starting or during the course of treatment.
- iv. **Treatment failed:** A TB patient whose sputum or culture is positive at month 5 or later during treatment.
- v. **Loss to follow-up:** A TB patient who did not start treatment or whose treatment was interrupted for two (2) consecutive months or more.
- vi. **Not evaluated:** A TB patient for whom no treatment outcome is assigned. This includes cases "transferred out" to another treatment unit as well as cases for whom the treatment outcome is unknown to the reporting unit.
- vii. **Treatment success:** A TB patient who is cured or who has completed treatment.

Patient's Name: _____
Updated April 12, 2024

Medical Record Number: _____
NSU, MOHW, Jamaica

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Appendix 37: Acute Flaccid Paralysis (AFP) Flow Chart – Poliomyelitis



National Surveillance Unit, October 2024

Appendix 38: Acute Flaccid Paralysis (AFP) Investigation Form



National Surveillance Unit, National Epidemiology Branch,
ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



ACUTE FLACCID PARALYSIS INVESTIGATION

Complete this form for any person aged <15 years with acute flaccid paralysis, and for a person of any age in whom polio is suspected.

INITIAL DIAGNOSIS			
Initial Diagnosis/Notification: Acute Flaccid Paralysis			
Date of Notification: DD/MM/YYYY		Date Investigation Started: DD/MM/YYYY	
PATIENT'S DEMOGRAPHICS			
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		Patient's ID Number:	
First Name:		Middle Name(s):	
Last Name:		Pet Name(s):	
Patient's Maiden Name (if applicable):			
Sex Assigned at Birth: ☐ Male ☐ Female		Date of Birth: DD/MM/YYYY	Age:
Name of Facility Where Patient Seen:		Medical Record Number:	
Country of Residence:	Community:	Parish (Jamaica):	
House Number, Street Name:			
Landmark or directions to address:			
Phone Number:	Email Address:	Occupation:	
Name of School/Workplace:			
Address of School/Workplace:			
Phone Number:		Email Address:	
NEXT OF KIN			
First Name:		Last Name:	Pet Name(s):
NOK ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		NOK ID Number:	
Address: Lot/ Street/Community/Parish:			
Name of School/ Workplace:			
Address of School/Workplace:			
Phone number:		Email address:	
Relationship to the Patient:			
☐ Wife	☐ Son	☐ Uncle	☐ Cousin
☐ Husband	☐ Daughter	☐ Aunt	☐ Friend
☐ Mother	☐ Sister	☐ Grandmother	☐ Guardian
☐ Father	☐ Brother	☐ Grandfather	☐ Other (specify):

Name of patient:

Medical Record Number:

CLINICAL PROFILE

Date Patient Seen: DD/MM/YYYY

Date of Home Visit: DD/MM/YYYY

Sign or Symptom	Presence			Date of Onset
Prodrome				
Fever (current)	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
History of fever	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Respiratory symptoms	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Diarrhoea	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Associated Symptoms/Signs				
Muscle pain	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Meningeal signs	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Cranial nerve pair involvement	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Weakness/Paralysis				
Weakness/Paralysis	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
Other Symptoms/Signs				
	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY
	☐ Yes	☐ No	☐ Unknown	DD/MM/YYYY

Time over which weakness/paralysis developed: _____ ☐ day(s) ☐ hour(s)Weakness/Paralysis Progression: ☐ Ascending ☐ Descending ☐ Other Nadir (date weakness was worst): DD/MM/YYYY

Name of patient:

Medical Record Number:

Examination Findings Per Limb

Limb	Location of Weakness/Paralysis	Presence of Weakness/Paralysis	Power (/5)	Tone	Reflex detail	Sensation
Right upper limb	☐ Proximal	☐ Yes ☐ No ☐ Unknown	____/5	☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
	☐ Distal	☐ Yes ☐ No ☐ Unknown	____/5	☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
Left upper limb	☐ Proximal	☐ Yes ☐ No ☐ Unknown	____/5	☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
	☐ Distal			☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
Right lower limb	☐ Proximal	☐ Yes ☐ No ☐ Unknown	____/5	☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
	☐ Distal			☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
Left lower limb	☐ Proximal	☐ Yes ☐ No ☐ Unknown	____/5	☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
	☐ Distal			☐ Increased ☐ Normal ☐ Decreased ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown

Name of patient:

Medical Record Number:

CASE MANAGEMENT AND INVESTIGATION

Did the patient come in contact with any foreign travelers? ☐ Yes ☐ No ☐ Unknown

Did the patient travel in the Last 7-23 weeks: ☐ Yes ☐ No ☐ Unknown

If travelled: ☐ Domestic Travel ☐ International Travel ☐ Domestic and International Travel

If travelled, fill out the table below (Fill out if contact with foreign traveller as well):

Country Visited	City Visited	Date Arrived in Country	Date Departed Country
		DD/MM/YYYY	DD/MM/YYYY
		DD/MM/YYYY	DD/MM/YYYY
		DD/MM/YYYY	DD/MM/YYYY

Date returned to Jamaica DD/MM/YYYY

VACCINATION STATUS

Type of Vaccine	Number of doses	Date of last dose	Source of vaccination Information
OPV		DD/MM/YYYY	
IPV		DD/MM/YYYY	
Polio Vaccine, Unknown Route of Administration		DD/MM/YYYY	

Summary of Clinical Management:

HOSPITAL SUMMARY

Admitted to Hospital? ☐ Yes ☐ No Date of Admission: DD/MM/YYYY Date of Discharge: DD/MM/YYYY

Hospital: Ward: Medical Record Number:

Vital Status (Outcome): ☐ Alive ☐ Dead Date of Death: DD/MM/YYYY Time of Death: __:__ AM/PM

Place of Death: ☐ Home ☐ Dead on Arrival ☐ Ward ☐ ICU ☐ Other, Specify: _____

SAMPLE AND LAB INFORMATION

Was a sample taken? ☐ Yes ☐ No ☐ Unsure

What date was the specimen taken? DD/MM/YYYY

Which lab is the sample being sent to?

☐ National Public Health Laboratory ☐ University of the West Indies ☐ Other

Specify Other Laboratory:

Name of patient:

Medical Record Number:

Two separate stool samples should be taken at least 24 hours apart and within 14 days of onset of weakness/paralysis.

DIAGNOSTIC INVESTIGATION SCREENING

Virus and Intratypic Isolation

Stool Specimen- Laboratory Details							
Specimen Number	Date Specimen Collected	Date Specimen Sent to Lab	Date/time received by laboratory	Adequacy for testing	Quantitative Test result	Test result (Virus Isolation)	Test result (Virus Isolation)
1	DD/MM/YYYY	DD/MM/YYYY	DD/MM/YYYY	☐ Satisfactory - analysed ☐ Unsatisfactory- analysed ☐ Unsatisfactory – not analysed		☐ Positive - poliovirus ☐ Positive – Non-Polio Enterovirus (NPEV) ☐ Positive - Poliovirus and NPEV ☐ Negative ☐ Indeterminate ☐ Unknown	☐ P1 Sabin ☐ P2 Sabin ☐ P3 Sabin ☐ P1 Vaccine-derived ☐ P2 Vaccine-derived ☐ P3 Vaccine-derived ☐ P1 Wild
2	DD/MM/YYYY	DD/MM/YYYY	DD/MM/YYYY	☐ Satisfactory - analysed ☐ Unsatisfactory- analysed ☐ Unsatisfactory – not analysed		☐ Positive - poliovirus ☐ Positive – Non-Polio Enterovirus (NPEV) ☐ Positive - Poliovirus and NPEV ☐ Negative ☐ Indeterminate ☐ Unknown	☐ P1 Sabin ☐ P2 Sabin ☐ P3 Sabin ☐ P1 Vaccine-derived ☐ P2 Vaccine-derived ☐ P3 Vaccine-derived ☐ P1 Wild

Laboratory Comments:

Name of patient:

Medical Record Number:

CONTACT TRACING		
Contacts should be <5 years of age and not vaccinated within 30 days. List additional contacts on separate page.		
1. First Name:		Last Name(s):
Age: YY/MM		Date of Birth: DD/MM/YYYY
House Number, Street Name:		
Parish:		Community:
Phone Number:		Email Address:
Symptoms: ☐ Yes ☐ No		Specify symptoms:
Number of doses Polio Vaccine:		Date of last dose: DD/MM/YYYY
Specimen Collected: ☐ Yes ☐ No		Date Specimen Collected: DD/MM/YYYY
2. First Name:		Last Name(s):
Age: YY/MM		Date of Birth: DD/MM/YYYY
House Number, Street Name:		
Parish:		Community:
Phone Number:		Email Address:
Symptoms: ☐ Yes ☐ No		Specify symptoms:
Number of doses Polio Vaccine:		Date of last dose: DD/MM/YYYY
Specimen Collected: ☐ Yes ☐ No		Date Specimen Collected: DD/MM/YYYY
3. First Name:		Last Name(s):
Age: YY/MM		Date of Birth: DD/MM/YYYY
House Number, Street Name:		
Parish:		Community:
Phone Number:		Email Address:
Symptoms: ☐ Yes ☐ No		Specify symptoms:
Number of doses Polio Vaccine:		Date of last dose: DD/MM/YYYY
Specimen Collected: ☐ Yes ☐ No		Date Specimen Collected: DD/MM/YYYY
Classification		
Initial Diagnosis: AFP/Polio		Final Diagnosis:
Case classification:		
☐ Laboratory Confirmed	☐ Polio Compatible	☐ Vaccine Associated
☐ Vaccine Derived	☐ Inconclusive	☐ Pending
☐ Discarded		
Reason for classification:		
Transmission Classification: ☐ Imported ☐ Import-related ☐ Endemic ☐ Local Transmission ☐ Unknown		
Source country (If Imported or Import-Related):		
DEATH CLASSIFICATION		
Polio related death: ☐ Yes ☐ No ☐ Unknown		
Immediate Cause:		
Intermediate Cause:		
Underlying Cause:		

Name of patient:

Medical Record Number:

Investigator's Comments:		
Investigator's Details		
Date Investigation Completed: <i>DD/MM/YYYY</i>		
First Name:		
Last Name		
Professional Group:		
Phone number:		
Email Address:		
Name of Institution:		
Community:		
Parish:		
Investigator's Office Number/Street Name:		
Investigator's Health Region:		
Name of Parish MO(H):	Signature of MO(H):	Date signed by MO(H): <i>DD/MM/YYYY</i>

Appendix 39: Acute Flaccid Paralysis (AFP) - 60-Day Follow-Up Investigation



National Surveillance Unit, National Epidemiology Branch,
ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



ACUTE FLACCID PARALYSIS 60 DAY FOLLOW-UP INVESTIGATION

Complete this form for any person aged <15 years with acute flaccid paralysis, and for a person of any age in whom polio is suspected.

INITIAL DIAGNOSIS		
Initial Diagnosis/Notification:		
Date of Notification: DD/MM/YYYY	Date Investigation Started: DD/MM/YYYY	
PATIENT'S DEMOGRAPHICS		
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		Patient's ID Number:
First Name:	Middle Name(s):	
Last Name:	Pet Name(s):	
Patient's Maiden Name (if applicable):		
Sex Assigned at Birth: ☐ Male ☐ Female	Medical Record Number:	
Date of Birth: DD/MM/YYYY	Age:	
Country of Residence:	Parish (Jamaica):	
House Number, Street Name:		
Landmark or directions to address:	Community:	
Phone Number:	Email Address:	
NEXT OF KIN		
First Name:	Last Name:	
Phone number:	Email address:	
Address: Lot/ Street/Community/Parish:		
CLINICAL PROFILE		
Date Patient Seen: DD/MM/YYYY	Date of Home Visit: DD/MM/YYYY	

Sign or Symptom	Presence	Date of Onset
Associated Symptoms/Signs		
Cranial Pair Involvement	☐ Yes ☐ No ☐ Unknown	DD/MM/YYYY
Respiratory Symptoms	☐ Yes ☐ No ☐ Unknown	DD/MM/YYYY

Duration of Weakness/Paralysis: _____	☐ day ☐ hour (s)
Weakness/Paralysis Progression: ☐ Ascending ☐ Descending ☐ Other	Nadir: DD/MM/YYYY

Weakness/Paralysis Findings Per Limb

Limb	Presence of Weakness/Paralysis	Power (/5)	Location of Weakness/Paralysis	Reflex detail	Sensation
Right upper limb	☐ Yes ☐ No ☐ Unknown	____/5	☐ Proximal ☐ Distal ☐ Both	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
Left upper limb	☐ Yes ☐ No ☐ Unknown	____/5	☐ Proximal ☐ Distal ☐ Both	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
Right lower limb	☐ Yes ☐ No ☐ Unknown	____/5	☐ Proximal ☐ Distal ☐ Both	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown
Left lower limb	☐ Yes ☐ No ☐ Unknown	____/5	☐ Proximal ☐ Distal ☐ Both	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown	☐ Increased ☐ Normal ☐ Decreased ☐ Absent ☐ Unknown

Summary of Clinical Management:

Classification

Initial Diagnosis: AFP/Polio

Final Diagnosis:

Investigator's Comments:

Investigator's Details

Date Investigation Completed: DD/MM/YYYY

First Name:

Last Name

Professional Group:

Phone number:

Email Address:

Name of Institution:

Parish:

Community:

Office Number/Street Name:

Health Region:

Name of Parish MO(H):

Signature of MO(H)

Date signed by MO(H): DD/MM/YYYY

Appendix 40: Fever and Rash Case Investigation Form

 NATIONAL EPIDEMIOLOGY BRANCH		NATIONAL SURVEILLANCE UNIT, NATIONAL EPIDEMIOLOGY BRANCH, ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew Email: surveillance@moh.gov.jm, Telephone 876-633-7937		 MINISTRY OF HEALTH & WELLNESS	
FEVER AND RASH INVESTIGATION					
INITIAL DIAGNOSIS					
☐ Measles ☐ Rubella ☐ Other Rash Illness Specify: _____					
Date of Notification: DD/MM/YYYY			Date Investigation Started: DD/MM/YYYY		
PATIENT'S DEMOGRAPHICS					
Patient's ID Type:			Patient's ID Number:		
☐ Driver's License ☐ TRN ☐ Passport ☐ NIN					
First Name:			Middle Name(s):		
Last Name:			Pet Name(s):		
Patient's Maiden Name (if applicable):					
Sex Assigned at Birth: ☐ Male ☐ Female			Medical Record Number:		
Date of Birth: DD/MM/YYYY			Age:		
Country of Residence:			Parish (Jamaica):		
House Number, Street Name:					
Landmark or directions to address:			Community:		
Phone Number:			Email Address:		
Occupation:					
Name of Workplace/School:					
Address of School/Workplace:					
Phone Number:			Email Address:		
NEXT OF KIN					
First Name:			Last Name:		
Pet Name(s):					
Address: Lot/ Street/ Community/Parish:					
Phone number:			Email address:		
Relationship to the Patient:					
☐ Wife	☐ Son	☐ Uncle	☐ Cousin		
☐ Husband	☐ Daughter	☐ Aunt	☐ Friend		
☐ Mother	☐ Sister	☐ Grandmother	☐ Guardian		
☐ Father	☐ Brother	☐ Grandfather	☐ Other (specify):		
VACCINATION HISTORY					
Type of Vaccine	Number of doses	Date of last dose	Source of vaccination Information (Vaccination Card, Medical Record, Verbal)		
Measles		DD/MM/YYYY			
Rubella		DD/MM/YYYY			
Measles, Rubella		DD/MM/YYYY			
Measles, Mumps, Rubella		DD/MM/YYYY			

FEVER AND RASH INVESTIGATION

MEDICAL RECORD NUMBER: _____

CLINICAL PROFILE

Date Seen: DD/MM/YYYY

Date of Home Visit: DD/MM/YYYY

☐ History of Fever ☐ Recorded Fever ☐ No History of Fever or Recorded Fever ☐ Unknown

Recorded temperature: _____

Duration (days): _____

Date of Onset: DD/MM/YYYY

☐ Maculopapular Rash☐ Vesicular Rash☐ No History of Rash☐ Unknown

If other Rash (Specify): _____

Duration (days): _____

Date of Onset: DD/MM/YYYY

☐ Cough☐ Coryza☐ Conjunctivitis☐ Koplik Spots☐ Lymphadenopathy☐ Arthralgia☐ Other Unlisted Symptoms: _____**MEDICAL HISTORY**

Pre-existing Conditions

☐ Asthma☐ Diabetes mellitus☐ HIV/AIDS☐ Malignancy☐ Autoimmune☐ Heart disease☐ Hypertension☐ Sickle cell disease☐ Chronic Obstructive
Pulmonary Disease (COPD)☐ Kidney disease☐ Liver dysfunction☐ Other (Specify): _____**PREGNANCY**Is Patient Pregnant: ☐ Yes ☐ No ☐ Unsure ☐ Not Applicable

Gestational Age: _____

Trimester: ☐ First ☐ Second ☐ Third**HOSPITAL SUMMARY**Admitted to Hospital? ☐ Yes ☐ No

Date of Admission: _____

Date of Discharge: _____

DD/MM/YYYY

DD/MM/YYYY

Hospital: _____

Ward: _____

Medical Record Number: _____

Vital Status(Outcome): ☐ Alive ☐ Dead

Date of Death: _____

Time of Death: __: __ AM/PM

DD/MM/YYYY

Place of Death: ☐ Home☐ Dead on Arrival☐ Ward☐ ICU☐ Other, Specify: _____**SAMPLE AND LAB INFORMATION**Was a sample taken? ☐ Yes ☐ No ☐ Unsure

If yes, what specimen was taken?

☐ Serum☐ Nasopharyngeal Swab☐ Urine

Other, Specify: _____

What date was the specimen taken? DD/MM/YYYY

Which lab is the sample being sent to?

☐ National Public Health Laboratory☐ University of the West Indies☐ Other

Specify Other Laboratory: _____

FEVER AND RASH INVESTIGATION

MEDICAL RECORD NUMBER:

LAB TEST(S) REQUESTED

LAB RESULT(S)

DATE RESULT RECEIVED

☐ Measles IgM	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY
☐ Measles IgG	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY
☐ Rubella IgM	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY
☐ Rubella IgG	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY
☐ Viral Culture	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY
☐ Viral Genotype	Specify Genotype:				DD/MM/YYYY
☐ RT-PCR	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY
☐ Other, Specify:	☐ Positive	☐ Negative	☐ Indeterminate	☐ Unknown	DD/MM/YYYY

CASE MANAGEMENT AND INVESTIGATION

Were active case searches conducted? ☐ Yes ☐ No ☐ Unknown

Was the patient in contact with any pregnant woman? ☐ Yes ☐ No ☐ Unknown

Are there other cases present in the case's community of residence? ☐ Yes ☐ No ☐ Unknown

Travelled in the Last 7-23 weeks: ☐ Yes ☐ No ☐ Unknown

If travelled: ☐ Domestic Travel ☐ International Travel ☐ Domestic and International Travel

If travelled, fill out the table below:

COUNTRY VISITED	CITY VISITED	DATE ARRIVED IN COUNTRY	DATE DEPARTED COUNTRY
		DD/MM/YYYY	DD/MM/YYYY
		DD/MM/YYYY	DD/MM/YYYY
		DD/MM/YYYY	DD/MM/YYYY

Date returned to Jamaica DD/MM/YYYY

Setting where patient was suspected to have been infected and names of ill contacts
(SETTING = Household Contact, Health Facility, Work, Community, Conveyance, Other)

NAME	DATE OF CONTACT	CONTACT SETTING
	DD/MM/YYYY	
	DD/MM/YYYY	
	DD/MM/YYYY	
	DD/MM/YYYY	
	DD/MM/YYYY	
	DD/MM/YYYY	
	DD/MM/YYYY	
	DD/MM/YYYY	

RESPONSE MEASURES

Ring vaccination completed? ☐ Yes ☐ No ☐ Unknown

Was rapid coverage monitoring done? ☐ Yes ☐ No ☐ Unknown

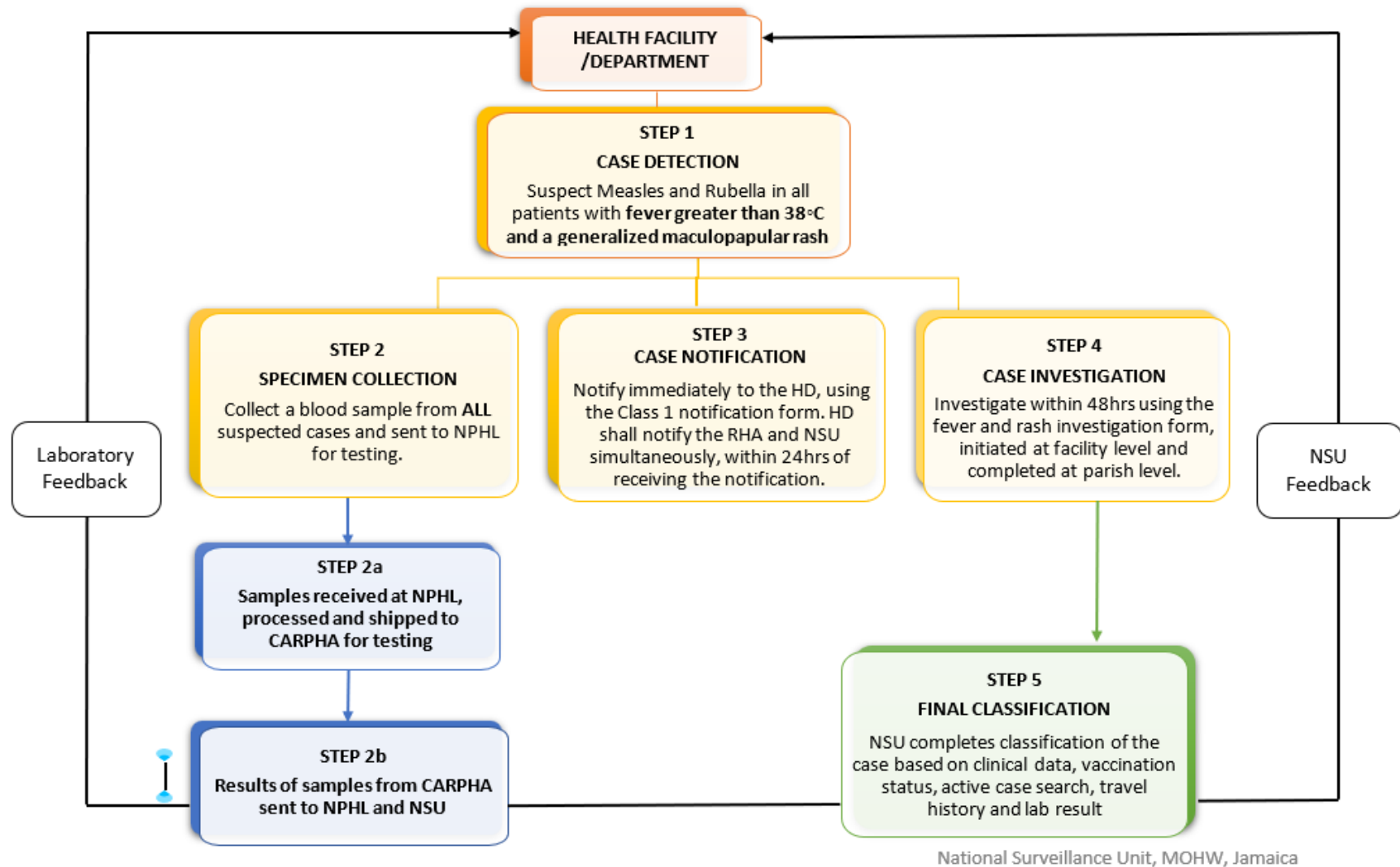
Cases followed up for 30 days after rash onset? ☐ Yes ☐ No ☐ Unknown

FEVER AND RASH INVESTIGATION

MEDICAL RECORD NUMBER:

CLASSIFICATION		
INDEX DIAGNOSIS: MEASLES/RUBELLA		FINAL DIAGNOSIS:
VARIANTS (IF APPLICABLE):		
Case classification:		
☐ Laboratory Confirmed	☐ Probable	☐ Pending
☐ Inconclusive	☐ Discarded	
Reason for classification:		
Is the case epidemiologically linked? ☐ Yes ☐ No ☐ Unknown		
Aetiology Category: ☐ Imported case ☐ Import-related cases ☐ Endemic ☐ Vaccine Associated		
☐ Unknown		
Source country (If Imported or Import-Related):		
DEATH CLASSIFICATION		
Measles/Rubella related death: ☐ Yes ☐ No ☐ Unknown		
Immediate Cause:		
Intermediate Cause:		
Underlying Cause:		
Investigator's Comments:		
Investigator's Details		
Date Investigation Completed: DD/MM/YYYY		
First Name:		
Last Name:		
Professional Group:		
Phone number:		
Email Address:		
Name of Institution:		
Community:		
Parish:		
Investigator's Office Number/Street Name:		
Investigator's Health Region:		
Name of Parish MO(H):	Signature of MO(H):	Date signed by MO(H): DD/MM/YYYY

Appendix 41: Fever and Rash Flow Chart – Measles and Rubella



Appendix 42: Hepatitis B Case Investigation Form

PATIENT INFORMATION									
Parish				Date report received at Health Department (dd/mm/yy)					
Name				Age			Gender M F		
If Child, Name of Parent / Guardian (<i>relationship to case</i>)				Date of Birth			Hosp name / Docket Number		
Telephone Number				Landmarks					
Home Address				Occupation					
CLINICAL DATA									
Date of onset of illness								Immunization History	
SYMPTOMS	Y	N	SYMPTOMS	Y	N	SYMPTOMS	Y	N	Number of doses of Hepatitis B: Date of last dose:
Fever			Nausea			Vomiting			
Anorexia			Itching/Rash			Jaundice			
Malaise			Arthralgia			Palpable liver			
Fatigue/Lethargy			Abd. Pain/discomfort			Dark urine, pale stools			
Is / Was this patient hospitalized?		Hosp & Date of Admission		Date of discharge		Outcome of illness			
						Survived			
						Died Date / /			
Was patient identified through blood donation testing?				Date of donation		Date result of test rec'd			
Date of repeat HbsAg (>6 mths)		Which Laboratory?		Lab number and Result of repeat HbsAg Test					
EXPOSURE HISTORY									
	Y	N	Dates	Details					
Sexual contact with Hepatitis Case									
Blood transfusion in past 6 months									
Haemodialysis? Injections? Injecting drug use?									
Hospitalized in past 6 months									
Tattooing, ear piercing, acupuncture									
Child of Hepatitis B surface antigen positive mother									
Resident in institution, attends daycare									
Healthcare worker with occupational sharps injury									
LABORATORY DATA									
Specimen	Date collected	Date rec'd	Condition	Test	Result	Date sent	Comment		
Blood				HBsAg					
Blood				Anti-HBc IgM					
Blood				HBeAg					
FINAL CASE CLASSIFICATION						Date reported:			
LABORATORY CONFIRMED HEPATITIS B CASE HEPATITIS						Signature: MO(H)			
INDETERMINATE						signature:			
						Date:			

CONTACT INVESTIGATION							
CONTACT'S NAME (Level of contact with case e.g. Household or sexual)	SEX	AGE	ADDRESS	Hepatitis B Vaccine status	Number of doses	HbsAg Test Date	RESULT
CONTACTS RECEIVING HEPATITIS B VACCINE							
NAME	DATE FIRST DOSE		DATE SECOND DOSE		DATE THIRD DOSE		
OTHER RELEVANT INFORMATION							

Appendix 43: Microcephaly Births Line Listing

Date of Birth	Maternal Record #	BBA	Multiple Birth	GA	Weight gms	Length cms	Head Circumference cms	Sex	Outcome			APGAR		Describe any anomalies/ malformations
									LB	SB	NND	1min	5min	
		Yes/No	Yes/No					Male/ Female						
		Yes/No	Yes/No					Male/ Female						
		Yes/No	Yes/No					Male/ Female						
		Yes/No	Yes/No					Male/ Female						
		Yes/No	Yes/No					Male/ Female						

Instructions: Use a separate line for babies in multiple births

Key: BBA: Born before arrival GA: Gestational Age LB: Live Birth SB:
Still Birth NND: Neonatal Death

Appendix 44: Congenital Syndrome Associated with Zika Virus Case Investigation Form

MINISTRY OF HEALTH



JAMAICA

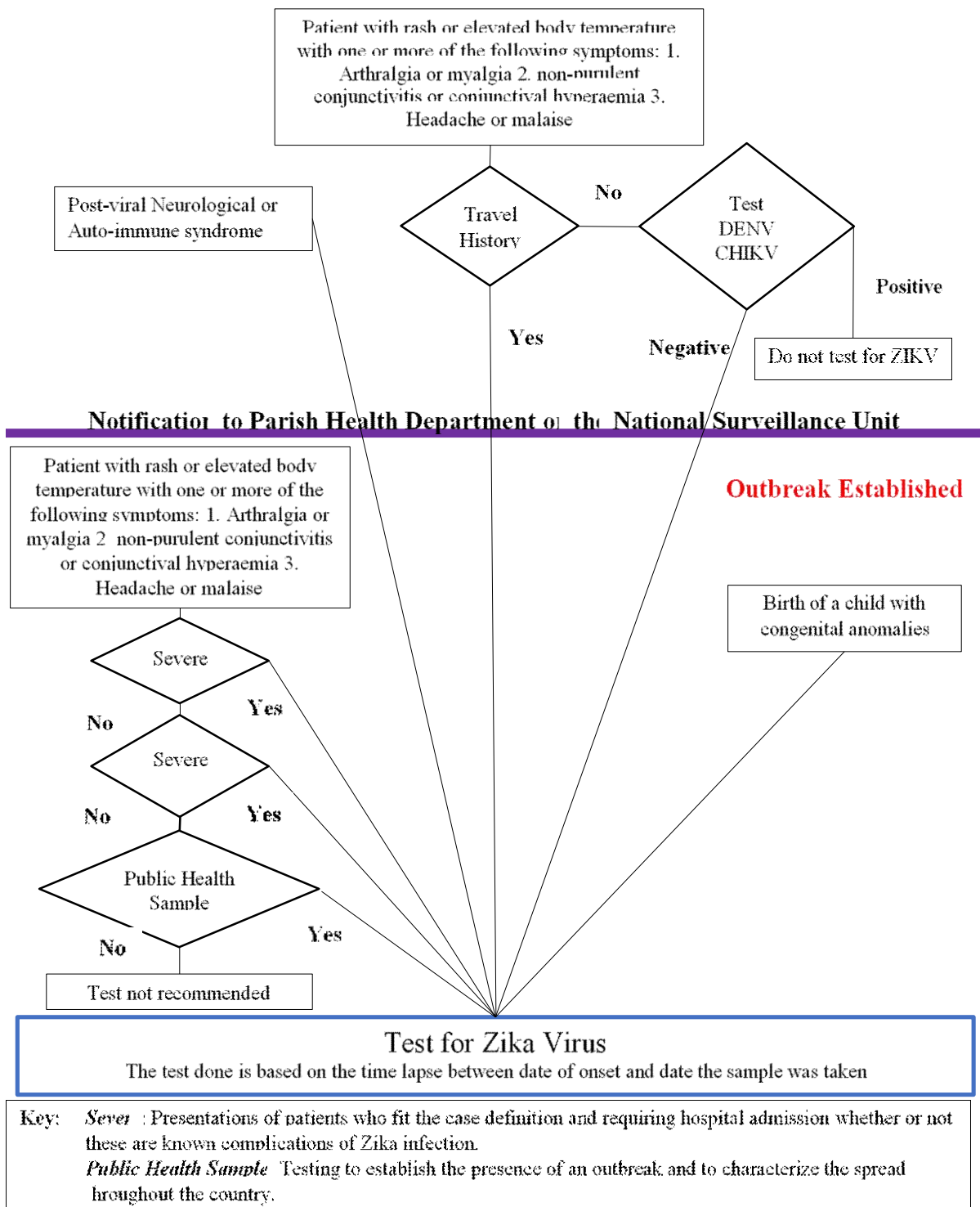
CONGENITAL SYNDROME ASSOCIATED WITH ZIKA VIRUS INFECTION CASE INVESTIGATION FORM			
Reporting Centre:		Date of Report / / (dd/mmm/yyyy)	
1. Patient (Infant) Information			
FIRST NAME:		Last Name:	
Sex: Male Female (circle the one that applies)		D.O.B: / / (dd/mmm/yyyy)	
2. Mother's Information			
FIRST NAME:		Last Name:	
Address		D.O.B: / / (dd/mmm/yyyy)	
Phone #		Age in years:	
Community (STATIN):		PARISH	
Occupation			
LMP: / / (dd/mmm/yyyy)		EDD: / / (dd/mmm/yyyy)	
Gravida: Para:			
Ultrasound: EDD		Gestational Age at Birth (Obstetrician's Recommendation): +---/40	
Multiple Pregnancy: Yes/ No		If yes, give number:	
A. Mother's Antenatal History (Date of onset: dd/mmm/yyyy, if exact date is unknown give the best estimate)			
Rash: Yes ☐ No ☐ Date of Onset / /		Describe the rash or other clinical features:	
Fever: Yes ☐ No ☐ Date of Onset / /			
Conjunctivitis: Yes ☐ No ☐ Date of Onset / /			
Joint Pain: Yes ☐ No ☐ Date of Onset / /			
Muscle Pain: Yes ☐ No ☐ Date of Onset / /			
HIV Status:		Syphilis Status:	
B. Infant's Anthropometric Findings			
Measurements	Date: / /	Date: / /	Date: / /
Gestational Age:	+---/40		
Compare to Standard Charts	Circle one that applies		
Head Circumference (cm)	____.____	<-3 <-2 Normal	____.____
Weight (kg)	____.____	<-3 <-2 Normal	____.____
Length (cm)	____.____	<-3 <-2 Normal	____.____
Classification (Circle one): Normal Small for Gestational Age Microcephaly Severe Microcephaly			
C. Infant's Clinical Findings (Describe)			
1. Cranial-facial disproportion	Yes No		
• Redundant scalp	Yes No		
2. Other Anthropometric Disproportion	Yes No		
3. Hypertonia or Spasticity	Yes No		
4. Irritability	Yes No		
5. Epileptic Seizures	Yes No		
6. Visual Impairment	Yes No		
7. Auditory Impairment	Yes No		
8. Club feet	Yes No		
9. Arthrogryposis	Yes No		
10. Others (specify)			

Name: _____

Date of Birth ____/____/____

D. INVESTIGATIONS & RESULTS			
	Y / N	Date done/ received (dd/mm/yyyy)	Describe Findings
1. Cranial Ultrasound	☐ ☐	____/____/____	
2. MRI Brain	☐ ☐	____/____/____	
3. CT Brain	☐ ☐	____/____/____	
4. Ophthalmologic evaluation	☐ ☐	____/____/____	
5. Auditory Test	☐ ☐	____/____/____	
Findings (answer all)			
-Cerebral hypoplasia	Yes No	-Lissencephaly	Yes No
-Hypoplasia or agenesis of the corpus callosum	Yes No	-Focal retinal pigment epithelium changes	Yes No
-Cerebral calcifications	Yes No	-Chorioretinal atrophy	Yes No
-Alterations in the cerebral ventricles	Yes No	-Optic nerve hypoplasia	Yes No
-Abnormalities of the posterior fossa	Yes No	- Iris colobomas	Yes No
	Date Collected dd/mm/yyyy	Results	
a. Toxoplasmosis	____/____/____		
b. Rubella	____/____/____		
c. Herpes Simplex Virus	____/____/____		
d. HIV	____/____/____		
e. Syphilis	____/____/____		
f. Zika virus	____/____/____		
1) Blood: Zika PCR	____/____/____		
2) Urine: Zika PCR	____/____/____		
3) Blood: Zika IgM	____/____/____		
g. Blood: Dengue IgM	____/____/____		
E. OUTCOME			
	Y / N	Date (dd/mm/yyyy)	Comments
a. Nursery Admission	☐ ☐	____/____/____	
b. NCU/ICU Admission	☐ ☐	____/____/____	
c. Discharge Home	☐ ☐	____/____/____	
d. Died	☐ ☐	____/____/____	
F. Classification: Congenital Syndrome Associated with Zika Virus Infection			
Circle One: Not Suspected Suspected Probable Confirmed Unable to Classify			
Reason for the classification:			
Medical Officer (Health)			
Name: _____		Signature: _____	
Date: _____			

Appendix 45: Algorithm for Surveillance of Zika Virus Infection and Laboratory Testing



Appendix 46: Zika Case Investigation Form



MINISTRY OF HEALTH JAMAICA

ZIKA FEVER CASE INVESTIGATION FORM											
Reporting Centre:					Date of Report / / (dd/mm/yy)						
1. Patient information											
Name				Age (yrs)		Sex: M F					
Address			Phone #		D.O.B: / / (dd/mm/yy)						
Community (STATIN):				Case #		Occupation					
2. Clinical Data											
Date of onset of illness / / (dd/mm/yy)				(dd/mm/yy)							
Clinical features		Y / N		Date of Onset		Clinical features		Y / N		Date of Onset	
Fever		☐ / ☐		_/_/_		Non-purulent Conjunctivitis		☐ / ☐		_/_/_	
Arthralgia		☐ / ☐		_/_/_		Lower Limb Oedema		☐ / ☐		_/_/_	
Is the Arthralgia Severe?		☐ / ☐				Periarticular oedema		☐ / ☐		_/_/_	
Arthralgia (joint pain) – Circle/list joints involved						Skin manifestations Describe:		☐ / ☐		_/_/_	
Hand: R L		Wrist R L		Foot: R L		Ankle: R L		Myalgia		☐ / ☐	
Others:						Back pain		☐ / ☐		_/_/_	
Clinical features		Y / N		Date of Onset		Headache		☐ / ☐		_/_/_	
Arthritis		☐ / ☐		_/_/_		Retro-orbital Pain		☐ / ☐		_/_/_	
Abdominal Pain		☐ / ☐		_/_/_		Asthenia (generalized weakness)		☐ / ☐		_/_/_	
Vomiting		☐ / ☐		_/_/_		Others: (Specify)		☐ / ☐		_/_/_	
Diarrhoea		☐ / ☐		_/_/_				☐ / ☐		_/_/_	
Additional Clinical Features:											
Risk Factors:											
Pregnancy		☐				Other:		☐		☐	
Diabetes		☐						☐		☐	
Cardiovascular diseases		☐						☐		☐	
Sickle Cell Disease		☐						☐		☐	
Clinical diagnosis:											
Is/was this patient hospitalised?		Y		N		Date (s)		Outcome of illness			
		☐		☐		_/_/_					
Resolution of symptoms		☐		☐		_/_/_					

3. Exposure and Travel History							
	Y	N	Date	Details			
Has the patient travelled to a Zika fever endemic/epidemic area within the past 2 weeks	☐	☐					
Has the patient been in contact with a Zika fever case within the past 2 weeks	☐	☐					
Country, Endemic/Epidemic Area	Arrival-Date & Time		Departure-Date & Time	Accommodations			
Visitors from abroad - Country	Date of Arrival		Date of Departure	Remarks			
Places visited in the past 2 weeks – emphasis on places visited in the daylight hours							
4. LABORATORY DATA							
Specimen	Date collected	Date rec'd	Condition	Test	Result	Date sent	Comment
First blood specimen				Virus Isolation			
				IgMELISA			
Second blood specimen				IgG			
				RT-PCR			
5. ENVIRONMENTAL SURVEY							
Community type	☐ Planned		☐ Unplanned				
Aedes population :	☐ Aegypti		☐ Albopictus				
Water supply:	☐ Piped		☐ Stored				
Solid waste collection:	☐ Frequent (at least once weekly)		☐ Infrequent				
AEDES INDICES SURROUNDING COMMUNITY:							
	Home		Workplace/School				
Premises index:							
Container index:							
Breteaues index:							
INVESTIGATOR:							
Name:		Signature:		Date:			
TO BE COMPLETED BY THE MEDICAL OFFICER OF HEALTH							
6. Final Case Classification:	☐ Suspected Case ☐ Imported ☐ Autochthonous			☐ Confirmed Case ☐ Imported ☐ Autochthonous ☐ Discarded			

Appendix 47: Contact Tracing and Community Fever Surveillance

The methodology for community fever surveillance is different based on the epidemiological situation. The following scenarios are considered in this document:

- A. Imported Zika fever
- B. One Autochthonous Case
- C. Cluster of Suspected Zika fever

A. Imported Zika Fever

Imported Zika fever cases (suspected and confirmed) should be investigated and a case investigation completed using the Case Investigation Form as a guide. The investigation is to be reviewed by the Parish Medical Officer of Health (MO(H)) and forwarded to the National Surveillance Unit (NSU)

Travel Companions: persons with similar exposure. That is, persons who travelled with the case should be followed for at least 12 days (maximum intrinsic incubation period) for the development of symptoms of Zika fever.

Household Contacts: household contacts should be followed for at least 22 days (total of the maximum extrinsic and intrinsic incubation period) after the onset of symptoms or after the person returned from overseas.

School and Work Contacts: If persons with the diagnosis of Zika fever during the likely period for transmission went to work or school the appropriate officer (nurse, HR Department, Manager, Principal, Guidance Counsellor) should be sensitized to report any new case or reported illness to the Health Department.

Community: Households/Premises within a **200m** radius from the case should be visited **at least** eight days after the onset of symptoms. Persons fitting the case definition should have a case investigation form completed at first contact. Blood should be taken from 1 in 10 persons that fit the case definition. That is a case investigation form should be completed for all suspected cases and blood from 1 in 10 of the cases. Community Health Alert Cards should be distributed to each household visited.

The following information should be obtained for each community surveyed:

- Total number of households in the **200m** radius
- Number of households visited
- Number of households interviewed

- Number of persons with fever
- Number with fever and joint pain
- Number of Community Health Alert Cards distributed

Note Well: The symptoms of Zika fever are non-specific and the following should be considered in the differential:

1. Dengue – often cases fit the definition for Zika virus infection, chikungunya fever and dengue fever.
2. Chikungunya - often cases fit the definition for Zika virus infection, Chikungunya fever and dengue fever.
3. Fever and rash – persons presenting fitting the Case Definition for this Class 1 Notifiable Disease and should be investigated on first contact and blood taken (first contact) for all.
4. Malaria
5. Leptospirosis

Community Health Alert Card

Sample

Front



Community Health Alert Card

To: The Resident

The _____ Health Department is conducting surveillance for a mosquito borne disease in your area. If you become ill with fever

In the next three (3) weeks, please visit your health center or healthcare provider and present this card.

Back

Community Health Alert Card

To: The Physician

The patient presenting this card may have been exposed to a communicable disease. Please contact the Medical Officer of Health for the Parish.

Medical Officer of Health

_____ Health Department

Address: _____

Telephone: _____

B. One Autochthonous Case

Zika fever cases (suspected and confirmed) should be investigated and a case investigation completed using the Case Investigation Form as a guide. The investigation is to be reviewed by the Parish Medical Officer of Health (MO(H)) and forwarded to the National Surveillance Unit (NSU).

Household Contacts: household contacts should be followed for at least 22 days after the onset of symptoms or after the person returned from overseas.

School and Work Contacts: If persons with the diagnosis of Zika fever during the likely period for transmission went to work or school the appropriate officer (nurse, HR Department, Manager, Principal, Guidance Counsellor) should be sensitized to report any new case or reported illness to the Health Department.

Community: Households/Premises within a **200m** radius from the case should be visited **at least** eight days after the onset of symptoms. Persons fitting the case definition should have a case investigation form completed at first contact. Blood should be taken from 1 in 10 persons that fit the case definition. That is a case investigation form should be completed for all suspected cases and blood for 1 in 10 of the cases. Community Health Alert Cards should be distributed to each household visited.

The following information should be obtained for each community surveyed:

- Total number of households in the **200m** radius
- Number of households visited
- Number of households interviewed
- Number of persons with fever
- Number with fever and joint pain
- Number of Community Health Alert Cards distributed

Note Well: The symptoms of Zika fever are non-specific and the following should be considered in the differential:

1. Dengue – often cases fit the definition for Zika virus infection, chikungunya fever and dengue fever.
2. Chikungunya - often cases fit the definition for Zika virus infection, Chikungunya fever and dengue fever.
3. Fever and rash – persons presenting fitting the Case Definition for this Class 1 Notifiable Disease and should be investigated on first contact and blood taken (first contact) for all.
4. Malaria
5. Leptospirosis

C. Cluster of Suspected Zika Fever

In the event that there is a cluster of cases (2 or more cases) with symptoms suggestive of Zika fever the parish team should investigate as an outbreak. An outbreak report (see Outbreak Reporting Form on pages 16 and 17) should be written and include a line listing, an epidemiological curve for the cases, hypothesis as to the cause and spread and interventions/ actions taken or to be taken. The investigation should be reviewed by the MO(H) and forwarded to the NSU.

Example of a Line Listing:

	Name	DOB	A	S	Address	Telephone	Fever	Rash	Joint pain	Where	Other	DOO	Treatment	Travel History	Visitors from Overseas	Comments
1																
2																
3																
4																
5																
6																
7																
8																
9																
10																
11																
12																
13																
14																
15																
16																
17																
18																
19																
20																
21																
22																
23																
24																
25																

Based on the report received about the cluster, a geographical area for community surveillance should be defined. A map of the area and the defined boundaries would be useful for the investigating team and for the reviewing team.

A house-to-house (premises-to-premises) fever surveillance should be conducted in the defined area. Persons fitting the case definition should have a case investigation Form completed at first contact. Blood should be taken from 1 in 10 persons that fit the case definition. Please note that a case investigation form is to be completed for all suspected cases and blood taken from 1 in 10 of the suspected cases. Community Health Alert cards should be distributed to each household visited.

The following information should be obtained for each community surveyed:

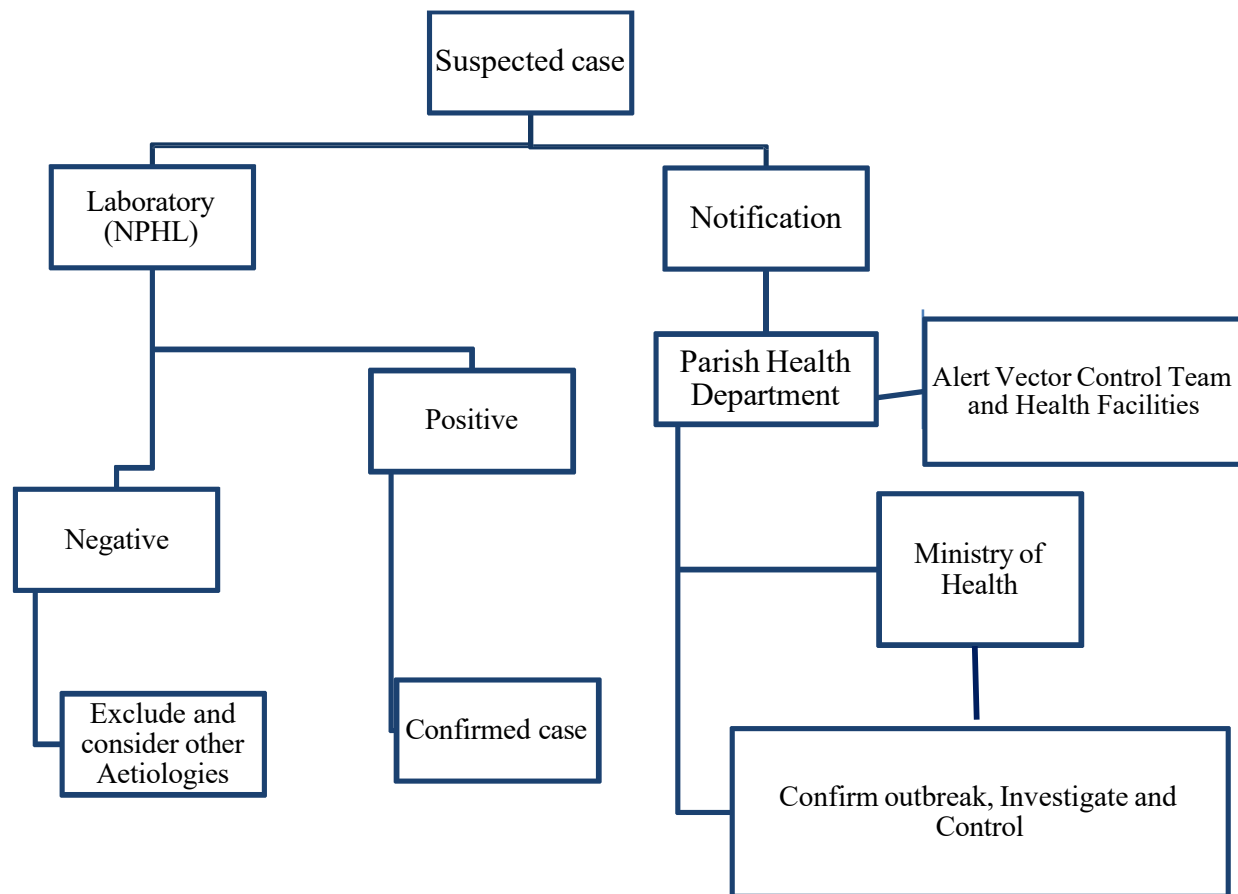
- Total number of households in the defined area
- Number of households visited
- Number of households interviewed
- Number of persons with fever
- Number with fever and joint pain
- Number of Community Health Alert Cards distributed

Given that the symptoms for Zika fever are non-specific and the following conditions should be considered in the differential:

1. Dengue – often cases fit the definition for Zika virus infection, chikungunya fever and dengue fever.
2. Chikungunya - often cases fit the definition for Zika virus infection, Chikungunya fever and dengue fever.
3. Fever and rash – persons presenting fitting the Case Definition for this Class 1 Notifiable Disease and should be investigated on first contact and blood taken (first contact) for all.
4. Malaria
5. Leptospirosis

It is likely that follow-up visits will need to be done in the communities with clusters.

Appendix 48: Algorithm for Notification and Intervention of a Suspected Case of Zika



Appendix 49: Accidental Poisoning Investigation Form

PATIENT INFORMATION						
Parish		Date report received at Health Department		Date Accidental poisoning occurred		
Name			Age YRS MTHS		Gender M F	
Name of Parent / Guardian (Relationship to Case)			Date of Birth		Hosp Name / Docket Number	
Telephone Number			Landmarks			
Home Address			Where did poisoning occur? <i>Circle appropriate site</i> Home Day Care center School Other, _____			
CLINICAL DATA						
Substance suspected of causing poisoning			Quantity ingested			
SYMPTOMS	Y	N	SYMPTOMS	Y	N	SYMPTOMS
Headache			Drowsiness			Flushing of skin
Vomiting			Shortness of breath			Dizziness
Diarrhoea			Convulsions			Delirium
Burning sensation in mouth			Loss of consciousness			Weakness
Was this patient hospitalized? Where? Y N Adm. Date: Disch. Date Hospital Name			Outcome of Illness Survived Resolved Died Sequelae present		If patient died, state date of death Post Mortem done? Y N DK	
Treatment given at Home (i.e. home remedy etc)				Treatment given at Health Care Facility		

EXPOSURE HISTORY					
Describe the circumstances under which the poisoning incident occurred					
LABORATORY DATA			ACTIONS TAKEN / RECOMMENDATIONS Name/Position of person completing the form: Signature: Date:		
SPECIMEN	DATE	RESULT			
FINAL CLASSIFICATION CONFIRMED CASE DISCARDED CASE					
SOCIAL AND ENVIRONMENTAL CONDITIONS					
GENERAL STATUS					
SAFETY FEATURES IN HOME					
	Y	N		Y	N
Presence of appropriate cupboards			Presence of doors		
Presence of appropriate storage areas for household chemicals			Inaccessible to children		
Presence of appropriate storage areas for pharmaceuticals			Presence of child-proof locks		
Presence of appropriate storage areas for other chemicals					
OTHER RELEVANT INFORMATION (Physical Description of agent implicated)					

Appendix 50: Congenital Syphilis Investigation Form

 MINISTRY OF HEALTH & WELLNESS	NATIONAL SURVEILLANCE UNIT, NATIONAL EPIDEMIOLOGY BRANCH, ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew Email: surveillance@moh.gov.jm, Telephone 876-633-7937	 NATIONAL EPIDEMIOLOGY BRANCH
SUSPECTED CONGENITAL SYPHILIS/SYPHILIS-EXPOSED INFANT		
Date of Notification: (DD/MM/YYYY)	Date Investigation Started: (DD/MM/YYYY)	
INFANTS'S DEMOGRAPHICS (Use mother's data if infant's details not yet known)		
Patient's ID Type:	Patient's ID Number:	
☐ National Identification Number (NIN) ☐ Other Specify:		
First Name:	Middle Name(s):	
Last Name:	Pet Name(s)	
Sex Assigned at Birth: ☐ Male ☐ Female	Date of Birth: (DD/MM/YYYY)	Age: Years/Months/Weeks/Days:
Name of Facility Where Infant Seen:	Medical Record Number:	
Country of Residence:	Parish (Jamaica):	
House Number, Street Name:		
Landmark or directions to address:	Community:	
MOTHER'S INFORMATION (no need to repeat if already in demographic data above)		
ID Type:	Mother's ID Number:	
☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		
First Name:	Middle Name(s):	
Last Name:	Pet Name(s)	
Mother's Maiden Name (If Applicable):		
Facility where mother seen:	Medical Record Number:	
House Number, Street Name:	Parish (Jamaica):	
Phone Number:	Email Address:	
Occupation:		
Phone Number:		
Name of School /Workplace:	Email Address:	
Address of School/Workplace:		
Marital Status:		
☐ Married	☐ Visiting Union	☐ Widowed
☐ Common Law	☐ Single	☐ Divorced
☐ Separated ☐ Other (specify):		
NEXT OF KIN OF MOTHER		
NOK's First Name:	NOK's Last Name:	
NOK's Address: Lot/ Street/Community/Parish:		
NOK's Phone number:	NOK's Email address:	

UPDATED FEBRUARY 9, 2024
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CONGENITAL SYPHILIS INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

MOTHER'S CLINICAL PROFILEWas the mother tested for syphilis? ☐ Yes ☐ No ☐ Unknown

Mother's Test (Current Pregnancy)	Result	Titre	Date Test Done (DD/MM/YYYY)
Treponemal Test			
SD Bioline	☐ Positive ☐ Negative ☐ Indeterminate		
Other Treponemal Test (e.g., TPPA, MHA-TP)	☐ Positive ☐ Negative ☐ Indeterminate		
Non-Treponemal Test			
VDRL	☐ Reactive ☐ Non-Reactive ☐ Unknown		
TRUST	☐ Reactive ☐ Non-Reactive ☐ Unknown		
RPR	☐ Reactive ☐ Non-Reactive ☐ Unknown		

Did mother have a syphilis test (VDRL/TRUST) done in her previous pregnancy? ☐ Yes ☐ No ☐ UnknownWas the VDRL/TRUST result reactive in her previous pregnancy? ☐ Yes ☐ No ☐ UnknownWas the mother treated for syphilis in previous pregnancy? ☐ Yes ☐ No ☐ Unknown**MOTHER'S MEDICATION HISTORY (including previous pregnancy)**

Medication	Dose	Frequency	Route of administration	Date administered (DD/MM/YYYY)	Number of doses
☐ Benzathine penicillin G					
Dose 1					
Dose 2					
Dose 3					
☐ Other (specify):					
☐ Other (specify):					

CONGENITAL SYPHILIS INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

MOTHER'S ANTENATAL HISTORY

Gravida:	Parity:
Stillbirths:	Pre-Term births:
Miscarriages:	Interview Record #:
Livebirths:	Number of living children:

LMP (DD/MM/YYYY):	EDD (DD/MM/YYYY):
Ultrasound EDD:	
Multiple Gestation: ☐ Yes ☐ No ☐ Unknown	Number of foetuses:
Place of delivery:	
Did the mother have a history of vaginal discharge in mother? ☐ Yes ☐ No ☐ Unknown	
Number of lifetime sex partners:	
Did the mother receive antenatal care? ☐ Yes ☐ No ☐ Unknown	
Where did mother receive antenatal care? ☐ Public ☐ Private ☐ Both	
What is the source of primary antenatal care?	

SITE	NUMBER OF VISITS
☐ Health Centre	
☐ High Risk Clinic	
☐ General Practitioner	
☐ Obstetrician/ Gynaecologist	
☐ Other (Specify):	
TOTAL VISITS	

Date of first visit (DD/MM/YYYY):	Gestational age at first visit:
Date of last visit (DD/MM/YYYY):	Gestational age at last visit:

NAME OF SEXUAL CONTACT	AGE	SEX	RELATIONSHIP TO PATIENT OR MOTHER	CONTACT INFORMATION	SYPHILIS TEST RESULT (SPECIFY TEST AND DATE OF TEST)	TYPE OF TREATMENT GIVEN	DATE(S) TREATED (DD/MM/YYYY)
			Father of Infant				

TRACING OF SEXUAL CONTACTS OF MOTHER

CONGENITAL SYPHILIS INVESTIGATIONMEDICAL RECORD NUMBER:
PATIENTS' NAME:

Date Patient Seen (DD/MM/YYYY):

Gestational Age at Birth:

Birth Outcome:

☐ Live Birth☐ Stillbirth

Birth Category by Gestation:

☐ Pre-Term☐ Term☐ Post-Term

Admitted to Hospital?

☐ Yes☐ No☐ Unknown

Date of Admission (DD/MM/YYYY):

Date of Discharge (DD/MM/YYYY):

Hospital:

Medical Record Number:

Ward:

Admitting Diagnosis:

Date of Diagnosis (DD/MM/YYYY):

Admitted to Special Care Nursery (SCN) or NICU:

☐ Yes☐ No☐ Unknown

Admitted to NICU:

☐ Yes☐ No☐ Unknown

Date of Admission to SCN/NICU (DD/MM/YYYY):

Date of Discharge from SCN/NICU (DD/MM/YYYY):

Vital Status (Outcome):

☐ Alive☐ Dead

Date of Death (DD/MM/YYYY):

Time of Death: _____ AM/PM

Place of Death:

☐ Home☐ DOA☐ Ward☐ ICU☐ Other (Specify): _____

Signs or Symptoms	Presence			Date of Onset (DD/MM/YYYY):
Generalised lymphadenopathy	☐ Yes	☐ No	☐ Unknown	
Vesiculo-bullous rash	☐ Yes	☐ No	☐ Unknown	
Pneumonitis/pneumonia	☐ Yes	☐ No	☐ Unknown	
Neurological symptoms	☐ Yes	☐ No	☐ Unknown	
Mucous patches	☐ Yes	☐ No	☐ Unknown	
Other rashes	☐ Yes	☐ No	☐ Unknown	
Snuffles	☐ Yes	☐ No	☐ Unknown	
Jaundice	☐ Yes	☐ No	☐ Unknown	
Anaemia	☐ Yes	☐ No	☐ Unknown	
Hepatosplenomegaly	☐ Yes	☐ No	☐ Unknown	
Failure to thrive	☐ Yes	☐ No	☐ Unknown	
Other (Specify):	☐ Yes	☐ No	☐ Unknown	

Infant's Measurements

Measure	Value (units)	Date Measurement Taken (DD/MM/YYYY)
Birth weight		
Current weight		
Weight for age Z-score		
Birth length		
Current length		
Length for age Z-score		
Birth head circumference		
Head circumference		
Head circumference for age Z-score		
Other problem or diagnosis:	Date of Diagnosis (DD/MM/YYYY):	

CONGENITAL SYPHILIS INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

DIAGNOSTIC INVESTIGATION SCREENING FOR NEONATE/INFANTWas a sample taken? ☐ Yes ☐ No ☐ Unsure

If yes, what specimen was taken?

☐ Blood Sample☐ CSF☐ Other, Specify: _____

What date was the specimen taken? (DD/MM/YYYY): _____

Which lab is the sample being sent to?

☐ National Public Health Laboratory☐ University of the West Indies☐ Other

Specify Other Laboratory: _____

INFANT'S LABORATORY TEST RESULTS

Test Type	Specimen Type	Date Specimen Collected (DD/MM/YYYY)	Test Result	
Treponemal Tests				
SD Bioline	Blood		☐ Negative ☐ Indeterminate	☐ Positive ☐ Unknown
TPPA	Blood		☐ Negative ☐ Indeterminate	☐ Positive ☐ Unknown
Other Treponemal test (please specify)	Blood		☐ Negative ☐ Indeterminate	☐ Positive ☐ Unknown
Non-Treponemal Tests				
VDRL	Blood		☐ Non-Reactive ☐ Reactive ☐ Unknown	Titre:
TRUST	Blood		☐ Non-Reactive ☐ Reactive ☐ Unknown	Titre:
VDRL	CSF		☐ Non-Reactive ☐ Reactive ☐ Unknown	
Other Test (Specify):				
Other Test (Specify):				
Other Test (Specify):				

CONGENITAL SYPHILIS INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

Imaging Examination for Infant

Study	Done	Date of Study (DD/MM/YYYY)	Findings
Long Bone X-rays	☐ Yes ☐ No ☐ Unknown		
Abdominal Ultrasound	☐ Yes ☐ No ☐ Unknown		
Other (specify):			

MEDICATION HISTORY FOR NEONATE/INFANT

Past or Current Medication	Dose	Frequency	Route of administration	Date Started (DD/MM/YYYY)	Number of doses	Duration
☐ Benzathine penicillin G						
☐ Crystalline penicillin						
☐ Other (specify):						
☐ Other (specify):						

Additional Notes on Clinical Management:

CONGENITAL SYPHILIS INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

Date Investigation Completed (DD/MM/YYYY):

CLASSIFICATION

Initial Diagnosis: Congenital Syphilis

Final Diagnosis:

Case classification:

☐ Laboratory Confirmed	☐ Probable	☐ Discarded
☐ Pending	☐ Inconclusive	

Reason for classification:

DEATH CLASSIFICATIONCongenital Syphilis related death? ☐ Yes ☐ No ☐ Unknown

Immediate Cause:

Intermediate Cause:

Underlying Cause:

Comments:

Notifier's /Investigator's Details:

First Name:	Last Name
Professional Group:	
Phone number:	Email Address:
Name of Institution:	
Parish:	Community:
Investigator's Office Number/Street Name:	

Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA

Investigator's Signature:	Date Signed by Investigator (DD/MM/YYYY):	
Name of Parish MO(H):	Signature of MO(H):	Date signed by MO(H) (DD/MM/YYYY):

Appendix 51: Events Supposedly Attributed to Vaccines and Immunizations (ESAVI) Notification Form



MINISTRY OF
HEALTH &
WELLNESS

ESAVI reporting ID number:

MINISTRY OF HEALTH AND WELLNESS REPORTING FORM FOR EVENTS SUPPOSEDLY ATTRIBUTED TO VACCINATION OR IMMUNIZATION (ESAVI)

*Patient Name: _____ *Patient's full Address: _____ Telephone: _____ Sex: ☐ M ☒ F Currently pregnant: ☐ Yes ☐ No *Date of birth (DD/MM/YYYY): _____ OR Age at onset: _____ Years _____ Months _____ Days	*Reporter's Name: _____ Institution: _____ Post & Department: _____ Address: _____ Telephone & E-mail: _____ Date patient notified event to health system (DD/MM/YYYY): _____ Today's date (DD/MM/YYYY): _____
---	---

Health Facility (place or vaccination centre) name & address:									
Vaccine						Diluent (if applicable)			
*Name of vaccine	*Date of vaccination	*Time of vaccination	Dose (1 st , 2 nd , etc.)	*Batch /Lot number	Expiry date	Name of diluent	*Batch /Lot number	Expiry date	Date and time of reconstitution

*Adverse event(s): ☐ Local reaction ☐ > 3 days ☐ beyond nearest joint ☐ pain ☐ redness ☐ swelling ☐ Seizures ☐ febrile ☐ afebrile ☐ Headache ☐ Sepsis ☐ Fever ≥ 38°C ☐ Encephalopathy ☐ Abscess ☐ Thrombocytopenia ☐ Myalgia ☐ Anaphylaxis Other (specify) _____ ☐ _____ ☐ _____	Date ESAVI started (DD/MM/YYYY): _____ Time _____:____ Describe ESAVI (Signs & Symptoms) and Treatment (if any):
--	---

***Serious:** ☐ Yes ☐ No; → If Yes ☐ Death ☐ Life threatening ☐ Persistent or significant disability ☐ Hospitalization ☐ Congenital anomaly

☐ Other important medical event (specify): _____

***Outcome:** ☐ Recovering ☐ Recovered ☐ Recovered with sequelae ☐ Not Recovered ☐ Unknown

☐ Died If Died, date of death (DD/MM/YYYY): _____ Autopsy done: ☐ Yes ☐ No ☐ Unknown

Past medical history (including history of similar reaction or other allergies), concomitant medication and other relevant information (e.g. other cases). Use additional sheets if needed:

Parish level to complete:

Investigation needed: ☐ Yes ☐ No	If Yes, date investigation planned (DD/MM/YYYY): _____
--	--

Reviewed and signed off by MO(H): _____ Date (DD/MM/YYYY): _____

Name and Signature

National level to complete

Date report received at National level (DD/MM/YYYY): _____	ESAVI worldwide unique ID : _____
Comments:	

* Compulsory field

Appendix 52: HIV Confidential Reporting Form



National Surveillance Unit, National Epidemiology Branch,
Arc Building, 15 Knutsford Boulevard, Kingston 5, St. Andrew
Email: surveillance@moh.gov.jm, Telephone 876-633-7937



HIV Confidential Reporting Form (CRF)

REASON FOR SUBMITTING CONFIDENTIAL REPORTING FORM				
☐ Initial HIV Diagnosis	☐ Pregnancy of HIV positive woman	☐ Risk History Update		
☐ Advanced HIV Diagnosis	☐ ART Initiation	☐ Migration		
☐ AIDS Diagnosis	☐ Treatment Regimen Change			
☐ Death Notification	☐ Viral suppression			
PATIENT'S DEMOGRAPHICS				
Patient's ID Type: ☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		Patient's ID Number:		
First Name:		Middle Name(s):		
Last Name:		Pet Name(s):		
Patient's Maiden Name (if applicable):				
Sex Assigned at Birth: ☐ Male ☐ Female		Gender: ☐ Male ☐ Female ☐ FTM ☐ MTF ☐ Other		
Date of Birth: DD/MM/YYYY		Age:		
Medical Record Number:		Is patient employed? ☐ Yes ☐ No		
Occupation:				
Marital Status:				
☐ Married	☐ Visiting union	☐ Widowed	☐ Separated	
☐ Common law	☐ Single	☐ Divorced	☐ Other (specify):	
Is patient homeless? ☐ Yes ☐ No				
Country of Residence:		Parish (Jamaica):		Community:
House Number, Street Name:				
Landmark or directions to address:				
Phone Number:		Email Address:		
Highest level of Education completed:				
☐ No formal schooling	☐ Primary/All Age/Preparatory	☐ Tertiary /Post-secondary	☐ Other	
☐ Basic	☐ Secondary/High School	☐ Vocational Skills Training		
Number of children under 15 years of age:				
Was client/patient/person ever deported? ☐ Yes ☐ No				
If yes, which country was the person deported from?			If yes, date deported (MM/YYYY):	
MOTHER'S NAME:				
NEXT OF KIN:				
First Name:		Last Name:		
Address: Lot/ Street/Community/Parish:				
Phone number of Next of Kin:		Email address of Next of Kin:		
Relationship to the Patient:				
☐ Wife	☐ Father	☐ Sister	☐ Aunt	☐ Cousin
☐ Husband	☐ Son	☐ Brother	☐ Grandmother	☐ Friend
☐ Mother	☐ Daughter	☐ Uncle	☐ Grandfather	☐ Other

Patient's Name: _____
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PATIENT'S RISK HISTORY						
	Ever			12 Months		
	Yes	No	Unknown	Yes	No	Unknown
Date of Assessment: DD/MM/YYYY						
Was patient incarcerated/ in prison?	☐	☐	☐	☐	☐	☐
Has patient used crack/cocaine?	☐	☐	☐	☐	☐	☐
Has patient used intravenous drugs?	☐	☐	☐	☐	☐	☐
Has patient had a sexually transmitted infection?	☐	☐	☐	☐	☐	☐
Has patient had genital ulcers/sores?	☐	☐	☐	☐	☐	☐
Has patient had sex with a commercial sex worker?	☐	☐	☐	☐	☐	☐
Has patient been a commercial sex worker?	☐	☐	☐	☐	☐	☐
Has patient had unprotected anal sex?	☐	☐	☐	☐	☐	☐
Has patient had multiple sex partners?	☐	☐	☐	☐	☐	☐
Was patient a victim of sexual assault?	☐	☐	☐	☐	☐	☐
Has patient had sex with a known person living with HIV (PLHIV)?	☐	☐	☐	☐	☐	☐
Has patient had transactional sex?	☐	☐	☐	☐	☐	☐
Did patient have perinatal exposure to HIV?	☐	☐	☐	☐	☐	☐
Has patient received a blood transfusion?	☐	☐	☐	☐	☐	☐
Blood Transfusion Date: DD/MM/YYYY						
Hospital where blood transfusion was received:						

STATUS OF PATIENT			
Date of Assessment: DD/MM/YYYY	Yes	No	Unknown
Does patient have generalised lymphadenopathy?	☐	☐	☐
Does patient have generalised dermatitis?	☐	☐	☐
Has patient had shingles?	☐	☐	☐
Has patient experienced persistent cough for more than 4 weeks?	☐	☐	☐
Has patient experienced fever for more than 1 month?	☐	☐	☐
Does patient have chronic diarrhoea?	☐	☐	☐
Has patient experienced recurrent pneumonia?	☐	☐	☐
Does patient have Tuberculosis?	☐	☐	☐
Pulmonary tuberculosis	☐	☐	☐
Extra-pulmonary/disseminated tuberculosis	☐	☐	☐
Does patient have central nervous system (CNS) disease/involvement?	☐	☐	☐
Has patient had unintentional weight loss of more than 10% of body weight?	☐	☐	☐
Has patient had Kaposi's Sarcoma?	☐	☐	☐
Has patient been diagnosed with Pneumocystis carinii (jiroveci) pneumonia (PCP)?	☐	☐	☐
Has patient had oral candidiasis?	☐	☐	☐
Has patient had oesophageal candidiasis	☐	☐	☐
Has patient had vaginal candidiasis?	☐	☐	☐
Has patient had invasive cervical cancer?	☐	☐	☐
Has patient had severe bacterial infection?	☐	☐	☐
If Yes, please specify:			
Other clinical issues:			

Patient's Name: _____

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Post-test counselling:		
Patient received post-test counselling: ☐ Yes ☐ No	Performed by: _____ (Enter Provider's Name)	Date of post-test counselling: DD/MM/YYYY

HIV SCREENING TEST

Date of Screening Test: DD/MM/YYYY

HIV Screening Test Type:

☐ Determine	☐ Colloidal Gold	☐ OraQuick	☐ Other {please specify}
☐ Uni-Gold	☐ ELISA	☐ Nucleic acid {NAT, PCR; viral load}	

HIV Screening Test Results:

☐ Positive	☐ Negative	☐ Indeterminate
--------------------------------	--------------------------------	-------------------------------------

How was test initiated?

☐ Provider-initiated	☐ Self-referral
--	-------------------------------------

Where was the test done?

☐ Antenatal Clinic	☐ Hospital	☐ Other Public Clinic	☐ STI Clinic
☐ Blood Bank	☐ Insurance	☐ Private Antenatal	☐ Private Pharmacy
☐ Home-based/self-test	☐ Mobile Clinic/Outreach	☐ Private Doctor	☐ Private Lab
Other {please specify}: _____			

Date of Confirmatory Test: DD/MM/YYYY

Confirmatory HIV Test Type?

☐ Determine	☐ Colloidal Gold	☐ OraQuick	☐ Other {please specify}
☐ Uni-Gold	☐ ELISA	☐ Nucleic acid {NAT, PCR; viral load}	

Confirmatory HIV Test Result:

☐ Positive	☐ Negative	☐ Indeterminate
--------------------------------	--------------------------------	-------------------------------------

Laboratory where confirmatory HIV test performed:

☐ National Public Health Laboratory	☐ University Hospital of the West Indies	☐ Other {Specify}: _____
---	--	--

BEHAVIOURAL PRACTICE OF PATIENT

With whom does patient have sex?

☐ Men only	☐ Women only
☐ Both men and women	☐ Refuses to answer

Key and vulnerable population characteristics of patient {Check all that apply}:

☐ Men who have sex with men	☐ Prisoner/Prison history	☐ Intravenous Drug User
☐ Commercial sex worker	☐ Transgender	

Transmission Category:

☐ Sexual	☐ Blood Transfusion	☐ Haemophiliac
☐ IV Drug Use	☐ Vertical	

Patient's Sexual Contacts: Please complete contact listing on page 5.

Was this patient transferred from another treatment site?

☐ Yes ☐ No

Contact Tracing:

☐ Trace	☐ Do not contact trace	☐ Contact partners only
-----------------------------	--	---

Patient's Name: _____

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Detailed Test Results		
CD4 Count (cells/ μ l):	Date of CD4 Count: DD/MM/YYYY	CD4/CD8 Ratio:
Viral Load (copies/ml):	Date viral load sample taken DD/MM/YYYY	
Has patient ever taken antiretroviral medications (ARVs)? ☐ Yes ☐ No	ARV Start Date DD/MM/YYYY	
Has patient ever taken pre-exposure prophylaxis (PrEP)? ☐ Yes ☐ No	If yes, what is the last date patient took (PrEP) DD/MM/YYYY	
Has patient ever taken post-exposure prophylaxis (PEP)? ☐ Yes ☐ No	If yes, what is the last date patient took (PEP) DD/MM/YYYY	
Other (Specify):		
ARV Line:		
☐ First line	☐ Third line	☐ Unknown
☐ Second line	☐ Salvage therapy	
Patient's Current Category:		
☐ HIV	☐ Advanced HIV	
☐ AIDS	☐ AIDS Death	
Date of Onset of Symptoms: DD/MM/YYYY	Date diagnosed as Advanced HIV: DD/MM/YYYY	
Date diagnosed as AIDS: DD/MM/YYYY	Date of death: DD/MM/YYYY	
Cause of death:		
ii.	iii.	iiii.
Immediate cause	Intermediate cause	Underlying cause
FOR PREGNANT WOMEN ONLY		
Is Patient Pregnant: ☐ Yes ☐ Unsure	Last Menstrual Period: DD/MM/YYYY	
Gravida:	Parity:	
Estimated Gestational Age ----- weeks	Estimated Date of Delivery: DD/MM/YYYY	
Clinic Medical Record Number:	Clinic Site:	Parish:
Status of Referral		
☐ Not Referred	☐ Referred	☐ Refused Referral
Facility Referred To:	Date of First Antenatal Referral Appointment: DD/MM/YYYY	
PHYSICIAN ATTENDING TO PATIENT		
Name of Patient's Doctor:		
Hospital/Clinic/Doctor's Office Address:		
Phone Number:		
Source(s) of information:		
☐ Patient	☐ Next of Kin	☐ Other
☐ Medical record	☐ Relative	

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Comments:

Investigator's Details		
First Name:		Last Name
Professional Group:		
Phone number:		Email Address:
Name of Institution:		
Parish:		Community:
Office Number/Street Name:		
Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA		
Date Investigation Completed: <i>DD/MM/YYYY</i>		Investigator's Signature:
Name of Parish MO(H):	Signature of MO(H):	Date signed by MO(H): <i>DD/MM/YYYY</i>

Patient's Name: _____
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Patient's Sexual Contacts:					
First Name (Include alias)	Last Name	Sex (M/F)	Relation	Address (include Parish)	Phone Number

Patient's Name: _____

Medical Record Number: -----

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Definitions and Notes

- **Antiretroviral (ARV) Line:** Indicate ARV line patient is currently on/or last took - 1st, 2nd, 3rd, salvage therapy
- **Candidiasis:** Fungal infection with *Candida* spp that can be oral, oesophageal, and/or vaginal
- **Cause of death:** Recorded as i. Immediate, ii. Intermediate or iii. underlying causes.
- **Central Nervous System (CNS) Involvement:** Unexplained recent onset of seizures, dementia, toxoplasmosis, cytomegalovirus (CMV), cryptococcus, encephalopathy
- **Chronic Herpes Simplex:** Reported history of recurrent genital ulcer, blisters or crusts
- **Clinical Status:** Diagnosis should be confirmed by a physician
- **Commercial sex worker (CSW):** Exchange of sex for cash as a main source of income
- **Cough (> 4 weeks):** May be intermittent or persistent, productive or not
- **Date of Assessment:** Date on which assessment is being done for risk history or clinical status
- **Dermatitis:** Unexplained rash to multiple parts of the body
- **Diarrhoea (> 1 month):** Three (3) or more watery stools in a 24-hour period
- **Education:** No formal schooling, Basic, Primary/All Age, Secondary/High School, Tertiary, Skills training, Other (specify)
- **Fever (> 1 month):** On and off temperatures of 100.4° F (38° C) or more than is temporarily alleviated by antipyretics or other medications
- **Generalised Lymphadenopathy:** Two or more sites with enlarged lymph nodes
- **Invasive Cervical Cancer:** History of or current cytological diagnosis of cervical cancer that has spread beyond the cervix – Stage 1a or greater
- **Kaposi's Sarcoma (KS):** HIV-related Kaposi's sarcoma
- **Marital Status:** Married, common law, visiting union, single
- **Multiple Partners:** Persons who report having sex with more than one person within the past 12 months
- **PCP/PJP:** Current or past diagnosis of *Pneumocystis carinii* (jiroveci) pneumonia
- **Recurrent Pneumonia:** Repeated lower respiratory tract infections; Two or more episodes within a 1-year period
- **Severe Bacterial Infection:** Infection at any site of the body for which the patients had to be hospitalised for any period of time for treatment
- **Sexual Assault:** Anal or vaginal intercourse without consent; incident involved intimidation or threat or fear of violence
- **Shingles:** Recent or current diagnosis of the skin infection caused by the Varicella zoster virus
- **Transactional Sex:** Exchange of sex for food, goods or cash (but not as main source of income)
- **Transfer:** Indicate if the patient transferred in from another treatment site or private physician
- **Tuberculosis:** Confirmed TB case; If Yes: Pulmonary/Extra Pulmonary/Disseminated – select the appropriate condition(s)
- **Weight Loss (> 10%):** Unintentional weight loss, i.e. patient has made no direct efforts to lose weight e.g., exercise or dieting

Patient's Name: _____ Medical Record Number: -----Updated
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Appendix 53: HIV-Exposed Infants and Paediatric Cases (Ages Less Than 12 Years)



HIV-Exposed Infants And Pediatric Cases - Case Investigation

National Surveillance Unit, National Epidemiology Branch,
ARC Building, 15 Knutsford Boulevard, 2nd Floor,
Kingston 5, St. Andrew, Jamaica
Email: surveillance@moh.gov.jm; Tel.: 876-633-7973



CASE INVESTIGATION REPORT

HIV-Exposed Infants and Paediatric Cases (Age Less Than 12 Years)

CONTACT TRACING	☐ Trace	☐ Do not trace
-----------------	-----------------------------	------------------------------------

REASON FOR SUBMITTING CONFIDENTIAL REPORTING FORM

☐ HIV-Exposed	☐ AIDS Diagnosis	☐ ART Initiation
☐ Initial HIV Diagnosis	☐ Death Notification	☐ Viral suppression
☐ Advanced HIV Diagnosis	☐ Treatment Regimen Change	☐ Risk History Update

INFANT'S/CHILD'S DEMOGRAPHICS

First Name:	Middle Name(s):
Last Name:	Pet Name(s):
Country of Residence:	Parish (Jamaica):
House Number, Street Name, Community:	
Landmark or directions to address:	
Phone Number:	Email Address:
Sex Assigned at Birth: ☐ Male ☐ Female	Infant's/Child's Medical Record Number:
Date of Birth: DD/MM/YYYY	

RISK FACTORS

Date of Assessment: DD/MM/YYYY	Yes	No	Unknown
☐ Infant born to HIV positive mother	☐	☐	☐
☐ Breastfeeding	☐	☐	☐
☐ Mother had High Viral Load in Pregnancy	☐	☐	☐
☐ Late HIV Diagnosis in Pregnant Mother	☐	☐	☐
☐ Non-Adherence to ART by Mother	☐	☐	☐
☐ Sexual Contact of Child	☐	☐	☐
☐ Lost to Follow-up Child	☐	☐	☐
☐ Lost to Follow-up Mother	☐	☐	☐
☐ Non-Disclosure of HIV Status by Mother to Partner/Relative	☐	☐	☐
☐ Vaginal Delivery	☐	☐	☐
☐ Infant/Child Received Transfusion of Blood Products	☐	☐	☐
☐ Other (specify):	☐	☐	☐

Nutrition: ☐ Exclusively Breastfed ☐ Replacement Feed Only ☐ Both Breastfed and Replacement Fed

CLINICAL STATUS (DIAGNOSIS BY PHYSICIAN)

Date of Assessment: DD/MM/YYYY	Yes	No	Unknown
☐ Asymptomatic	☐	☐	☐
☐ Generalised lymphadenopathy	☐	☐	☐
☐ Generalised dermatitis	☐	☐	☐
☐ Unexplained persistent parotid enlargement	☐	☐	☐
☐ Hepatosplenomegaly	☐	☐	☐
☐ Recurrent diarrhoea	☐	☐	☐
☐ Recurrent/chronic upper respiratory tract infections	☐	☐	324 ☐

CLINICAL STATUS (DIAGNOSIS BY PHYSICIAN)			
Date of Assessment: DD/MM/YYYY	Yes	No	Unknown
☐ Pulmonary tuberculosis	☐	☐	☐
☐ Failure to thrive	☐	☐	☐
☐ Multiple or recurrent bacterial infections	☐	☐	☐
☐ Neurological involvement	☐	☐	☐
☐ Oesophageal candidiasis	☐	☐	☐
☐ Recurrent lower respiratory tract infections	☐	☐	☐
☐ Cytomegalovirus retinitis	☐	☐	☐
☐ Extrapulmonary tuberculosis	☐	☐	☐
☐ Other (specify):	☐	☐	☐

Clinical Notes:

ANTIRETROVIRAL THERAPY (INFANT/CHILD)

ARV	Dosage	Actual Duration Received (weeks)	Comments
☐ Nevirapine			
☐ Zidovudine			
☐ Lamivudine			
☐ Abacavir			
☐ Lopinavir/Ritonavir			
☐ Dolutegravir			
☐ Tenofovir			
☐ Other (specify):			

Current Status of Patient:

☐ Pending	☐ Advanced HIV	☐ AIDS	☐ AIDS Death
☐ HIV	☐ HIV Negative		

Date of Onset of Symptoms: DD/MM/YYYY

Date diagnosed as Advanced HIV:

DD/MM/YYYY

Date diagnosed as AIDS: DD/MM/YYYY

Date of death: DD/MM/YYYY

Cause of death:

ii.	iii.	iiii.
(immediate cause)	(intermediate cause)	(underlying cause)

MOTHER'S INFORMATION

Mother's ID Type: ☐ Driver's License ☐ TRN ☐ Passport Number ☐ NIDS	Mother's ID Number:
Mother's First Name:	Mother's Middle Name(s):
Mother's Last Name:	Mother's Pet Name(s):
Mother's Maiden Name (if applicable):	
Mother's Date of Birth: DD/MM/YYYY	Mother's Age:
Mother's Country of Residence:	Parish (Jamaica):
House Number, Street Name, Community:	

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Name of Patient:-----

Medical Record Number:-----

Landmark or directions to address:			
Phone Number:		Email Address:	
Mother's Medical Record Number:		Treatment Site:	
HIV Status of Mother: ☐ Positive ☐ Negative		Date of Mother's Confirmatory HIV Test: <i>DD/MM/YYYY</i>	
Laboratory where Mother's Confirmatory HIV Test Performed:			
☐ National Public Health Laboratory	☐ University Hospital of the West Indies	☐ Other Government Laboratory Facility	☐ Other
Site of Delivery:			
☐ Public Hospital	☐ Private Doctor	☐ Health Centre	☐ Other (Specify):
☐ Private Hospital	☐ Private Obstetrician	☐ Home	
Gravida:		Parity:	
Treatment During Pregnancy			
Antenatal ARV Treatment	ART Start Date (DD/MM/YYYY)	Antenatal ARV Treatment	ART Start Date (DD/MM/YYYY)
☐ Tenofovir/Lamivudine		☐ Atazanavir/Ritonavir	
☐ Tenofovir/Emtricitabine		☐ Darunavir/Ritonavir	
☐ Abacavir/Lamivudine		☐ Zidovudine/Lamivudine	
☐ Dolutegravir		☐ Nevirapine	
☐ Lopinavir/Ritonavir		☐ Other (specify):	
Antenatal Care (this pregnancy):			
☐ Public Hospital	☐ Private Doctor	☐ Health Centre	
☐ Private Hospital	☐ Private Obstetrician	☐ None	
Mother's Next of Kin			
First Name:		Last Name:	
Phone number:		Email address:	
Address: Lot/ Street/Community/Parish:			
Relationship to the Mother:			
☐ Husband	☐ Daughter	☐ Brother	☐ Aunt
☐ Mother	☐ Son	☐ Grandmother	☐ Uncle
☐ Father	☐ Sister	☐ Grandfather	☐ Cousin
FATHER'S INFORMATION			
Father's First Name:		Father's Middle Name(s):	
Father's Last Name:		Father's Pet Name(s):	
Father's Date of Birth: <i>DD/MM/YYYY</i>		Father's Age:	
HIV Status of Father ☐ Positive ☐ Negative		Date of Father's Confirmatory HIV Test: <i>DD/MM/YYYY</i>	

Laboratory where Father's Confirmatory HIV Test Performed:

☐ National Public Health Laboratory	☐ University Hospital of the West Indies	☐ Other Government Laboratory Facility	☐ Other
---	--	--	-----------------------------

SAMPLE AND LAB INFORMATION

HIV Testing for Infant/Child	Date (DD/MM/YYYY)	Result	Comment
☐ PCR at birth	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ 2-Week PCR (if high risk)	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ 6-Week PCR	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ 4-Month PCR	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ 18-Month ELISA	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ Determine Rapid Antibody	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ Uni-Gold Rapid Antibody	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
☐ Other	DD/MM/YYYY	☐ Positive ☐ Negative ☐ Indeterminate	
Comments: 			

Notifier's Details		
Date of Reporting (DD/MM/YYYY):		
First Name:	Last Name	
Professional Group:		
Phone number:	Email Address:	
Name of Institution:		
Parish:	Community:	
Office Number/Street Name:		
Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA		
Name of Parish MO(H):	Signature of MO(H):	Date signed by MO(H): DD/MM/YYYY

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Name of Patient:-----

Medical Record Number:-----

Definitions (Diagnosis by physician)

Asymptomatic	No symptoms or signs suggestive of HIV infection.
Hepatosplenomegaly	Enlargement of liver and/or spleen.
Generalised Lymphadenopathy	Enlargement of more than 1 cm of more than 2 noncontiguous lymph node groups. Common sites are head, neck, post- or pre- auricular, supraclavicular, axillary, epitrochlear, and inguinal.
Cytomegalovirus retinitis	Commonly asymptomatic; fundoscopy usually reveals bilateral changes, often affecting the macula ranging from white flecks giving a granular appearance to more fulminant disease affecting the full thickness of the retina with retinal oedema, retinal vascular sheathing, or haemorrhages. CMV can also affect other organ systems.
Extrapulmonary tuberculosis	Systemic illness that may include prolonged fever, night sweats, weight loss; presentation dependent on organ system involved; culture, microscopy or other laboratory evidence of Mycobacterium tuberculosis in specimen.
Failure to thrive	Weight for age and sex below 3 rd to 5 th percentile; loss of weight associated with crossing two or more percentiles on a standardised growth curve; weight for height z-score below minus 2 (-2).
Generalised dermatitis	Unexplained rash to multiple parts of the body.
Recurrent diarrhoea	Three (3) or more watery stools in a 24-hour period, recurring on two or more days during a 14-day period.
Unexplained persistent parotid enlargement	Unexplained enlargement of the parotid glands, usually bilateral and painless.
Recurrent/chronic upper respiratory tract infections	Two or more episodes of upper respiratory tract infection such as ear, nose, throat, sinuses within a six-month period.
Oesophageal candidiasis (or candidiasis of trachea, bronchi or lungs)	Candidiasis of the mouth in infants above 3 months of age; may be associated with decreased feeding or crying during feeding.
Neurological involvement	Unexplained recent onset of seizures, toxoplasmosis, cytomegalovirus (CMV), cryptococcus, encephalopathy. Delayed development and/or loss of developmental milestones.
Recurrent lower respiratory tract infections	Two or more episodes of lower respiratory tract infections within a 6-month period presenting with respiratory distress or positive findings on examination e.g., crackles, consolidation; and/or positive findings on imaging or laboratory investigation.
Multiple or recurrent bacterial infections	Two or more episodes of bacterial infection requiring hospitalisation within a 6-month period.
Pulmonary tuberculosis	Symptoms of chronic cough, fever, night sweats, weight loss; abnormal findings on imaging e.g., chest X-ray and/or demonstration of the Mycobacterium tuberculosis pathogen by microscopy, culture or molecular (e.g., polymerase chain reaction) methods.

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Name of Patient:-----

Medical Record Number:-----

Appendix 54: Leprosy (Hansen's Disease) Case Investigation Form

LEPROSY (HANSEN'S DISEASE) CASE INVESTIGATION FORM								
Reporting Centre:				Date of report / /				
1. Patient information								
Name				Age (yrs)		Sex M F		
Address				Phone		Occupation		
2. Clinical data								
Date of onset / /						Immunization history		
Symptom	Y	N	Symptom	Y	N	Symptom	Y	N
Skin rash								
Hypopigmentation			Nerve Thickening					
Sensory loss								
Nodules								
Infiltration								
Is/was this patient hospitalised?				Y	N	Date(s)	Outcome of illness	
							Survived	
							Died Date:	
3. Exposure history								
				Y	N	Date	Details	
Childhood exposure								
Household contact								
Other prolonged contact								
Droplet exposure								
Other								
4. Laboratory data								
Specimen	Date collected	Date rec'd	Condition	Test	Result	Date sent	Comment	
Skin smear				Histology				
Biopsy				Histology				
5. Final case classification:				Clinically confirmed:		Date reported:		
				Laboratory confirmed:		To whom:		
						Route:		
						Signature:		

Appendix 55: Leptospirosis Case Investigation Form

 MINISTRY OF HEALTH & WELLNESS		NATIONAL SURVEILLANCE UNIT, NATIONAL EPIDEMIOLOGY BRANCH, ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew Email: surveillance@moh.gov.jm, Telephone 876-633-7937		 NATIONAL EPIDEMIOLOGY BRANCH	
LEPTOSPIROSIS CASE INVESTIGATION FORM					
Date of Notification (DD/MM/YYYY):			Date Investigation Started: (DD/MM/YYYY)		
PATIENT'S DEMOGRAPHICS					
Patient's ID Type:			Patient's ID Number:		
☐ Driver's License ☐ TRN ☐ Passport ☐ NIN					
First Name:			Middle Name(s):		
Last Name:			Pet Name(s)		
Maiden Name (if applicable):					
Name of Facility Where Patient Seen:			Medical Record Number:		
Sex Assigned at Birth: ☐ Male ☐ Female			Date of Birth: (DD/MM/YYYY)		Age:
Country of Residence:			Parish of Residence (Jamaica):		
House Number, Street Name:			Community of Residence:		
Landmark or directions to address:					
Phone Number:			Email Address:		
NEXT OF KIN (NOK)					
First Name:			Last Name:		
Phone Number:			Email address:		
Address: Lot/ Street/Community/Parish:					
Relationship to the Patient:					
☐ Wife	☐ Father	☐ Sister	☐ Aunt	☐ Cousin	
☐ Husband	☐ Son	☐ Brother	☐ Grandmother	☐ Friend	
☐ Mother	☐ Daughter	☐ Uncle	☐ Grandfather	☐ Guardian	
CLINICAL PROFILE					
Symptom/Sign	Duration (Days)	Date of Onset (DD/MM/YYYY)	Symptom/Sign	Duration (Days)	Date of Onset (DD/MM/YYYY)
☐ Fever			☐ Conjunctival suffusion		
• History of fever			☐ Anuria or oliguria		
• Recorded temperature (° C):			☐ Jaundice		
☐ Headache			☐ Cough, haemoptysis and/or shortness of breath		
☐ Myalgia			☐ Haemorrhage (e.g., gastrointestinal bleed)		
• Calves and/or thighs			☐ Meningeal irritation		
• Other areas			☐ Skin rash		
☐ Weakness			☐ Cardiac arrhythmia/failure		
☐ Other (Specify):					
Clinical summary:					
EPOSURE HISTORY					
During the 3 weeks prior to onset:	Yes	No	Date(DD/MM/YYYY)	Details	
Contact with animals (including pets) or their excreta at home or during travel	☐	☐			
Contact with known or possibly contaminated water	☐	☐			
Ingested possibly rodent-contaminated food/drink	☐	☐			
Contact with case of leptospirosis	☐	☐			

LEPTOSPIROSIS CASE INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

Travelled in the past six (6) weeks? ☐ Yes ☐ No ☐ Unknown Country/Area Visited:**MEDICAL HISTORY**

Pre-existing Conditions:

☐ Asthma	☐ Diabetes mellitus	☐ HIV/AIDS	☐ Malignancy
☐ Autoimmune	☐ Heart disease	☐ Hypertension	☐ Sickle cell disease
☐ Chronic Obstructive Pulmonary Disease(COPD)	☐ Kidney disease	☐ Liver dysfunction	☐ Specify Other:

PREGNANCYIs Patient Pregnant: ☐ Yes ☐ No ☐ Unsure ☐ Not applicableGestational Age: Trimester: ☐ First ☐ Second ☐ Third**HOSPITAL SUMMARY**Admitted to Hospital? ☐ Yes ☐ No

Date of Admission (DD/MM/YYYY):

Date of Discharge (DD/MM/YYYY):

Hospital:

Medical Record Number:

Vital Status (Outcome):

Date of Death (DD/MM/YYYY):

Time of Death: ____ : ____ AM/PM

☐ Alive ☐ Dead

Place of Death:

☐ Home☐ DOA☐ Ward☐ ICU**SAMPLE AND LAB INFORMATION**Was a specimen collected? ☐ Yes ☐ No

Date Specimen Taken (DD/MM/YYYY):

Specimen Type: ☐ Blood, with anticoagulant (e.g., purple top tube)☐ Blood, NO anticoagulant/clotted (e.g., red top tube)☐ CSF☐ Urine☐ Other (Specify):

Which lab is the sample being sent to?

☐ National Public Health Laboratory☐ Other (Specify):

Test Type	Date Test Done (DD/MM/YYYY)	Results	Interpretation
☐ PCR		☐ Positive ☐ Negative ☐ Indeterminate	
☐ IgM		☐ Positive ☐ Negative ☐ Indeterminate	
☐ Other (Specify):			

Classification

Initial Diagnosis:

☐ Laboratory Confirmed☐ Probable☐ Discarded☐ Pending

Final Diagnosis:

☐ Suspected

Reason for classification:

Aetiology Category:

☐ Imported case☐ Endemic☐ Unknown

Source country (If Imported or Import-Related):

DEATH CLASSIFICATIONLeptospirosis-related death: ☐ Confirmed ☐ Probable ☐ Suspected ☐ Discarded ☐ Under Investigation ☐ Cannot be classified**Investigator's Details**

Date Investigation Completed (DD/MM/YYYY):

First Name:

Last Name

Professional Group:

Phone number:

Email Address:

Name of Institution:

Parish:

Community:

Office Number/Street Name:

Health Region:

Investigator's Signature:

Date signed by Investigator (DD/MM/YYYY):

Name of Parish MO(H):

Signature of MO(H)

Date signed by MO(H), (DD/MM/YYYY)

Appendix 56: Meningitis/ Encephalitis Investigation Form

 MINISTRY OF HEALTH & WELLNESS	NATIONAL SURVEILLANCE UNIT, NATIONAL EPIDEMIOLOGY BRANCH, ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew Email: surveillance@moh.gov.jm, Telephone 876-633-7937	 NATIONAL EPIDEMIOLOGY BRANCH							
MENINGITIS/ENCEPHALITIS INVESTIGATION FORM									
INITIAL DIAGNOSIS									
☐ Meningitis ☐ Encephalitis ☐ Meningoencephalitis ☐ Other (Specify): _____									
Date of Notification (DD/MM/YYYY):		Date Investigation Started: (DD/MM/YYYY)							
PATIENT'S DEMOGRAPHICS									
Patient's ID Type:		Patient's ID Number:							
☐ Driver's License ☐ TRN ☐ Passport ☐ NIN									
First Name:		Middle Name(s):							
Last Name:		Pet Name(s):							
Patient's Maiden Name (if applicable):									
Sex Assigned at Birth: ☐ Male ☐ Female		Date of Birth: (DD/MM/YYYY) Age:							
Country of Residence:		Parish/State/Province:							
House Number, Street Name:									
Landmark or directions to address:		Community:							
Phone Number:		Email Address:							
Occupation:									
Name of Workplace/School:									
Address of Workplace/School:									
Phone Number:		Email Address:							
NEXT OF KIN									
First Name:		Last Name:							
Phone number:									
Address: Lot/ Street/Community/Parish:									
NOK's Phone number:		NOK's Email address:							
Relationship to the Patient:									
☐ Wife	☐ Son	☐ Uncle							
☐ Husband	☐ Daughter	☐ Aunt							
☐ Mother	☐ Sister	☐ Grandmother							
☐ Father	☐ Brother	☐ Grandfather							
		☐ Cousin							
		☐ Friend							
		☐ Guardian							
		☐ Other (specify):							
CLINICAL PROFILE									
Sign or Symptom	Presence			Date of Onset (DD/MM/YYYY)	Sign or Symptom	Presence			Date of Onset (DD/MM/YYYY)
	Yes	No	Unknown			Yes	No	Unknown	
Fever	☐	☐	☐		Confusion	☐	☐	☐	
Headache	☐	☐	☐		Paresis/Paralysis	☐	☐	☐	
Nausea	☐	☐	☐		Para-/Hyperesthesia	☐	☐	☐	
Vomiting	☐	☐	☐		Vesicles – hands/ feet	☐	☐	☐	
Neck stiffness	☐	☐	☐		Delirium	☐	☐	☐	
Disorientation	☐	☐	☐		Tremors	☐	☐	☐	
Itching	☐	☐	☐		Convulsions	☐	☐	☐	
Rash	☐	☐	☐		Spasticity	☐	☐	☐	
Ecchymoses	☐	☐	☐		Coma	☐	☐	☐	
Drowsiness	☐	☐	☐		Neckache/backache	☐	☐	☐	
Irritability	☐	☐	☐		Visual disturbances	☐	☐	☐	
Cough	☐	☐	☐		Other (Specify):	☐	☐	☐	
Stupor	☐	☐	☐		Other (Specify):	☐	☐	☐	

MENINGITIS/ENCEPHALITIS

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

Medical History

Pre-existing Conditions

☐ None	☐ Asthma	☐ Diabetes mellitus	☐ HIV/AIDS
☐ Malignancy	☐ Autoimmune	☐ Heart disease	☐ Hypertension
☐ Sickle cell disease	☐ Chronic Obstructive Pulmonary Disease (COPD)	☐ Kidney disease	☐ Liver dysfunction
☐ Other (Specify):			

PregnancyIs Patient Pregnant: ☐ Yes ☐ No ☐ Unsure ☐ Not Applicable Gestational Age:**VACCINATION HISTORY**

Type of Vaccine	Number of doses	Date of last dose (DD/MM/YYYY)	Source of vaccination information (Vaccination Card, Medical Record, Verbal)
BCG			
Diphtheria			
Pertussis			
Tetanus			
Haemophilus influenzae B			
Hepatitis B			
Polio - inactivated (IPV)			
Polio - oral (OPV)			
Measles			
Mumps			
Rubella			
Meningococcus ACYW			
Meningococcus B			
Pneumococcus (conjugate)			
Pneumococcus (polysaccharide)			
Other (Specify):			
Other (Specify):			

HOSPITALISATION

Admitted to Hospital? ☐ Yes ☐ No ☐ Unknown	Date of Admission (DD/MM/YYYY):	Date of Discharge (DD/MM/YYYY):
Hospital:	Medical Record Number	Ward:
Vital Status (Outcome): ☐ Alive ☐ Dead	Date of Death (DD/MM/YYYY):	Time of Death: ____ AM/PM
Place of Death: ☐ Home ☐ Dead on Arrival ☐ Ward ☐ ICU ☐ Other (Specify):		

MEDICATION HISTORYHas patient received antibiotics? ☐ Yes ☐ No ☐ Unknown

Past or Current Medication	Dosage	Frequency	Route of administration	Date of first administration	Number of doses

MENINGITIS/ENCEPHALITIS

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

SAMPLE AND LABORATORY DATAWas a sample taken? ☐ Yes ☐ No ☐ Unsure

If yes, what specimen was taken?

☐ Acute Blood Sample	☐ Convalescent Blood	☐ CSF
☐ Stool	☐ Vesicle Swab	☐ Brain Tissue
☐ Urine	☐ Other (Specify): _____	

What date was the specimen taken? (DD/MM/YYYY):

Which lab is the sample being sent to? ☐ National Public Health Laboratory ☐ University of the West Indies ☐ Other

Specify Other Laboratory:

HAEMATOLOGY

	Blood Sample	CSF	Other (Specify):	Other (Specify):
	Date of Study (DD/MM/YYYY)	Date of Study (DD/MM/YYYY)	Date of Study (DD/MM/YYYY)	Date of Study (DD/MM/YYYY)
Red cell count (RCC)				
White cell count (WCC)				
Neutrophil count				
Lymphocyte count				
Monocyte count				
Eosinophil count				
Basophil count				
Blast count				
Platelet count				
Neutrophil %				
Lymphocyte %				
Monocyte %				
Eosinophil %				
Basophil %				
Blast %				
Other:				

Microscopy

Study	Done	Date of Study (DD/MM/YYYY)	Findings			
			CSF	Blood	Other (specify):	Other (specify):
Gram Stain	☐ Yes ☐ No					
Ziehl Neelsen	☐ Yes ☐ No					
India Ink	☐ Yes ☐ No					

MENINGITIS/ENCEPHALITIS

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

Culture and Molecular						
	Done		CSF	Blood	Urine	Other (specify):
Bacterial Culture	☐ Yes ☐ No					
Antimicrobial Sensitivity:	☐ Yes ☐ No	Sensitive to:				
		Resistant to:				
RT-PCR/PCR	☐ Yes ☐ No					
Viral Culture	☐ Yes ☐ No					
Antigen Detection	☐ Yes ☐ No					
Other test (specify):						
Other test (specify):						

Biochemistry				
Study	Done	Date of Study (DD/MM/YYYY)	Results	
			CSF	Blood
Protein	☐ Yes ☐ No			
Glucose	☐ Yes ☐ No			

Parasitology						
Study	Done	Date of Study (DD/MM/YYYY)	Findings			
			CSF	Blood	Stool	Other (specify):
Larvae/ Worms	☐ Yes ☐ No					

EXPOSURE PROFILE				
Case or Institutional Contact	☐ Yes	☐ No	☐ Unknown	Details:
Recent History of Viral Illness	☐ Yes	☐ No	☐ Unknown	
Consumption of Lettuce, Cabbage, Snails, Shrimp, etc.	☐ Yes	☐ No	☐ Unknown	
Snail Infestation of Premises	☐ Yes	☐ No	☐ Unknown	
Dead Birds on Premise	☐ Yes	☐ No	☐ Unknown	
Other Exposure (Specify):	☐ Yes	☐ No	☐ Unknown	
Are there other cases present in the case's community of residence? ☐ Yes ☐ No				

TRAVEL HISTORY				
Travelled in the Last 23 days: ☐ Yes ☐ No ☐ Unknown				
If travelled: ☐ Domestic Travel ☐ International Travel				
Country/Parish Visited	City/Community Visited	Accommodation	Date Arrived (DD/MM/YYYY)	Date Departed (DD/MM/YYYY)

Date returned to Jamaica/Date returned to home address (DD/MM/YYYY):

MENINGITIS/ENCEPHALITIS

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

SOCIAL ASSESSMENT

(To be completed for Haemophilus influenzae and Meningococcal meningitis)

LIVING ARRANGEMENT

☐ Lives alone	☐ Lives with partner	☐ Lives with family
☐ Lives with friends	☐ Other (Specify Other):	
Number of household members: _____		Number of household members under 5 years: _____

Are there any ill persons in the household? ☐ Yes ☐ No ☐ UnknownHave there been any deaths in the case's community of residence in the last 3 months? ☐ Yes ☐ No ☐ UnknownHave any family members travelled in the Last 3 months? ☐ Yes ☐ No ☐ Unknown

If Yes, where to?

Have there been any visitors from abroad in the last 3 months? ☐ Yes ☐ No ☐ Unknown

If yes, where from?

(NB. Check for the following symptoms: fever, sore throat, cough, runny nose, post nasal draining)

List of Contacts

Name	Age	Sex	Meningococcus Immunisation Status	Symptoms	Prophylaxis Given	Date Prophylaxis given (DD/MM/YYYY)	Immunisation Administered	Date immunisation given (DD/MM/YYYY)
					☐ Yes ☐ No ☐ Unkown		☐ Yes ☐ No	
					☐ Yes ☐ No ☐ Unkown		☐ Yes ☐ No	
					☐ Yes ☐ No ☐ Unkown		☐ Yes ☐ No	
					☐ Yes ☐ No ☐ Unkown		☐ Yes ☐ No	

MENINGITIS/ENCEPHALITIS

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

CLASSIFICATION

Initial Diagnosis: Meningitis/Encephalitis

Final Diagnosis:

Case classification:

☐ Laboratory Confirmed	☐ Probable	☐ Suspected
☐ Discarded	☐ Pending	☐ Inconclusive

Reason for classification:

Is the case epidemiologically linked? ☐ Yes ☐ No ☐ UnknownAetiology Category: ☐ Imported case ☐ Import-related case ☐ Endemic ☐ Unknown

Source country (If Imported or Import-Related):

DEATH CLASSIFICATIONMeningitis/Encephalitis related death? ☐ Confirmed ☐ Probable ☐ Suspected ☐ Discarded ☐ Under Investigation
☐ Cannot be classified

Date Investigation Completed (DD/MM/YYYY):

Additional Information/Action Taken/Control Measures Implemented:

Hypothesis as to source, method of transmission and existence of reservoirs or carriers:

Other Comments:

Notifier's /Investigator's Details:

First Name:	Last Name
Professional Group:	
Phone number:	Email Address:
Name of Institution:	
Parish:	Community:
Investigator's Office Number/Street Name:	

Health Region: ☐ SERHA ☐ NERHA ☐ SRHA ☐ WRHA

Investigator's Signature:

Date Signed by Investigator (DD/MM/YYYY):

Name of Parish MO(H):

Signature of MO(H):

Date signed by MO(H) (DD/MM/YYYY):

Appendix 57: Needle Stick, Sharp Object Injury and Fluid Exposure Form

NEEDLE STICK, SHARP OBJECT INJURY AND FLUID EXPOSURE REPORT

Parish: _____ DOB: _____ Sex: M F Occupation: _____

Date/Time of Exposure/Injury: _____

Reported by: _____ Date: _____

1. Institution where exposure/injury occurred: _____
2. Where did the exposure/injury occur?

a. Ward (specify) _____	☐	g. Operating Theatre	☐
b. Dressing Room	☐	h. Dialysis Unit	☐
c. Phlebotomy Room	☐	i. Labour & Delivery Room	☐
d. Outpatient Clinic	☐	j. Service/Utility Area (laundry, garage, disposal, etc.)	☐
e. ICU	☐	k. Other (specify) _____	☐
f. A&E/Casualty	☐		
3. Name of the source patient: _____ Source Unknown
4. Docket No: _____ Not Applicable
5. Source patient HIV Status: Positive NegativeUnknown Source patient tests positive for other blood-borne pathogen (specify): _____
6. Type of exposure: Sharp item Body fluid exposure (specify type and volume): _____
7. In the case of body fluid exposure, was the skin of the exposed person intact? (if there is no body fluid exposure, skip this question): Yes No (explain) _____
8. Specify sharp item (if not sharp item, skip to question 11):
 - ☐ Needle (specify gauge) _____ ☐ Blade
 - ☐ Branula (specify gauge) _____ ☐ Glass (specify, broken test tube, etc.) _____
 - ☐ Other needle (suture needle, etc.) specify type & size _____
 - ☐ Other (specify) _____
9. Was the injury: ☐ Superficial (little or no bleeding) ☐ Moderate (skin punctured, some bleeding) ☐ Severe (deep stick/cut, or profuse bleeding)
10. If the injury was to the hands, did the sharp item penetrate: (check one)
 - ☐ single pair gloves ☐ no gloves ☐ other (specify) _____
11. Did the injury/exposure occur:
 - ☐ Restraining Patient
 - ☐ Disassembling device or equipment
 - ☐ In preparation for reuse of reusable instruments (sorting, disinfecting, sterilizing, etc.)
 - ☐ While recapping a used needle
 - ☐ Withdrawing a needle from rubber or other resistant material (rubber stopper, I.V. port, etc.)
 - ☐ Device left on floor, table, bed or other inappropriate place
 - ☐ Other after use, before disposal (in transit to trash, cleaning, sorting, etc.)
 - ☐ From items left near or on disposal container
 - ☐ While putting the item in a disposal container
 - ☐ While putting the item in a disposal container
 - ☐ After disposal, stuck by an item protruding from the opening of the disposal container
 - ☐ Item placed on the side of disposal container
 - ☐ After disposal, item protruded from trash bag or inappropriate waste container
 - ☐ Other (describe) _____

12. Describe the circumstances leading to this injury: *(please note if a device malfunction was involved)*

13. State the location of the exposure/injury: _____

14. Hepatitis B immunization? ☐None ☐Yes Date(s): _____

15. Immunization card seen? ☐Yes ☐No

16. Has the injured person had any previous needle stick injuries? ☐Yes ☐No

17. If yes, were the previous incidents reported? ☐No ☐Yes Date(s): _____

18. Risk Category: ☐Low ☐Moderate ☐High

19. State the location of the exposure/injury: _____

20. Was the area bled/flushed/washed? ☐Yes ☐No

21. Was disinfectant used? ☐Yes ☐No

****NOTE:** The use of bleach, alcohol, Savlon or other disinfectants is not recommended.

22. Action taken by the head of department

a. Counselling ☐Yes ☐No

b. Blood taken for HIV testing ☐Yes ☐No

c. Blood taken for Hepatitis B Antigen ☐Yes ☐No (if "No", explain) _____

d. PEP medication given? *(see last page of this form for PEP Guidelines)*

☐Yes Type _____ Date/Time started _____

☐No (if "no", explain) _____

☐ Low Risk ☐ Not Available ☐ Exposed person refusal ☐ Other (specify) _____

***In the case of refusal, the exposed person must sign the attached waiver form to be sent to the Medical Officer of**

Health for surveillance.

Form completed by:

Name: _____

Designation: _____

Signature: _____

Post Exposure Prophylaxis (PEP) Dosages:

All of the following are to be given within 1-2 hours or at most 24-36 hours after exposure* and continued for four weeks:

Either:

a. Zidovudine (AZT) 300 mg bid or 200 mg po tid after meals AND
Lamivudine (3TC) 150 mg po bid after meals

OR

b. Combivir (AZT + 3TC) 1 tablet po bid with or without food

Indinavir should be used in addition to either a. or b. when there is a very high risk to the exposed person. 800 mg po q8h on an empty stomach.
Drink at least 48 oz of fluid/24 hours.

***Studies in animals (no human studies done) suggest that treatment is not effective when started more than 24-36 hours after exposure.**

Commencement of treatment later is recommended if considered highest risk.

PEP Refusal form:

I, _____, hereby waive my right to take the PE Prophylaxis to prevent possible infection of HIV. I understand that by refusing to take the medication I am putting myself at greater risk for infection.

Signed:

Date:

Witness signature:

Witness (print name neatly):

Appendix 58: Ophthalmia Neonatorum Investigation Form

 MINISTRY OF HEALTH & WELLNESS	NATIONAL SURVEILLANCE UNIT, NATIONAL EPIDEMIOLOGY BRANCH, ARC Building, 15 Knutsford Boulevard, 2nd Floor, Kingston 5, St. Andrew Email: surveillance@moh.gov.jm, Telephone 876-633-7937	 NATIONAL EPIDEMIOLOGY BRANCH
OPHTHALMIA NEONATORUM INVESTIGATION		
Initial Diagnosis(-es): ☐ Ophthalmia neonatorum ☐ Other (Specify): _____		
Date of Notification: (DD/MM/YYYY)		Date Investigation Started: (DD/MM/YYYY)
INFANTS'S DEMOGRAPHICS (Use mother's data if infant's details not yet known)		
Patient's ID Type:		Patient's ID Number:
☐ National Identification Number (NIN) ☐ Other Specify: _____		
First Name:		Middle Name(s):
Last Name:		Pet Name(s)
Sex Assigned at Birth: ☐ Male ☐ Female		Date of Birth: (DD/MM/YYYY)
		Age: _____ Years/Months/Weeks/Days:
Name of Facility Where Infant Seen:		Medical Record Number:
Country of Residence:		Parish (Jamaica):
House Number, Street Name:		
Landmark or directions to address:		Community:
MOTHER'S INFORMATION (no need to repeat if already in demographic data above)		
ID Type:		Mother's ID Number:
☐ Driver's License ☐ TRN ☐ Passport ☐ NIN		
First Name:		Middle Name(s):
Last Name:		Pet Name(s)
Mother's Maiden Name (If Applicable):		
Facility where mother seen:		Medical Record Number:
House Number, Street Name:		Parish (Jamaica):
Phone Number:		Email Address:
Occupation:		
Phone Number:		
Name of School /Workplace:		Email Address:
Address of School/Workplace:		
Marital Status:		
☐ Married	☐ Visiting Union	☐ Widowed
☐ Common Law	☐ Single	☐ Divorced
☐ Separated		☐ Other (specify): _____
NEXT OF KIN OF MOTHER		
NOK's First Name:		NOK's Last Name:
NOK's Address: Lot/ Street/Community/Parish:		
NOK's Phone number:		NOK's Email address:
CLINICAL PROFILE		
Date Patient Seen (DD/MM/YYYY):		
Gestational age at birth:		☐ Birth Outcome: ☐ Live Birth ☐ Stillbirth
Birth category by gestation: ☐ Pre-Term ☐ Term		☐ Post-Term

OPHTHALMIA NEONATORUM INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

SIGNS OR SYMPTOMS	PRESENCE	DATE OF ONSET/DIAGNOSIS (DD/MM/YYYY)
Muco-purulent or purulent conjunctivitis	☐ Yes ☐ No ☐ Unknown	
Redness of conjunctivae and palpebrae	☐ Yes ☐ No ☐ Unknown	
Eyelids sticking together	☐ Yes ☐ No ☐ Unknown	
Oedema and swelling of eyelids	☐ Yes ☐ No ☐ Unknown	
Chemosis of conjunctivae	☐ Yes ☐ No ☐ Unknown	
Other (Specify):	☐ Yes ☐ No ☐ Unknown	
Other (Specify):	☐ Yes ☐ No ☐ Unknown	
Other (Specify):	☐ Yes ☐ No ☐ Unknown	

HOSPITAL SUMMARY

Admitted to Hospital? ☐ Yes ☐ No ☐ Unknown	Date of Admission (DD/MM/YYYY):	Date of Discharge (DD/MM/YYYY):
Hospital:	Medical Record Number:	Ward:
Admitting Diagnosis:	Date of Diagnosis (DD/MM/YYYY):	
Admitted to Special Care Nursery (SCN) or NICU: ☐ Yes ☐ No ☐ Unknown		
Admitted to NICU: ☐ Yes ☐ No ☐ Unknown		
Date of Admission to SCN/NICU (DD/MM/YYYY):		
Date of Discharge from SCN/NICU (DD/MM/YYYY):		
Vital Status (Outcome): ☐ Alive ☐ Dead	Date of Death (DD/MM/YYYY):	Time of Death: ____ AM/PM
Place of Death: ☐ Home ☐ DOA ☐ Ward ☐ ICU ☐ Other (Specify): _____		

MEDICATION HISTORY

Was the patient given any prophylaxis? Yes No Unknown

PAST OR CURRENT MEDICATION	DOSAGE	FREQUENCY	ROUTE OF ADMINISTRATION	DATE OF FIRST ADMINISTRATION (DD/MM/YYYY)	NUMBER OF DOSES
☐ Silver Nitrate					
☐ Tetracycline					
☐ Ceftriaxone					
☐ Other (specify):					
☐ Other (specify):					

DIAGNOSTIC INVESTIGATION SCREENING

Was a sample taken? ☐ Yes ☐ No ☐ Unsure

If yes, what specimen was taken?

☐ Blood Sample	☐ Eye Swab
☐ Other (Specify): _____	

What date was the specimen taken? (DD/MM/YYYY):

Which lab is the sample being sent to?

☐ National Public Health Laboratory ☐ University of the West Indies ☐ Other

Specify Other Laboratory: _____

OPHTHALMIA NEONATORUM INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

RESULTS OF LABORATORY INVESTIGATION			
TEST TYPE	SPECIMEN TYPE	DATE SPECIMEN COLLECTED (DD/MM/YYYY)	TEST RESULT
GRAM STAIN			
CULTURE			
RT-PCR			
OTHER (SPECIFY):			

MOTHER'S ANTENATAL HISTORY

Gravida:	Parity:
Stillbirths:	Pre-Term births:
Livebirths:	Number of living children:

LMP (DD/MM/YYYY):	EDD (DD/MM/YYYY):
Multiple pregnancy: ☐ Yes ☐ No ☐ Unknown	Number of foetuses:
Place of delivery:	
Did the mother have a history of vaginal discharge in mother? ☐ Yes ☐ No ☐ Unknown	
Number of lifetime sex partners:	
Did the mother receive antenatal care? ☐ Yes ☐ No ☐ Unknown	
Where did mother receive antenatal care? ☐ Public ☐ Private ☐ Both	
What is the source of primary antenatal care?	

SITE	NUMBER OF VISITS
☐ Health Centre	
☐ High Risk Clinic	
☐ General Practitioner	
☐ Obstetrician/ Gynaecologist	
☐ Other:	
TOTAL VISITS	

OPHTHALMIA NEONATORUM INVESTIGATION

MEDICAL RECORD NUMBER:

PATIENTS' NAME:

CLASSIFICATION**INITIAL DIAGNOSIS: OPHTHALMIA NEONATORUM****FINAL DIAGNOSIS:**

Case classification:

☐ Laboratory Confirmed☐ Probable/ Clinically diagnose☐ Discarded☐ Pending☐ Inconclusive

Reason for classification:

DEATH CLASSIFICATIONOphthalmia Neonatorum related death? ☐ Yes ☐ No ☐ Unknown

Immediate Cause:

Intermediate Cause:

Underlying Cause:

Date Investigation Completed (DD/MM/YYYY):

Comments:

NOTIFIER'S/ INVESTIGATOR'S DETAILS

First Name:

Last Name:

Professional Group:

Phone number:

Email Address:

Name of Institution:

Parish:

Community:

Investigator's Office Number/Street Name:

Health Region: ☐ SERHA ☐ SRHA ☐ WRHA ☐ NERHA

Investigator's Signature:

Date signed by Investigator (DD/MM/YYYY):

Name of Parish MO(H):

Signature of MO(H):

Date signed by MO(H) (DD/MM/YYYY):

Appendix 59: Pharmwatch Drug Monitoring Form

"PHARMWATCH" DRUG MONITORING FORM

A. PATIENT DETAILS					
Patient Initials: (First, Last)	Gender: M F	Date of Birth:	Ethnicity	Weight: (kg)	Height: (cm)
B. SUSPECTED DRUG EVENT					
Outcomes attributed to use of drug (check all that apply): ☐ Failure of therapy ☐ Allergy ☐ Disability ☐ Life threatening ☐ Hospitalization ☐ Death _____ (mm/dd/yyyy) ☐ Other (describe) _____		Describe event or problem		Date event started (mm/dd/yyyy)	
				Date event ended (mm/dd/yyyy)	
Describe action taken in response (e.g. drug changed, prolonged therapy, increased dose)		Describe other relevant history including abnormal laboratory test results, days of hospitalization.			
C. DRUG INFORMATION					
Name of suspected drug (give specific name on package)	Dose & Route	Indication		Batch number if known	
Name of other drugs taken (give specific name on package)	Dose & Route	Indication		Batch number if known	
D. REPORTING HEALTH PROFESSIONAL INFORMATION					
Profession: _____ Name: _____ Address: _____			Telephone: _____ Fax: _____ Email: _____		
Also reported to:					
Signature			Date (mm/dd/yyyy)		
FOR OFFICIAL USE ONLY					
Received by:			Action taken:		
Date Received:					

Appendix 60: Rheumatic Fever/ Rheumatic Heart Disease Investigation Form

Notes 1. Date example: 25-JUNE-01, 2. Please write in Block, 3. Revised: September 18, 2001 L:/LINK/COMMUNICATION/WARD_DOC/SCHEMA/RFEVER.PPT
PLEASE FAX INVESTIGATION FORMS TO EPIDEMIOLOGY UNIT AT 967-1280

RHEUMATIC FEVER/RHEUMATIC HEART DISEASE INVESTIGATION FORM

Case identification Notification Site: _____ Parish: _____

Patient Name _____ Pet Name _____ Sex _____ Birth Date _____ Age _____ Address _____ Community _____ Telephone _____		Next of Kin / Guardian/ Contact Name _____ Address _____ Relationship _____ Telephone _____		School or Work Information Name _____ Address _____ Telephone _____	
Hospital _____ Date Admitted _____ Date Discharged _____ Docket# _____ Hospital _____ Date Admitted _____ Date Discharged _____ Docket# _____					
Clinical Information (Circle)					
Major Manifestation		Date		Minor Manifestation	
Carditis: Yes No Unk _____ Polyarthritis: Yes No Unk _____ Chorea: Yes No Unk _____ Erythema marginatum: Yes No Unk _____ Subcutaneous Nodules: Yes No Unk _____		_____ _____ _____ _____ _____		Joint Pains: Yes No Unk _____ High Temperature (Fever): Yes No Unk _____ History of Rheumatic Fever: Yes No Unk _____ Previous Rheumatic Heart Disease: Yes No Unk _____ Positive C-reactive protein test: Yes No Unk _____ Increase erythrocyte sedimentation rate: Yes No Unk _____ Increase in the PR interval (ECG): Yes No Unk _____ PR Measurement _____ (mm)	
Laboratory evidence of recent streptococcal infection (Circle) Date _____ Elevated ASTO: Yes No Unk _____ Throat swab pos for group A. Beta haemolytic strep: Yes No Unk _____					
Cardiac Involvement (Circle)					
Mitral Stenosis (MS): Yes No Unk _____ Mitral Regurgitation (MR): Yes No Unk _____ Aortic Stenosis (AS): Yes No Unk _____ Aortic regurgitation (AR): Yes No Unk _____ Tricuspid Stenosis (TS): Yes No Unk _____ Tricuspid Regurgitation (TR): Yes No Unk _____ Heart Failure: Yes No Unk _____ Bacterial Endocarditis: Yes No Unk _____		Severity of Heart Dam None _____ Very Little _____ Moderate _____ Serious _____ Not Determined _____			
Echocardiogram Findings Circle VALVULAR LESION MS MR AR AS TR MS+MR MR+AR MS+TR MS+AS MS+TS MS+MR+TR MR+TR+AR MR+TR+AS OTHER NONE comments _____					
History and Prophylaxis Circle Initial attack: Yes No Unk _____ If No, year of initial attack: _____ # of recurrences since initial attack _____		Prophylaxis Received Yes No _____ Prophylaxis Type Injection Oral _____ Frequency Regular Irregular _____ No. of Benzathine penicillin injection received in the last 12 months _____ Drug Reaction? Yes No _____ Specify Reaction: _____ Prophylaxis Site: _____			
Status (circle) Suspected Confirmed Discarded Dead Date of change in Status _____ Local for medical exam _____ Next Medical Date _____					

Investigators Comments: _____

Signature _____ Date _____

MOH Comments: _____

Signature _____ Date _____

Appendix 61: Typhoid Investigation Form

Reporting Centre: _____				Date of Report: YYYY/MM/DD			
I. Patient information							
Name:				Age (yrs)	Sex M F		
Home address:				Phone	Occupation		
Work address:							
Nature of duties:							
2. Clinical data							
Date of onset:						Immunization history	
Symptoms	Y	N	Symptoms	Y	N	Number of doses	
Fever			Cough				
Malaise						Date of last dose	
Anorexia							
Constipation							
Diarrhoea							
Is/was this patient hospitalized?			Y	N	Date(s)	Outcome of illness	
						Survived	
						Died	Date:
3. Exposure history							
During the 3 weeks prior to onset:			Y	N	Date:	Details	
Known case							
Food							
Water							
Carrier							
In 4 weeks prior to onset:							
Travelling/visiting if considered risk							
Change of work environment							
Bathing in rivers/pools							
4. Laboratory data							
Specimen	Date collected	Date received	Condition	Test	Result	Date sent	Comment
Blood			Culture				
Stool			Culture				
Stool			Culture				
Urine			Culture				
5. Final case classification				Date reported: To Whom: Route: Signature:			
Laboratory confirmed							
Discarded							
Phage type:							